Gynecology Clinical Textbook

Bonus Question Bank
ABOG-Style Board Review Questions with Full Explanations

550

single-best-answer questions, five options each

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How to Use This Question Bank

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This bank holds 550 additional single-best-answer questions, each with five options and a full explanation. The questions follow the content areas of the American Board of Obstetrics and Gynecology gynecology and office-practice outline, and they cover material beyond the chapter review questions in the book.

To study a topic, choose a subject in the contents and open its questions. Commit to one answer, then press Answer under the question. The explanation opens with the answer letter, the reasoning, and a one-line key principle. Press Back to question to return, or Contents to choose another subject.

For a mixed set, work the questions in number order, or pick numbers at random, and check each answer as you go.

Footer links. Every page carries links to Contents, Questions, Answers and How to Use.

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Questions

Part 1. Gynecology

G1. Preoperative Care

1. A 61-year-old woman with a body mass index of 33 and treated hypertension is scheduled for abdominal hysterectomy for a large fibroid uterus. She has no chest pain, dyspnea or syncope. At the preoperative visit she completes the Duke Activity Status Index and scores 29. Her surgeon asks how the 2024 AHA/ACC perioperative guideline treats scores of this kind. Which of the following statements is most accurate?

(A) Scores of 34 or lower were associated with increased odds of death or myocardial infarction at 30 days
(B) Scores of 34 or higher were associated with increased odds of death or myocardial infarction at 30 days
(C) The score has not been linked to outcomes in noncardiac surgery, so it carries no weight
(D) A low score is an indication for routine stress testing before the date is confirmed
(E) The score is informative in known coronary disease but not in patients with hypertension

Answer Contents

2. A 58-year-old woman is scheduled for staging surgery for grade 1 endometrial cancer. She has hypertension controlled with one agent and no cardiac symptoms. She walks briskly for 40 minutes most days and climbs two flights of stairs without stopping. The anesthesia clinic asks whether a pharmacologic stress test should be done before surgery. Which of the following is the best next step under the 2024 AHA/ACC perioperative guideline?

(A) Proceed to surgery without routine preoperative stress testing
(B) Obtain pharmacologic stress myocardial perfusion imaging before scheduling
(C) Obtain a treadmill exercise stress test before scheduling
(D) Obtain coronary CT angiography to exclude occult disease
(E) Obtain a resting transthoracic echocardiogram to grade ventricular function first

Answer Contents

3. A gynecologic surgery committee is updating its preoperative template to match the 2024 AHA/ACC perioperative guideline. The template asks the surgeon to document whether each patient is at elevated risk of major adverse cardiovascular events (MACE) before operations such as radical hysterectomy, using a validated risk tool. The committee needs the numeric threshold the guideline generally uses. Which of the following estimated MACE risks defines elevated risk?

(A) A risk of 0.1% or higher
(B) A risk of 0.5% or higher
(C) A risk of 5% or higher
(D) A risk of 1% or higher
(E) A risk of 10% or higher

Answer Contents

4. A 56-year-old woman with long-standing hypertension is scheduled for laparoscopic hysterectomy next week for symptomatic fibroids. She takes one antihypertensive daily and has no angina, heart failure, kidney disease or diabetes. Blood pressure in the preoperative clinic is 168/98 mm Hg on two readings, and she feels well. The anesthesiologist asks whether the date should change. Which of the following is the best management according to the 2024 AHA/ACC perioperative guideline?

(A) Postpone surgery until pressure is under 140/90 mm Hg on two visits
(B) Add a second agent and delay the operation by two weeks
(C) Admit her for intravenous antihypertensive therapy before the date
(D) Cancel the operation and refer her for cardiac stress testing
(E) Proceed on the date and continue her antihypertensive

Answer Contents

5. A 36-year-old woman with no medical history and a normal examination is scheduled for operative hysteroscopy to resect a submucous myoma. She has no palpitations, syncope, chest pain or family history of sudden death, and her exercise tolerance is excellent. A trainee proposes a routine preoperative 12-lead electrocardiogram for completeness. Which of the following is the best response under the 2024 AHA/ACC perioperative guideline?

(A) Obtain a resting electrocardiogram to serve as a baseline
(B) Obtain an electrocardiogram and a high-sensitivity troponin level
(C) Obtain an electrocardiogram and a transthoracic echocardiogram
(D) Obtain a natriuretic peptide level in place of the electrocardiogram
(E) Omit the electrocardiogram from her preoperative evaluation

Answer Contents

6. A 54-year-old woman with type 2 diabetes takes an SGLT2 inhibitor, metformin and basal insulin. She is scheduled for elective abdominal hysterectomy for adenomyosis in 8 days. Her last hemoglobin A1c was 7.2% and she has no history of ketoacidosis. The pharmacist asks when the SGLT2 inhibitor should be stopped. Which of the following is the best answer under the 2024 AHA/ACC perioperative guideline?

(A) Stop it 3 to 4 days before the operation
(B) Stop it on the morning of the operation
(C) Stop it 24 hours before the operation
(D) Stop it 2 weeks before the operation
(E) Continue it through the operation and restart with the first meal

Answer Contents

7. A 44-year-old woman has taken a weekly GLP-1 receptor agonist for weight loss at a stable dose for 9 months. She is scheduled for elective laparoscopic hysterectomy next week. She has no nausea, vomiting, abdominal pain or bloating, and no condition that slows gastric emptying. Her surgeon asks what the October 2024 multisociety guidance recommends about the drug. Which of the following is the best answer?

(A) Hold the weekly dose for one week before surgery in every patient
(B) Stop it four weeks before surgery to restore normal gastric emptying
(C) Substitute a different weight-loss drug for the two weeks before surgery
(D) Continue it in many cases and adjust the anesthesia plan
(E) Continue it and shorten the usual fasting period to offset the effect

Answer Contents

8. A 39-year-old woman has taken a weekly GLP-1 receptor agonist for 10 months and has been on the highest dose for the past 3 months. She has no nausea, vomiting, abdominal pain or constipation. She is scheduled for elective hysterectomy in 6 days. The team applies the October 2024 multisociety guidance, which notes that patients on higher doses typically have more gastrointestinal side effects. Which of the following is the best way to reduce her aspiration risk?

(A) A 24-hour liquid diet before surgery
(B) Hold the drug for four weeks before surgery
(C) Nasogastric tube decompression on arrival to the operating room
(D) A prokinetic agent taken the night before surgery
(E) Fasting from solid food for 48 hours before surgery

Answer Contents

9. A 47-year-old woman on a weekly GLP-1 receptor agonist arrives on the morning of elective hysterectomy. She followed the fasting instructions. She reports nausea with two episodes of vomiting yesterday and abdominal pain today. Vital signs are normal and the abdomen is soft. The anesthesiologist applies the October 2024 multisociety guidance. Which of the following is the best next step?

(A) Proceed with rapid sequence induction and a cuffed tube
(B) Postpone the elective case until symptoms resolve
(C) Give an antiemetic and proceed with the usual induction
(D) Proceed under neuraxial anesthesia alone
(E) Proceed after two additional hours of fasting

Answer Contents

10. A 50-year-old woman without diabetes or reflux is scheduled for an 8 am elective hysterectomy under general anesthesia. She asks whether she may drink a carbohydrate-containing sports drink in the morning to ease hunger and thirst. According to the 2023 American Society of Anesthesiologists fasting guideline for healthy adults, which of the following is the latest time she may drink it?

(A) Until 4 hours before the procedure
(B) Until 6 hours before the procedure
(C) Until 8 hours before the procedure
(D) Until 2 hours before the procedure
(E) Until midnight before the procedure

Answer Contents

11. A 33-year-old woman waits in the holding area for diagnostic laparoscopy for infertility under general anesthesia. She has followed the fasting rules but has been chewing gum for the past hour. She has no reflux, nausea or gastroparesis. The anesthesiologist reviews the 2023 American Society of Anesthesiologists fasting guideline on chewing gum. Which of the following is the best management?

(A) Postpone the case for six hours
(B) Cancel the case and reschedule it for the next day
(C) Give a proton pump inhibitor and delay the case by two hours
(D) Proceed with rapid sequence induction and leave the gum in place
(E) Have her remove the gum and proceed without delay

Answer Contents

12. A 45-year-old woman is in the preoperative area before abdominal hysterectomy. The resident notes pubic and lower abdominal hair near the planned incision and offers to remove it with a razor to improve exposure. The attending asks what the WHO surgical site infection guideline recommends for hair at the operative site. Which of the following is the best approach?

(A) Shave it with a razor in the preoperative area
(B) Shave it in the operating room just before incision
(C) Shave it after skin antisepsis is complete
(D) Remove it with a razor the evening before surgery
(E) Leave the hair, or use a clipper if it must go

Answer Contents

13. A 48-year-old woman with a body mass index of 29 is scheduled for abdominal hysterectomy for symptomatic fibroids. She has no drug allergies and has not taken antibiotics recently. The anesthesiologist plans cefazolin prophylaxis and asks when the first dose should be started. Which of the following is consistent with the ASHP, IDSA, SIS and SHEA guideline on antimicrobial prophylaxis in surgery?

(A) The evening before the day of surgery
(B) Within 60 minutes before surgical incision
(C) Three hours before the surgical incision
(D) At the time of skin closure after the case
(E) In the recovery room after the operation

Answer Contents

14. A 38-year-old woman with Crohn disease in remission takes azathioprine and is scheduled for elective hysterectomy for leiomyomas. Her gastroenterologist asks the surgical team whether the immunosuppressant should be stopped before surgery to lower infection risk. The team consults the WHO surgical site infection guideline, which graded its position conditional with very low quality evidence. Which of the following does the panel suggest?

(A) Stop it two weeks before surgery and restart after the wound heals
(B) Stop it one week before surgery and restart on the third postoperative day
(C) Replace it with a systemic corticosteroid for the month before surgery
(D) Do not discontinue it before surgery to prevent infection
(E) Hold the dose on the morning of surgery and restart with the first meal

Answer Contents

15. A 52-year-old woman will have laparoscopic hysterectomy in 2 days. At the preoperative visit she asks whether she must buy a special antiseptic soap for her shower the night before, because a neighbor was told it was required. The surgeon reviews the WHO guideline on preoperative measures to prevent surgical site infection. Which of the following is the best answer?

(A) Shower or bathe before surgery, using either plain or antimicrobial soap
(B) Shower or bathe before surgery, using antimicrobial soap, since plain soap does not count
(C) Skip bathing at home, because skin antisepsis in the operating room makes it unnecessary
(D) Use a povidone-iodine scrub at home for ten minutes before surgery
(E) Bathe in plain water, since soaps raise skin colonization

Answer Contents

16. A 41-year-old woman who weighs 128 kg is scheduled for abdominal hysterectomy for a large fibroid uterus. She has no allergies and normal kidney function. The pharmacy asks for the cefazolin prophylaxis dose under the ASHP, IDSA, SIS and SHEA guideline on antimicrobial prophylaxis in surgery. Which of the following is the best choice?

(A) Cefazolin 3 g IV once, started within 60 minutes before incision
(B) Cefazolin 2 g IV once, started within 60 minutes before incision
(C) Cefazolin 1 g IV once, started within 60 minutes before incision
(D) Cefazolin 3 g IV once, then every 8 hours for 72 hours after surgery
(E) Cefazolin 3 g IV once, started 3 hours before incision

Answer Contents

G2. Perioperative Care

17. A 44-year-old woman with a body mass index of 31 will have an open hysterectomy under general anesthesia with endotracheal intubation. The surgical team is building a bundle to reduce surgical site infection and asks the anesthesiologist about the inspired oxygen fraction. Which of the following reflects the WHO guideline recommendation, which its 2018 second edition grades as conditional?

(A) 30% inspired oxygen throughout surgery to avoid oxygen toxicity
(B) 50% inspired oxygen at induction, then room air for the rest of surgery
(C) 100% inspired oxygen continued for 24 hours after surgery
(D) Room air during maintenance, titrated to a saturation of 90%
(E) 80% oxygen during surgery and, if feasible, for 2 to 6 hours after

Answer Contents

18. A 71-year-old woman is undergoing a 4-hour laparotomy for ovarian cancer debulking. The operating room is cool, her abdomen is open, and her core temperature has fallen from 36.4 to 35.1 degrees C. The anesthesiologist asks which infection prevention step in the 2017 CDC surgical site infection guideline applies, a recommendation graded Category IA. Which of the following is the best action?

(A) Accept the temperature, since mild hypothermia protects tissue
(B) Lower the room temperature further to slow bacterial growth
(C) Give a second antibiotic dose now to offset the heat loss
(D) Start active warming to maintain perioperative normothermia
(E) Switch to cold irrigation fluid to limit heat production

Answer Contents

19. A 57-year-old woman with type 2 diabetes is having an abdominal hysterectomy. Her intraoperative blood glucose is 246 mg/dL on two checks. The anesthesiologist wants to follow the 2017 CDC surgical site infection guideline, which gives a Category IA recommendation on perioperative glycemic control. Which of the following is the blood glucose target the guideline names?

(A) 80 to 110 mg/dL
(B) Less than 200 mg/dL
(C) Less than 110 mg/dL
(D) Less than 250 mg/dL
(E) Less than 300 mg/dL

Answer Contents

20. The anesthesia department at a hospital that performs many abdominal hysterectomies is revising its intraoperative fluid protocol to match WHO guidance on preventing surgical site infection. A 60-year-old woman with ASA class II status is scheduled for a 4-hour laparotomy. The department head asks which fluid strategy the WHO panel suggested for lowering infection risk, a conditional recommendation with low quality evidence. Which of the following is the best answer?

(A) A fixed liberal crystalloid rate set before incision
(B) A restrictive crystalloid regimen aiming at zero net balance
(C) A hydroxyethyl starch colloid in place of crystalloid
(D) Goal-directed fluid therapy guided by monitored measures
(E) Fluid boluses triggered by urine output alone

Answer Contents

21. A 49-year-old woman will have an open abdominal hysterectomy for a large fibroid uterus. The vaginal cuff will be opened, so the case is classed as clean-contaminated. The surgeon wants to lower the risk of wound infection at the incision. Which of the following measures is supported by the WHO guideline on intraoperative measures to prevent surgical site infection?

(A) An advanced silver-impregnated dressing in place of a standard dressing
(B) A wound-protector device
(C) Laminar airflow ventilation in the operating room
(D) A plastic adhesive incise drape with antimicrobial properties
(E) Continued antibiotic prophylaxis for as long as a wound drain remains

Answer Contents

22. A 53-year-old woman is having an open abdominal hysterectomy. The surgeon is closing the fascia and skin and asks the scrub nurse about suture selection. The hospital infection committee has asked all surgeons to follow the WHO guideline on intraoperative measures to prevent surgical site infection, which includes a conditional recommendation with moderate quality evidence about suture type. Which of the following does the panel suggest?

(A) Uncoated monofilament sutures, since coatings do not change infection risk
(B) Silver-coated sutures
(C) Triclosan-coated sutures
(D) Sutures soaked in antibiotic solution in the operating room
(E) Skin staples in place of sutures

Answer Contents

23. A 43-year-old woman is draped for laparoscopic hysterectomy. She has no skin allergies and no open wounds. The circulating nurse asks which abdominal skin preparation is consistent with the 2017 CDC surgical site infection guideline, which gives a Category IA recommendation on intraoperative skin preparation. Which of the following is the best choice?

(A) Aqueous povidone-iodine solution
(B) Plain soap and water scrub
(C) Aqueous chlorhexidine solution
(D) Saline irrigation of the abdominal wall
(E) An alcohol-based antiseptic agent

Answer Contents

24. A 52-year-old woman who weighs 80 kg receives cefazolin before abdominal hysterectomy. Dense adhesions prolong the operation, and blood loss so far is 400 mL. The first cefazolin dose was started 4 hours and 10 minutes ago. The circulating nurse asks whether another dose is needed under the ASHP, IDSA, SIS and SHEA guideline. Which of the following is the best response?

(A) Give a second cefazolin dose now, as 4 hours have passed

(B) Give no further dose, since one dose covers a long operation

(C) Wait until blood loss exceeds 1500 mL before giving another dose

(D) Switch to clindamycin for the rest of the case

(E) Give the second dose at the 8-hour mark

Answer Contents

25. A 38-year-old woman is undergoing abdominal myomectomy for multiple large fibroids. The case is going well on time, and the cefazolin was started 2 hours ago. The uterus bleeds briskly, and estimated blood loss reaches 1800 mL. The anesthesiologist asks whether the antibiotic should be redosed before the 4-hour interval. Under the ASHP, IDSA, SIS and SHEA guideline, which of the following is the best response?

(A) Wait until 4 hours have passed from the first dose

(B) Give another dose now because blood loss is excessive

(C) Give no further dose, since the first dose covers the whole case

(D) Give a dose of a different antibiotic class instead

(E) Give another dose once the transfusion is complete

Answer Contents

26. A 50-year-old woman had an uncomplicated abdominal hysterectomy for fibroids. She received a single dose of cefazolin before incision and the case was classed as clean-contaminated. In the recovery area the nurse asks whether to order additional antibiotic doses for the next 24 hours, as the former surgeon did. Which of the following is the position of the 2017 CDC surgical site infection guideline, graded Category IA?

(A) Give a second dose every 8 hours for 24 hours

(B) Give doses until the urinary catheter is removed

(C) Give doses until the first bowel movement

(D) Give a 7-day oral course after discharge

(E) Give no extra doses after incision closure

Answer Contents

27. At the end of an abdominal hysterectomy, the surgeon asks the scrub nurse for a bowl of gentamicin solution to irrigate the subcutaneous tissue before skin closure, to be safe. The patient is a 46-year-old woman with no wound complications so far. The nurse recalls that the hospital follows the WHO guideline on intraoperative measures to prevent surgical site infection. Which of the following is the best response?

(A) Irrigate with the antibiotic solution to lower infection risk
(B) Irrigate with the antibiotic solution if the patient has diabetes
(C) Omit antibiotic irrigation of the incision before closure
(D) Instill antibiotic powder into the subcutaneous space instead
(E) Irrigate with the antibiotic solution and extend systemic prophylaxis for 24 hours

Answer Contents

28. A 64-year-old woman with nonvalvular atrial fibrillation takes warfarin and is scheduled for elective hysterectomy for atypical endometrial hyperplasia. She has no mechanical valve or prior stroke, and the team judges her thrombotic risk as ordinary. The plan is to stop warfarin before surgery, and the resident asks whether to bridge with heparin. Which of the following is the best answer under the 2024 AHA/ACC perioperative guideline?

(A) Bridge with therapeutic low-molecular-weight heparin from the day warfarin is stopped
(B) Continue warfarin at the usual dose through the operation
(C) Bridge with an intravenous unfractionated heparin infusion in hospital
(D) Interrupt warfarin and do not bridge with heparin
(E) Place a retrievable inferior vena cava filter instead of stopping warfarin

Answer Contents

29. A 47-year-old woman with a mechanical mitral valve prosthesis takes warfarin and needs hysterectomy for symptomatic fibroids. The cardiology team classifies her thrombotic risk as high. The surgeon plans to stop warfarin before the operation and asks about bridging. Which of the following approaches fits the 2024 AHA/ACC perioperative guideline for patients at high thrombotic risk?

(A) Interrupt warfarin and bridge with parenteral heparin
(B) Interrupt warfarin and give no bridging
(C) Replace warfarin with a direct oral anticoagulant before the operation
(D) Continue warfarin at the usual dose through the operation
(E) Place an inferior vena cava filter before stopping warfarin

Answer Contents

30. A 70-year-old woman takes apixaban for atrial fibrillation and is scheduled for laparoscopic staging of endometrial cancer. She is anxious about stroke and asks to keep taking the drug through the operation. Her kidney function is normal and she has no history of stroke. The surgeon consults the 2024 AHA/ACC perioperative guideline. Which of the following is the best plan?

(A) Continue the usual dose through the morning of surgery
(B) Stop it and start enoxaparin as a bridge until the day of surgery
(C) Withhold it before surgery on a written schedule
(D) Halve the dose for the week before surgery
(E) Switch to aspirin until the operation

Answer Contents

31. A 66-year-old woman is on postoperative day 2 after radical hysterectomy with lymphadenectomy. She has new atrial fibrillation with a ventricular rate of 138 per minute, blood pressure 108/68 mm Hg and mild dyspnea. Her hemoglobin has fallen to 7.2 g/dL from 11.0 g/dL, her temperature is 38.6 degrees C and her white blood cell count is 16,000 per microliter. Which of the following is the best initial step under the 2024 AHA/ACC perioperative guideline?

(A) Proceed to immediate electrical cardioversion regardless of the cause
(B) Start an amiodarone infusion and defer further evaluation
(C) Begin therapeutic anticoagulation and observe the rhythm
(D) Treat triggers such as anemia and sepsis along with rate control
(E) Stop rate-controlling drugs until the rhythm converts

Answer Contents

32. A 66-year-old woman has a routine troponin drawn on postoperative day 1 after radical hysterectomy because of her cardiac risk factors. The level exceeds the reference range. She has no chest pain or dyspnea, her electrocardiogram shows no ischemic change and her vital signs are normal. The team diagnoses myocardial injury after noncardiac surgery. Which of the following is supported by the 2024 AHA/ACC perioperative guideline as reasonable for this patient?

(A) Reassurance and no follow-up because she has no symptoms
(B) Outpatient cardiovascular follow-up after discharge
(C) Coronary angiography before discharge in every such patient
(D) Long-term therapeutic anticoagulation with warfarin
(E) Repeat troponin testing at each gynecologic follow-up visit

Answer Contents

G3. Surgical Complications

33. A 47-year-old woman is evaluated on postoperative day 7 after an uncomplicated total laparoscopic hysterectomy for fibroids. She has left flank discomfort and a temperature of 37.8 C. Creatinine is unchanged from baseline. CT urogram with delayed images shows mild left hydronephrosis and extravasation near the distal ureter, with some contrast also reaching the bladder, consistent with an incomplete ureteral injury. Which of the following is the best next step?

(A) Cystoscopic retrograde ureteral stent attempt
(B) Immediate laparotomy with ureteroneocystostomy
(C) Percutaneous nephrostomy as the first intervention
(D) Observation with a repeat CT in six weeks
(E) Indwelling urethral catheter drainage alone

Answer Contents

34. A 53-year-old woman has left flank pain and mild hydronephrosis nine days after a radical hysterectomy. CT urogram confirms a left ureteral injury with extravasation. At cystoscopy, the urologist cannot pass a guidewire beyond the injury despite repeated attempts. She is afebrile, hemodynamically normal, and her creatinine is 1.0 mg/dL. Which of the following is the best next step in management?

(A) Repeated retrograde attempts on following days
(B) Emergent nephrectomy of the affected kidney
(C) Oral antibiotics with observation of the leak
(D) Immediate ileal ureter replacement with bowel
(E) Percutaneous nephrostomy with delayed repair

Answer Contents

35. A 45-year-old woman reports constant watery vaginal leakage that began 18 days after an abdominal hysterectomy for adenomyosis. She voids normally. A tampon placed after instilling dye into the bladder stays unstained. CT urogram with delayed images shows contrast entering the vagina from the left distal ureter, with mild hydronephrosis. Which of the following is the most appropriate initial management?

(A) Ureteral stent placement when feasible
(B) Transvaginal repair with a Martius flap
(C) Prolonged urethral catheter drainage alone
(D) Vaginal packing with topical estrogen
(E) Left nephrectomy

Answer Contents

36. During an open hysterectomy for a large fibroid uterus, a 50-year-old woman has clear fluid welling in the pelvis. The surgeon finds a clean transection of the left ureter just proximal to the bladder, distal to the iliac vessels. Blood pressure is stable, temperature is normal, and the cut ends are well vascularized. Which of the following is the best management at this time?

(A) Ligation of the ureter with delayed nephrectomy
(B) Ileal ureter interposition
(C) Ureteral reimplantation or primary repair over a stent
(D) Nephrostomy alone with repair deferred for three months
(E) Renal autotransplantation

Answer Contents

37. A 60-year-old woman undergoing debulking for ovarian cancer develops massive pelvic venous bleeding. Systolic pressure is 70 mm Hg despite transfusion, she is hypothermic and coagulopathic, and the team has moved to damage control. A transected right distal ureter is found. Applying the AUA urotrauma guideline, which of the following is the best approach to the ureteral injury?

(A) Immediate ureteroneocystostomy with psoas hitch
(B) Primary spatulated anastomosis over a stent now
(C) Ileal ureter substitution during this operation
(D) Temporary drainage with delayed repair
(E) Right nephrectomy during this operation

Answer Contents

38. A 41-year-old woman is having laparoscopic excision of deep endometriosis. After ureterolysis, a 2 cm segment of the left pelvic ureter is bruised and edematous, but the ureter is in continuity and no urine leaks. She is stable. Which of the following is the best way to manage this ureteral contusion at the time of surgery?

(A) Closure of the abdomen without ureteral intervention
(B) Ligation of the ureter with a proximal nephrostomy
(C) Placement of a pelvic drain and close observation for days
(D) Left nephrectomy with ureteral excision
(E) Stenting or resection with primary repair, per viability

Answer Contents

39. A 38-year-old woman has a 2 cm cystotomy at the bladder dome recognized during abdominal hysterectomy. It is closed in two layers and a retrograde fill test shows no leak. The team asks how to drain the bladder afterward, and the surgeon wants to follow the AUA urotrauma guideline's position on drainage after repair of a bladder injury. Which of the following is the best approach?

(A) Urethral catheter plus suprapubic cystostomy
(B) Urethral catheter drainage without suprapubic cystostomy
(C) Suprapubic catheter alone
(D) Immediate catheter removal and voiding in the recovery room
(E) Bilateral ureteral stents with no bladder catheter

Answer Contents

40. A 42-year-old woman is seen on postoperative day 7 after an abdominal hysterectomy through a midline incision. The lower incision is red and indurated over 3 cm with purulent drainage and a fluctuant area. Temperature is 37.8 C, heart rate 84 beats/min, and white blood cell count 9,800/microL. She is eating and ambulating. Which of the following is the best next step?

(A) Oral antibiotic therapy alone
(B) Intravenous broad-spectrum antibiotics alone
(C) Suture removal with incision and drainage
(D) Needle aspiration through intact skin
(E) Warm compresses with a repeat examination in one week

Answer Contents

41. A 56-year-old woman with obesity had a laparotomy five days ago. The incision was opened and drained for a superficial abscess today. After drainage, erythema extends 2 cm from the wound edge, temperature is 37.4 C, heart rate is 80 beats/min, and white blood cell count is 8,900/microL. The resident asks about adding antibiotics. Which of the following is most consistent with the IDSA guideline?

(A) No routine adjunctive systemic antimicrobial therapy
(B) A prolonged course of intravenous broad-spectrum antibiotics
(C) Oral antibiotics for each surgical site infection
(D) Topical antibiotic ointment to the open wound
(E) Negative pressure therapy with systemic antibiotics

Answer Contents

42. A 34-year-old woman had laparoscopic ovarian cystectomy four days ago. A periumbilical port site is now erythematous for 6 cm with induration and a small fluctuant area. Temperature is 38.9 C and heart rate is 112 beats/min. The site is opened and drained. The operation was clean, performed on the trunk. Which of the following is the best additional treatment?

(A) No antimicrobial therapy
(B) Topical mupirocin alone
(C) Delayed antibiotics if fever persists for one week
(D) A brief course of systemic antimicrobial therapy
(E) Hyperbaric oxygen therapy

Answer Contents

43. A 49-year-old woman, known to be nasally colonized with methicillin-resistant Staphylococcus aureus after a skin infection last year, has a purulent port-site infection five days after a laparoscopic ovarian cystectomy. The site was drained. Temperature is 38.8 C and heart rate is 114 beats/min. The wound is on the trunk and the operation was clean. Which of the following antimicrobial choices is most appropriate?

(A) Cefazolin
(B) Intravenous vancomycin
(C) Nafcillin
(D) Ciprofloxacin with metronidazole
(E) Ceftriaxone alone

Answer Contents

44. A 51-year-old woman has a wound infection of the lower abdominal incision eight days after hysterectomy for fibroids. The wound was opened and drained. Erythema extends 7 cm, temperature is 38.8 C, heart rate is 116 beats/min, and white blood cell count is 15,200/microL. Systemic therapy is planned. Which of the following regimens best fits the IDSA recommendation for infections after operations on the female genital tract?

(A) Vancomycin as a single agent
(B) Dicloxacillin as a single agent
(C) Doxycycline as a single agent
(D) Linezolid as a single agent
(E) Ceftriaxone plus metronidazole

Answer Contents

45. A 48-year-old woman with no drug allergies is scheduled for abdominal hysterectomy for symptomatic fibroids. She has had no recent antibiotics and has no history of methicillin-resistant Staphylococcus aureus infection. The anesthesia team follows the ASHP, IDSA, SIS and SHEA guideline on antimicrobial prophylaxis in surgery and asks which agent the guideline lists for hysterectomy. Which of the following is the recommended agent?

(A) Intravenous gentamicin alone
(B) Intravenous vancomycin alone
(C) Intravenous cefoxitin
(D) Intravenous metronidazole alone
(E) Oral ciprofloxacin alone

Answer Contents

46. A 52-year-old woman receives prophylactic enoxaparin starting the morning after laparotomy for endometrial cancer. On postoperative day 6, her platelet count is 90,000/mm3, down from 280,000/mm3 at baseline. She has no bleeding and no new leg symptoms. Which of the following is the best management according to the enoxaparin labeling?

(A) Continue the same dose and recheck the platelet count in one week
(B) Add aspirin and continue enoxaparin at the same dose for now
(C) Reduce enoxaparin to half the dose and continue with monitoring
(D) Stop enoxaparin and test for heparin-induced thrombocytopenia
(E) Transfuse platelets and continue enoxaparin at the planned dose

Answer Contents

47. A 57-year-old woman with endometrial cancer is scheduled for hysterectomy. Six weeks ago she was hospitalized for pneumonia and received heparin, and she was diagnosed with immune-mediated heparin-induced thrombocytopenia at that time. The team is choosing postoperative venous thromboembolism prophylaxis. Which of the following is contraindicated by the enoxaparin labeling?

(A) Intermittent pneumatic compression
(B) Graduated compression stockings
(C) Early ambulation
(D) Leg elevation with ankle exercises
(E) Prophylactic enoxaparin

Answer Contents

48. A 63-year-old woman with a body mass index of 38, hypertension, and type 2 diabetes is scheduled for abdominal hysterectomy. The anesthesiologist plans an epidural catheter for postoperative analgesia, and the surgeon plans enoxaparin prophylaxis. Which of the following features most increases her risk of an epidural or spinal hematoma according to the enoxaparin labeling?

(A) Indwelling epidural catheter
(B) A body mass index of 38 kg/m2
(C) Treated chronic hypertension
(D) Long-standing type 2 diabetes
(E) Age in the seventh decade

Answer Contents

49. A 70-year-old woman with chronic kidney disease and a creatinine clearance of 24 mL/min is on postoperative day 1 after cytoreductive surgery. Her platelet count is normal and she has no bleeding. The team wants to start enoxaparin for venous thromboembolism prophylaxis. Which of the following is the best approach according to the enoxaparin labeling?

(A) Use the standard dosage without extra monitoring
(B) Adjust the dose and watch closely for bleeding
(C) Add aspirin to the standard dosage
(D) Withhold pharmacologic prophylaxis
(E) Switch to oral warfarin without bridging

Answer Contents

50. A 54-year-old woman is on postoperative day 1 after myomectomy. Her closed-suction drain has returned 300 mL of bloody fluid over the past hour, hemoglobin has fallen from 11.2 to 8.0 g/dL, and her blood pressure is 88/50 mm Hg. A scheduled enoxaparin dose is due. Which of the following is the best next step regarding venous thromboembolism prophylaxis?

(A) Give the scheduled dose and recheck hemoglobin in 6 hours
(B) Give the dose with a platelet transfusion
(C) Increase to a treatment-range regimen
(D) Give half of the scheduled dose
(E) Hold enoxaparin and use mechanical prophylaxis

Answer Contents

51. A 45-year-old woman is on postoperative day 1 after hysterectomy complicated by 900 mL of blood loss that has been controlled. She has no cardiovascular disease, is hemodynamically stable, and feels well. Hemoglobin is 6.7 g/dL. The team follows the 2023 AABB guideline for hospitalized adults. Which of the following is the best management?

(A) Observation until she becomes symptomatic
(B) Oral iron alone
(C) Erythropoietin alone
(D) Red blood cell transfusion
(E) Transfusion once hemoglobin falls under 5 g/dL

Answer Contents

52. A 68-year-old woman with known coronary artery disease is on postoperative day 2 after cytoreductive surgery. She is hemodynamically stable and asymptomatic, with no chest pain. Hemoglobin is 7.6 g/dL. The team wants the AABB guideline's suggested threshold for patients with preexisting cardiovascular disease. Which of the following is the best management?

(A) Observe, because the threshold is 7 g/dL for everyone
(B) Observe until hemoglobin is under 6 g/dL
(C) Transfuse red blood cells
(D) Start oral iron and discharge
(E) Order erythropoietin

Answer Contents

53. A 62-year-old woman with ovarian cancer is hospitalized for a small bowel obstruction that is managed without surgery. She has no cardiovascular disease, no active bleeding, and is hemodynamically stable. Hemoglobin is 7.4 g/dL. The team follows the AABB guideline recommendation for hospitalized adults with hematologic and oncologic disorders. Which of the following is the best management?

(A) Transfuse now to reach a hemoglobin over 10 g/dL
(B) Observe and transfuse if hemoglobin falls under 7 g/dL
(C) Transfuse now because oncology patients need a higher threshold
(D) Start erythropoietin and withhold transfusion regardless of value
(E) Transfuse when symptomatic regardless of hemoglobin

Answer Contents

54. A 56-year-old woman is on postoperative day 1 after radical hysterectomy. She is tachycardic at 118 beats/min, lightheaded on standing, and hemoglobin is 6.4 g/dL after a 1,200 mL blood loss. The surgeon hesitates to transfuse, worried that blood products will increase her risk of wound infection. Which of the following is most consistent with the CDC surgical site infection guideline?

(A) Transfuse as indicated, since withholding gives no protection
(B) Delay transfusion for three days to protect the wound
(C) Replace blood products with crystalloid fluid alone
(D) Give prophylactic antibiotics for the length of the transfusion
(E) Postpone transfusion until the incision has fully healed

Answer Contents

55. A 33-year-old woman is under sevoflurane and succinylcholine for laparoscopic salpingectomy. Twenty minutes in, end-tidal CO2 rises from 38 to 68 mm Hg despite doubled minute ventilation, heart rate is 142 beats/min, her jaw is rigid, and temperature is climbing to 38.9 C. Which of the following is the best immediate step?

(A) Deepen the volatile anesthetic and proceed with the case
(B) Give acetaminophen and apply surface cooling measures
(C) Give oral dantrolene through a gastric tube and wait
(D) Stop triggering agents and give intravenous dantrolene
(E) Treat presumed sepsis with antibiotics and continue

Answer Contents

56. A 33-year-old woman received dantrolene for a malignant hyperthermia crisis during laparoscopic salpingectomy, and her temperature, heart rate, and end-tidal CO2 normalized. Two hours later in the recovery unit, her temperature rises to 38.6 C, her jaw stiffens, and the creatine kinase is climbing. Which of the following is the best next step?

(A) Observe with acetaminophen alone
(B) Resume the volatile anesthetic for sedation
(C) Repeat the dantrolene regimen
(D) Start oral antibiotics
(E) Discharge to the ward with routine checks

Answer Contents

57. A 40-year-old woman is treated with intravenous dantrolene for malignant hyperthermia during hysterectomy. On postoperative day 2 she reports nausea and right upper quadrant discomfort, and laboratory testing shows new elevations of serum transaminases. Her muscle symptoms have resolved. Which of the following complications of the drug best explains these findings?

(A) Nephrotoxicity
(B) Ototoxicity
(C) Pancreatitis
(D) QT prolongation
(E) Hepatotoxicity

Answer Contents

58. During a malignant hyperthermia crisis in a 29-year-old woman undergoing laparoscopic ovarian cystectomy, an initial intravenous push of dantrolene lowers her temperature slightly but end-tidal CO2 and rigidity persist. The anesthesiologist asks about the labeled endpoint for dantrolene dosing. Which of the following is correct?

(A) Continue until symptoms resolve or the maximum dose is reached
(B) Give one bolus and observe for two hours before redosing
(C) Continue for a fixed 24-hour course regardless of response
(D) Stop once the temperature has fallen by 1 degree Celsius from peak
(E) Continue until the creatine kinase level normalizes

Answer Contents

59. A 36-year-old woman undergoing laparoscopic hysterectomy under sevoflurane has end-tidal CO2 of 75 mm Hg, rigid jaw and limbs, temperature of 39.4 C, potassium of 6.3 mEq/L, and a mixed respiratory and metabolic acidosis. Other causes have been excluded. Which of the following agents is specifically indicated for this condition?

(A) Bromocriptine
(B) Dantrolene
(C) Cyproheptadine
(D) Acetaminophen
(E) Naloxone

Answer Contents

G4. Postoperative Care

60. A 61-year-old woman is scheduled for debulking surgery for ovarian cancer that is expected to include a partial small bowel resection with primary anastomosis. She has had no opioid exposure before admission, and the plan includes opioid-based patient-controlled analgesia. The team wants a labeled drug to accelerate recovery of upper and lower gastrointestinal function. Which of the following is the best choice?

(A) Metoclopramide
(B) Alvimopan
(C) Neostigmine
(D) Erythromycin
(E) Bisacodyl

Answer Contents

61. A 66-year-old woman with ovarian cancer has taken morphine daily for cancer pain for three weeks and is scheduled for bowel resection with anastomosis during cytoreduction. The anesthesiologist asks whether to give alvimopan before surgery to speed gastrointestinal recovery. Which of the following is the best answer according to the alvimopan labeling?

(A) Give alvimopan at half of the usual dose before surgery
(B) Delay alvimopan until the first postoperative day
(C) Give alvimopan alongside a low-dose intravenous naloxone infusion
(D) Give alvimopan together with oral metoclopramide
(E) Omit alvimopan, given opioid use for over 7 consecutive days

Answer Contents

62. A 58-year-old woman recovers after bowel resection and hysterectomy for endometrial cancer. She received alvimopan in the hospital. At discharge, she asks to continue the capsules at home for constipation from her opioid pain medicine, saying that they worked well in the hospital. Which of the following is the best response according to the alvimopan labeling?

(A) Continue it at home with monthly liver enzyme monitoring
(B) Continue it at a higher dose to treat the constipation
(C) Continue it at home with daily aspirin for cardiac protection
(D) Stop it at discharge because of myocardial infarction risk
(E) Continue it if she has no family history of heart disease

Answer Contents

63. A 70-year-old woman with recurrent ovarian cancer has a complete bowel obstruction and is going to surgery for diversion. A resident suggests alvimopan to speed bowel recovery afterward. The attending asks what the labeling says about use in patients with complete gastrointestinal obstruction. Which of the following is correct?

(A) It is recommended, with the usual dose halved

(B) It is required before a bowel operation of this kind

(C) It is not recommended, as it is unstudied here

(D) It is recommended when given by intravenous infusion

(E) It is preferred in place of nasogastric decompression

Answer Contents

64. A 52-year-old woman is on opioid patient-controlled analgesia after a hysterectomy with bowel resection and is given a drug that the team says prevents opioid-induced slowing of the gut without reversing her pain relief. The resident asks why analgesia is preserved. Which of the following mechanisms best explains this property of alvimopan?

(A) Peripheral antagonism of opioid effects on gut motility

(B) Central blockade of opioid receptors in the brain and spinal cord

(C) Agonism of serotonin 5-HT4 receptors in the intestinal wall

(D) Agonism of motilin receptors in the stomach and small bowel

(E) Inhibition of acetylcholinesterase at the neuromuscular junction

Answer Contents

65. A 44-year-old woman is in the recovery unit after laparoscopic myomectomy. She has moderate incisional and shoulder-tip pain and asks for a strong opioid. She has no allergies or contraindications to acetaminophen or NSAIDs. Which of the following is the best approach according to the 2022 CDC opioid prescribing guideline?

(A) Start with an opioid first and add other agents if it fails

(B) Maximize nonopioid therapies and reserve opioids for favorable risk-benefit

(C) Use extended-release opioids to cover the first week of pain

(D) Give scheduled opioids around the clock for the first 72 hours after surgery

(E) Delay analgesic treatment until the pain becomes severe

Answer Contents

66. A 49-year-old woman is ready for discharge on postoperative day 1 after laparoscopic hysterectomy. Her pain is controlled with acetaminophen and ibuprofen, and she needs a rescue opioid prescription. The surgeon is considering a default one-month supply. Which of the following is most consistent with the CDC guideline?

(A) Prescribe no greater quantity than needed for the expected duration of severe pain
(B) Prescribe a standard 30-day supply to avoid refills
(C) Prescribe an extended-release formulation for convenience
(D) Allow unlimited automatic refills
(E) Prescribe the amount she used on the day of discharge multiplied by the number of days until follow-up

Answer Contents

67. A 52-year-old woman is prescribed a short course of an immediate-release opioid after hysterectomy. At the discharge visit she mentions that she survived a nonfatal opioid overdose four years ago. She has no current substance use and takes no other sedating drugs. The surgeon wants to follow the CDC guideline on managing opioid risk. Which of the following is the best additional step?

(A) Cancel the opioid prescription and prescribe nothing
(B) Add a benzodiazepine for sleep
(C) Switch to an extended-release opioid
(D) Require a urine drug screen before discharge
(E) Offer naloxone

Answer Contents

68. A 47-year-old woman who takes alprazolam nightly for anxiety is discharged after hysterectomy. Her surgeon plans a brief opioid prescription for breakthrough pain. She asks whether the two medications can be taken together. Which of the following is most consistent with the CDC guideline on this combination?

(A) Prescribe both without counseling because the opioid course is short
(B) Stop the benzodiazepine abruptly on the day of discharge
(C) Use caution and counsel her about sedation and breathing risk
(D) Double the opioid quantity to allow for tolerance
(E) Add a second sedating agent for sleep

Answer Contents

69. A 55-year-old woman is receiving intravenous ketorolac on postoperative day 2 after myomectomy and wants to continue it at home for two weeks because it works better than her other medications. She has normal renal function and no history of ulcers. Which of the following is the best counseling based on the ketorolac labeling?

(A) Continue ketorolac for two weeks if taken with food
(B) Continue ketorolac while renal function stays normal
(C) Continue ketorolac with a proton pump inhibitor added
(D) Limit combined injection and oral ketorolac to 5 days
(E) Continue ketorolac at the lowest dose as chronic therapy

Answer Contents

70. A 50-year-old woman is waiting in the preoperative holding area for an open myomectomy. A resident proposes giving intravenous ketorolac now as a prophylactic analgesic to reduce pain after the operation. The attending asks whether the ketorolac labeling supports this plan. Which of the following is correct?

(A) Contraindicated as prophylactic analgesia before major surgery
(B) Preferred as preemptive analgesia before open abdominal surgery
(C) Allowed before surgery if the dose is reduced by half
(D) Allowed before surgery if given 30 minutes before incision
(E) Allowed before surgery in patients under 65 years of age

Answer Contents

71. A 64-year-old woman with stage 4 chronic kidney disease has severe pain on postoperative day 1 after open hysterectomy. Her estimated glomerular filtration rate is 22 mL/min/1.73 m2. The surgeon proposes ketorolac as a rescue analgesic. Which of the following is the best response according to the ketorolac labeling?

(A) Use ketorolac at a lower dose with daily creatinine checks
(B) Use ketorolac because it does not affect renal blood flow
(C) Use ketorolac with extra intravenous fluids for renal protection
(D) Use ketorolac by the oral route rather than intravenously
(E) Avoid ketorolac, a contraindication in advanced renal impairment

Answer Contents

72. A 43-year-old woman undergoes laparoscopic sacrocolpopexy and the surgeon wants to avoid postoperative vomiting because straining could disrupt the repair. Her baseline risk of nausea is low. Anesthesia asks whether ondansetron prophylaxis is appropriate. Which of the following is most consistent with the ondansetron injection labeling?

(A) Withhold prophylaxis in low-risk patients regardless of the surgery
(B) Give prophylaxis when vomiting must be avoided, even if risk is low
(C) Wait to give treatment until nausea has actually started
(D) Give oral ondansetron in place of the injectable form
(E) Replace antiemetic prophylaxis with routine nasogastric suction

Answer Contents

73. A 62-year-old woman is on postoperative day 2 after bowel resection with anastomosis. Nausea eases each time she receives intravenous ondansetron, but her abdomen is becoming more distended and bowel sounds are quiet. She has not passed flatus. Which of the following is the best interpretation and next step according to the ondansetron labeling?

(A) Increase the ondansetron frequency and continue to observe
(B) Treat the distention as a drug effect and plan discharge
(C) Evaluate for ileus or obstruction before more antiemetic
(D) Add a second serotonin antagonist for better control
(E) Stop the antiemetic and advance her to a regular diet

Answer Contents

74. A 67-year-old woman with Parkinson disease takes subcutaneous apomorphine at home. She has nausea on postoperative day 1 after hysterectomy and a resident asks which antiemetic to avoid. Her blood pressure is 118/70 mm Hg and QTc is normal. Which of the following antiemetics is contraindicated with her home medication?

(A) Dexamethasone
(B) Aprepitant
(C) Dimenhydrinate
(D) Ondansetron
(E) Acupressure at the P6 point

Answer Contents

75. A 36-year-old woman has persistent vomiting in the recovery unit after laparoscopic myomectomy. The nurse proposes giving promethazine as an undiluted intravenous push into a hand vein. The pharmacist reviews the promethazine injection boxed warning. Which of the following is the preferred route of administration according to the label?

(A) Subcutaneous injection
(B) Intra-arterial injection
(C) Deep intramuscular injection
(D) Undiluted intravenous push into a small hand vein
(E) Intradermal injection

Answer Contents

76. A 41-year-old woman has refractory nausea after laparoscopic hysterectomy. The nurse draws promethazine from a vial whose concentration is greater than 1 mg/mL and plans to give it intravenously without dilution. The pharmacist objects. Which of the following is the best explanation based on the promethazine labeling?

(A) Intravenous use is contraindicated in patients who have just had surgery
(B) Intravenous use over 1 mg/mL is contraindicated due to extravasation risk
(C) Intravenous use is acceptable undiluted through a peripheral hand vein
(D) Intravenous use requires premedication with atropine to prevent bradycardia
(E) Intravenous use is limited to patients with a normal QT interval on ECG

Answer Contents

77. A 63-year-old woman on postoperative day 1 after hysterectomy receives intravenous ondansetron for persistent nausea. Telemetry later shows a transient prolonged QT interval, and she has no chest pain, syncope, or electrolyte abnormality. The resident asks what the drug labeling says about this finding. Which of the following best describes the ondansetron labeling regarding this finding?

(A) Transient QT prolongation occurs with oral dosing but not intravenous
(B) QT prolongation carries a boxed warning on the injection label
(C) No electrocardiographic changes have been reported with the drug
(D) Rare, transient QT prolongation reported mainly with intravenous use
(E) QT prolongation is expected in most patients who receive it

Answer Contents

78. An 80-year-old woman is on postoperative day 2 after cytoreductive surgery and is intermittently confused and restless. Her family reports that donepezil is not part of her home regimen, but they ask that it be started to treat the delirium. She is hemodynamically stable and her pain is controlled. According to the American Geriatrics Society guideline, which of the following is the best response?

(A) Avoid a new cholinesterase inhibitor and address precipitants
(B) Start a cholinesterase inhibitor to treat the delirium now
(C) Start a cholinesterase inhibitor at half the usual dose today
(D) Start a cholinesterase inhibitor to prevent recurrence
(E) Start one if she has a documented diagnosis of dementia

Answer Contents

79. A 76-year-old woman is on postoperative day 1 after hysterectomy. She is agitated and pulling at her intravenous line but is not an immediate danger to herself, and she has no history of alcohol use or benzodiazepine dependence. The covering resident orders lorazepam. Which of the following is the best response according to the American Geriatrics Society guideline?

(A) Give the lorazepam and repeat it as needed for agitation
(B) Give a higher dose of lorazepam so she sleeps overnight
(C) Add a second benzodiazepine for breakthrough agitation
(D) Apply physical restraints to protect the intravenous line
(E) Avoid benzodiazepines as first-line and look for causes

Answer Contents

80. An 82-year-old woman with postoperative delirium after laparotomy has severe agitation. Despite reorientation, treatment of her pain, and removal of unnecessary lines, she is trying to climb out of bed and has pulled out her drain, putting herself at risk of harm. Which of the following is most consistent with the American Geriatrics Society guideline?

(A) An antipsychotic at the lowest effective dose, briefly
(B) Scheduled lorazepam for the remainder of the hospitalization
(C) A cholinesterase inhibitor to reverse the delirium quickly
(D) Physical restraints applied for the rest of the admission
(E) A higher-dose antipsychotic to ensure sedation overnight

Answer Contents

81. A 79-year-old woman is on postoperative day 3 after radical hysterectomy. She is unusually quiet, drowsy, and inattentive, does not engage when addressed, and has a fluctuating level of alertness. Her family says she is normally active and sharp. Delirium of the hypoactive type is diagnosed. Which of the following is most consistent with the American Geriatrics Society guideline?

(A) Start a scheduled benzodiazepine to improve her sleep at night
(B) Start a scheduled antipsychotic to improve her level of alertness
(C) Start a cholinesterase inhibitor to improve her attention span
(D) Do not treat it with antipsychotic or benzodiazepine drugs
(E) Start a stimulant medication each morning to increase wakefulness

Answer Contents

82. A 77-year-old woman has had an open hysterectomy and is at high risk for postoperative delirium. Her pain is moderate on postoperative day 1, and the team is choosing her analgesic regimen. According to the American Geriatrics Society guideline, which of the following is the best approach?

(A) Use scheduled intravenous morphine to ensure complete pain relief
(B) Avoid analgesics to prevent confusion
(C) Use meperidine as the first-line opioid
(D) Rely on a benzodiazepine for pain and anxiety
(E) Optimize pain control, preferably with nonopioid medications

Answer Contents

83. A 48-year-old woman had an uncomplicated abdominal hysterectomy yesterday. An indwelling urethral catheter was placed in the operating room. She has no epidural, no urinary retention history, is ambulating, and urine output is adequate. The nurse asks when the catheter should come out, applying the CDC catheter-associated infection guideline. Which of the following is the best answer?

(A) After her first postoperative bowel movement occurs
(B) Five days after surgery to ensure bladder recovery
(C) As soon as possible, preferably within 24 hours
(D) At discharge, whichever day that turns out to be
(E) Once a repeat urine culture has returned negative

Answer Contents

84. A 35-year-old woman is scheduled for a 40-minute laparoscopic bilateral tubal salpingectomy under general anesthesia. The circulating nurse asks whether to place an indwelling urinary catheter routinely before the case. There is no bladder pathology, no anticipated prolonged duration, and no need for strict output monitoring. Which of the following is most consistent with the CDC guideline for operative patients?

(A) Place a catheter routinely for each operation
(B) Use a catheter when a specific indication exists
(C) Place a catheter for each general anesthetic
(D) Place a suprapubic catheter in place of a urethral one
(E) Place a catheter and leave it in for 48 hours

Answer Contents

85. A 56-year-old woman needs bladder drainage for several days after pelvic reconstructive surgery because of persistent urinary retention. She is alert, has good hand function, and is willing to learn self-catheterization. The team compares indwelling urethral, suprapubic, and intermittent catheterization according to the CDC guideline. Which of the following is the best choice?

(A) Indwelling urethral catheter
(B) Suprapubic catheter
(C) Condom-type external device
(D) Intermittent catheterization
(E) Continuous bladder irrigation

Answer Contents

86. A 52-year-old woman is on intermittent catheterization every 6 hours after pelvic surgery for postoperative urinary retention. Nursing staff catheterize on schedule even when she reports she has voided. The unit wants to reduce unnecessary catheterizations while staying within the CDC guideline. Which of the following is the best step?

(A) Switch to a continuous indwelling urethral catheter for comfort
(B) Increase the catheterization schedule to every 2 hours day and night
(C) Send a urine culture before each scheduled catheterization episode
(D) Order daily prophylactic antibiotics during the catheterization period
(E) Consider a portable bladder ultrasound to assess urine volume

Answer Contents

87. A 62-year-old woman has had an indwelling urethral catheter for three days after a radical hysterectomy. She is afebrile and has no dysuria, flank pain, or suprapubic tenderness. The resident wants a urine culture before the catheter is removed to screen for bacteriuria. Which of the following is most consistent with the CDC guideline?

(A) Send a urine culture and treat a positive result
(B) Send a culture when the urine appears cloudy or dark
(C) Do not screen routinely for asymptomatic bacteriuria
(D) Treat empirically with antibiotics at the time of removal
(E) Replace the catheter, then send the culture from the new one

Answer Contents

88. A 58-year-old woman is having her urinary catheter removed on postoperative day 1 after hysterectomy for fibroids. She has no urinary symptoms, no fever, and no history of recurrent urinary infections. The resident asks whether to give a dose of an antibiotic at the time of removal to prevent infection. Which of the following is most consistent with the CDC guideline?

(A) Give a single oral antibiotic dose at the time of removal
(B) Do not routinely use systemic antimicrobials to prevent infection
(C) Give a three-day course of oral antibiotics once the catheter is out
(D) Give a fluoroquinolone to patients who were catheterized
(E) Give an intravenous antibiotic 30 minutes before removal

Answer Contents

89. A 62-year-old woman is having laparotomy for ovarian cancer and is at high risk for thromboembolism. The surgical team plans fondaparinux prophylaxis. Hemostasis is secure at the end of the case. According to the fondaparinux labeling for abdominal surgery, which of the following is the correct timing for the initial dose?

(A) No earlier than 6 to 8 hours after surgery
(B) Two hours before the surgical incision is made
(C) At induction of anesthesia in the operating room
(D) 12 to 24 hours after the operation ends
(E) At 72 hours, once the bleeding risk has passed

Answer Contents

90. A 66-year-old woman had an open radical hysterectomy for cervical cancer, and fondaparinux prophylaxis was started on the evening of surgery. The resident asks about the usual duration of administration stated in the fondaparinux labeling for patients undergoing abdominal surgery. Which of the following is correct?

(A) 1 day
(B) 2 to 3 days
(C) 14 to 21 days
(D) 5 to 9 days
(E) Until discharge, with no specified duration

Answer Contents

91. A 74-year-old woman weighing 46 kg is scheduled for laparotomy for endometrial cancer. Her creatinine clearance is 58 mL/min and she has no active bleeding. The surgical team proposes fondaparinux for venous thromboembolism prophylaxis. Which of the following features makes fondaparinux contraindicated for her according to the label?

(A) Age over 70 years
(B) Creatinine clearance of 58 mL/min
(C) Body weight less than 50 kg
(D) Planned laparotomy
(E) Endometrial cancer

Answer Contents

92. A 60-year-old woman has an open hysterectomy for a large fibroid uterus. The surgeon orders enoxaparin prophylaxis to start the evening of surgery and asks the resident for the usual duration of administration after abdominal surgery stated in the enoxaparin labeling. Which of the following is the correct duration?

(A) 1 to 2 days
(B) 7 to 10 days
(C) 3 days
(D) 14 to 21 days
(E) Until the drain is removed

Answer Contents

93. A 59-year-old woman has a thoracic epidural catheter in place after laparotomy. She received a prophylactic dose of enoxaparin at 18:00 the previous evening. The anesthesia team asks when the catheter may be removed relative to that lower dose according to the enoxaparin labeling. Which of the following is correct?

(A) Within 2 hours of the dose
(B) Immediately, because the effect is brief
(C) At least 48 hours after the dose
(D) After protamine is given
(E) At least 12 hours after the dose

Answer Contents

94. A 46-year-old woman had an abdominal hysterectomy through a midline incision, closed primarily, and the wound is clean and dry on postoperative day 1. A nurse manager proposes a silver-impregnated advanced dressing for every closed incision to reduce infection. The infection committee asks what the WHO guideline suggests about dressing choice for primarily closed wounds. Which of the following is the best response?

(A) Use a standard dressing rather than an advanced dressing
(B) Use a silver-impregnated dressing for all closed incisions
(C) Use a hydrocolloid dressing for all closed incisions
(D) Leave the incision uncovered from the first day
(E) Apply a topical antibiotic ointment under the dressing

Answer Contents

95. A 40-year-old woman with healthy lungs is recovering from general endotracheal anesthesia after an abdominal hysterectomy. The anesthesiologist asks about an intervention in the immediate postoperative period that the CDC guideline recommends to reduce surgical site infection. Which of the following is the best answer?

(A) Give increased inspired oxygen after extubation
(B) Continue room air until she is awake and alert
(C) Give prophylactic antibiotics until discharge
(D) Apply an antibiotic ointment to the incision
(E) Leave the first dressing in place for two weeks

Answer Contents

96. A 52-year-old woman is discharged after abdominal hysterectomy with a closed, clean incision. Her surgeon routinely tells patients to apply triple antibiotic ointment to the incision twice daily to prevent infection. A nursing educator asks whether this is supported by the CDC surgical site infection guideline. Which of the following is correct?

(A) Apply the ointment to the incision for the first two weeks
(B) Do not apply antimicrobial agents to the incision
(C) Apply antiseptic powder if the incision is moist
(D) Apply ointment to the incision in patients with diabetes
(E) Apply ointment to the incision in patients who are obese

Answer Contents

G5. Gynecologic Emergencies

97. A 14-year-old girl comes to the emergency department with 5 hours of sudden right lower quadrant pain and two episodes of vomiting. Temperature is 37.2 C and heart rate is 104/min. Urine hCG is negative. Pelvic ultrasonography shows an enlarged right ovary with peripheral follicles and no cyst, mass or free fluid. Her resident suggests that the absence of a mass makes torsion unlikely. Which of the following is the most appropriate next step?

(A) Measure serum CA-125 and inhibin B before any decision
(B) Obtain pelvic MRI to confirm torsion before surgery
(C) Admit for 24 hours of observation with serial abdominal examinations and labs
(D) Discharge with analgesia and repeat ultrasonography in 6 weeks
(E) Proceed to diagnostic laparoscopy with detorsion if the ovary is twisted

Answer Contents

98. A 15-year-old girl underwent urgent laparoscopy for sudden left pelvic pain and vomiting with an enlarged left ovary on ultrasonography. Both ovaries and tubes were found to be normal and untwisted, and no procedure was performed. Her mother asks whether the surgeon made a mistake by operating. Which of the following is the best explanation to give?

(A) Torsion was likely present earlier and resolved, which proves the ovary is permanently normal
(B) A normal laparoscopy shows the preoperative imaging was misread and should be reported
(C) No preoperative test confirms torsion, so about half of suspected cases show none at laparoscopy
(D) Ultrasonography with Doppler is accurate enough that this finding indicates a technical error
(E) Laparoscopy should have been delayed until tumor markers returned to avoid a negative exploration

Answer Contents

99. A 16-year-old girl had laparoscopic detorsion of the left ovary for acute pain. The ovary regained a pink color after untwisting. A 5-cm simple-appearing cyst was present, and the surgeon chose not to remove it. She is recovering well, afebrile and tolerating a diet on postoperative day 1. Which of the following is the most appropriate follow-up plan?

(A) Pelvic MRI at 1 week to confirm ovarian viability
(B) Repeat laparoscopy at 6 weeks for elective cystectomy
(C) Serial CA-125 measurements every week until normal
(D) Repeat ultrasonography of the cyst at 6 to 12 weeks
(E) No imaging because functional cysts resolve on their own

Answer Contents

100. A 27-year-old woman has had 4 days of lower abdominal pain, fever to 38.9 C and purulent vaginal discharge. Cervical motion tenderness is present, and transvaginal ultrasonography shows a 6-cm complex right adnexal mass consistent with a tubo-ovarian abscess. She is hemodynamically stable and tolerating oral intake. Which of the following is the most appropriate disposition according to the CDC guideline?

(A) Outpatient oral therapy with reevaluation in 72 hours
(B) A single intramuscular dose of ceftriaxone and discharge from the clinic
(C) Admission for more than 24 hours of inpatient observation
(D) Same-day discharge after 4 hours of emergency department monitoring
(E) Immediate laparotomy for drainage regardless of response

Answer Contents

101. A 24-year-old woman with a copper intrauterine device placed 2 years ago has 3 days of pelvic pain and cervical motion tenderness. Nucleic acid testing is pending and she is started on a recommended outpatient regimen for mild-to-moderate pelvic inflammatory disease. She is afebrile and wants to keep the device if possible. Which of the following is the best management of the device?

(A) Remove it immediately and place a new device after treatment is completed
(B) Leave it in place until all test results are negative, then remove it and switch methods
(C) Remove it and insert a levonorgestrel system the same day
(D) Remove it if gonorrhea or chlamydia is confirmed
(E) Leave it in place and consider removal if there is no improvement within 48 to 72 hours

Answer Contents

102. A 31-year-old woman with well-controlled HIV infection on antiretroviral therapy has 2 days of lower abdominal pain and cervical motion tenderness. Temperature is 37.8 C, she tolerates oral intake, and examination shows no peritoneal signs or adnexal mass. The clinician is unsure whether HIV changes the treatment choice. Which of the following is the most appropriate management?

(A) Use the same regimens as for women without HIV, because response was equivalent
(B) Admit for parenteral therapy because HIV infection mandates inpatient treatment
(C) Extend the antibiotic course to 28 days
(D) Delay antibiotics until a viral load is available
(E) Add empiric antifungal therapy to every regimen

Answer Contents

103. A 22-year-old woman presents with 6 days of extremely painful vulvar ulcers, fever to 38.7 C, vomiting and poor oral intake. Examination shows confluent vulvar and perianal ulcerations, and she cannot tolerate oral medication. Herpes simplex virus type 2 is confirmed by polymerase chain reaction of a lesion swab. She is admitted for severe disease. Which of the following is the most appropriate antiviral approach?

(A) Topical antiviral cream applied to the ulcers twice daily for 10 days
(B) Intravenous acyclovir, then oral therapy for more than 10 days total
(C) Oral antiviral for 3 days started once serologic confirmation returns
(D) Withhold antivirals because the episode is self-limited
(E) Suppressive oral therapy starting after the lesions heal

Answer Contents

104. A 52-year-old postmenopausal woman reports a painless swelling at the right posterior vestibule for 4 months. Examination shows a firm, nontender 3-cm mass at the site of the Bartholin gland with no erythema, warmth or fluctuance. She has no fever and no discharge. According to the Greater Glasgow and Clyde guideline, which of the following is the most appropriate next step?

(A) Incision and drainage with Word catheter placement
(B) Marsupialization in the operating room without tissue diagnosis
(C) A 5-day course of empiric oral antibiotics
(D) Biopsy of the mass because she is over 40 years old
(E) Reassurance with return if the mass grows

Answer Contents

105. A 29-year-old woman has a 4-cm tender, fluctuant right Bartholin abscess incised and drained under local anesthesia, and a Word catheter is placed. She asks why the catheter must stay in place for several weeks rather than being removed the next day. Which of the following is the best explanation?

(A) It lets an epithelialized fistula tract form so the cavity cannot refill
(B) It delivers antibiotic continuously to the infected gland tissue
(C) It compresses the cavity wall to control bleeding from the cyst bed
(D) It destroys the gland by pressure necrosis so the cyst cannot return
(E) It keeps the cavity open so the clinician can irrigate daily with antiseptic

Answer Contents

106. A 24-year-old woman falls onto a bicycle crossbar and presents 3 hours later with severe vulvar pain. Examination shows a 6-cm tense left labial hematoma that is not enlarging over 2 hours of observation, with no active bleeding or laceration. Blood pressure is 118/72 mm Hg, hemoglobin is stable, and she is voiding spontaneously. Which of the following is the most appropriate management?

(A) Immediate incision with clot evacuation in the operating room
(B) Needle aspiration of the hematoma at the bedside
(C) Pelvic angiographic embolization
(D) Ice packs, analgesia and serial examinations
(E) Broad-spectrum intravenous antibiotics with a closed drain

Answer Contents

107. A 46-year-old woman with diabetes has 2 days of vulvar pain that has progressed to spreading perineal erythema, dusky skin with bullae and pain out of proportion to the findings. Temperature is 39.2 C, heart rate 118/min, blood pressure 88/50 mm Hg, glucose 380 mg/dL and white blood cell count 24,000/microliter. A CT scan could be done in about 90 minutes. Which of the following is the most appropriate next step?

(A) Delay surgery until CT confirms gas in the tissue
(B) Emergent debridement with broad-spectrum antibiotics
(C) Oral antibiotics with reexamination in 24 hours
(D) Bedside incision of a single fluctuant area and discharge
(E) Hyperbaric oxygen therapy before any operation

Answer Contents

108. A 28-year-old woman at about 5 weeks by last menstrual period has mild left pelvic discomfort and light spotting. She is hemodynamically stable and desires the pregnancy. Serum hCG is 2,800 mIU/mL. Transvaginal ultrasonography shows an empty uterus, no adnexal mass and no free fluid. A colleague proposes dilation and curettage to look for villi. Which of the following is the most appropriate next step?

(A) Empiric methotrexate today for a presumed ectopic pregnancy
(B) Diagnostic laparoscopy today to inspect the tubes
(C) Repeat hCG in 48 hours with clinical review
(D) Dilation and curettage today to look for villi
(E) Reassurance with no further testing planned

Answer Contents

109. A 30-year-old woman with an unruptured tubal ectopic pregnancy received single-dose methotrexate. Her day 1 hCG was 2,000 mIU/mL, day 4 was 2,200 mIU/mL and day 7 is 2,050 mIU/mL, a 7% decline from day 4. She has mild unchanged pain, normal vital signs and no free fluid on ultrasonography. Which of the following is the most appropriate interpretation and next step?

(A) Treatment is working, so continue weekly hCG until undetectable
(B) Start leucovorin rescue and observe for a further week
(C) Treatment has failed in every case, so proceed to immediate salpingectomy
(D) Stop monitoring because the hCG has fallen from the day 4 level
(E) The decline is inadequate, so reassess and plan a second dose or surgery

Answer Contents

110. A 26-year-old woman has a positive pregnancy test, light bleeding and minimal discomfort. She is stable, and transvaginal ultrasonography shows no pregnancy in the uterus or adnexa. Serum hCG is 600 IU/L, and 48 hours later it is 1,200 IU/L. According to the NICE guideline, which of the following is the most appropriate next step?

(A) Give methotrexate today for presumed ectopic pregnancy
(B) Clinical review in the early pregnancy service within 24 hours
(C) Repeat hCG in 4 weeks without further imaging
(D) Diagnostic laparoscopy to inspect the fallopian tubes
(E) Transvaginal ultrasonography 7 to 14 days later

Answer Contents

111. A 33-year-old woman at about 6 weeks has severe left lower quadrant pain that required intravenous analgesia. Her blood pressure is 116/70 mm Hg and heart rate is 88/min. Transvaginal ultrasonography shows a 2-cm left adnexal mass with no cardiac activity and a small amount of free fluid. Serum hCG is 1,300 IU/L. Under the NICE guideline, which of the following is the most appropriate management?

(A) Expectant management with weekly hCG
(B) Surgical management because she has significant pain
(C) Repeat hCG in 48 hours before deciding
(D) Single-dose methotrexate because the hCG is under 1,500 IU/L
(E) Intramuscular progesterone with outpatient review

Answer Contents

112. A 29-year-old woman has a 2.5-cm unruptured left tubal ectopic pregnancy with no cardiac activity. She has minimal pain, normal vital signs and a serum hCG of 3,200 IU/L. She would prefer to avoid surgery and asks about the downsides of methotrexate. Which of the following is the most appropriate counseling according to the NICE guideline?

(A) Methotrexate is an option but may need further treatment
(B) Methotrexate is contraindicated once hCG exceeds 1,500 IU/L
(C) Expectant management is the recommended choice at this hCG level
(D) Surgery is required for every hCG over 1,500 IU/L
(E) Methotrexate removes the need for any follow-up hCG testing

Answer Contents

113. A 31-year-old woman has an unruptured 18-mm tubal ectopic pregnancy with no cardiac activity, no pain and normal vital signs. Serum hCG is 780 IU/L. She lives nearby, understands the warning signs, can attend follow-up and prefers to avoid drugs and surgery if that is safe. Under the NICE guideline, which of the following is the most appropriate management?

(A) Immediate laparotomy with salpingectomy
(B) Laparoscopic salpingostomy within the week
(C) Expectant management with serial hCG
(D) Dilation and curettage with villous examination
(E) Two-dose methotrexate with leucovorin rescue

Answer Contents

114. A 15-year-old girl, 14 months after menarche, has had 10 days of heavy vaginal bleeding soaking a pad every hour. Heart rate is 114/min, blood pressure is 98/58 mm Hg with orthostatic dizziness, and hemoglobin is 6.8 g/dL. She is vomiting and cannot keep oral medication down. Pregnancy test is negative and a bleeding disorder evaluation has been sent. Which of the following is the most appropriate acute medical therapy?

(A) Oral medroxyprogesterone acetate once daily
(B) Placement of a levonorgestrel intrauterine system
(C) Endometrial ablation
(D) Intravenous conjugated equine estrogen
(E) Depot medroxyprogesterone acetate injection

Answer Contents

115. A 17-year-old girl with newly diagnosed von Willebrand disease was admitted with life-threatening bleeding, transfused, and stabilized on hormonal therapy and tranexamic acid. Her hemoglobin is now 9.4 g/dL and bleeding has nearly stopped. Her family asks about definitive surgery to prevent future episodes. Which of the following is the most appropriate recommendation?

(A) Endometrial ablation before discharge from the hospital today
(B) Total hysterectomy with preservation of ovaries
(C) Continuous hormonal contraception with hematology follow-up
(D) Scheduled dilation and curettage every few months
(E) Uterine artery embolization by interventional radiology

Answer Contents

116. A 15-year-old girl with heavy menses for a year has fatigue, a hemoglobin of 8.4 g/dL and a mean corpuscular volume of 72 fL. She has had a febrile respiratory illness this week, and her C-reactive protein is elevated. Serum ferritin is reported as 52 ng/mL. Her mother says the ferritin is normal, so iron is not the cause. Which of the following is the best interpretation?

(A) Iron deficiency is excluded and a hemoglobinopathy is the probable cause
(B) Ferritin can be raised by inflammation, so iron deficiency is not excluded
(C) A normal ferritin proves the anemia is from von Willebrand disease alone
(D) Ferritin falls with acute infection, so the value underestimates her stores
(E) Ferritin is irrelevant and transferrin saturation should be measured instead

Answer Contents

117. A 38-year-old woman has cyclic heavy menstrual bleeding and is offered tranexamic acid tablets. She has chronic kidney disease with a serum creatinine of 1.9 mg/dL. She has no history of venous thromboembolism and uses no hormonal contraception. According to the FDA label, which of the following is the most appropriate approach?

(A) Prescribe the standard dose, since renal function does not matter
(B) Avoid the drug entirely, because chronic kidney disease is a labeled contraindication
(C) Prescribe the standard dose and add a combined oral contraceptive
(D) Extend the course beyond the usual limit to offset reduced clearance
(E) Reduce the dose, since adjustment is needed over a creatinine of 1.4 mg/dL

Answer Contents

118. A 29-year-old breastfeeding woman had a levonorgestrel intrauterine system placed 5 weeks after delivery. Three weeks later she has pelvic pain. She is afebrile and hemodynamically stable, with no peritoneal signs. Transvaginal ultrasonography does not show the device in the uterine cavity, and a CT scan shows it lying free in the pelvis near the sigmoid colon. According to the product label, which of the following is the most appropriate next step?

(A) Leave it in place because the progestin continues to act
(B) Reassure her and repeat the imaging in 6 months
(C) Treat with oral antibiotics and observe for fever
(D) Locate and remove the device, with surgery if needed
(E) Insert a second system to preserve her contraception

Answer Contents

119. A 36-year-old woman is admitted with fever, right flank pain and pyuria. Blood cultures are negative. She receives intravenous ceftriaxone, and urine culture grows Escherichia coli susceptible to an oral non-fluoroquinolone agent. She has been afebrile for 48 hours and is tolerating a diet. Which of the following is the most appropriate total duration of therapy according to the IDSA guideline?

(A) 3 days
(B) 7 days
(C) 10 to 14 days
(D) 21 days
(E) 28 days

Answer Contents

120. A 44-year-old woman was hospitalized 3 days ago with a complicated urinary tract infection and has been receiving an intravenous beta-lactam. She is now afebrile and hemodynamically normal, tolerates oral intake, and the urine isolate is susceptible to an oral agent with good tissue penetration. Which of the following is the most appropriate management?

(A) Transition to an oral agent for the remaining duration
(B) Stop antibiotics now because she is afebrile
(C) Continue intravenous therapy until the full course is complete
(D) Repeat urine culture daily before changing the route
(E) Switch to a broader intravenous agent as a precaution

Answer Contents

121. A 43-year-old woman has constant clear vaginal leakage beginning 9 days after total laparoscopic hysterectomy. Intravesical methylene blue does not stain a vaginal tampon, but the tampon is wet. CT urography shows contrast leaking from the left distal ureter into the vagina, with mild left hydronephrosis and contrast still reaching the bladder. She is afebrile and her creatinine is normal. According to the AUA urotrauma guideline, which of the following is the most appropriate initial management?

(A) Open left ureteral reimplantation within the week
(B) Cystoscopic left ureteral stent placement
(C) Left nephrectomy because of the fistula
(D) Vaginal pessary and observation for 3 months
(E) Delayed repair at 6 months without urinary drainage

Answer Contents

122. A 41-year-old woman has a laparoscopic ovarian cystectomy and no catheter is in place afterward. Four hours after surgery she voids 120 mL. A bladder ultrasound immediately afterward shows a residual of 90 mL, or 75% of the voided volume. She has suprapubic fullness and no fever, flank pain or hematuria. Using the Greater Glasgow and Clyde guideline, which of the following is the best interpretation?

(A) Postoperative retention, because the residual exceeds 50% of the voided volume
(B) Normal emptying, because she passed more than 100 mL of urine without assistance today
(C) Ureteral injury requiring urgent imaging
(D) Urinary tract infection requiring empiric antibiotics
(E) Detrusor overactivity requiring an anticholinergic drug

Answer Contents

123. A 63-year-old woman has urinary retention after an autologous fascial sling for stress incontinence. She has repeated residuals over 50% of voided volumes, and after teaching she cannot perform clean intermittent self-catheterization because of severe hand arthritis. She is otherwise well and ready for discharge. Using the Greater Glasgow and Clyde postoperative bladder care guideline, which of the following is the most appropriate plan?

(A) Keep her in hospital until she voids spontaneously
(B) Discharge without a catheter and advise increased fluids
(C) Perform immediate cystoscopy and ureteral stenting
(D) Discharge with oxybutynin for bladder spasm
(E) Discharge with an indwelling catheter and follow-up

Answer Contents

G6.1. Minimally Invasive Procedures

124. A 42-year-old nulligravid woman requires hysterectomy for symptomatic adenomyosis. The uterus is mobile and about 10 weeks in size, the adnexa are normal, and vaginal access is adequate on examination. She asks her surgeon whether nulliparity rules out a vaginal operation. Which of the following is the best response according to ACOG?

(A) Nulliparity is an absolute contraindication to the vaginal route
(B) Hysterectomy should be delayed until she has had a vaginal delivery
(C) An abdominal approach should be used in nulligravid women
(D) Vaginal hysterectomy can usually be done safely even in nulliparous women
(E) A robot-assisted approach is required whenever the vaginal introitus is narrow

Answer Contents

125. A 45-year-old woman with pelvic pain has severe endometriosis and dense adhesions documented at a prior laparoscopy, and a 14-week-size uterus. Her surgeon, an experienced minimally invasive surgeon, has assumed that these findings require an open abdominal hysterectomy. Which of the following statements about route selection is most accurate?

(A) Each of these findings is an absolute contraindication to a laparoscopic or robotic approach
(B) Vaginal hysterectomy is required before any laparoscopic attempt can be made
(C) These findings may exclude a vaginal route but do not exclude minimally invasive surgery
(D) An enlarged uterus mandates a robot-assisted approach
(E) Endometriosis mandates supracervical rather than total hysterectomy

Answer Contents

126. A hospital gynecology department plans to introduce robot-assisted hysterectomy. A surgeon who has performed only vaginal hysterectomies asks to begin robotic cases first, since the robot will make the technique easy. According to the ACOG committee opinion, which of the following is the most appropriate expectation?

(A) Vaginal hysterectomy skill alone is a sufficient prerequisite
(B) No prior laparoscopic experience is needed because the robot replaces hand skill
(C) Skill in abdominal and laparoscopic hysterectomy should come first
(D) Robotic privileges should be granted after observing a single case
(E) Robotic hysterectomy should replace laparoscopic hysterectomy in all departments

Answer Contents

127. A 41-year-old woman with a 12-week-size fibroid uterus is choosing between laparoscopic and abdominal hysterectomy for benign disease. She values a quick return to work and asks about the main tradeoff. According to the Cochrane review of surgical approaches to hysterectomy, which of the following counseling statements is most accurate?

(A) Laparoscopic hysterectomy carries a lower ureteral injury risk than the abdominal route
(B) Recovery time is identical for the two routes
(C) Abdominal hysterectomy returns patients to activity sooner
(D) Laparoscopic hysterectomy has a shorter operating time than the abdominal route
(E) Laparoscopic gives quicker recovery but more ureteral injury risk

Answer Contents

128. A 44-year-old premenopausal woman has a 7-cm intramural fibroid and requests a laparoscopic approach with power morcellation to avoid an incision. Imaging does not suggest malignancy. As part of informed consent, the surgeon quotes the FDA's updated estimate of the prevalence of unsuspected uterine sarcoma among women undergoing surgery for presumed fibroids. Which of the following is that estimate?

(A) About 1 in 225 to 1 in 580 women
(B) About 1 in 22 to 1 in 58 women
(C) About 1 in 2,250 to 1 in 5,800 women
(D) About 1 in 22,500 to 1 in 58,000 women
(E) About 1 in 2 to 1 in 5 women

Answer Contents

129. A 54-year-old postmenopausal woman has symptomatic fibroids and elects hysterectomy. Her surgeon plans a total laparoscopic procedure and intends to use a power morcellator to remove the uterus in pieces. Endometrial biopsy is benign. Which of the following is most consistent with the FDA safety communication?

(A) Proceed, because a benign biopsy excludes sarcoma
(B) Remove the specimen intact, vaginally or by mini-laparotomy
(C) Proceed with uncontained morcellation if the patient consents
(D) Use a higher-power morcellator to shorten tissue exposure
(E) Convert to subtotal hysterectomy and morcellate the fundus

Answer Contents

130. A 33-year-old woman with a body mass index of 24 kg/m2 and no prior abdominal surgery is scheduled for laparoscopic ovarian cystectomy. The surgeon is deciding between closed entry with a Veress needle and direct trocar entry and asks what the Cochrane review of entry techniques found about failed entry. Which of the following is the most accurate summary?

(A) Direct trocar entry caused more deaths than the Veress needle
(B) Veress needle entry had fewer failed entries than direct trocar entry
(C) Failed entry was identical in all comparisons
(D) Direct trocar entry had fewer failed entries than Veress needle entry
(E) Open entry eliminated failed entry in all studies

Answer Contents

131. A 29-year-old gynecology resident asks her attending which laparoscopic entry technique has been proven to prevent major vascular injury during gynecologic surgery in women of reproductive age. The attending cites the most recent Cochrane review of laparoscopic entry techniques. Which of the following is the best answer?

(A) Open entry has been proven to prevent all vascular injury
(B) Evidence is insufficient to favor one entry technique over another
(C) Direct trocar entry has been shown to cause more bowel injury than open entry
(D) The Veress needle has proven safety over all other techniques
(E) Entry under direct vision is proven less safe than open entry

Answer Contents

132. A 48-year-old woman with two prior midline laparotomies and a suspected pelvic adhesive burden is scheduled for laparoscopic salpingo-oophorectomy. The surgeon is concerned that adhesions at the umbilicus make the usual primary port site unsafe. According to the RCOG guideline, which of the following is the preferred alternative entry point?

(A) Suprapubic midline, 2 cm over the pubic symphysis, with the bladder emptied
(B) Left lower quadrant, lateral to the inferior epigastric vessels, near the iliac crest
(C) Right lower quadrant, 2 cm medial to the anterior superior iliac spine
(D) Left upper quadrant, 3 cm under the left costal margin, mid-clavicular line
(E) Epigastrium, 5 cm under the xiphoid process in the midline

Answer Contents

133. A 36-year-old woman undergoes laparoscopic cystectomy with a 5-mm umbilical port, a 12-mm port in the right lower quadrant, a 5-mm port in the left lower quadrant and a 5-mm suprapubic midline port. The procedure ends, and the surgeon reviews fascial closure to prevent port-site hernia. According to the RCOG guideline, which of the following ports requires formal deep sheath closure?

(A) The 5-mm port at the umbilicus
(B) No port requires fascial closure at these sizes
(C) The 5-mm left lower quadrant port
(D) The 5-mm suprapubic midline port
(E) The 12-mm right lower quadrant port

Answer Contents

134. A 37-year-old healthy woman is scheduled for hysteroscopic resection of a submucous fibroid using monopolar energy and a glycine solution as the hypotonic distension medium. The anesthesiologist asks for the maximum fluid deficit at which the team should stop. According to the BSGE/ESGE guideline, which of the following is the correct limit for a healthy woman?

(A) 500 mL
(B) 1000 mL
(C) 1500 mL
(D) 2500 mL
(E) 3000 mL

Answer Contents

135. A 71-year-old woman with ischemic heart disease and a prior myocardial infarction is undergoing bipolar hysteroscopic polypectomy using normal saline. The team sets a limit for the maximum fluid deficit for this patient. According to the BSGE/ESGE guideline, which of the following is the suggested upper limit for an older woman with comorbidity using an isotonic solution?

(A) 750 mL
(B) 1000 mL
(C) 1500 mL
(D) 2500 mL
(E) 3500 mL

Answer Contents

136. A 45-year-old woman develops nausea, headache and confusion 30 minutes after hysteroscopic myomectomy with a glycine solution. The recorded fluid deficit was 1,800 mL. Blood pressure is 156/94 mm Hg, heart rate 58/min, oxygen saturation is normal, and there are no skin or airway findings. Which of the following is the most likely cause?

(A) Gas embolism from air entering the distension tubing system
(B) Anaphylactic reaction to the glycine distension medium
(C) Dilutional hyponatremia from hypotonic fluid absorption
(D) Residual effect of the general anesthetic
(E) Uterine perforation with concealed hemorrhage

Answer Contents

137. A 45-year-old woman is diagnosed with symptomatic hyponatremia after hysteroscopic surgery with a hypotonic medium. Serum sodium is 121 mmol/L, and she is being treated in the intensive care unit. The team asks for the target rate of sodium correction. According to the BSGE/ESGE guideline, which of the following is the recommended target?

(A) Raise sodium by 6 mmol/L over 24 hours
(B) Raise sodium by 18 mmol/L over 24 hours
(C) Raise sodium to 140 mmol/L within 6 hours
(D) Raise sodium by 12 mmol/L over 4 hours
(E) Raise sodium by 24 mmol/L over 24 hours

Answer Contents

138. During operative hysteroscopy with a pump, the fluid deficit is rising faster than expected in a 40-year-old woman. The pump is set to a high inflow pressure, and the view is excellent. The surgeon asks which single change would most reduce systemic fluid absorption without stopping the case. Which of the following is the best answer?

(A) Increase the inflow pressure to improve the view further
(B) Control the intrauterine pressure at a lower level
(C) Add a second inflow line to speed distension
(D) Switch to a larger resectoscope sheath
(E) Pause monitoring until the end of the case

Answer Contents

139. A surgeon plans hysteroscopic resection of a submucous fibroid with a monopolar loop and proposes using 0.9% saline as the distension medium because it is isotonic and safer for sodium. The anesthesiologist questions whether it can be used. According to the BSGE/ESGE guideline, which of the following is the best response?

(A) Saline is the required medium for monopolar energy
(B) Saline is acceptable if the inflow pressure is doubled
(C) Saline can be used with monopolar energy if a vasopressin injection is given first
(D) Glycine causes no risk of hyponatremia with monopolar energy
(E) Electrolyte fluids do not work with monopolar energy, so use bipolar equipment

Answer Contents

G6.2. Minor Gynecologic Procedures

140. A 56-year-old woman is scheduled for total vaginal hysterectomy combined with anterior and apical prolapse repair for stage III uterovaginal prolapse. She has no urinary symptoms, and her urinalysis and renal function are normal. Before the case, the team discusses how to confirm lower urinary tract integrity before she leaves the operating room. Which of the following is best supported by ACOG guidance on cystourethroscopy?

(A) No assessment because the vaginal route lowers the injury risk
(B) Palpation of both ureters and a bladder-filling test alone
(C) Cystoscopy after every hysterectomy with reconstructive surgery
(D) Cystoscopy if a bladder injury is seen during the dissection itself
(E) Cystoscopy if hematuria or leakage appears after the operation ends

Answer Contents

141. A 61-year-old woman has gross hematuria 6 months after a hysterectomy. During office cystoscopy, a 5 mm papillary lesion is seen on the posterior bladder wall, and the surgeon decides to fulgurate it with a monopolar electrode. The nurse asks which fluid to connect for bladder distension during the cautery. Which of the following is the most appropriate choice?

(A) Glycine solution
(B) Half-normal saline
(C) Lactated Ringer solution
(D) Three percent hypertonic saline
(E) Normal saline

Answer Contents

142. A 47-year-old woman undergoes laparoscopic hysterectomy for fibroids with dense bladder adhesions from two prior cesarean deliveries. The bladder flap was difficult, and a large pedicle was taken near the cervix. Cystoscopy is performed before closing, and the bladder wall appears intact, with clear urine in the catheter bag. Which of the following cystoscopic findings best confirms ureteral integrity?

(A) Clear urine draining into the catheter bag
(B) Normal mucosa across the whole bladder dome
(C) A bladder that holds 300 mL without visible leak
(D) Urine efflux seen from both ureteral orifices
(E) No gross blood in the bladder lumen

Answer Contents

143. A 38-year-old never-smoker has a routine urinalysis showing 6 red blood cells per high-power field. She has no history of gross hematuria, no urinary symptoms, no pelvic radiation, and no exposure to dyes or chemicals. She is not menstruating, and the specimen was properly collected. The patient asks whether she needs a bladder scope now. According to the AUA/SUFU microhematuria guideline, which of the following is the best next step?

(A) Repeat urinalysis within six months
(B) Immediate office cystoscopy
(C) Multiphasic CT urography
(D) Urine cytology and urinary tumor marker testing
(E) Renal and bladder ultrasound with cystoscopy

Answer Contents

144. A 63-year-old woman with a 40 pack-year smoking history has 30 red blood cells per high-power field on two urinalyses. Her gynecologist sent urine cytology, and the result is negative. She asks whether the negative result means she can skip further workup. Renal function is normal. Which of the following is the best next step?

(A) Repeat urine cytology in six months
(B) No further workup because cytology was negative
(C) Renal ultrasound alone
(D) Cystoscopy with axial upper tract imaging
(E) Repeat urinalysis in 12 months

Answer Contents

145. A 67-year-old woman has gross hematuria and a 35 pack-year smoking history. Cystoscopy is planned. She has a documented severe reaction to iodinated contrast, with normal renal function, and has no implants or other contraindications to magnetic resonance imaging. The team needs to choose upper tract imaging to pair with cystoscopy. Which of the following is the best choice?

(A) Urine cytology
(B) Repeat imaging in 12 months
(C) MR urography
(D) Intravenous pyelography
(E) Renal ultrasound alone

Answer Contents

146. A 40-year-old woman is having hysteroscopic resection of a submucous myoma with a bipolar system and saline delivered by an automated pump. The surgeon asks the nurse to raise the intrauterine pressure to 150 mm Hg for a sharper view. Her blood pressure is 118/72 mm Hg, which gives a mean arterial pressure of about 87 mm Hg. Which of the following is the best response based on the BSGE/ESGE guideline?

(A) Raise the pressure because higher pressure lowers fluid absorption in all cases
(B) Keep the pressure as low as the view allows and under mean arterial pressure
(C) Raise the pressure because saline absorption cannot cause any harm to her
(D) Set the pressure over the systolic pressure to stop the bleeding completely
(E) Ignore the pressure because measuring the deficit makes pressure irrelevant

Answer Contents

147. A 36-year-old woman is scheduled for hysteroscopic resection of a 2.5-cm type 1 submucous myoma. Her surgeon, worried about hyponatremia, proposes carbon dioxide gas as the distension medium so that no fluid deficit will accumulate. Her sodium and renal function are normal. Which of the following is the best response based on the BSGE/ESGE guideline?

(A) Carbon dioxide is the preferred medium with bipolar energy
(B) Carbon dioxide is acceptable for operative work if the pressure is low
(C) Carbon dioxide is preferred when the patient has kidney disease
(D) Carbon dioxide is acceptable for operative work on type 0 myomas only
(E) Carbon dioxide should not be used for operative hysteroscopy

Answer Contents

148. A 41-year-old premenopausal woman will have hysteroscopic resection of a 3-cm type 1 submucous myoma next month using a bipolar system and saline. Her surgeon asks which preoperative measure the BSGE/ESGE guideline suggests considering to reduce fluid absorption and electrolyte disturbance. She has no contraindication to any of the drugs listed.

(A) Oral furosemide given the evening before
(B) High-dose estrogen given for the two weeks before
(C) A GnRH agonist given before the resection
(D) Tranexamic acid given intravenously at induction
(E) Oral iron therapy given for the month before

Answer Contents

149. A hospital gynecology unit is writing a protocol for operative hysteroscopy in which fluid absorption is anticipated, such as resection of a large submucous myoma expected to last 45 minutes. The nurse manager asks the surgeons how often the fluid deficit must be recorded during surgery. Based on the BSGE/ESGE guideline, which of the following is the correct monitoring interval?

(A) At least every 30 minutes
(B) At least every 10 minutes
(C) At the end of the procedure
(D) At least every hour
(E) At least every 20 minutes

Answer Contents

150. A resident asks why the department has moved most resections from monopolar glycine systems to bipolar saline systems. A 45-year-old woman is about to undergo hysteroscopic removal of a type 2 myoma with saline, and she asks whether the change makes the operation safer if some fluid is absorbed. Which of the following best explains why isotonic media are considered safer than hypotonic media?

(A) Isotonic fluid is absorbed in smaller volumes
(B) Isotonic fluid carries no limit on fluid deficit
(C) Isotonic fluid permits higher uterine pressure without absorption
(D) Isotonic fluid is required for monopolar energy
(E) Absorbed isotonic fluid does not cause hyponatremia

Answer Contents

151. A 44-year-old woman with heavy menstrual bleeding and iron deficiency anemia has completed childbearing. Endometrial biopsy is benign, and ultrasound shows a normal cavity. She uses a copper intrauterine device for contraception and requests endometrial ablation next month with an impedance-controlled system. Which of the following statements about her plan is correct based on the device labeling?

(A) The IUD protects the lining and allows a lower energy setting
(B) The IUD must stay in place to prevent pregnancy afterward
(C) Ablation can proceed with the IUD if the strings are trimmed
(D) Ablation is contraindicated while the IUD remains in place
(E) Ablation can follow IUD insertion on the same day

Answer Contents

152. A 38-year-old woman with heavy menstrual bleeding has completed childbearing and has a benign endometrial biopsy. She had treatment for intrauterine adhesions years ago and is now planned for impedance-controlled endometrial ablation. Hysteroscopy shows a small, scarred cavity, and the measured cavity length is 3.5 cm. Which of the following best describes the effect of this finding on the planned procedure?

(A) It is a labeled contraindication to the procedure
(B) It lowers the energy needed and has no other effect
(C) It is acceptable if the biopsy is benign
(D) It is an indication to add progestin after treatment
(E) It requires a second pass to complete treatment

Answer Contents

153. A 49-year-old woman with heavy menstrual bleeding is scheduled for endometrial ablation. Her office biopsy 3 months ago showed adenomatous hyperplasia, and after a course of hormonal therapy a repeat biopsy shows the same finding. She has completed childbearing and asks to go ahead because she prefers a short procedure. Which of the following best explains why ablation is not appropriate now?

(A) Hormonal therapy has already thinned the endometrium too much
(B) Unresolved premalignant change is a labeled contraindication
(C) Hyperplasia contraindicates ablation only in women over age 50
(D) Ablation would worsen hyperplasia by stimulating endometrial regrowth
(E) A biopsy within the prior year rules out ablation for any patient

Answer Contents

154. A 40-year-old woman with heavy menstrual bleeding is planning endometrial ablation. She has completed childbearing, and her partner has not had a vasectomy. She tells you that she assumes the procedure will make pregnancy impossible, so she plans to stop using condoms afterward. Which of the following is the most appropriate counseling point based on the device labeling?

(A) Pregnancy after ablation needs no contraception after one year
(B) Ablation prevents implantation by removing the entire lining
(C) Pregnancy after ablation can be dangerous for mother and fetus
(D) Ablation provides sterilization equal to tubal occlusion
(E) Pregnancy after ablation is rare but carries no added risk

Answer Contents

155. A 41-year-old woman with heavy menstrual bleeding is scheduled for endometrial ablation tomorrow. At the preoperative visit she reports 3 days of pelvic pain and purulent vaginal discharge. Temperature is 38.2 C, the cervix is tender with mucopurulent drainage, and a nucleic acid test returned positive for chlamydia. Which of the following is the best next step?

(A) Convert the plan to hysterectomy tomorrow
(B) Proceed with ablation and add intraoperative antibiotics
(C) Proceed with ablation using a lower energy setting
(D) Cancel the case and offer no further endometrial treatment
(E) Postpone the ablation and treat the infection

Answer Contents

156. A 42-year-old woman had a loop excision 14 months ago for CIN 3 with negative margins. She has had her initial post-treatment HPV tests and all were negative. She asks how long she must keep having cervical cancer screening, and says she would like to stop once the first tests are normal. According to the 2019 ASCCP guideline, which of the following is the correct long-term plan?

(A) Colposcopy every year for 5 years, then stop screening
(B) Routine 5-year cotesting resumed right after the first normal tests
(C) Annual cytology alone continued for the next 10 years
(D) HPV testing or cotesting at 3-year intervals for at least 25 years
(E) Screening stopped once the initial post-treatment tests are negative

Answer Contents

157. A 33-year-old nonpregnant woman has a biopsy-proven CIN 2 lesion after an HSIL cytology result. Colposcopy was adequate, the squamocolumnar junction was fully seen, and endocervical sampling was negative. She asks which treatment the guideline favors and whether freezing is acceptable. She has no history of cervical treatment. Which of the following best reflects the 2019 ASCCP guideline position?

(A) Excision is preferred, and ablation is acceptable
(B) Hysterectomy is preferred, and excision is acceptable
(C) Observation is preferred, and excision is acceptable
(D) Excision is required, and ablation is not acceptable
(E) Ablation is preferred, and excision is acceptable

Answer Contents

158. A 36-year-old woman has cytology showing atypical glandular cells that favor neoplasia. Colposcopy, endocervical sampling and endometrial sampling show no high-grade or invasive disease. A diagnostic excisional procedure is planned, and the gynecologist wants the specimen to be useful for pathologists. According to the 2019 ASCCP guideline, which of the following is the best approach?

(A) Excision in several fragments to shorten operating time
(B) Cryotherapy ablation of the whole transformation zone
(C) Excision of the transformation zone as one intact specimen
(D) A single punch biopsy taken at the squamocolumnar junction
(E) Endocervical curettage as the only tissue sampling

Answer Contents

159. A 28-year-old woman at 12 weeks of gestation has cytology reported as atypical glandular cells, not otherwise specified, and is referred for colposcopy. The colposcopist sees an acetowhite lesion at the squamocolumnar junction and takes a directed biopsy. A resident suggests also performing endocervical curettage, as is done at initial colposcopy in nonpregnant women with this cytology. Which of the following is the best response based on the 2019 ASCCP guideline?

(A) Omit endocervical sampling because she is pregnant
(B) Repeat colposcopy at 24 weeks without any biopsy
(C) Perform endocervical curettage with a larger curette
(D) Obtain an endocervical brush only after delivery of the biopsy
(E) Perform a cold knife cone to avoid sampling

Answer Contents

160. A 31-year-old nonpregnant woman has a biopsy showing CIN 3 after HSIL cytology, with adequate colposcopy and a negative endocervical sample. She would like to delay treatment for a year because she is trying to conceive and wants repeat colposcopy every 6 months in the meantime. Which of the following statements best reflects the 2019 ASCCP guideline?

(A) Observation is acceptable if cytology stays normal
(B) Observation is acceptable for 12 months with colposcopy
(C) Observation is acceptable if the patient is under age 35
(D) Treatment is recommended and observation is unacceptable
(E) Treatment is optional if the lesion is small

Answer Contents

G6.3. Major Open Procedures

161. A 48-year-old woman with a 14-week-size fibroid uterus and heavy menstrual bleeding is scheduled for hysterectomy. The enlarged uterus and a prior myomectomy with pelvic adhesions make a vaginal hysterectomy not feasible. She asks whether laparoscopy offers any recovery benefit over open abdominal surgery. Which of the following is a documented advantage of laparoscopic over open abdominal hysterectomy?

(A) Lower rates of vaginal cuff dehiscence
(B) Lower rates of bladder and ureteral injury than open surgery
(C) Shorter operating time and lower hospital cost overall
(D) Easier removal of a very large uterus
(E) Shorter hospital stay and fewer wound infections

Answer Contents

162. A 45-year-old woman with symptomatic fibroids is scheduled for hysterectomy. She asks to keep her cervix, thinking it would lower the chance of bleeding, infection, and injury to her bladder or bowel. She has normal cervical screening results and has no history of cervical dysplasia. Which of the following is the most accurate counseling based on ACOG guidance?

(A) Higher bowel injury rate with total hysterectomy
(B) No clinically significant difference in complications overall
(C) Fewer infections and less blood loss with supracervical surgery overall
(D) Lower ureteral injury and bowel injury risk with supracervical surgery
(E) Better sexual function after total hysterectomy

Answer Contents

163. A 47-year-old woman has heavy menstrual bleeding unresponsive to medical therapy and requests hysterectomy. Examination shows a mobile 8-week-size uterus and a normal vagina, ultrasonography shows normal adnexa, and she has no history of pelvic surgery. Vaginal and laparoscopic approaches are both available, and her surgeon is skilled in each. Which of the following does the ACOG committee opinion on route of hysterectomy state?

(A) Laparoscopic hysterectomy is the approach of choice whenever feasible
(B) Robot-assisted hysterectomy is preferred whenever the platform is available
(C) Abdominal hysterectomy is preferred when the indication is uterine bleeding
(D) Vaginal hysterectomy is the approach of choice whenever feasible
(E) All routes are equivalent, so the choice rests on surgeon preference alone

Answer Contents

164. A 52-year-old woman with fibroids and heavy menstrual bleeding is deciding between conventional laparoscopic and robot-assisted laparoscopic hysterectomy. Both are available, and her gynecologist is experienced in each. She asks what professional guidance says about the role of robotic assistance. Which of the following best describes the position of the ACOG opinion on route of hysterectomy?

(A) Robotic assistance should replace the vaginal route
(B) Robotic assistance has an undetermined role and needs more data
(C) Robotic assistance is not recommended for any hysterectomy patient
(D) Robotic assistance is the clearly preferred route for benign disease
(E) Robotic assistance is preferred for any enlarged uterus

Answer Contents

165. A 44-year-old woman is scheduled for vaginal hysterectomy for symptomatic prolapse and heavy bleeding. She has read about removing the fallopian tubes to reduce ovarian cancer risk and asks whether this requires a laparoscopic approach instead. Her surgeon performs vaginal salpingectomy regularly. Which of the following is the best response based on the ACOG opinion?

(A) Vaginal salpingectomy can usually be done safely
(B) Salpingectomy should be done at a separate laparoscopy
(C) Salpingectomy requires conversion to an open abdominal approach first
(D) Salpingectomy adds nothing at the time of hysterectomy
(E) Vaginal salpingectomy is unsafe and should not be attempted at all

Answer Contents

166. A surgeon is reviewing five women ages 40 to 55 who are scheduled for hysterectomy for benign indications and wants to identify the one most favorable for a vaginal approach. All are healthy and have similar vaginal access, and each option describes one woman's pathology. Which of the following patients is the most favorable candidate for the vaginal route according to ACOG guidance?

(A) A suspected ovarian malignancy needing staging
(B) A fibroid uterus that reaches the umbilicus
(C) Stage IV endometriosis with suspected dense pelvic adhesions present
(D) A known adnexal mass needing removal, with suspected adhesions
(E) Uterine-only pathology, no adnexal disease, no suspected adhesions

Answer Contents

167. A 34-year-old woman with intramural fibroids and heavy bleeding plans abdominal myomectomy to preserve fertility. She asks whether a laparoscopic or robotic approach would make a repeat operation less likely than the open approach. Her surgeon is skilled in all three. According to ACOG Practice Bulletin 228, which of the following is the best counseling statement?

(A) Reintervention is lower after a robotic myomectomy
(B) Reintervention is lower after a laparoscopic myomectomy than open
(C) Reintervention is about 10 to 12 percent whichever route is used
(D) Reintervention is higher after any minimally invasive route chosen
(E) Reintervention is lower after an abdominal myomectomy

Answer Contents

168. A 36-year-old woman had an abdominal myomectomy 2 years ago for five fibroids and is considering a second pregnancy. She asks how likely fibroids are to come back after her operation. Ultrasound now shows two new small intramural fibroids. According to ACOG Practice Bulletin 228, which of the following is the reported recurrence rate after myomectomy?

(A) 25 percent by 40 months
(B) 10 percent by 60 months
(C) 75 percent by 40 months
(D) 50 percent by 24 months
(E) 5 percent by 12 months

Answer Contents

169. A 38-year-old woman with heavy menstrual bleeding and anemia has a single 3 cm type 2 submucosal fibroid on saline infusion sonography. She wants to keep her uterus and may try to conceive. Medical therapy has failed. Which of the following is the most appropriate procedure according to ACOG Practice Bulletin 228?

(A) Long-term GnRH agonist therapy
(B) Hysteroscopic myomectomy
(C) Abdominal myomectomy
(D) Hysterectomy
(E) Uterine artery embolization

Answer Contents

170. A 43-year-old woman with a 12-week-size fibroid uterus, hemoglobin of 8.6 g/dL, and heavy bleeding is scheduled for myomectomy in 4 months. Her surgeon prescribes a GnRH agonist to improve hemoglobin and shrink the fibroids first. The patient asks how long this drug can be used without added hormone therapy. Which of the following is the maximum duration given in ACOG Practice Bulletin 228?

(A) 3 months without add-back therapy
(B) Unlimited with careful monitoring
(C) 6 months without add-back therapy
(D) 12 months without add-back therapy
(E) 24 months without add-back therapy

Answer Contents

171. A 45-year-old woman with a 14-week-size fibroid uterus is offered a 6-month course of a GnRH agonist. She asks whether the drug could be used to avoid surgery, since her fibroids shrank by a third on a previous short course. Her symptoms returned a few months after stopping. Which of the following best explains why the GnRH agonist is not a definitive treatment?

(A) The drug shrinks submucosal fibroids and no other type
(B) The drug permanently damages fibroid tissue after one course
(C) The drug stops working after the first month of use
(D) Fibroids regrow within 6 to 9 months after stopping the drug
(E) Fibroids regrow when add-back estrogen therapy is started

Answer Contents

172. A 34-year-old woman who carries a BRCA1 pathogenic variant has finished childbearing and has had normal pelvic examinations. Her mother died of ovarian cancer at age 48. She asks at what age ACOG typically recommends risk-reducing salpingo-oophorectomy for BRCA1 carriers. Which of the following is the best answer?

(A) Age 35 to 40 years
(B) Age 45 to 50 years
(C) Age 40 to 45 years
(D) Age 50 to 55 years
(E) Age 30 to 33 years

Answer Contents

173. A 36-year-old woman with a BRCA2 pathogenic variant has two children and asks about risk-reducing salpingo-oophorectomy. She learns that BRCA1 carriers are advised surgery between 35 and 40 and asks why her gynecologist says she may consider waiting until age 40 to 45. Which of the following is the best explanation?

(A) BRCA2 carriers have no ovarian cancer risk
(B) BRCA2 carriers need preoperative chemotherapy
(C) Surgery is less effective in BRCA2 carriers
(D) Surgery before age 40 raises breast cancer risk
(E) Ovarian cancer begins later in BRCA2 carriers

Answer Contents

174. A 39-year-old BRCA1 carrier has finished childbearing. She asks whether twice-yearly transvaginal ultrasound and CA-125 testing could replace surgery, since she dislikes the thought of menopause. Her sister died of ovarian cancer at 44. According to ACOG Practice Bulletin 182, which of the following is the best statement?

(A) Surgery is the only proven way to cut ovarian cancer mortality
(B) Ultrasound with CA-125 is proven to cut ovarian cancer mortality
(C) Salpingectomy alone is proven to cut ovarian cancer mortality
(D) Prophylactic hysterectomy is proven to cut ovarian cancer mortality
(E) Oral contraceptives are proven to cut ovarian cancer mortality

Answer Contents

175. A 41-year-old BRCA1 carrier is having laparoscopic risk-reducing bilateral salpingo-oophorectomy. The resident asks what should be done at surgery to make the operation complete and to look for occult disease. Which of the following is the best surgical principle based on ACOG Practice Bulletin 182?

(A) Leave a rim of ovarian tissue on the pedicle to preserve hormone output
(B) Take pelvic washings only if ascites is found
(C) Remove the ovaries and leave the tubes for later pathology sampling
(D) Skip peritoneal inspection when the pelvis looks normal
(E) Remove all ovarian and tubal tissue and take washings

Answer Contents

176. A 46-year-old woman with average ovarian cancer risk is having hysterectomy for fibroids. She asks the surgeon to remove her ovaries to prevent ovarian cancer. She is premenopausal, has no family history of cancer, and has normal bone density. Which of the following statements is the best counseling point based on ACOG guidance?

(A) Oophorectomy is required to lower the ovarian cancer risk at all
(B) Oophorectomy lowers the risk of osteoporosis
(C) Salpingectomy at hysterectomy alone causes surgical menopause
(D) Surgical menopause raises cardiovascular and osteoporosis risk
(E) Oophorectomy has no effect on cardiovascular risk

Answer Contents

177. A 33-year-old woman has two children and requests permanent contraception, to be done at a laparoscopic operation next month. She has no family history of cancer and asks which tubal procedure is preferred, since she has read about ovarian cancer prevention. Based on ACOG guidance summarized by ObG Project, which of the following is the best recommendation?

(A) Partial salpingectomy with a Pomeroy technique
(B) Bilateral tubal ligation with clips
(C) Complete bilateral salpingectomy
(D) Hysteroscopic tubal occlusion
(E) Bilateral fimbriectomy

Answer Contents

178. A 28-year-old woman at 7 weeks of gestation has had a day of right lower quadrant pain. She is mildly tender but stable, with a heart rate of 88/min and blood pressure of 118/72 mm Hg. Serum hCG is 3,100 mIU/mL. Transvaginal ultrasonography shows a 2.5-cm right tubal mass without cardiac activity and a large amount of free fluid with echogenic material extending to Morison pouch. Which of the following is the most appropriate management according to ACOG Practice Bulletin 193?

(A) Single-dose intramuscular methotrexate with weekly hCG monitoring
(B) Surgical management, because intraperitoneal bleeding is present
(C) Two-dose methotrexate protocol with hCG on day 4 and day 7
(D) Expectant management with a repeat hCG in 48 hours
(E) Dilation and curettage to exclude a failed intrauterine pregnancy

Answer Contents

179. A 30-year-old woman, gravida 2 para 1, has a ruptured right tubal ectopic pregnancy and is going for laparoscopy. Her left tube has hydrosalpinx on prior hysterosalpingography, and she strongly wants future pregnancy. The surgeon asks the resident how the choice between salpingostomy and salpingectomy should be made. Which of the following best reflects the guideline principle?

(A) Individualize by clinical status and desire for future fertility
(B) Choose salpingostomy because it clearly improves later fertility
(C) Decide by the gestational sac size alone
(D) Decide by the initial hCG level alone
(E) Choose salpingectomy because of the persistent trophoblast risk of salpingostomy

Answer Contents

180. A 26-year-old woman with a 2 cm unruptured tubal pregnancy, no pain, and an initial hCG of 1,400 mIU/mL is hemodynamically stable and declines surgery. She has no contraindication to methotrexate. The resident asks which protocol the ACOG bulletin suggests may be most appropriate for her. Which of the following is the best answer?

(A) A two-dose protocol
(B) A single-dose protocol
(C) Salpingostomy instead of methotrexate
(D) A fixed multidose protocol with leucovorin
(E) A three-dose protocol

Answer Contents

181. A 27-year-old woman received single-dose methotrexate for a tubal ectopic pregnancy 7 days ago. Her hCG has dropped by 40 percent from day 4 to day 7 and she has no pain. She asks if she can stop follow-up now that her levels are falling and she feels well. Which of the following is the best response?

(A) Stop monitoring now because the level has fallen
(B) Schedule laparoscopy to confirm that the pregnancy has resolved
(C) Monitor hCG serially until it reaches a non-pregnancy level
(D) Repeat the dose now to hasten the decline of hCG
(E) Test again after her next period

Answer Contents

G6.4. Pelvic Floor Surgery

182. A 71-year-old woman at an annual visit has an anterior vaginal wall that descends to within 1 cm of the hymen on Valsalva, consistent with stage II prolapse. She reports no bulge, pelvic pressure, or voiding, bowel, or sexual complaints. She is not sexually active and is content with her health. Her daughter asks whether a repair should be scheduled. Which of the following is the most appropriate management?

(A) Anterior colporrhaphy with apical suspension, to prevent progression of the stage
(B) Laparoscopic sacrocolpopexy, to treat the stage II defect before it progresses
(C) Fitting of a ring pessary, to support the anterior wall
(D) Partial colpocleisis, to close the vaginal canal
(E) Observation with reassurance and a return visit for symptoms

Answer Contents

183. A 63-year-old woman is staged with the POP-Q system before surgical planning. Total vaginal length is 9 cm. On maximal Valsalva the leading edge of the prolapse, the anterior vaginal wall, lies 3 cm beyond the hymen. The posterior wall and the cervix are well supported. Which of the following is the correct POP-Q stage of her prolapse?

(A) Stage 0, normal support of the anterior wall
(B) Stage I, descent more than 1 cm proximal to the plane of the hymen
(C) Stage II, descent within 1 cm of either side of the hymen
(D) Stage III, descent beyond the hymen but short of eversion
(E) Stage IV, complete eversion of the vagina through the introitus

Answer Contents

184. A 66-year-old woman has a stage III cystocele that she reports as a heavy vaginal bulge. She has no urinary leakage and a normal urinalysis. She is scheduled for a vaginal repair. During counseling, which of the following is the most appropriate step before surgery?

(A) Reassure her that without leakage now, postoperative stress incontinence will not occur
(B) Perform cystoscopy to exclude a urethral abnormality
(C) Start an antimuscarinic to prevent postoperative urgency
(D) Test for latent stress incontinence with the prolapse reduced and counsel her about it
(E) Order a pelvic MRI to define the extent of the cystocele

Answer Contents

185. A 78-year-old woman with a long-standing, unreduced stage IV uterine prolapse reports fatigue and mild left flank discomfort. Blood pressure is 142/84 mm Hg and she is afebrile. Serum creatinine is 1.6 mg/dL, compared with 0.9 mg/dL two years ago, and urinalysis is bland. Which of the following is the most likely explanation for her renal findings?

(A) Ureteral obstruction from the advanced prolapse
(B) Vesicovaginal fistula formation
(C) Detrusor overactivity with reflux
(D) Renal artery stenosis
(E) Chronic pyelonephritis from a urethral diverticulum

Answer Contents

186. A 44-year-old sexually active woman has symptomatic stage III uterovaginal prolapse and wants a definitive operation. She plans to remain sexually active and has no major comorbidity. The surgeon reviews the general aim of prolapse surgery for a woman like her. Which of the following best describes that aim?

(A) Close the vaginal canal to prevent any recurrence
(B) Remove the vaginal epithelium and plicate the levator muscles
(C) Narrow the introitus with perineorrhaphy as the main step
(D) Restore support while preserving vaginal length and axis
(E) Shorten the vagina by excising redundant vaginal tissue

Answer Contents

187. A 55-year-old woman with a body mass index of 34 kg/m2, four vaginal deliveries, and a recently diagnosed stage II cystocele asks which of her features contributed to the prolapse and which she can change. Which of the following is both an established risk factor for prolapse and modifiable?

(A) Age
(B) Obesity
(C) Parity
(D) Vaginal delivery history
(E) Cystocele stage

Answer Contents

188. A 58-year-old woman has stage III uterine prolapse with the cervix 4 cm beyond the hymen, a heavy bulge, and no uterine pathology. She chooses vaginal hysterectomy. Her surgeon asks what should accompany the hysterectomy to lower the risk of recurrence. Which of the following is the best answer according to ACOG and AUGS guidance?

(A) Hysterectomy alone, because uterine removal restores apical support
(B) Anterior colporrhaphy alone
(C) Vaginal apex suspension at the time of hysterectomy
(D) Sacrospinous fixation as a separate staged operation
(E) Perineorrhaphy to narrow the genital hiatus

Answer Contents

189. A 62-year-old woman has a stage III anterior wall prolapse with the cervix descending to 2 cm beyond the hymen on Valsalva. The surgeon proposes anterior colporrhaphy alone because the bulge she notices is anterior. Which of the following is the best recommendation?

(A) Perform anterior colporrhaphy alone and reassess in one year
(B) Add posterior colporrhaphy prophylactically
(C) Add a sacrospinous ligament mesh arm
(D) Add a midurethral sling prophylactically
(E) Correct the apical defect together with the anterior defect

Answer Contents

190. A 49-year-old woman has a symptomatic stage III posterior wall prolapse with a bulge on defecation and no fecal incontinence. She has completed childbearing and wants a surgical repair. The surgeon is choosing the route. According to ACOG and AUGS guidance, which of the following is the best approach?

(A) Transanal rectocele repair
(B) Abdominal sacrocolpopexy alone
(C) Posterior repair with a synthetic mesh insert
(D) Vaginal posterior wall repair
(E) Fixation of the perineal body to the pubic bone

Answer Contents

191. A surgeon reviews several women with post-hysterectomy or uterine prolapse to choose who is the best candidate for abdominal sacrocolpopexy, using the factors ACOG and AUGS link to higher recurrence risk. Each woman is otherwise healthy and has completed childbearing. Which of the following women is the best candidate?

(A) A 71-year-old with stage II prolapse and a body mass index of 24 kg/m2
(B) A 54-year-old with stage IV prolapse and a body mass index of 29 kg/m2
(C) A 64-year-old with stage III prolapse and a body mass index of 24 kg/m2
(D) A 69-year-old with stage II prolapse and a body mass index of 28 kg/m2
(E) A 76-year-old with stage III prolapse and a body mass index of 22 kg/m2

Answer Contents

192. A 46-year-old woman with stage III uterine prolapse wishes to keep her uterus and prefers a vaginal operation. Her surgeon describes a uterus-preserving apical procedure in which the cervix is suspended to a pelvic ligament through a vaginal incision. Which of the following structures is the cervix suspended to?

(A) Anterior longitudinal ligament at the sacral promontory
(B) Iliopectineal ligament
(C) Arcus tendineus fasciae pelvis
(D) Obturator internus fascia
(E) Sacrospinous ligament

Answer Contents

193. A 49-year-old woman with stress urinary incontinence is scheduled for a midurethral sling. After seeing a news report about the 2019 FDA order on vaginal mesh, she asks whether her sling operation was banned. Which of the following is the best response?

(A) The order applied to transvaginal prolapse mesh, not to slings or abdominal mesh
(B) The order removed vaginal mesh products, so her sling would use native tissue instead
(C) The order halted mesh used in both sacrocolpopexy and slings in equal measure across the market
(D) The order applied to mesh slings passed through the obturator foramen
(E) The order has no relevance to any mesh procedure performed in the United States today

Answer Contents

194. A hospital pharmacy and therapeutics committee is reviewing surgical mesh products. The chair asks a urogynecologist what the FDA required of manufacturers of surgical mesh for transvaginal prolapse repair after the agency reclassified these devices in 2016. Which of the following is the most accurate description of the 2016 regulatory action?

(A) Reclassification to class II, allowing clearance based on a similar existing device
(B) A voluntary manufacturer recall with no new regulatory requirement for the devices
(C) Reclassification to class III, with clinical data needed for premarket approval
(D) Removal of the devices for prolapse and for stress incontinence alike
(E) Reclassification to class I, exempting the mesh from premarket review altogether

Answer Contents

195. A 45-year-old woman has a symptomatic stage III rectocele with no prior surgery. The surgeon suggests placing a synthetic mesh or graft to reinforce a posterior colporrhaphy so that recurrence is less likely. Which of the following is the best response according to ACOG and AUGS guidance?

(A) Use mesh, because grafts are the accepted standard for primary posterior wall repair in this setting
(B) Use mesh if the rectocele is smaller than stage III, and a graft if it is larger
(C) Perform native tissue repair, since mesh is not for routine primary repair
(D) Use a biologic graft, because biologic material carries no complication risk of its own
(E) Use mesh together with routine cystoscopy as a standard surgical bundle

Answer Contents

196. A 61-year-old woman has recurrent stage III vault prolapse 3 years after a native tissue vaginal repair. She is healthy and wants a durable correction. Her surgeon performs mostly vaginal hysterectomies and offers to place a graft at an abdominal sacrocolpopexy after watching a short manufacturer video. The patient asks what ACOG and AUGS expect of a surgeon who places mesh or biologic grafts for prolapse. Which of the following is the most accurate response?

(A) Any gynecologic surgeon with general operative privileges is qualified
(B) A manufacturer product video is enough preparation for a first case
(C) Only surgeons who completed a urology fellowship may place a graft
(D) Surgeons should have training specific to mesh and graft procedures
(E) Graft placement is restricted to surgeons working in academic centers

Answer Contents

197. A 57-year-old woman is scheduled for laparoscopic sacrocolpopexy for post-hysterectomy vault prolapse. After reading about the FDA warning on vaginal mesh, she asks whether her abdominal mesh operation is now restricted. According to the 2013 American Urogynecologic Society position statement, which of the following is the best response?

(A) There is no justification for restricting mesh placed abdominally
(B) Abdominal mesh carries the same restriction as transvaginal prolapse mesh
(C) Abdominal mesh is permitted but the robotic approach is prohibited
(D) Abdominal mesh is restricted until a national registry exists
(E) Abdominal mesh is reserved for women with prior failed native tissue repair

Answer Contents

198. A hospital credentialing committee proposes banning all surgical mesh for pelvic floor disorders after hearing about the FDA warning. The chair asks a urogynecologist for the alternative recommended in the 2013 AUGS position statement that protects patients while keeping options available. Which of the following is the best recommendation?

(A) A complete ban on mesh for stress incontinence and prolapse at the hospital level
(B) A ban limited to surgeons performing fewer than ten cases yearly
(C) Abandonment of sacrocolpopexy in favor of vaginal procedures across the department
(D) Self-attestation of competence by surgeons who request new privileges
(E) Credentialing guidelines, with proctoring for new sacrocolpopexy privileges

Answer Contents

199. A 56-year-old woman has had pelvic pain and dyspareunia for 18 months since a transvaginal mesh prolapse repair elsewhere. Examination shows a tender band in the anterior vaginal wall and no exposed mesh. She asks whether mesh removal will definitely cure her pain. According to the 2011 ACOG and AUGS statement on vaginal mesh, which of the following is the most appropriate counseling?

(A) Complete removal is feasible and relieves the pain in the large majority
(B) Complete removal may not be possible, and some pain may persist after surgery
(C) Removal is contraindicated because it worsens prolapse outcomes in most women
(D) Pain after a mesh repair resolves on its own within about five years
(E) Mesh pain is not a recognized complication of vaginal prolapse mesh repair

Answer Contents

200. An 81-year-old frail woman with stage IV vaginal vault prolapse is not sexually active and does not wish to have intercourse. She prefers a single, durable operation under regional anesthesia. The surgeon is considering colpocleisis and begins consent. Which of the following must be included in counseling?

(A) Vaginal length is preserved
(B) Future vaginal mesh augmentation remains easy
(C) Hysteroscopy remains possible
(D) Vaginal length will be lost
(E) A pessary can still be fitted later

Answer Contents

201. A 79-year-old woman with hypertension and coronary disease had a prior failed sacrospinous suspension and now has recurrent stage III vault prolapse. She is not sexually active. She wants the option with the best chance of avoiding another operation. Which of the following is the best choice, based on the available review evidence?

(A) Repeat sacrospinous suspension, which is more durable than obliterative surgery
(B) Colpocleisis, which tends to be more durable than reconstruction
(C) Vaginal mesh kit, which has the lowest reoperation rate of the available options
(D) Anterior colporrhaphy alone, which restores apical support adequately in this case
(E) Observation, because recurrence after obliterative surgery cannot be prevented

Answer Contents

202. A 77-year-old woman is on postoperative day 3 after LeFort partial colpocleisis. She has dysuria and urinary frequency, a temperature of 38.1 C, and is otherwise stable with normal oxygen saturation. Urinalysis shows positive nitrite and leukocyte esterase. The incision is clean. Which of the following is the most likely diagnosis?

(A) Postoperative pulmonary embolism with pleuritic pain
(B) Postoperative ureteral obstruction or kinking
(C) Postoperative urinary tract infection
(D) Postoperative pelvic hematoma
(E) Postoperative vesicovaginal fistula formation

Answer Contents

203. A 76-year-old woman has a stage IV vault prolapse and bothersome stress urinary incontinence when the prolapse is reduced. She chooses colpocleisis with a concomitant midurethral sling. She asks what proportion of women are continent afterward. According to the narrative review of colpocleisis series, which of the following ranges is closest?

(A) 40.2% to 52%
(B) 55.5% to 66%
(C) 86.8% to 94%
(D) 68.2% to 78%
(E) 98.1% to 100%

Answer Contents

204. A 60-year-old woman has stage III cystocele and bothersome stress urinary incontinence demonstrated on cough stress testing. She is planning vaginal prolapse repair and wants to treat both problems at one operation. According to the AUA/SUFU guideline, which of the following statements is correct?

(A) A Burch colposuspension is the required choice at prolapse surgery
(B) A midurethral sling is contraindicated at the time of prolapse surgery in most women
(C) A bulking agent is the required option, to protect the prolapse repair
(D) Incontinence surgery must be delayed for 12 months to avoid postoperative urinary retention
(E) Any of a midurethral sling, pubovaginal sling, or Burch colposuspension

Answer Contents

205. A 70-year-old woman reports leakage with coughing and has stage III prolapse. Her cough stress test is negative with a full bladder. After the prolapse is reduced with a speculum blade, repeat cough testing still shows no leakage. She is considering a sling at the time of her prolapse repair. According to the AUA/SUFU guideline, which of the following is the best next step?

(A) Additional evaluation, such as urodynamics, before a sling
(B) Proceed with a retropubic sling because her history is classic for stress leakage
(C) Perform cystoscopy and place a sling in the same session as the prolapse repair
(D) Start duloxetine for stress incontinence and defer the sling decision
(E) Place a urethral bulking agent empirically at the time of the repair

Answer Contents

206. A 72-year-old woman was dry before surgery. On postoperative day 1 after anterior repair with sacrocolpopexy, she has new leakage with coughing and standing. The urinalysis is normal, the postvoid residual is 30 mL, and the incisions are intact. Which of the following is the most likely mechanism?

(A) Direct injury to the urethral sphincter by the retractor blade during surgery
(B) A vesicovaginal fistula from the anterior repair sutures placed yesterday
(C) Overflow incontinence from postoperative urinary retention after the repair
(D) Detrusor overactivity triggered by the indwelling urinary catheter used overnight
(E) Relief of urethral outlet obstruction that unmasked latent stress incontinence

Answer Contents

207. A 59-year-old woman is having a uterosacral ligament suspension at the time of a vaginal hysterectomy for stage III prolapse. The surgeon asks whether to perform cystoscopy at the end. According to ACOG and AUGS guidance, which of the following is the best answer?

(A) Perform cystoscopy, since the bladder or ureter is at risk
(B) Omit cystoscopy because ureteral injury cannot occur at suspension
(C) Perform cystoscopy when gross hematuria is seen in the operative field
(D) Replace cystoscopy with a postoperative renal ultrasound in the recovery room
(E) Perform cystoscopy if mesh is used in the operation

Answer Contents

208. During a retropubic midurethral sling, the surgeon sees a urethral laceration at cystoscopy after trocar passage. The patient is a 44-year-old woman who still has bothersome stress incontinence. According to the AUA/SUFU guideline, which of the following is the best management?

(A) Leave the mesh sling in place and keep the catheter for 7 days after surgery
(B) Remove the mesh sling, repair the injury, and use a non-synthetic sling
(C) Place a bulking agent through the laceration to seal the urethral defect
(D) Convert to a synthetic single-incision sling of smaller size and lower tension
(E) Abandon incontinence surgery for this patient after the urethral injury

Answer Contents

209. A 48-year-old woman has a symptomatic urethral diverticulum and stress urinary incontinence. She is scheduled for urethral diverticulectomy and wants the incontinence treated at the same time. According to the AUA/SUFU guideline, which of the following is the best approach?

(A) Place a retropubic synthetic midurethral sling through the same surgical field
(B) Place a transobturator synthetic sling because that route spares the urethra
(C) Use a non-synthetic sling rather than a synthetic midurethral sling
(D) Place a synthetic single-incision sling for a shorter operative time
(E) Insert a synthetic sling after waiting 48 hours from the diverticulectomy

Answer Contents

210. A 55-year-old woman with stress urinary incontinence declines synthetic mesh and asks which non-mesh operation has the lower retreatment rate. She is choosing between an autologous fascial pubovaginal sling and open Burch colposuspension. Based on the discussion in the AUA/SUFU guideline, which of the following is the best statement?

(A) Burch colposuspension had lower retreatment than the sling, 13% versus 4%, in the guideline discussion
(B) The two operations had identical retreatment rates in the comparative data reviewed
(C) Burch colposuspension has fewer wound problems than the sling
(D) The autologous sling had lower retreatment than Burch colposuspension, 4% versus 13%
(E) Neither can be offered without synthetic mesh augmentation

Answer Contents

211. A 52-year-old woman with stress urinary incontinence wants the least invasive sling possible and asks about a single-incision mini-sling instead of a standard midurethral sling. She is otherwise healthy and has no prior pelvic surgery. Based on the AUA/SUFU guideline discussion, which of the following should she be told?

(A) It cures more patients than standard slings with fewer complications overall
(B) It matches retropubic slings in head-to-head data and carries no mesh risk
(C) It is the preferred sling for women who have had prior radiation
(D) Objective cure was inferior, with more exposure and repeat surgery
(E) It removes the need for postoperative follow-up visits and examination

Answer Contents

212. A 62-year-old woman had an autologous rectus fascial pubovaginal sling three months ago for stress incontinence. She now notes a bulge at the lower abdominal incision that enlarges when she coughs or stands. She is continent. The incision is healed. Which of the following is the most likely diagnosis?

(A) Mesh extrusion into the bladder from the sling
(B) Incisional hernia at the fascial harvest site
(C) Urethral diverticulum from sling pressure
(D) Inguinal lymphadenopathy after the procedure
(E) Pubic osteitis from the harvest dissection

Answer Contents

213. An 82-year-old frail woman with stress urinary incontinence has not improved with pelvic floor muscle training. She has chronic heart failure, wants the least invasive procedure, and is considering urethral bulking. Which of the following is the most important counseling point about this treatment?

(A) A single injection gives a lifelong cure
(B) Bulking agents have the highest cure rate of any option
(C) She is likely to need repeat injections
(D) A mesh sling is placed with the injection
(E) Postoperative follow-up is not required

Answer Contents

Part 2. Office Practice

OP1.1. Routine Care and Prevention

214. A healthy 76-year-old woman comes for her annual visit. She has had normal biennial screening mammograms since age 50, has no personal or family history of breast cancer, and has no known pathogenic variant. She walks daily and has no major comorbidity. She asks whether to continue mammography. Which of the following best describes the USPSTF position on screening mammography for women her age?

(A) The evidence is insufficient to assess the balance of benefits and harms
(B) Annual screening is recommended through age 85 in healthy women
(C) Biennial screening is recommended, with the same grade as for women in their sixties
(D) Screening is recommended against after the age of 74 years
(E) Screening is recommended when prior mammograms showed dense breasts

Answer Contents

215. A 52-year-old woman has just had her biennial screening mammogram, which was read as negative and showed heterogeneously dense breasts. She has no family history, no prior biopsy and no personal risk factors beyond density. She asks whether she should add breast ultrasonography or magnetic resonance imaging every year to be safe. Which of the following is the USPSTF position on supplemental imaging for women with dense breasts and a negative mammogram?

(A) Supplemental imaging is recommended against with a grade D rating
(B) Biennial whole-breast ultrasonography is recommended with a grade B rating
(C) Supplemental imaging is recommended if she is younger than 50 years
(D) Annual magnetic resonance imaging is recommended with a grade B rating
(E) Evidence is insufficient to assess the balance of benefits and harms

Answer Contents

216. A 45-year-old woman is due for her first screening mammogram. The imaging center offers both conventional digital mammography and digital breast tomosynthesis, often called three-dimensional mammography, at the same price. She has average risk and asks which modality the Task Force prefers. Which of the following statements about the two modalities is consistent with the USPSTF screening statement?

(A) Tomosynthesis should be reserved for women with dense breasts on a prior study
(B) Both modalities are effective forms of mammographic screening
(C) Tomosynthesis is preferred and carries a separate grade A recommendation
(D) Conventional digital mammography is preferred because tomosynthesis lacks evidence
(E) Either modality is acceptable, but tomosynthesis should be repeated annually

Answer Contents

217. A 51-year-old postmenopausal woman has two first-degree relatives with breast cancer and a prior breast biopsy showing atypical hyperplasia. A validated model estimates her 5-year invasive breast cancer risk at 4 percent. She has no history of venous thromboembolism and no other contraindication. She asks about prevention. Which of the following is the USPSTF recommendation for a woman like her?

(A) Offer annual breast magnetic resonance imaging in place of any medication
(B) Offer estrogen-progestin therapy for its protective effect on breast tissue
(C) Offer bilateral prophylactic mastectomy as first-line prevention
(D) Offer no intervention beyond biennial mammography, since medication is reserved for carriers of pathogenic variants
(E) Offer a risk-reducing medication such as tamoxifen, raloxifene or an aromatase inhibitor

Answer Contents

218. A 47-year-old woman has read about breast cancer prevention and asks for tamoxifen. She has no family history, no prior breast biopsy, and a model-estimated 5-year invasive breast cancer risk of about 1 percent. Her mammograms have been normal. She understands the medication has side effects but wants maximum protection. Which of the following is the USPSTF position on risk-reducing medication for women who are not at increased risk?

(A) Routine use is recommended against, with a grade D rating
(B) Use is recommended for women who still menstruate, with a grade C rating
(C) Use is recommended if she also has dense breasts on mammography
(D) Evidence is insufficient to recommend for or against routine use
(E) Use is recommended for women older than 40 years, with a grade B rating

Answer Contents

219. A 46-year-old woman with a body mass index of 27 comes for a preventive visit. She has no bowel symptoms, no family history of colorectal cancer, no inflammatory bowel disease and has never been screened. She believes screening begins at 50 and asks to wait. Which of the following is the most appropriate advice under the USPSTF colorectal cancer screening recommendation?

(A) Screening is recommended at her age if a relative had polyps
(B) Screening should begin at age 50 for average-risk adults
(C) Evidence is insufficient to recommend screening before age 50
(D) Screening is recommended now, with a grade B rating
(E) Screening should wait until rectal bleeding or anemia develops

Answer Contents

220. A 58-year-old woman at average risk declines colonoscopy because of the bowel preparation. She is willing to collect stool at home and prefers the stool-based test with the longest interval between tests. She has no symptoms. Which of the following stool-based strategies and intervals is included in the USPSTF colorectal cancer screening recommendation?

(A) Stool DNA with fecal immunochemical test every 5 years
(B) High-sensitivity guaiac fecal blood test every 5 years
(C) Stool DNA with fecal immunochemical test every 1 to 3 years
(D) Fecal immunochemical test alone, repeated every 3 years
(E) Stool DNA with fecal immunochemical test once every 10 years

Answer Contents

221. A 62-year-old woman at average risk has never been screened and does not want sedation or a stool kit. She asks about an imaging-based test done without sedation. She has no gastrointestinal symptoms and no personal history of polyps. Which of the following is the USPSTF-listed interval for the imaging test that fits her preference?

(A) Computed tomography colonography every 7 years
(B) Computed tomography colonography every 10 years
(C) Computed tomography colonography every 3 years
(D) Computed tomography colonography every year
(E) Computed tomography colonography every 5 years

Answer Contents

222. A 78-year-old woman in good health, with no gastrointestinal symptoms and a negative colonoscopy at age 66, asks if she should be screened again. She lives independently and has no family history of colorectal cancer. Which of the following best describes the USPSTF recommendation for adults aged 76 to 85 years?

(A) Screening continues by default through age 85 at the usual interval
(B) Evidence is insufficient to assess benefits and harms
(C) Clinicians selectively offer screening, a grade C recommendation
(D) Screening is recommended against with a grade D rating
(E) Screening is recommended with a grade A rating

Answer Contents

223. A 56-year-old woman has a 25 pack-year smoking history and quit 6 years ago. She has no cough, weight loss or hemoptysis, and her chest examination is normal. She asks what lung cancer screening, if any, applies to her. Which of the following is the USPSTF recommendation for a woman like her?

(A) No screening, since she quit smoking 6 years ago
(B) Low-dose computed tomography repeated every 2 years
(C) Annual screening with posteroanterior chest radiography
(D) Annual screening with low-dose computed tomography
(E) Annual sputum cytology combined with a chest examination

Answer Contents

224. A 67-year-old woman has been screened with annual low-dose computed tomography for 8 years because of a 30 pack-year smoking history. She quit smoking 17 years ago. Her studies have been negative and she has no symptoms. She asks if she should keep having a scan each year. Which of the following is the most appropriate recommendation based on the USPSTF statement?

(A) Stop screening because she has not smoked for 15 years
(B) Continue screening because her pack-year history exceeds 20
(C) Continue annual screening until age 80 as originally planned
(D) Change to chest radiography every 2 years instead of CT
(E) Lengthen the low-dose CT interval to every 3 years

Answer Contents

225. A 38-year-old woman with a regular partner has never received human papillomavirus vaccine. She is healthy, not pregnant, and has had normal cervical screening. At her annual visit she asks whether the vaccine is worth getting at her age. Which of the following best describes the CDC adult schedule recommendation for HPV vaccination at ages 27 through 45 years?

(A) Vaccination is not an option after the age of 26 years
(B) Vaccination is offered based on shared clinical decision-making
(C) Vaccination is offered when she has a history of abnormal cytology
(D) Vaccination is routinely recommended for every adult in this age range
(E) Vaccination is offered when her partner has genital warts

Answer Contents

226. A 31-year-old woman completed her childhood vaccine series. She does not recall ever receiving a tetanus, diphtheria and pertussis booster as an adolescent or adult, has no wound, and is not pregnant. She asks what is recommended now and how often afterward. Which of the following is the most appropriate plan according to the CDC adult schedule?

(A) One dose of Td now, then Td every 5 years
(B) Two doses of Tdap 4 weeks apart, then every 10 years
(C) One dose of Tdap during each future pregnancy and no routine booster
(D) One dose of Tdap now and no later boosters
(E) One dose of Tdap now, then Td or Tdap every 10 years

Answer Contents

227. A 52-year-old woman has no immunocompromising condition and has never received a zoster vaccine. She recalls having chickenpox as a child and is otherwise healthy, with an up-to-date vaccine record for influenza and Tdap. She asks whether she needs shingles prevention before age 65. Which of the following is the recommendation in the CDC adult schedule?

(A) Defer zoster vaccination until she reaches age 65 or older
(B) Recombinant zoster vaccine, 2 doses, 2 to 6 months apart
(C) Recombinant zoster vaccine, 3 doses over 6 months total
(D) Recombinant zoster vaccine as a single dose given today
(E) Live zoster vaccine, 2 doses given 4 weeks apart now

Answer Contents

228. A 45-year-old woman comes for a preventive visit. She has no record of hepatitis B vaccination and has no chronic liver disease, injection drug use, health care occupation or known exposure. She is in a monogamous relationship. Her clinician reviews the CDC adult schedule. Which of the following is the most appropriate recommendation?

(A) Give vaccine if her partner tests positive for surface antigen
(B) Check for hepatitis B virus DNA before considering any vaccine
(C) Give vaccine if she has chronic liver disease or an occupational exposure risk
(D) Complete a hepatitis B vaccine series, as routine at ages 19 through 59
(E) Defer vaccination until she is 60 years old or older

Answer Contents

229. A 29-year-old woman has a negative urine pregnancy test at her preventive visit. Rubella IgG testing ordered at her preconception visit shows no evidence of immunity. She has not received measles, mumps and rubella vaccine and is not planning a pregnancy this month. Which of the following is the best next step according to the CDC adult schedule?

(A) Give varicella vaccine in place of measles, mumps and rubella vaccine
(B) Repeat the rubella IgG every year until she conceives
(C) Give measles, mumps and rubella vaccine now, since she is not pregnant
(D) Give measles, mumps and rubella vaccine in the third trimester of a future pregnancy
(E) Defer the vaccine until after her next pregnancy and vaccinate postpartum

Answer Contents

230. A 76-year-old woman has never received respiratory syncytial virus vaccine. She is otherwise healthy, has stable hypertension, and received influenza vaccine this season. At her annual visit she asks what is advised for respiratory syncytial virus and whether she will need yearly doses. Which of the following is the recommendation in the CDC adult schedule?

(A) One dose of a licensed respiratory syncytial virus vaccine, no boosters
(B) A single dose that is delayed until she reaches age 85 years
(C) A two-dose respiratory syncytial virus vaccine series given 8 weeks apart
(D) One dose each autumn, repeated every year with influenza vaccine
(E) One dose if she has chronic lung disease or heart failure

Answer Contents

231. A 24-year-old woman requests a contraceptive implant. Her last normal menses began 8 days ago. She has had no intercourse since that menses began, has no symptoms of pregnancy and has no medical conditions. The implant is placed today. Which of the following is the most appropriate advice about additional protection after placement, based on the CDC selected practice recommendations?

(A) Abstain or use condoms for the next 14 days
(B) Abstain or use condoms for the next 3 months
(C) No additional protection is needed because she was not pregnant
(D) Abstain or use condoms for the next 2 days
(E) Abstain or use condoms for the next 7 days

Answer Contents

232. A 27-year-old woman receives depot medroxyprogesterone acetate by injection and returns for repeat injection 14 weeks after her last dose. She has had no unprotected intercourse since the last injection, has no symptoms of pregnancy, and is otherwise well. Which of the following is the most appropriate plan according to the CDC selected practice recommendations?

(A) Give the repeat injection today, since she is within the grace period
(B) Restart with two injections 4 weeks apart
(C) Switch to a combined oral contraceptive instead of injecting
(D) Obtain a pregnancy test and repeat it 2 weeks later before any injection
(E) Give the injection and require abstinence or condoms for the next 7 days

Answer Contents

233. A 22-year-old woman wants to begin a combined oral contraceptive today. Her last normal menses began 3 weeks ago, and she has not had intercourse since it began. She has no symptoms of pregnancy and her blood pressure is normal. Which of the following is the most appropriate management based on the CDC selected practice recommendations?

(A) Start the pill today and require no backup method at any point
(B) Give emergency contraception first, then start the pill in 5 days
(C) Start the pill today with no pregnancy test, and use backup for 7 days
(D) Delay the pill until a clinic visit for cervical cytology is completed
(E) Obtain a urine pregnancy test and defer the pill until her next menses

Answer Contents

234. A 23-year-old nulligravid woman requests a copper intrauterine device. She is asymptomatic, her examination shows no cervical discharge or motion tenderness, and her last chlamydia and gonorrhea screening was 14 months ago. She has had a new partner in the past 3 months. Which of the following is the most appropriate management based on the CDC selected practice recommendations?

(A) Decline placement because of her age and nulligravid status
(B) Test for chlamydia and gonorrhea and place the device today
(C) Prescribe prophylactic antibiotics before placing the device
(D) Delay placement until the test results return negative
(E) Treat empirically for chlamydia before placing the device

Answer Contents

235. A 29-year-old woman with migraine preceded by visual aura asks to start a combined hormonal contraceptive for cycle control. She has normal blood pressure, does not smoke and has no other risk factors. Her headaches began at age 17. She is not interested in a copper device. Which of the following is the most appropriate method?

(A) Transdermal combined contraceptive patch
(B) Combined vaginal contraceptive ring
(C) Low-dose combined pill taken at bedtime
(D) Etonogestrel contraceptive implant
(E) Combined oral contraceptive with extended cycling

Answer Contents

236. A 38-year-old woman who smokes one pack of cigarettes daily asks to start a combined oral contraceptive. Her blood pressure is 118/74 mm Hg, her body mass index is 24, and she has no personal history of thrombosis. She is advised against it. Which of the following is the primary risk behind the U.S. MEC category 4 classification for her?

(A) Hormone-sensitive malignancy such as breast cancer
(B) Arterial events such as myocardial infarction and stroke
(C) Functional ovarian cysts with torsion or rupture
(D) Cervical neoplasia progression in persistent HPV infection
(E) Biliary disease with cholelithiasis and cholecystitis

Answer Contents

237. A 24-year-old woman takes a combined oral contraceptive. She missed two consecutive active pills, 52 hours late, in the second week of the pack, and had unprotected intercourse 2 days ago. She asks what to do. Based on the CDC selected practice recommendations, which of the following is the most appropriate instruction?

(A) Double the dose of the next 3 pills and use no backup method
(B) Take the latest missed pill now and add backup for just 2 days
(C) Take both missed pills together now and continue with no backup at all
(D) Stop the pack, wait for withdrawal bleeding, then start a new pack
(E) Take the latest missed pill, discard the other, add backup for 7 days

Answer Contents

238. A 44-year-old woman with a desk job reports no regular exercise. Her blood pressure, glucose and lipids are normal and her body mass index is 26. She asks how much weekly aerobic activity she should aim for to gain health benefits. Which of the following targets is consistent with the Physical Activity Guidelines for Americans for adults?

(A) At least 300 to 450 minutes of moderate-intensity aerobic activity each week
(B) At least 60 to 90 minutes of moderate-intensity aerobic activity each week
(C) At least 30 minutes of moderate-intensity aerobic activity on 1 day each week
(D) At least 150 to 300 minutes of moderate-intensity aerobic activity each week
(E) At least 150 to 300 minutes of vigorous-intensity aerobic activity each week

Answer Contents

239. A 68-year-old woman walks briskly for 30 minutes five days a week. She has osteopenia, normal blood pressure and no cardiovascular disease. She asks what else to add to her routine to meet the guidelines. Which of the following additions is consistent with the Physical Activity Guidelines for Americans for older adults?

(A) Add gentle stretching on 1 day a week, with no resistance exercise
(B) Add bed rest after any muscle soreness until symptoms resolve fully
(C) Muscle strengthening 2 or more days a week, plus balance training
(D) Add a further 150 minutes of vigorous-intensity activity each week
(E) Continue aerobic activity alone, since balance training follows a first fall

Answer Contents

240. A 26-year-old woman is sexually active and uses condoms inconsistently. She is not trying to conceive but would not be upset by a pregnancy. She has no personal or family history of neural tube defect and takes no antiepileptic drug. She asks about a vitamin to take. Which of the following is the USPSTF recommendation for a person like her?

(A) Folic acid 0.4 to 0.8 mg orally daily, started now
(B) Folic acid 4 mg orally daily, begun when pregnancy is confirmed
(C) Folic acid 0.4 to 0.8 mg orally daily, begun after the first trimester
(D) Folic acid 4 mg orally weekly, begun 3 months before conception
(E) Folic acid held until a positive pregnancy test, then a prenatal vitamin

Answer Contents

241. A 67-year-old woman with no prior fracture, no glucocorticoid use, a body mass index of 24 and no first-degree relative with hip fracture asks if she needs a bone density test. She is postmenopausal and healthy. Which of the following is the most appropriate recommendation based on the USPSTF osteoporosis screening statement?

(A) Screen if she has a first-degree relative with hip fracture
(B) Begin a bisphosphonate without testing
(C) Defer screening until she has a fragility fracture
(D) Check a serum 25-hydroxyvitamin D level
(E) Screen with bone mineral density testing at a central site

Answer Contents

242. A 58-year-old postmenopausal woman smokes, weighs 49 kg, and has a mother who sustained a hip fracture at 72 years. She has had no fractures herself. She asks whether she should have bone density testing now or wait until 65. Which of the following is the most appropriate next step based on the USPSTF osteoporosis screening statement?

(A) Wait until age 65 years before any assessment
(B) Order bone density testing for all postmenopausal women regardless of risk
(C) Use a clinical risk assessment tool and screen if she is at increased risk
(D) Start calcium and vitamin D supplementation in place of testing
(E) Order a bone turnover marker panel as the screening test

Answer Contents

243. A 62-year-old community-dwelling postmenopausal woman is healthy, has no osteoporosis, and has never had a fracture. She asks whether to take daily vitamin D 400 IU and calcium 1000 mg to prevent a first fracture. Which of the following is the 2018 USPSTF position on supplementation at these doses or lower in postmenopausal women?

(A) Recommended when the 25-hydroxyvitamin D level is low for the laboratory
(B) Recommended against for primary fracture prevention, grade D
(C) Evidence is insufficient to assess benefits and harms at these doses
(D) Recommended for postmenopausal women generally, with a grade B rating
(E) Recommended against in women with a history of kidney stones, with grade D

Answer Contents

244. A 24-year-old woman has a positive home pregnancy test and a confirmed intrauterine pregnancy at 6 weeks by ultrasonography. She tells her gynecologist she feels ambivalent. She is employed, has no children and is unsure whether she wants to continue. She asks the clinician what she would do. Which of the following is the most appropriate counseling approach according to the ACOG committee statement on pregnancy options counseling?

(A) Share the clinician's own preference to help her decide quickly
(B) Postpone any mention of abortion unless she raises the topic
(C) Refer her to another practice without discussing the options
(D) Explore her values and give balanced information on all options
(E) Ask her to reach a decision before she leaves the visit

Answer Contents

245. A 31-year-old woman at 7 weeks of gestation has reflected for several days and states clearly that she has decided to end the pregnancy. She has been counseled once on all options and shows no sign of coercion. A resident suggests reviewing parenting and adoption again at each contact until the procedure date. Which of the following is the best response according to the ACOG committee statement?

(A) Respect the decision; do not reopen options unless something changes
(B) Review every alternative again at each contact to make sure she is certain
(C) Require a second counseling visit before she may proceed with the plan
(D) Refer her for a mental health evaluation before scheduling the procedure now
(E) Ask a family member to confirm that she agrees with the plan

Answer Contents

246. A 27-year-old woman at about 6 weeks by last menstrual period requests medication abortion. She has right lower quadrant pain and spotting. Her serum hCG is 2400 mIU/mL, and transvaginal ultrasonography shows an empty uterus and a 3-cm right adnexal mass. She is hemodynamically stable. Which of the following is the most appropriate next step?

(A) Give mifepristone and discharge with a 7-to-14-day follow-up visit
(B) Give mifepristone now because gestation is under 70 days
(C) Give mifepristone after placing an intrauterine device
(D) Give misoprostol alone and repeat ultrasonography in 2 weeks
(E) Withhold mifepristone and evaluate for ectopic pregnancy

Answer Contents

247. A 25-year-old woman at 8 weeks of confirmed intrauterine pregnancy takes mifepristone in the clinic for medication abortion. She asks when to take the misoprostol tablets she has been given. She lives 30 minutes from the clinic and has no contraindication. Which of the following is the interval between mifepristone and misoprostol stated in the FDA label?

(A) At 7 days after mifepristone
(B) Within 24 to 48 hours after mifepristone
(C) At the follow-up visit 7 to 14 days later
(D) Within 1 to 2 hours after mifepristone
(E) At 72 to 96 hours after mifepristone

Answer Contents

248. A 27-year-old woman has had one mutually monogamous partner for 3 years. She uses a contraceptive implant, has no vaginal discharge or pelvic pain, and has never had a sexually transmitted infection. Her partner has no symptoms. She asks whether she needs annual chlamydia testing. Which of the following is the best answer based on the USPSTF statement for sexually active women?

(A) Annual screening is recommended for every sexually active woman at any age
(B) Screening should be done every 6 months while she uses an implant
(C) Screening is advised in place of any cervical cancer testing she needs
(D) Screening is for women 25 and older at increased risk, which she is not
(E) Screening is advised when symptoms appear, such as discharge or dysuria

Answer Contents

249. A 26-year-old woman had a negative chlamydia and gonorrhea nucleic acid amplification test 5 months ago at her annual visit. She now reports a new sexual partner 6 weeks ago and inconsistent condom use. She has no symptoms and a normal examination. Which of the following is the most appropriate approach based on the USPSTF statement on screening intervals?

(A) Test if vaginal discharge or dysuria develops in the coming months ahead
(B) Treat both partners empirically and skip nucleic acid testing
(C) Screen again now, given new risk since the last negative test
(D) Do not repeat testing because the earlier result was negative
(E) Wait until 12 months have passed since the last negative test result

Answer Contents

250. A 34-year-old woman has never been tested for HIV. She is married, has had one lifetime partner, has no history of injection drug use or transfusion, and is not pregnant. She has no symptoms. At her preventive visit she asks if testing is necessary. Which of the following is the 2019 USPSTF recommendation about HIV screening for her?

(A) Screen as routine care, since the statement covers ages 15 to 65
(B) Screen after a diagnosis of another sexually transmitted infection
(C) Screen during a pregnancy, when HIV testing is routinely offered to her
(D) Defer screening until she reaches age 40 years or older
(E) Screen if she has a history of injection drug use or a new partner

Answer Contents

251. A 44-year-old woman has normal liver enzymes and no history of injection drug use, transfusion before 1992 or incarceration. She has never been tested for hepatitis C virus. She is not pregnant and has no symptoms. Which of the following is the USPSTF recommendation for her?

(A) Annual hepatitis C screening beginning at the present visit
(B) Screening limited to adults with a history of injection drug use
(C) Screening beginning at age 50 years, once average risk is confirmed here
(D) Screening if her transaminases become elevated on later testing
(E) One-time hepatitis C screening, as the statement covers ages 18 to 79

Answer Contents

252. A 31-year-old woman with ongoing injection drug use had a negative hepatitis C antibody test 18 months ago. She is not pregnant, has no symptoms, and is interested in harm reduction services. Which of the following is the most appropriate approach to hepatitis C screening according to the USPSTF statement?

(A) Screen periodically while her risk continues
(B) Screen if her transaminases rise
(C) Consider her screening complete after the single negative test
(D) Delay repeat testing until she stops injecting
(E) Screen once more at age 79 years

Answer Contents

253. A 28-year-old woman at her annual well-woman visit answers a validated intimate partner violence screening tool in a way that indicates ongoing physical and emotional abuse. She has no acute injuries and says she is not ready to leave. She is not pregnant. Which of the following is the most appropriate next step based on the 2025 USPSTF recommendation?

(A) Repeat the screening tool at her next annual visit before taking action
(B) Evaluate her and provide or refer for interventions with support
(C) Prescribe an anxiolytic and schedule joint couples counseling sessions
(D) Give her a pamphlet and arrange no further contact until next year
(E) Report to law enforcement over her objection as the first step

Answer Contents

254. A clinic that serves adult women plans a new intake form for preventive visits. The medical director asks which depression and suicide screening practice is supported by the USPSTF. A 38-year-old patient with no symptoms or psychiatric history will be seen under this protocol. Which of the following best reflects the 2023 USPSTF position?

(A) Screen for depression in women with a prior psychiatric diagnosis
(B) Screen adults for suicide risk with a grade B recommendation and omit depression screening
(C) Screen adults for depression; evidence on screening for suicide risk is insufficient
(D) Recommend against depression screening in adults who have no symptoms
(E) Screen for depression and suicide risk, each with a grade A recommendation

Answer Contents

255. A 49-year-old woman has repeated office blood pressures of 142/88 mm Hg, an elevated LDL cholesterol, and an estimated 10-year cardiovascular risk of 8 percent. She does not have diabetes and her body mass index is 27. She is not currently taking any medication. Which of the following is the 2020 USPSTF recommendation about lifestyle intervention for her?

(A) Give written dietary advice without referral or follow-up
(B) Counsel on physical activity alone, since diet counseling lacks benefit
(C) Recommend against counseling because she has not yet been diagnosed with dyslipidemia
(D) Offer or refer her to behavioral counseling for a healthy diet and physical activity
(E) Defer counseling until she develops cardiovascular disease

Answer Contents

256. A 35-year-old woman completes an alcohol screening questionnaire at a preventive visit. Her score is consistent with risky drinking, with about two drinks most evenings. She has no withdrawal symptoms, no liver disease, and no evidence of alcohol use disorder, and she is not pregnant. Which of the following is the best management according to the 2018 USPSTF recommendation?

(A) Refer to inpatient detoxification
(B) Provide brief behavioral counseling intervention
(C) Repeat the questionnaire at the next annual visit
(D) Prescribe naltrexone at this visit
(E) Advise abstinence and discharge without follow-up

Answer Contents

OP1.2. Patients with Unique Needs

257. An 18-year-old woman comes for contraception counseling. She has had one sexual partner for the past year, uses condoms most of the time, and has no discharge, pelvic pain, or dysuria. Her examination is normal and she has never been tested for sexually transmitted infections. She says she does not need testing because she has only one partner. Which of the following is the most appropriate recommendation under USPSTF guidance?

(A) Screen, because every sexually active woman 24 or younger qualifies
(B) Screen if she reports more than one partner in the past year
(C) Screen if she develops discharge or other symptoms of infection
(D) Screen if her partner has a positive test result for either infection
(E) Delay screening until she reaches the age of 25 years and is then retested

Answer Contents

258. A 16-year-old girl asks to be tested for sexually transmitted infections after unprotected sex last week. Her mother is in the waiting room and tells the nurse that she expects to be told all results and to sign every consent form. The patient says she does not want her mother informed. She has no symptoms and is not in danger. Which of the following is the best approach under the CDC guidelines?

(A) Require the mother's signature before any testing is done in clinic today
(B) Delay testing until she turns 18 years of age and can sign a form
(C) Provide care with her consent and review confidentiality and billing
(D) Test her but share all of the results with the mother who is waiting
(E) Refer her to a pediatrician for parent-led care and a parent-signed form

Answer Contents

259. A 19-year-old woman in a monogamous relationship for 2 years comes for a contraception refill. Her only prior chlamydia test was negative 13 months ago. She has no discharge, pelvic pain, or dysuria, and her partner has no symptoms. She uses condoms inconsistently. She asks whether she needs to be tested again. Which of the following is the best recommendation under CDC guidance?

(A) Repeat screening in 3 years, matching the cervical cytology interval
(B) Repeat screening if she develops discharge or other symptoms
(C) Repeat screening if her partner has a new sexual partner
(D) Repeat screening now, since annual screening is advised
(E) Stop screening because she has been monogamous for 2 years

Answer Contents

260. A 14-year-old girl attends her annual well visit with her father. He reports that her grades have dropped over the last semester. She has no suicidal statements, no complaints, and a normal examination. Alone with you, she admits to feeling tired and down most days. Which of the following is the most appropriate screening step according to the USPSTF?

(A) Screen for suicide risk alone with a validated tool because the USPSTF recommends it
(B) Defer all screening until she develops symptoms
(C) Obtain parental report and skip the adolescent interview
(D) Order thyroid function tests and a complete blood count as the workup
(E) Screen for major depressive disorder using a validated instrument

Answer Contents

261. A 72-year-old woman reports two falls in the past year, both when rising from a chair. She walks slowly and uses no assistive device. She takes no sedatives, has no orthostatic hypotension, and has normal vision. She asks what would best reduce her chance of another fall. Which of the following does the USPSTF recommend for community-dwelling adults 65 or older at increased risk of falls?

(A) Routine daily vitamin D supplementation
(B) Advice to restrict activity to avoid falls
(C) A low-dose benzodiazepine at night
(D) Exercise interventions
(E) Head computed tomography

Answer Contents

262. A 78-year-old woman is healthy, lives independently, and has a life expectancy well beyond 10 years. She had a normal colonoscopy at age 68 and has never had colorectal cancer. She asks whether she should have another screening test. Which of the following best describes the USPSTF recommendation for adults aged 76 to 85?

(A) Recommend screening every year with a fecal immunochemical test kit
(B) Recommend against any further screening after age 75 years of age
(C) Require colonoscopy every 5 years until she reaches age 85 years
(D) Recommend screening if she develops symptoms of rectal bleeding or anemia
(E) Selectively offer screening based on health, history, and wishes

Answer Contents

263. A 77-year-old woman has hypertension and an LDL cholesterol of 135 mg/dL. She has never had a cardiovascular event and has no diabetes. A risk calculator gives her a 10-year cardiovascular risk of 24%. She asks whether she should start a statin. Which of the following best reflects the USPSTF position on starting a statin for primary prevention at her age?

(A) Evidence is insufficient to weigh starting a statin
(B) A statin is recommended at her age with a grade B recommendation
(C) A statin is recommended against at her age with a grade D rating
(D) Aspirin is recommended instead of a statin at her age and risk
(E) A statin is recommended if her LDL cholesterol exceeds 190 mg/dL

Answer Contents

264. A clinic that serves older women wants to add universal screening for anxiety disorders to every visit for patients 65 and older. A 68-year-old patient with no complaints and no psychiatric history is due for her annual visit. The clinic manager asks you what the USPSTF says about anxiety screening in this age group. Which of the following is the most accurate statement?

(A) Screening is recommended in older adults with a grade B rating
(B) Evidence is insufficient to weigh benefits and harms of screening
(C) Screening is recommended against after age 65 with a grade D rating
(D) Screening is recommended each year with a validated tool
(E) Screening is recommended for those who have chronic pain

Answer Contents

265. A 29-year-old woman who has sex only with women has never had a cervical cytology test. She says she was told she does not need one because she has never had sex with men. She is healthy and has no symptoms. Which of the following is the best counseling statement?

(A) Cervical screening is unnecessary for women who have sex with women
(B) Cervical screening should start at 40 for women who have sex with women
(C) HPV likely spreads between women, so routine screening applies
(D) HPV spreads through penile-vaginal intercourse alone, so screening is unneeded
(E) A vaginal swab for herpes is the preferred screening test for her

Answer Contents

266. A 22-year-old woman reports that she has sex with women and is monogamous. She declines STI testing, saying that her risk is lower than that of women with male partners. She has no symptoms. You take a sexual history and learn she had two male partners in the past year. Which of the following is the most appropriate approach consistent with CDC guidance?

(A) Defer screening because women who have sex with women have negligible risk
(B) Do not assume lower risk and screen by the criteria used for other women
(C) Screen for syphilis alone and omit chlamydia and gonorrhea tests
(D) Screen for herpes simplex virus by culture and omit other tests
(E) Screen if her female partner reports symptoms of an infection

Answer Contents

267. A 32-year-old transgender man has taken testosterone for 4 years and has not had a hysterectomy. He has never had cervical cancer screening and has no symptoms. He asks whether hormone therapy means he no longer needs a Pap test. Which of the following is the best response?

(A) Screening is unnecessary while he takes testosterone therapy
(B) Screening is needed if he reports vaginal bleeding to the clinician
(C) Urine HPV testing replaces cervical sampling in all such patients
(D) Screen on the usual schedule because screening depends on anatomy
(E) A hysterectomy is required before any further care is given

Answer Contents

268. A 24-year-old transgender woman establishes care. She takes estradiol and spironolactone, has been in a monogamous relationship for 2 years, and says she has never been tested for HIV because she believes her risk is low. She has no symptoms. Which of the following is the most appropriate recommendation under CDC guidance?

(A) Defer HIV testing because she is in a monogamous relationship
(B) Test for HIV if she begins sex work in the future
(C) Discuss and offer HIV screening to all transgender persons
(D) Test for HIV after her estradiol therapy has been stopped
(E) Offer syphilis testing and omit HIV testing for now

Answer Contents

269. A 27-year-old woman visits for contraception. On routine screening with a validated tool, she answers yes to questions about being hit and feeling afraid of her partner. She has no acute injuries and says she is not ready to leave. She asks what you can do. According to the USPSTF, which of the following is the most appropriate next step?

(A) Provide or refer her to multicomponent care with ongoing support
(B) Document the answers and rescreen at her next annual visit
(C) Report the findings to the police without asking for her consent
(D) Refer the couple for joint counseling to resolve the conflict
(E) Prescribe an anxiolytic to relieve her fear and reassess later

Answer Contents

270. A 21-year-old woman presents 4 hours after an assault by a man she does not know. She is stable, has no injuries needing surgery, and agrees to examination. Initial testing is performed, and she is unlikely to return for follow-up because she lives far away. Which of the following approaches to sexually transmitted infection prophylaxis is consistent with CDC guidelines?

(A) Await culture results from all sites and treat if they are positive
(B) Treat chlamydia alone and omit coverage for the other infections
(C) Antiviral prophylaxis directed at genital herpes simplex virus
(D) Defer all treatment to a follow-up visit in 2 weeks
(E) An empiric regimen for chlamydia, gonorrhea, and trichomonas

Answer Contents

271. A 19-year-old woman is examined 6 hours after a sexual assault. She has never received HPV vaccine. She is not pregnant, has no injuries needing surgery, and has received presumptive STI treatment. She asks whether any vaccine should be given. Which of the following is the CDC recommendation?

(A) HPV vaccination after a positive HPV test result is returned
(B) No vaccination until 6 months have passed since the assault
(C) HPV vaccination restricted to assaults by an unknown assailant
(D) HPV vaccination for unvaccinated survivors aged 9 to 26
(E) Hepatitis A vaccination as the sole vaccine offered

Answer Contents

272. A 30-year-old woman is evaluated 12 hours after a sexual assault by a man she does not know. She has no history of hepatitis B vaccination, and the assailant's hepatitis B status is unknown. She has no symptoms, is not pregnant, and has no injuries needing surgery. Which of the following hepatitis B prophylaxis is recommended by the CDC?

(A) Hepatitis B vaccination without hepatitis B immune globulin
(B) Hepatitis B immune globulin given by itself without vaccine
(C) Hepatitis B vaccine given together with immune globulin
(D) No prophylaxis until the assailant has been tested
(E) A 4-week course of entecavir as antiviral prophylaxis

Answer Contents

273. A 23-year-old woman presents 36 hours after a sexual assault. She uses no contraception, her last menstrual period was 12 days ago, and a urine pregnancy test is negative. She is distressed and asks whether pregnancy can be prevented. Which of the following is the most appropriate step according to the CDC?

(A) Wait for her next menses before any intervention is offered
(B) Offer emergency contraception if she reports that ejaculation occurred
(C) Offer emergency contraception because pregnancy could result
(D) Repeat the pregnancy test in 2 weeks before deciding
(E) Defer the decision to a follow-up visit in 6 weeks

Answer Contents

274. A 36-year-old woman comes for her annual gynecologic visit. She reports no mood complaints and has no psychiatric history. The clinic uses a validated nine-item questionnaire for every adult. A new staff member asks whether depression screening is recommended in this population. Which of the following is the best answer under USPSTF guidance?

(A) Screening is recommended for patients with a prior diagnosis
(B) Screening is recommended for the adult population
(C) Screening is recommended after a suicide attempt
(D) Screening is not recommended and carries a grade D rating
(E) Screening is recommended for adults with a chronic medical illness

Answer Contents

275. A 33-year-old woman visits for contraception and mentions that she has felt restless and unable to relax for several months. She has no suicidal thoughts, normal thyroid function, and no substance use. A colleague asks whether the USPSTF recommends routine anxiety screening in a patient like her. Which of the following is the best answer?

(A) Screening is recommended against in young adults, grade D
(B) Evidence is insufficient for adults who are younger than 65
(C) Screening is recommended after the age of 65 years
(D) Screening is recommended after a positive depression screen
(E) Screening is recommended for adults 64 or younger

Answer Contents

276. A 30-year-old woman completes a depression questionnaire at a gynecology visit, and her answer to the item about thoughts of self-harm is zero. Her total score is low. The practice manager asks whether the USPSTF recommends adding a separate universal suicide risk screening tool for all adults at each visit. Which of the following is the most accurate response?

(A) The evidence is insufficient on suicide risk screening in adults
(B) The USPSTF recommends universal suicide risk screening at every visit
(C) The USPSTF recommends against suicide risk screening, grade D
(D) The USPSTF recommends suicide risk screening for adults over 65 years
(E) The USPSTF recommends suicide risk screening for pregnant adults

Answer Contents

277. A 29-year-old woman with schizophrenia has taken risperidone for 2 years. Over the past 5 months she has had no menses and reports milky nipple discharge. Urine hCG is negative, thyroid-stimulating hormone is normal, and serum prolactin is elevated. Her examination is otherwise normal. Which of the following is the most likely cause of her amenorrhea?

(A) Premature ovarian insufficiency from follicle depletion
(B) Polycystic ovary syndrome with chronic anovulation
(C) Primary hypothyroidism with elevated prolactin
(D) Drug-induced hyperprolactinemia
(E) Asherman syndrome after uterine instrumentation

Answer Contents

278. A 38-year-old woman has cirrhosis with ascites and jaundice. Her liver function is impaired, and she is not pregnant. She wants a reliable contraceptive and asks about options. According to the 2024 US Medical Eligibility Criteria for Contraceptive Use, which of the following represents an unacceptable health risk (category 4)?

(A) Copper intrauterine device
(B) Combined hormonal contraception
(C) Levonorgestrel intrauterine device
(D) Etonogestrel implant
(E) Progestin-pill

Answer Contents

279. A 29-year-old woman with a body mass index of 36 kg/m2 has no hypertension, diabetes, smoking, migraine, or thrombosis history. She requests a combined oral contraceptive for cycle control. She is concerned because a pamphlet says that obesity is a contraindication. According to the 2024 US Medical Eligibility Criteria, which of the following is the best classification for her?

(A) Category 1: no restriction applies to her situation
(B) Category 3: risks usually outweigh the advantages
(C) Category 2: advantages generally outweigh the risks
(D) Category 4: unacceptable health risk if she takes it
(E) Category 4 when body mass index exceeds 40 kg/m2

Answer Contents

280. A 41-year-old woman who takes a combined oral contraceptive is scheduled for a total hip arthroplasty, after which she is expected to have prolonged immobilization. She has no personal history of thrombosis. She asks how to manage her contraception around surgery. According to the 2024 US Medical Eligibility Criteria, which of the following is the best classification for combined hormonal contraception with prolonged immobilization after major surgery?

(A) Category 2, so she can continue with compression stockings
(B) Category 1, so no restriction applies to her
(C) Category 4, so she should switch to a nonestrogen method
(D) Category 3, so she can continue if aspirin is added
(E) Category 2, so she should switch to a transdermal patch

Answer Contents

281. A 35-year-old woman who uses a power wheelchair calls to establish care for a pelvic examination and contraception. The office has a fixed-height examination table and no lift. The receptionist suggests that she go to the hospital clinic instead because the practice cannot examine wheelchair users. Which of the following is the best statement of the practice's obligation?

(A) The practice may refer her away if it lacks accessible equipment
(B) The practice may require her to bring an assistant to lift her
(C) The practice must examine her in the wheelchair without transfer
(D) The practice may schedule her at the end of the day in a quiet room
(E) The practice cannot deny her a service it would otherwise provide

Answer Contents

282. A 29-year-old woman with cerebral palsy and mild dysarthria arrives for a contraception visit with her personal care aide. She makes her own decisions and has no guardian. The clinician asks the aide about her sexual history, medications, and what she wants. The patient looks frustrated. Which of the following is the best way to proceed?

(A) Continue to direct the questions to the aide who accompanies her today
(B) Require a guardian to be present before proceeding with any care
(C) Address her directly, allow extra time, and involve the aide if asked
(D) Reschedule the visit with a clinician who has dysarthria experience
(E) Complete the history from the chart without asking her any questions

Answer Contents

283. A 52-year-old woman with a spinal cord injury who uses a wheelchair needs a pelvic examination and weighing at her annual visit. The practice has a standard fixed-height table and a bathroom scale. Staff propose that two employees lift her onto the table. Which of the following is the best approach consistent with federal guidance?

(A) Have two staff members lift her manually onto the standard table today
(B) Ask her family members to lift her onto the examination table
(C) Use a lowering table or a lift, and an accessible scale
(D) Omit the weighing and the pelvic examination at this visit
(E) Examine her on a floor mat while she remains in her clothing

Answer Contents

OP2.1. Primary Care Problems

284. A healthy 76-year-old woman comes for her annual visit. She has no breast symptoms, no family history of breast cancer, and no prior abnormal imaging. Her last three biennial screening mammograms were normal, and she is active with no major medical conditions. She asks whether she should keep having mammograms. Which of the following best reflects the USPSTF position on screening mammography at her age?

(A) The evidence is insufficient to weigh benefits and harms at 75 or older
(B) Biennial screening keeps a grade B recommendation through age 79
(C) Annual screening is advised until life expectancy falls under 10 years
(D) The Task Force recommends against screening once a woman reaches age 75
(E) Breast MRI should replace mammography for women past age 75

Answer Contents

285. A 52-year-old woman has a normal screening mammogram that the radiologist reports as heterogeneously dense. She has no symptoms, no family history, and no prior biopsy. She read that dense tissue can hide cancers and asks whether she should add a breast ultrasound or MRI to her routine screening. Which of the following is the USPSTF position on supplemental imaging for women with dense breasts and a negative mammogram?

(A) Added MRI every year is recommended for dense breasts with a grade B
(B) Added ultrasonography every 2 years is recommended for dense breasts, grade B
(C) Added imaging for dense breasts is recommended against with a grade D
(D) Evidence is insufficient to weigh benefits and harms of added imaging
(E) Mammography should be repeated every 6 months until the density declines

Answer Contents

286. A 55-year-old postmenopausal woman had a breast biopsy last year that showed atypical ductal hyperplasia. She has no history of venous thromboembolism, no stroke, and no osteoporosis. Her clinician estimates her risk of breast cancer as increased and her risk of adverse medication effects as low. She is anxious and asks whether any medication can reduce her chance of developing breast cancer. Which of the following is the best next step according to the USPSTF?

(A) Recommend against any risk-reducing drug because she does not have cancer yet
(B) Offer a drug such as tamoxifen, raloxifene, or an aromatase inhibitor
(C) Prescribe daily low-dose aspirin to reduce her risk of breast cancer
(D) Start menopausal estrogen therapy to lower her future risk of breast cancer
(E) Refer for prophylactic bilateral mastectomy before discussing any medication

Answer Contents

287. A 44-year-old woman has an average risk of breast cancer by a validated tool. She has no family history, no prior biopsy, and no chest irradiation. She read that tamoxifen can prevent breast cancer and asks her gynecologist to prescribe it as protection, saying she would accept the side effects. She has normal mammography and a normal examination. Which of the following is the best response according to the USPSTF?

(A) Prescribe tamoxifen for 5 years because she is premenopausal
(B) Prescribe an aromatase inhibitor, which suits premenopausal women
(C) Prescribe raloxifene, which is approved for her age group
(D) Offer low-dose tamoxifen until menopause for added safety
(E) Routine use of risk-reducing medication is not advised for her

Answer Contents

288. A 30-year-old woman asks for BRCA1 and BRCA2 testing after reading about the genes online. She has no personal history of cancer. Neither parent nor any sibling, grandparent, aunt, or uncle has had breast, ovarian, tubal, or peritoneal cancer, and her ancestry is not associated with a high frequency of harmful BRCA variants. She is otherwise well. Which of the following is the best response according to the USPSTF?

(A) Order the sequencing because testing is recommended for every woman by age 30
(B) Offer testing if dense breast tissue appears on future screening mammograms
(C) Recommend annual breast MRI starting now in place of any genetic testing
(D) Advise prophylactic salpingo-oophorectomy by age 35 as a precaution
(E) Explain that routine testing is not recommended without a relevant history

Answer Contents

289. A 58-year-old woman with treated hypertension and no diabetes visits for a preventive examination. She has never had a cardiovascular event and does not smoke. Her LDL cholesterol is 134 mg/dL, and her estimated 10-year risk of a cardiovascular event is 13%. She asks whether a medication could lower her risk. Which of the following is the most appropriate recommendation under the USPSTF statin guidance?

(A) Offer low-dose aspirin in place of a statin
(B) Recheck lipids yearly and treat if LDL cholesterol exceeds 190 mg/dL
(C) Prescribe ezetimibe as first-line therapy
(D) Prescribe a statin for primary prevention
(E) Recommend lifestyle change alone for five years before reassessment

Answer Contents

290. A 62-year-old woman has dyslipidemia and a family history of premature coronary disease but no prior cardiovascular event. Her calculated 10-year cardiovascular risk is 8.5%. She is a nonsmoker with normal blood pressure and no diabetes. She asks whether she must take a statin. Which of the following best describes the USPSTF recommendation for a person with a risk of 7.5% to less than 10%?

(A) Selectively offer a statin after discussing the small net benefit
(B) Prescribe a statin to every patient who falls within this risk range
(C) Recommend against statin use whenever the risk is under the 10% mark
(D) Defer all drug treatment until she reaches the age of 65 years
(E) Prescribe a statin if her coronary calcium score is zero on testing

Answer Contents

291. A 64-year-old woman has hypertension controlled on one drug, a 10-year cardiovascular risk of 12%, and no prior cardiovascular event. She has never had a gastrointestinal bleed and takes no anticoagulant. A friend told her to take daily low-dose aspirin to prevent a heart attack, and she asks for your advice. Which of the following is the best recommendation under the 2022 USPSTF guidance?

(A) Start daily low-dose aspirin because her risk exceeds 10%
(B) Start aspirin if she also takes a statin
(C) Do not initiate low-dose aspirin for primary prevention
(D) Start aspirin with a proton pump inhibitor
(E) Start aspirin if her LDL cholesterol is over 130 mg/dL

Answer Contents

292. A 52-year-old woman with type 2 diabetes controlled with diet has a 10-year cardiovascular risk of 11% and no prior cardiovascular event. She has no history of bleeding or ulcers and is not at increased bleeding risk. She asks whether she should begin daily low-dose aspirin. Which of the following is the best description of the USPSTF recommendation for her?

(A) Aspirin is recommended for all adults 40 to 59 with a risk of 10% or greater
(B) The decision to start low-dose aspirin should be an individual one
(C) Aspirin is recommended against for all adults younger than 60
(D) Aspirin is recommended when she has a family history of stroke
(E) Aspirin should be started now and reassessed at age 70

Answer Contents

293. A 47-year-old woman with no known medical problems has an office blood pressure of 146/92 mm Hg at a routine visit, with repeat measurement in the same visit giving a similar value. She has no symptoms, a normal examination, and no known kidney disease. Which of the following is the most appropriate next step according to the USPSTF?

(A) Start an antihypertensive drug at this visit given the elevated reading
(B) Order an echocardiogram and renal arteriography to look for causes
(C) Recheck the office blood pressure at a routine visit in 3 years
(D) Confirm with out-of-clinic measurements before starting treatment
(E) Start a thiazide diuretic and confirm the diagnosis if no response occurs

Answer Contents

294. A 37-year-old woman with a body mass index of 32 comes for a well-woman visit. She has no symptoms of hyperglycemia, no history of gestational diabetes, and no prior glucose testing. Her blood pressure is normal. She asks whether she needs any blood tests for diabetes. Which of the following is the most appropriate recommendation under USPSTF guidance?

(A) Defer screening until age 45
(B) Screen for prediabetes and type 2 diabetes now
(C) Screen if she develops polyuria or polydipsia
(D) Screen if her body mass index exceeds 40
(E) Screen after a fasting insulin level is elevated

Answer Contents

295. A 46-year-old woman has an average-risk history with no family history of colorectal cancer, no inflammatory bowel disease, and no gastrointestinal symptoms. She has never had colorectal screening. She asks whether she should wait until age 50 to begin. Which of the following is the best response according to the USPSTF?

(A) Wait until age 50 because screening before that age is not recommended
(B) Screen if she has a first-degree relative with colorectal cancer
(C) Begin with colonoscopy at age 40 and repeat it every year
(D) Screen after fecal blood is detected or symptoms of bleeding appear
(E) Begin screening now because it is recommended for adults aged 45 to 49

Answer Contents

296. A 34-year-old woman comes for a routine visit. She has no history of injection drug use, transfusion, or tattoos, and her liver enzymes were normal when last checked. She has never been tested for hepatitis C. She is not pregnant. She asks whether testing is necessary since she feels well. Which of the following is the best recommendation under USPSTF guidance?

(A) Testing if alanine aminotransferase becomes elevated at some point
(B) Testing for persons with a history of injection drug use
(C) One-time hepatitis C screening for all adults 18 to 79
(D) Annual hepatitis C antibody testing for all adults at checkups
(E) Testing deferred until she reaches the age of 50 years

Answer Contents

297. A 38-year-old woman has persistent dyspepsia and a positive stool antigen test for Helicobacter pylori obtained after she had stopped her proton pump inhibitor for 2 weeks. She has had no prior treatment for H pylori, no penicillin allergy, and no known antibiotic susceptibility data. Which of the following is the preferred regimen under the 2024 American College of Gastroenterology guideline?

(A) Bismuth quadruple therapy for 14 days
(B) Clarithromycin triple therapy for 7 days
(C) A proton pump inhibitor and amoxicillin alone for 7 days
(D) Levofloxacin triple therapy for 10 days
(E) A proton pump inhibitor alone for 8 weeks

Answer Contents

298. A 32-year-old woman has 9 months of abdominal pain related to defecation with altered stool form and frequency. She has no weight loss, rectal bleeding, nocturnal symptoms, anemia, or family history of colorectal cancer or celiac disease. Physical examination and a complete blood count are normal. She asks for a colonoscopy to be sure nothing is missed. Which of the following is the best next step in management?

(A) Perform colonoscopy with random mucosal biopsies at this visit
(B) Order computed tomography of the abdomen and pelvis with contrast
(C) Diagnose from the history and skip routine colonoscopy
(D) Obtain a full hydrogen breath test panel before making any diagnosis
(E) Perform diagnostic laparoscopy to look for pelvic or bowel disease

Answer Contents

299. A 33-year-old woman has irritable bowel syndrome with diarrhea. Celiac serology is negative, and she has not responded adequately to dietary changes and loperamide. She has no alarm features. She wants a prescription option aimed at the diarrhea and abdominal pain. Which of the following is recommended for irritable bowel syndrome with diarrhea in the American College of Gastroenterology guideline?

(A) A guanylate cyclase activator taken for diarrhea
(B) A daily probiotic supplement taken as treatment
(C) A systemic fluoroquinolone given for 14 days
(D) A daily osmotic laxative taken for symptoms
(E) A gut-selective, minimally absorbed antibiotic

Answer Contents

300. A 34-year-old woman has had 10 days of low back pain after lifting a box. The pain is worse with bending and eases when she rests. She has no leg weakness, numbness, fever, bowel or bladder changes, or history of cancer. Neurologic examination is normal. She would like to avoid medication if possible. Which of the following is the most appropriate initial approach according to the American College of Physicians?

(A) Superficial heat
(B) Lumbar spine magnetic resonance imaging
(C) A short course of an oral opioid
(D) Bed rest for 2 weeks
(E) Epidural corticosteroid injection

Answer Contents

301. A 49-year-old woman has had nonradicular low back pain for 5 months. She has completed a structured exercise program and a course of physical therapy with partial relief and now asks for a medication. She has no gastrointestinal disease, kidney disease, or cardiovascular disease, and imaging showed only mild degenerative changes. Which of the following is the best pharmacologic choice for chronic low back pain per the American College of Physicians?

(A) An extended-release opioid
(B) An NSAID as first-line therapy
(C) Systemic corticosteroids
(D) A skeletal muscle relaxant for long-term use
(E) Repeated epidural corticosteroid injections

Answer Contents

302. A 54-year-old postmenopausal woman has had three gout flares in the past year and a palpable tophus. Her serum urate is 9.4 mg/dL and her estimated glomerular filtration rate is 42 mL/min. She has no allopurinol allergy and no HLA-B*5801 positivity. She is ready to start urate-lowering therapy. Which of the following is the preferred first-line agent and approach according to the 2020 American College of Rheumatology guideline?

(A) Febuxostat started at a high dose without any titration
(B) Probenecid as the preferred first-line urate-lowering agent
(C) Colchicine alone as long-term urate-lowering therapy
(D) Allopurinol started low and titrated to the urate target
(E) Pegloticase as the first urate-lowering agent used

Answer Contents

303. A 58-year-old woman with recurrent gout has just begun allopurinol, and her serum urate remains over target. She had two flares last year and fears that starting the drug will trigger more attacks. You plan to give concomitant anti-inflammatory prophylaxis. According to the 2020 American College of Rheumatology guideline, for how long should prophylaxis be continued after starting urate-lowering therapy?

(A) 1 to 2 weeks
(B) Until the current flare resolves
(C) Indefinitely for life
(D) Until the first normal serum urate value
(E) At least 3 to 6 months

Answer Contents

304. A 63-year-old woman with a body mass index of 34 has chronic pain in both knees that worsens with stairs and improves with rest. Radiographs show joint space narrowing and osteophytes. She asks about supplements and an osteoporosis drug her friend takes for joint health. Which of the following is strongly recommended for her according to the 2019 American College of Rheumatology and Arthritis Foundation guideline?

(A) Glucosamine daily
(B) Chondroitin sulfate daily
(C) Exercise and weight loss
(D) An oral bisphosphonate
(E) Opioid analgesic as first-line therapy

Answer Contents

305. A 31-year-old woman has had recurrent unilateral throbbing headaches with nausea and light sensitivity for 10 years. The pattern has not changed, attacks last about a day, and she has a normal neurologic examination, no new medications, and no red-flag features. She asks for a brain MRI to be certain nothing serious is present. Which of the following is the best management according to the American Headache Society?

(A) Order brain MRI with gadolinium now
(B) Diagnose migraine clinically and avoid neuroimaging
(C) Order head computed tomography as the preferred first test
(D) Order lumbar puncture to exclude infection
(E) Order carotid duplex imaging

Answer Contents

306. A 38-year-old woman has episodic migraine twice monthly and was given butalbital-acetaminophen-caffeine at an emergency visit. She now asks to keep it as her main attack treatment. She has no cardiovascular disease, no pregnancy, and no history of substance misuse. Which of the following is the best choice for her acute treatment according to the American Headache Society Choosing Wisely recommendation?

(A) An NSAID or triptan rather than butalbital or an opioid
(B) Butalbital combination tablets for each migraine attack
(C) An oral opioid for each attack that is severe
(D) Daily over-the-counter analgesics to prevent attacks
(E) An ergotamine tablet taken every night at bedtime

Answer Contents

307. A 44-year-old woman with a 15-year history of migraine now reports headache on 22 days per month for the past 5 months. She takes a combination analgesic on about 20 days each month and says it helps only for a few hours. Neurologic examination and recent brain imaging are normal, and the headaches are not attributed to another condition. Which of the following is the most likely diagnosis?

(A) New daily persistent headache
(B) Chronic tension-type headache without medication effect
(C) Idiopathic intracranial hypertension
(D) Medication overuse headache
(E) Trigeminal neuralgia

Answer Contents

308. A 46-year-old woman with chronic migraine has been diagnosed with medication overuse headache after taking an acute analgesic on most days for 6 months. She is willing to change treatment but fears her headaches will worsen. She has no medical contraindication to withdrawal. Which of the following is the best management plan?

(A) Continue the analgesic and add a second acute medication each day
(B) Increase the analgesic dose until the headaches are controlled
(C) Wait until 6 months after withdrawal to begin any preventive treatment
(D) Switch to a different opioid analgesic for the same daily use
(E) Withdraw the overused drug and begin prevention at once

Answer Contents

309. A 40-year-old woman presents to the emergency department 3 hours after a sudden severe headache that peaked within minutes. She is alert, with a normal neurologic examination and normal vital signs. A noncontrast head computed tomography scan, obtained on a third-generation or newer scanner, is read as normal. Under the 2019 American College of Emergency Physicians clinical policy, which of the following is the most appropriate statement about her evaluation?

(A) Lumbar puncture is required for every patient after a normal CT
(B) A normal neurologic examination alone excludes subarachnoid hemorrhage
(C) A normal noncontrast CT within 6 hours can rule out hemorrhage
(D) CT angiography should replace CT as the first imaging test
(E) A normal CT at any time after onset rules out hemorrhage

Answer Contents

310. A 27-year-old woman has asthma symptoms about twice a week and one night awakening in the past month. She has used albuterol alone for 2 years and had an oral corticosteroid course for an exacerbation last winter. Spirometry shows reversible obstruction. Which of the following is the best change in therapy according to the GINA recommendations?

(A) Continue as-needed albuterol by itself without other drugs
(B) As-needed low-dose inhaled corticosteroid with formoterol
(C) Daily long-acting beta agonist given as monotherapy
(D) Daily montelukast with albuterol used as needed
(E) Daily oral prednisone to control the symptoms

Answer Contents

311. A 35-year-old woman with persistent asthma takes a daily low-dose inhaled corticosteroid and uses albuterol as needed. Over the past 3 months she has used albuterol most days and has had two oral corticosteroid courses. Adherence and technique are good. According to the 2020 NAEPP focused update, which of the following is the preferred next approach for mild to moderate persistent asthma?

(A) Add daily montelukast and keep albuterol as the reliever
(B) Double the albuterol to four times daily on a fixed schedule
(C) Switch to daily oral prednisone for maintenance control
(D) Single maintenance and reliever therapy with ICS-formoterol
(E) Stop the inhaled corticosteroid and rely on albuterol alone

Answer Contents

312. A 29-year-old woman has persistent asthma that remains uncontrolled on a medium-dose inhaled corticosteroid-formoterol regimen, with night symptoms three times weekly and one oral corticosteroid course this year. Her inhaler technique has not been reviewed recently, and she has a cat at home. You are considering stepping up treatment. Before increasing therapy, which of the following is the best first step according to GINA?

(A) Check technique and adherence and treat modifiable factors
(B) Start an oral corticosteroid taper at this visit before other steps
(C) Add a biologic agent at this visit without further review
(D) Refer her for bronchial thermoplasty without other changes
(E) Double how often the reliever inhaler is used each day

Answer Contents

313. A 41-year-old woman has asthma that is poorly controlled despite good adherence and technique on a daily medium-dose inhaled corticosteroid-long-acting beta agonist combination. She has no smoking exposure and her comorbidities are controlled. Under the 2020 NAEPP focused update, which of the following is recommended as add-on therapy?

(A) A scheduled short-acting beta agonist
(B) Daily oral prednisone
(C) Daily long-acting beta agonist alone
(D) Bronchial thermoplasty as first add-on
(E) A long-acting muscarinic antagonist

Answer Contents

314. A 26-year-old woman has mild to moderate allergic asthma triggered by dust mite exposure. Skin testing is positive for dust mite, and symptoms improve when she avoids bedding exposure but flare again at home. She is on daily inhaled corticosteroid, adherent, and not pregnant. Which of the following additional treatments does the 2020 NAEPP focused update conditionally recommend for her?

(A) Bronchial thermoplasty
(B) Allergen immunotherapy as an adjunct
(C) Daily oral prednisone
(D) Chronic azithromycin
(E) Replacing inhaled corticosteroid with antihistamine alone

Answer Contents

315. A 67-year-old postmenopausal woman has never had bone density testing. She has no fractures, takes no glucocorticoids, does not smoke, and has a body mass index of 25. Her mother had no hip fracture. She asks whether she needs a bone density test. Which of the following is the best recommendation according to the 2025 USPSTF statement?

(A) Screen for osteoporosis because she is 65 or older
(B) Defer screening until she has a fragility fracture
(C) Screen if her FRAX score exceeds 20%
(D) Measure serum 25-hydroxyvitamin D instead of bone density
(E) Screen with heel ultrasonography every year

Answer Contents

316. A 58-year-old postmenopausal woman has a body mass index of 18.5, currently smokes, and reports that her mother had a hip fracture at age 72. She has had no fractures herself. Her clinician calculates an increased fracture risk by a clinical risk assessment tool. She is asking whether she should wait until 65 for bone density testing. Which of the following is the best recommendation?

(A) Wait until age 65 because screening is not recommended earlier
(B) Start alendronate without any bone density testing
(C) Screen now because she is under 65 and at increased risk
(D) Screen with a spine radiograph instead of bone density
(E) Check a serum calcitonin level as the screening test

Answer Contents

317. A 60-year-old postmenopausal woman is healthy and lives independently. She has no osteoporosis, no history of fracture, no vitamin D deficiency, and no increased risk for falls. She asks whether daily vitamin D at 400 IU and calcium at 1000 mg would prevent her from breaking a bone. Which of the following is the best response according to the USPSTF?

(A) Those doses are strongly recommended in her case for prevention
(B) Evidence is insufficient to say anything about any supplement dose
(C) Supplementation helps if she takes higher doses of calcium
(D) Those doses are recommended against for fracture prevention
(E) Supplementation is recommended during the winter months

Answer Contents

318. A 70-year-old woman has a spine T-score of -2.8 on bone density testing. She has no prior fracture, normal kidney function, no esophageal disease, and can sit upright for 30 minutes after taking a pill. She is not taking glucocorticoids. She wants to start treatment. Which of the following is the initial pharmacologic treatment recommended by the American College of Physicians?

(A) Denosumab
(B) Romosozumab
(C) Raloxifene
(D) Calcitonin nasal spray
(E) A bisphosphonate

Answer Contents

319. A 72-year-old woman has osteoporosis with a femoral neck T-score of -2.9 and no prior fracture. She is a candidate for treatment, but she has severe esophageal strictures from reflux and cannot take or tolerate an oral bisphosphonate. Her kidney function permits intravenous therapy, but she declines infusions. Which of the following is the best next choice according to the American College of Physicians?

(A) Calcitonin nasal spray
(B) Estrogen-progestin therapy
(C) Calcium supplementation alone
(D) Denosumab
(E) Daily teriparatide as the second-line choice

Answer Contents

320. A 74-year-old woman has completed 5 years of oral bisphosphonate treatment for osteoporosis. Her most recent hip T-score is -2.1, improved from baseline, and she has had no fractures during treatment. She has no other major risk factors and tolerates the drug. She asks whether she should keep taking it. Which of the following is the best recommendation from the American College of Physicians?

(A) Continue the drug without any interruption for life
(B) Consider stopping the bisphosphonate after 5 years
(C) Switch to teriparatide for five more years of therapy
(D) Add denosumab to the current bisphosphonate
(E) Repeat bone density every 6 months to guide timing

Answer Contents

OP3.1. General Office Gynecology

321. A 38-year-old woman has had noncyclic lower abdominal pain for 14 months. Pelvic examination, urinalysis, cervical testing and transvaginal ultrasonography are normal. Naproxen and a course of pelvic floor physical therapy gave partial relief. She asks for an opioid prescription because the pain disturbs her sleep. Which of the following is the best response under the ACOG guidance on chronic pelvic pain?

(A) Decline the opioid and add cognitive behavioral therapy to her plan
(B) Prescribe a short opioid course for flares with a capped monthly supply
(C) Prescribe a low-dose long-acting opioid with monthly review visits
(D) Proceed to diagnostic laparoscopy to search for a cause of the pain
(E) Add a weak opioid to naproxen because combination therapy is better studied

Answer Contents

322. A 41-year-old woman has had noncyclic pelvic pain for 3 years. Diagnostic laparoscopy 2 years ago showed filmy adhesions between the sigmoid colon and the left pelvic sidewall, with no endometriosis. Pain has not changed. She has heard that cutting adhesions can cure pelvic pain and asks to schedule it. Which of the following is the best counsel based on the ACOG guidance on chronic pelvic pain?

(A) Adhesiolysis relieves pain in most women whose adhesions involve the bowel
(B) Adhesiolysis helps only when it is combined with removal of the ovaries
(C) Laparoscopic adhesiolysis has not been shown to help and may worsen outcomes
(D) Adhesiolysis is reasonable if oral medicines have failed for at least 6 months
(E) Adhesiolysis is best done by laparotomy because the view is more complete

Answer Contents

323. A 33-year-old woman has had constant pelvic pain for 2 years. Laparoscopy, pelvic MRI and cystoscopy found no pathology, and she has no allodynia or dermatomal pain pattern to suggest a neuropathic component. Her gynecologist proposes gabapentin and asks what randomized evidence shows. In the GaPP2 trial of 306 women with chronic pelvic pain, which of the following was the finding?

(A) Worst pain fell about twice as much as with placebo, and adverse reactions were similar in both arms
(B) Worst pain did not differ significantly from placebo, and adverse reactions were more frequent
(C) Pain fell significantly only in the subgroup of women who had a normal laparoscopy result earlier
(D) Pain was lower than with placebo at 3 months, but the difference disappeared by the end of follow-up
(E) Adverse reactions were fewer than with placebo, and pain scores were lower in the gabapentin arm

Answer Contents

324. A 35-year-old woman has had 8 months of urinary frequency, urgency and suprapubic pain that eases after she voids. Urine cultures are repeatedly negative, and cystoscopy shows no Hunner lesion. She was told to do daily pelvic floor strengthening exercises with a pelvic floor trainer, and her pain has increased. She asks what to change. Which of the following is the best advice under the AUA guideline for interstitial cystitis and bladder pain syndrome?

(A) Increase the strengthening exercises to three daily sessions
(B) Add vaginal electrical stimulation to deepen the muscle contraction
(C) Continue the exercises and add a short course of oral antibiotics
(D) Continue the exercises and repeat urodynamic testing every 3 months
(E) Stop the pelvic floor strengthening exercises

Answer Contents

325. A 44-year-old woman with a 2-year diagnosis of interstitial cystitis and bladder pain syndrome has pain and urgency despite behavioral measures and physical therapy. Cultures have been negative on every occasion, and cystoscopy found no Hunner lesion. She asks which treatment can be offered next. Which of the following is an option that the AUA guideline allows?

(A) Oral amitriptyline
(B) Long-term oral antibiotic therapy
(C) Long-term oral glucocorticoid therapy
(D) Intravesical bacillus Calmette-Guerin outside a study
(E) High-pressure, long-duration hydrodistension

Answer Contents

326. A 19-year-old woman has painful periods that begin the day before bleeding and last 2 days. Pelvic examination and ultrasonography are normal. She prefers to avoid medicines and asks whether a heating pad is worth trying. Based on the 2026 systematic review of 57 randomized trials, which of the following is the best answer?

(A) Heat gives no benefit beyond placebo in the randomized trials of primary dysmenorrhea pain
(B) Heat works only when it is applied together with a nonsteroidal drug taken in the same cycle
(C) Heat likely reduces pain more than no treatment, in prevention and in acute episodes
(D) Heat relieves pain in acute episodes but not when used for prophylaxis over several months
(E) Heat reduces pain at 3 months but makes acute pain worse within the first 24 hours of use

Answer Contents

327. A 24-year-old woman with primary dysmenorrhea has stopped ibuprofen because it causes dyspepsia on three occasions. Her examination is normal. She asks whether heat is as good as the tablets. In the 2026 systematic review of heat therapy, which of the following was the conclusion when heat was compared with NSAIDs?

(A) Clearly greater pain relief than NSAIDs with a similar adverse effect rate
(B) Clearly less pain relief than NSAIDs, so it is suitable only as an add-on
(C) Comparable analgesic efficacy but more adverse effects, mainly skin burns
(D) Comparable analgesic efficacy with a better safety profile
(E) Comparable efficacy at 3 months but no data on adverse effects

Answer Contents

328. A 22-year-old woman with primary dysmenorrhea gets partial relief from ibuprofen. A friend told her that a different NSAID works much better, and she wants to switch through several agents to find the best one. The gynecologist cites the Cochrane review of NSAIDs for dysmenorrhea, which included 80 randomized trials and 5820 women. Which of the following is the best summary of that review's finding on comparing agents?

(A) Newer agents were clearly better for pain relief than older agents
(B) One agent was clearly safer than the others in head-to-head trials
(C) Agents with a longer half-life gave clearly better pain relief
(D) There was little evidence that any single NSAID is better for pain relief or safety
(E) Agents given only after bleeding starts were clearly better than those taken before

Answer Contents

329. A 20-year-old woman with primary dysmenorrhea wants contraception. She asks whether the combined pill relieves period pain better than an NSAID and wants an answer based on trials. The gynecologist reviews the Cochrane review of the combined oral contraceptive pill for primary dysmenorrhea. Which of the following is the review's statement about trials against NSAIDs?

(A) Pooled trials showed the pill to be superior to NSAIDs for pain relief
(B) Pooled trials showed NSAIDs to be superior to the pill for pain relief
(C) Small trials showed equal pain relief with the pill and NSAIDs
(D) Trials compared them but the results could not be pooled because of heterogeneity
(E) No trials compared the combined pill with NSAIDs

Answer Contents

330. A 26-year-old woman has primary dysmenorrhea that persists despite scheduled NSAID use. She has declined hormonal treatment and asks about a transcutaneous electrical nerve stimulation unit that she has seen advertised. The gynecologist cites the 2024 Cochrane review of TENS for primary dysmenorrhea. Which of the following best states the review's conclusion for TENS compared with placebo or no treatment?

(A) TENS reduces pain, based on high-certainty evidence from large trials
(B) TENS does not reduce pain more than placebo in any trial
(C) TENS may reduce pain, based on low-certainty evidence
(D) TENS reduces pain only if combined with an NSAID
(E) TENS reduces pain but causes frequent skin burns that limit use

Answer Contents

331. A 33-year-old woman has had 2 years of infertility with stage III endometriosis. Her semen analysis and tubal assessment are otherwise normal. She had laparoscopic excision of endometriosis last week. Her surgeon wants to add 6 months of a combined hormonal contraceptive to improve her chance of pregnancy, and she wants to conceive now. What does the ESHRE endometriosis guideline recommend?

(A) Do not prescribe postoperative suppression solely to improve pregnancy rates
(B) Prescribe 6 months of suppression because it doubles the spontaneous pregnancy rate
(C) Prescribe a gonadotropin-releasing hormone agonist for 3 months instead
(D) Prescribe the combined contraceptive only if the surgery left residual disease
(E) Prescribe a levonorgestrel intrauterine system to be removed at 12 months

Answer Contents

332. A 28-year-old woman had laparoscopic cystectomy for a 4-cm endometrioma 6 weeks ago. Her cyclic pain has resolved and she does not plan to conceive for at least 2 years. She asks how to reduce the chance that the cyst and the pain return. Which of the following does the ESHRE endometriosis guideline recommend clinicians offer?

(A) Repeat laparoscopy at 12 months to look for early recurrence of the cyst
(B) Long-term hormone treatment such as a combined hormonal contraceptive
(C) Short-term hormone treatment for 3 months followed by observation alone
(D) Expectant management with pelvic ultrasonography every 6 months
(E) Oral analgesics taken on days with pain and no hormone treatment

Answer Contents

333. A 46-year-old woman had hysterectomy and bilateral salpingo-oophorectomy for severe endometriosis 2 years ago. She now has frequent hot flashes that disturb her sleep and no contraindication to hormone therapy. Her gynecologist plans estradiol alone because she has no uterus. What does the ESHRE guideline advise for menopausal symptoms in a woman with a history of endometriosis?

(A) Estrogen alone, since removal of the uterus removes the risk
(B) Combined estrogen and progestogen at least until the age of natural menopause
(C) Avoid all hormone therapy for life because it could reactivate residual disease
(D) Tibolone alone, stopped at the age of 50
(E) Vaginal estrogen alone, because it treats the vasomotor symptoms

Answer Contents

334. A 36-year-old woman with endometriosis diagnosed 5 years ago worries about ovarian cancer after reading an article. She has no family history of cancer and no new symptoms. She asks for annual CA-125 testing and transvaginal ultrasonography. Which of the following is the best response under the ESHRE endometriosis guideline?

(A) Offer annual CA-125 and ultrasonography because the risk is raised
(B) Offer annual pelvic MRI to look for malignant change
(C) Offer risk-reducing bilateral salpingo-oophorectomy at about age 40 years
(D) Reassure her and follow population-based screening without added tests
(E) Offer diagnostic laparoscopy every 3 years for surveillance

Answer Contents

335. A 31-year-old woman with endometriosis has severe dyspareunia that did not respond to NSAIDs and a progestin. She has moderate hepatic impairment (Child-Pugh B) from an unrelated cause and no osteoporosis. A pregnancy test is negative. Her gynecologist starts elagolix 150 mg orally once daily. According to the US label, what is the maximum duration of treatment for her?

(A) 3 months
(B) 12 months
(C) 18 months
(D) 6 months
(E) 24 months

Answer Contents

336. A 29-year-old woman with endometriosis is starting elagolix after a negative pregnancy test, and she has just had a menstrual period. She is sexually active with a male partner and uses condoms inconsistently. She asks whether the pain medicine will also prevent pregnancy. Which of the following is the best advice according to the US label?

(A) Use an effective nonhormonal contraceptive during treatment
(B) Rely on the missing periods, since elagolix stops ovulation completely
(C) Add a combined oral contraceptive pill for dependable protection
(D) Use emergency contraception only after unprotected intercourse
(E) Stop condoms after the first month of treatment

Answer Contents

337. A 36-year-old premenopausal woman has had distressing acquired, generalized low sexual desire for 2 years, with no pain, depression or medicine cause. Her blood pressure is 118/74 mm Hg. After counseling, she chooses bremelanotide. Which of the following is the correct instruction for using the autoinjector, according to the US label?

(A) Inject each evening at bedtime and continue the injection every day
(B) Inject 15 minutes before sex and repeat the dose in 6 hours if the first one was missed
(C) Inject at least 45 minutes before sex and give no more than one dose in 24 hours
(D) Inject at least 45 minutes before sex and give as many as 8 doses in one week
(E) Inject 2 hours after sex and give no more than one dose in a week

Answer Contents

338. A 40-year-old premenopausal woman has acquired, generalized hypoactive sexual desire disorder that distresses her. Her blood pressure is 168/102 mm Hg on repeated measurements, and she takes no antihypertensive. She requests bremelanotide because she prefers an as-needed treatment. What is the best next step according to the US label?

(A) Start bremelanotide with home blood pressure monitoring for the first month of use
(B) Start bremelanotide but give the first dose in the office with observation
(C) Start bremelanotide at the lowest monthly dose limit of 4 doses
(D) Switch to ospemifene because it does not affect blood pressure
(E) Withhold bremelanotide because uncontrolled hypertension is a contraindication

Answer Contents

339. A 38-year-old woman with acquired, generalized hypoactive sexual desire disorder has taken flibanserin 100 mg orally once daily at bedtime for 8 weeks. She tolerates it well but reports no change in her desire or distress. She asks whether she should double the dose or keep taking it longer. What is the best next step according to the US label?

(A) Increase the dose to twice daily and reassess at 8 weeks
(B) Discontinue flibanserin
(C) Continue the same dose for 8 more weeks
(D) Move the dose to the morning to improve effect
(E) Continue and add a daily phosphodiesterase inhibitor

Answer Contents

340. A 45-year-old woman with acquired, generalized hypoactive sexual desire disorder asks to start flibanserin. She takes oral ketoconazole for a fungal nail infection, expected to continue for 6 more weeks. She has normal liver tests and drinks no alcohol. Which of the following is the best response according to the US label?

(A) Start flibanserin at half the usual dose while the antifungal continues
(B) Start flibanserin and take it at bedtime on antifungal days only
(C) Start flibanserin with a blood pressure check one week later
(D) Do not start flibanserin while she takes the strong CYP3A4 inhibitor
(E) Start flibanserin and hold the antifungal on the evenings she takes it

Answer Contents

341. A 63-year-old woman has a newly diagnosed, biopsy-proven estrogen receptor-positive breast cancer and is awaiting surgery next month. She has moderate to severe dyspareunia from vulvovaginal atrophy, and vaginal moisturizers and lubricants have not helped. Her gynecologist considers ospemifene because it is an oral drug. Which of the following is the best response according to the US label?

(A) Prescribe ospemifene because its antiestrogen effect in breast tissue makes it safe
(B) Prescribe ospemifene with an annual mammogram as the added safeguard
(C) Prescribe ospemifene together with a progestin to protect the breast
(D) Prescribe ospemifene only if she is also taking tamoxifen
(E) Avoid ospemifene, since it should not be used with known or suspected breast cancer

Answer Contents

342. A 52-year-old woman with bothersome hot flashes cannot take estrogen because of a prior venous thromboembolism. She chooses fezolinetant. Baseline blood tests show ALT 95 U/L (upper limit of normal 35 U/L), AST 88 U/L (upper limit of normal 35 U/L) and total bilirubin 0.8 mg/dL. She drinks no alcohol and takes no other drugs. According to the US label, which of the following is the best next step?

(A) Start fezolinetant and repeat liver tests monthly for the first 3 months
(B) Start fezolinetant at a lower daily dose and repeat liver tests at 6 months of therapy
(C) Do not start fezolinetant, because ALT or AST is at least twice the upper limit
(D) Start fezolinetant and stop it only if bilirubin becomes elevated
(E) Start fezolinetant because the elevations are under 5 times the upper limit

Answer Contents

343. A 49-year-old woman starts fezolinetant 45 mg orally once daily for hot flashes after normal baseline liver tests. She asks when blood tests will be repeated. She has no liver symptoms. Which of the following is the schedule for follow-up hepatic laboratory testing recommended in the US label?

(A) Monthly for the first 3 months, then at 6 and 9 months
(B) At baseline only, since normal results predict later safety
(C) Monthly for the first 12 months, then yearly
(D) Every 6 months for as long as she takes the drug
(E) At 3 months only, then when symptoms occur

Answer Contents

344. A 54-year-old woman has moderate to severe hot flashes and night sweats. She has breast cancer in remission and does not want hormone therapy. Baseline transaminases are normal, and her gynecologist prescribes elinzanetant 120 mg orally once daily at bedtime. When should follow-up hepatic transaminase testing be done according to the US label?

(A) Weekly for the first month of therapy
(B) Monthly for the first 3 months of therapy
(C) At 6 and 9 months after starting therapy
(D) 3 months after starting therapy
(E) Once a year after starting therapy

Answer Contents

345. A 51-year-old woman with bothersome hot flashes wants a nondrug approach. She has no contraindication to any of the options she has read about. Her gynecologist reviews the 2023 NAMS nonhormone therapy position statement. Which of the following nonpharmacologic interventions does the statement not recommend for vasomotor symptoms?

(A) Paced respiration
(B) Cognitive behavioral therapy
(C) Clinical hypnosis
(D) Weight loss for women with overweight
(E) Stellate ganglion block in select women

Answer Contents

346. A 55-year-old woman with bothersome hot flashes and a uterus is offered hormone therapy. She says she would prefer 'bioidentical' estradiol and progesterone mixed by a compounding pharmacy, because her friend says they are safer than approved products. Which of the following is the best response according to the 2022 NAMS hormone therapy position statement?

(A) Compounded products are safer because dose is individualized by saliva testing
(B) Their purity and dose are poorly regulated, so an approved product is preferred
(C) Compounded products have equal safety because the ingredients are identical
(D) Compounded products are preferred only when she has a uterus
(E) Compounded products offer better endometrial protection than approved progesterone

Answer Contents

347. A 53-year-old woman with bothersome hot flashes has no contraindication to treatment. She asks about pregabalin because she takes it for a friend's nerve pain and was told it also helps hot flashes. The gynecologist reviews the 2023 NAMS nonhormone therapy position statement. Which of the following is the statement's position on pregabalin for vasomotor symptoms?

(A) It is recommended at Level I as first-line therapy
(B) It is recommended at Level II for women with a history of breast cancer
(C) It is not recommended
(D) It is recommended in place of gabapentin when sedation is a concern
(E) It is recommended only for women who cannot take paroxetine

Answer Contents

348. A 41-year-old woman and her husband have had regular unprotected intercourse for 4 months without pregnancy. Her cycles are regular and every 28 days. She has no known pelvic disease. She says her sister was told to wait a year before evaluation and asks whether she should do the same. Which of the following is the best advice according to ACOG?

(A) Wait until she has tried for 6 months, then evaluate
(B) Wait until 12 months, as for younger women
(C) Start clomiphene without any testing and evaluate only after 3 treatment cycles
(D) Evaluate only the male partner until 12 months have passed
(E) Begin the evaluation and treatment now because she is older than 40

Answer Contents

349. A 30-year-old woman has tried to conceive for 15 months. Her semen analysis is normal. Hysterosalpingography shows no filling of either tube, and she has no history of pelvic infection or surgery. She asks whether this proves both tubes are blocked. The reported positive predictive value of hysterosalpingography for tubal occlusion is 38% and the negative predictive value is 94%. Which of the following is the best interpretation?

(A) Nonpatency is diagnostic, so she should proceed directly to in vitro fertilization
(B) A normal film would still leave about 40% of tubes occluded
(C) The test excludes intrauterine adhesions and polyps when tubes fill
(D) Nonpatency is often a false positive, so confirm it before choosing treatment
(E) Repeat the study at the same visit with higher injection pressure

Answer Contents

350. A 29-year-old woman has not conceived after 18 months. Her cycles come every 45 to 60 days. A semen analysis for her husband is normal, and a hysterosalpingogram shows normal tubes and cavity. Her body mass index is 27 kg/m2, and the thyroid is not palpable. Which of the following is the most appropriate additional laboratory test according to ACOG?

(A) Thyroid peroxidase antibodies
(B) Antisperm antibodies in serum
(C) Serum thyrotropin
(D) Late luteal phase endometrial biopsy
(E) Serum progesterone on day 3 of the cycle

Answer Contents

351. A 32-year-old woman and her husband have been trying to conceive for 14 months. Her cycles are regular, she has no pelvic pain, and her gynecologist plans hysterosalpingography. The husband has not been evaluated, and the couple asks whether he must be seen by a urologist before any testing. Which of the following is the best next step according to ACOG?

(A) Delay the male evaluation until her tubal assessment has been completed and reviewed
(B) Refer him to a urologist before any testing is ordered
(C) Order a postcoital test to assess sperm function
(D) Order serum testosterone and karyotype as first-line tests
(E) Obtain his history and order a semen analysis from the women's health clinic

Answer Contents

352. A 37-year-old woman has not conceived after 7 months of trying. Her cycles are regular. A cycle day 3 FSH is 14 mIU/mL and estradiol is 58 pg/mL. She asks whether this means she cannot have a baby with her own eggs. Which of the following is the best response according to ACOG?

(A) The result confirms that donor oocytes are the only route to pregnancy
(B) Poor ovarian reserve results do not necessarily predict inability to achieve a live birth
(C) The result is inconclusive until the FSH is repeated on cycle day 21
(D) The result excludes her from intrauterine insemination but not from in vitro fertilization
(E) The result means she should wait for another year before further testing

Answer Contents

353. A 44-year-old woman is undergoing chemotherapy for current, hormone receptor-positive breast cancer. She is sexually active with a male partner, and her oncologist wants reliable contraception during treatment. She has regular, light periods and prefers a long-acting method that does not release hormones. Which of the following has a US Medical Eligibility Criteria category of 1 for her condition?

(A) Copper intrauterine device
(B) Levonorgestrel 52 mg intrauterine system
(C) Etonogestrel implant
(D) Depot medroxyprogesterone acetate injection
(E) Combined oral contraceptive pill

Answer Contents

354. A 24-year-old woman took a single oral 30 mg dose of ulipristal acetate 2 days ago after unprotected intercourse. She wants to start a combined oral contraceptive pill and asks when to begin and whether she needs backup. Pregnancy is reasonably excluded. Which of the following is the best advice according to the US Selected Practice Recommendations?

(A) Start the pill today and use condoms for 7 days
(B) Start the pill today with no backup, since ulipristal does not interact
(C) Start the pill tomorrow and use backup for 3 days
(D) Start the pill 14 days after ulipristal and use backup for 7 days
(E) Start 5 days after ulipristal, then use condoms or abstain for 7 days

Answer Contents

355. A 27-year-old woman received her last depot medroxyprogesterone acetate injection 14 weeks ago. She has had no intercourse since the last injection and is otherwise well. She asks whether she can still have the next injection and whether she needs a backup method afterward. Which of the following is the best response according to the US Selected Practice Recommendations?

(A) Give the injection now and require backup for 7 days
(B) Give the injection now, with no additional contraceptive protection needed
(C) Give the injection after a negative pregnancy test and 7 days of abstinence
(D) Restart with a new injection series after the next menses
(E) Switch to a different method because the injection is overdue

Answer Contents

356. A 27-year-old nulliparous woman requests a levonorgestrel intrauterine system. She has one partner, a recent negative chlamydia and gonorrhea screen, and a normal pelvic examination. She is anxious about infection and asks for a prophylactic antibiotic before placement. Which of the following is the best response according to the US Selected Practice Recommendations?

(A) Give a single oral dose of azithromycin before placement
(B) Give doxycycline for 7 days starting before placement
(C) Give a single intramuscular dose of ceftriaxone before placement
(D) Place the device without prophylactic antibiotics
(E) Delay placement until a pelvic ultrasound has been done

Answer Contents

357. A 22-year-old woman wants to start a combined oral contraceptive pill. Her last menstrual period began 9 days ago, she has had no intercourse since it began, and her urine pregnancy test is negative. She has no contraindications. She asks whether she can start today and what protection she needs. Which of the following is the best advice according to the US Selected Practice Recommendations?

(A) Start today and use condoms or abstain for the next 7 days
(B) Start today and rely on the pill from the very first tablet onward
(C) Wait until the next menstrual period to start
(D) Start today and use backup for 21 days
(E) Start today and use backup only for the first 3 days

Answer Contents

358. A 29-year-old woman with a body mass index of 36 kg/m2 had unprotected intercourse 4 days ago and does not want to become pregnant. A copper intrauterine device cannot be placed this week. She asks which emergency contraceptive pill works best for her. Which of the following is the best choice according to the US Selected Practice Recommendations?

(A) Levonorgestrel 1.5 mg orally in a single dose
(B) Levonorgestrel 0.75 mg orally twice, 12 hours apart
(C) Ulipristal acetate 30 mg orally in a single dose
(D) Ethinyl estradiol 100 micrograms with levonorgestrel 0.50 mg orally twice, 12 hours apart
(E) Levonorgestrel 3 mg orally in a single dose

Answer Contents

359. A 24-year-old woman has lower abdominal pain, fever of 38.2 C and cervical motion tenderness. A copper intrauterine device was placed 18 months ago. A pregnancy test is negative and nucleic acid testing has been sent. Outpatient therapy with intramuscular ceftriaxone, oral doxycycline and oral metronidazole is started. She asks whether the device must come out today. Which of the following is the best response according to the CDC guidelines?

(A) Remove the device now and replace it with a new one after 14 days of antibiotic treatment
(B) Remove the device before the first antibiotic dose to prevent spread
(C) Leave it in place, reassess in 48 to 72 hours, and remove it if she has not improved
(D) Remove the device after the test results return positive
(E) Leave the device in place and reassess only if the fever returns

Answer Contents

360. A 22-year-old woman is diagnosed with chlamydial cervicitis on nucleic acid testing. Her symptoms began 3 weeks ago. She had sex with partner A 2 weeks ago, partner B 6 weeks ago, partner C 4 months ago and partner D 8 months ago. She asks which partners need to be evaluated or treated. Which of the following is the best answer according to the CDC guidelines?

(A) Partner A only, as the most recent partner
(B) Partners A and B, who had sex within the 60 days before symptoms or diagnosis
(C) Partners A, B and C, because the lookback period is 3 months before diagnosis
(D) Partners A, B, C and D, because all partners in the past year need treatment
(E) No partners, because she is being treated

Answer Contents

361. A 26-year-old woman has mucopurulent cervicitis. A nucleic acid test for Neisseria gonorrhoeae is positive and the chlamydia result is not yet available. She weighs 70 kg and has no drug allergy. Which of the following is the best treatment according to the CDC guidelines?

(A) Ceftriaxone 500 mg intramuscularly once alone
(B) Ceftriaxone 250 mg intramuscularly once plus azithromycin 1 g orally once
(C) Cefixime 800 mg orally once alone
(D) Ceftriaxone 500 mg intramuscularly plus doxycycline 100 mg orally twice daily for 7 days
(E) Gentamicin 240 mg intramuscularly once plus azithromycin 2 g orally once, with no doxycycline

Answer Contents

362. A 28-year-old woman with no symptoms or genital lesions has a new male partner. She requests a full panel of sexually transmitted infection tests, including herpes antibody testing, because she wants to know her status. Cervical nucleic acid tests for chlamydia and gonorrhea and HIV testing are planned. Which of the following is the best response according to the CDC guidelines on genital herpes?

(A) Do not order HSV serology, because screening the general population is not recommended
(B) Order HSV-2 type-specific serology and treat her with antivirals if it is positive
(C) Order HSV-1 and HSV-2 IgM antibodies and treat her with antivirals if they are positive
(D) Order HSV culture from the cervix
(E) Order HSV polymerase chain reaction testing of the vagina

Answer Contents

363. A 33-year-old woman was treated for secondary syphilis 6 months ago. Her nontreponemal titer before treatment was 1:64. Today's rapid plasma reagin titer is 1:16. She has no symptoms and no new partners. Her clinician is deciding whether treatment succeeded. Which of the following is the best interpretation according to the CDC guidelines?

(A) A one-dilution decline is the smallest change that counts as a response
(B) The titer must be negative by 6 months or the treatment is judged to have failed
(C) The result requires cerebrospinal fluid examination before it can be read
(D) The result is a serofast state that needs retreatment
(E) A fourfold fall, equal to two dilutions, is a significant change

Answer Contents

364. A 31-year-old woman was treated for trichomoniasis with oral metronidazole 500 mg twice daily for 7 days. Her symptoms resolved, and her partner was treated at the same time. She asks whether she needs any follow-up testing. Which of the following is the best response according to the CDC guidelines?

(A) No retesting is needed once symptoms have resolved
(B) Repeat a test of cure 1 week after finishing treatment
(C) Retest at 4 weeks with a wet mount
(D) Retest approximately 3 months after treatment
(E) Retest in 12 months with her annual examination

Answer Contents

365. A 7-year-8-month-old girl is brought in because her mother noticed breast budding 2 months ago. Examination shows Tanner stage 2 breasts, no pubic hair, no neurologic findings, and a height at the 70th percentile. Growth velocity by her pediatric records is normal. Her mother asks for a brain scan and hormone tests. Which of the following is the best step according to the 2026 Endocrine Society guideline?

(A) Brain MRI with contrast now
(B) Gonadotropin-releasing hormone stimulation testing now
(C) Start a gonadotropin-releasing hormone agonist now
(D) Order a bone age radiograph and begin treatment if it is advanced
(E) Watchful waiting with periodic physical examinations

Answer Contents

366. A 6-year-6-month-old girl has Tanner stage 3 breast development, pubic hair, and a growth rate of 9 cm in the past year. Bone age is advanced by 2 years. Her pediatrician suspects central precocious puberty and asks which hormone test to order first. Which of the following is the best initial test according to the 2026 Endocrine Society guideline?

(A) Basal luteinizing hormone by ultrasensitive assay
(B) Gonadotropin-releasing hormone stimulation test first
(C) Random follicle-stimulating hormone alone
(D) Serum estradiol alone
(E) Overnight prolactin profile

Answer Contents

367. A 7-year-2-month-old girl has Tanner stage 3 breasts and a basal luteinizing hormone level in the pubertal range, which confirms central precocious puberty. She has no headaches, visual changes, seizures or focal neurologic signs, and her examination is otherwise normal. Her parents ask whether she needs a brain MRI. Which of the following is the best response according to the 2026 Endocrine Society guideline?

(A) Brain MRI is required in every girl who has confirmed central precocious puberty
(B) Brain MRI is needed only if the bone age is advanced by more than 2 years at diagnosis
(C) Routine brain MRI is not suggested at ages 6 to 8 years without neurologic findings
(D) Brain MRI should be repeated yearly during treatment
(E) Head CT should replace MRI in this age group

Answer Contents

368. A 10-year-6-month-old girl has been treated with a gonadotropin-releasing hormone agonist for central precocious puberty since age 6. Her bone age is 11 years, growth velocity is slowing, and she has had no adverse effects. Her mother asks to continue the injections until age 12 to delay menarche further. Which of the following is the best response according to the 2026 Endocrine Society guideline?

(A) Continue until age 12, because every additional year of delay improves adult height
(B) Stop the agonist, as treatment past age 10 to 11 is not routine
(C) Continue until bone age reaches 13 years
(D) Add growth hormone and continue for one more year
(E) Double the dose to improve suppression at this age

Answer Contents

369. A 15-year-old girl with short stature and absent breast development has a 45,X karyotype on a 30-cell peripheral blood analysis, consistent with Turner syndrome. Over the past year she has developed clitoral enlargement and coarse facial hair, and her serum testosterone is elevated for her age. Pelvic ultrasonography shows small gonads. Which of the following is the best next step according to the 2024 international Turner syndrome guideline?

(A) Repeat karyotype on a skin fibroblast culture only
(B) Dexamethasone suppression testing for an adrenal tumor
(C) Immediate bilateral gonadectomy without further testing
(D) Molecular testing for Y chromosomal material by PCR
(E) Reassurance, because virilization is expected in Turner syndrome

Answer Contents

370. A 52-year-old woman with a uterus has a high 5-year breast cancer risk by the Gail model, a prior biopsy showing atypical hyperplasia, and no history of venous thromboembolism. She is willing to take a risk-reducing medicine, and her gynecologist compares tamoxifen with raloxifene. Which of the following statements about their adverse effects is correct according to the USPSTF?

(A) Raloxifene, but not tamoxifen, increases the risk of venous thromboembolism in women with a uterus
(B) Raloxifene, but not tamoxifen, increases the risk of endometrial cancer in women with a uterus
(C) Tamoxifen and raloxifene increase the risk of endometrial cancer to the same degree in women
(D) Neither tamoxifen nor raloxifene increases the risk of venous thromboembolism in any woman
(E) Tamoxifen, but not raloxifene, increases the risk of endometrial cancer in women with a uterus

Answer Contents

371. A 47-year-old woman with average breast cancer risk has heterogeneously dense breasts on her screening mammogram. She asks whether she should add an annual breast MRI or ultrasonography, as a news article suggested. Which of the following best describes the 2024 USPSTF position on supplemental screening for women with dense breasts?

(A) The evidence is insufficient to assess the balance of benefits and harms
(B) Supplemental MRI is recommended every year
(C) Supplemental ultrasonography is recommended every 2 years
(D) Supplemental imaging is recommended against because harms outweigh benefits
(E) Supplemental imaging replaces mammography at her age

Answer Contents

372. A 52-year-old woman noticed spontaneous bloody discharge from a single duct of the left nipple for 6 weeks. There is no palpable mass, no skin change and no lymphadenopathy. She has not had imaging in 14 months. Her gynecologist orders an initial imaging study. Which of the following is rated usually appropriate by the ACR for pathologic nipple discharge at her age?

(A) Ductography of the affected duct as the first and only imaging study
(B) Contrast-enhanced breast MRI as the first and only imaging study
(C) Diagnostic mammography or tomosynthesis with breast ultrasonography
(D) FDG positron emission tomography of the breast as the initial study
(E) No imaging until the discharge has persisted for 3 months

Answer Contents

373. A 26-year-old woman has spontaneous bloody discharge from a single duct of the right nipple for 3 weeks. She is not pregnant or lactating and has no palpable mass. She asks for a mammogram because her mother had breast cancer at 60. Which of the following is the usually appropriate initial imaging for a female patient younger than 30 with pathologic nipple discharge according to the ACR?

(A) Diagnostic mammography
(B) Breast ultrasonography
(C) Digital breast tomosynthesis
(D) Breast MRI
(E) Ductography

Answer Contents

374. A 34-year-old woman notices milky to greenish fluid from both nipples only when she squeezes them. The fluid comes from several ducts, and there is no blood, no mass and no spontaneous leakage. Her pregnancy test is negative and her breast examination is normal. She asks whether she needs breast imaging. Which of the following is the ACR rating for imaging in physiologic nipple discharge?

(A) Ultrasonography is usually appropriate
(B) Imaging is usually not appropriate
(C) Diagnostic mammography is usually appropriate
(D) Breast MRI is usually appropriate
(E) Ductography is usually appropriate

Answer Contents

OP3.2. Endocrine Disorders

375. A 31-year-old woman is evaluated during an infertility workup. Her cycles are regular at 28 to 30 days with documented midluteal progesterone elevation. She has no hirsutism, acne or hair loss, and total and free testosterone are normal. Serum anti-Müllerian hormone (AMH) is markedly elevated, and a trainee proposes labeling her with polycystic ovary syndrome (PCOS) on that result. Which of the following best reflects the 2023 international PCOS guideline?

(A) AMH is not a single diagnostic test, and she fulfills no clinical criterion
(B) An elevated AMH alone is sufficient to diagnose PCOS in any adult woman
(C) AMH replaces the need to assess ovulation and androgens in adult women
(D) AMH is a valid stand-alone test in adults but not valid in adolescents
(E) A markedly elevated AMH confirms PCOS when pelvic ultrasound is not available

Answer Contents

376. A 24-year-old woman has had cycles every 50 to 70 days since age 15. Her modified Ferriman-Gallwey score is 12, and free testosterone is elevated. TSH, prolactin and 17-hydroxyprogesterone are normal, and she is not pregnant. She has not had pelvic imaging and asks whether she must have an ultrasound before a diagnosis can be made. Which of the following is the best response according to the 2023 international PCOS guideline?

(A) Obtain transvaginal ultrasound first to document at least 20 follicles per ovary
(B) Obtain an AMH level and diagnose PCOS if it is elevated
(C) Obtain pelvic MRI to confirm polycystic ovarian morphology
(D) Diagnose PCOS now because ovarian ultrasound is not necessary
(E) Repeat the androgen panel in 6 months before making any diagnosis

Answer Contents

377. A 15-year-old girl had menarche at age 12 and has had cycles every 40 to 90 days since. BMI is 27. She has mild acne but no hirsutism, and total testosterone is normal. TSH and prolactin are normal. Her mother has PCOS and asks whether her daughter has it. Which of the following is the best management according to the 2023 international PCOS guideline?

(A) Diagnose PCOS now on the basis of irregular cycles alone, given her family history
(B) Diagnose PCOS after obtaining a transvaginal pelvic ultrasound to count follicles
(C) Note increased risk, give lifestyle advice, and reassess by 8 years after menarche
(D) Reassure her that irregular cycles are expected until age 25 and plan no follow-up
(E) Start metformin now to prevent progression to PCOS, with no further assessment

Answer Contents

378. A 22-year-old woman has just been diagnosed with PCOS on the basis of oligomenorrhea and biochemical hyperandrogenism. Her BMI is 22, and she has no family history of diabetes. A fasting glucose last month was normal. She asks whether a normal weight and normal fasting glucose mean she can skip further glycemic testing. Which of the following does the 2023 international PCOS guideline name as the most accurate test of her glycemic status?

(A) Fasting plasma glucose repeated annually
(B) Hemoglobin A1c alone
(C) Random plasma glucose at the next visit
(D) Fasting insulin with homeostatic model assessment
(E) Two-hour 75-g oral glucose tolerance test

Answer Contents

379. A 19-year-old woman is seen for oligomenorrhea and hirsutism and is diagnosed with PCOS. She focuses on her skin and hair and does not mention her mood. She is a college student with no psychiatric history, takes no medications, and answers all questions politely. Examination is otherwise unremarkable. Which of the following is the most appropriate addition to her initial assessment according to the 2023 international PCOS guideline?

(A) Psychiatric referral if her BMI exceeds 30 kg/m2 at any later visit
(B) Starting an antidepressant before any assessment of her mood symptoms
(C) Deferring mood screening until she raises a concern about her mood
(D) Screening for depression and anxiety with regionally validated tools
(E) Screening for eating disorders instead of screening for depression

Answer Contents

380. A 26-year-old woman with PCOS has had hirsutism despite 4 months of a combined oral contraceptive and regular laser hair removal. Her score has improved only slightly, and she is frustrated and asks to add spironolactone today. She is using the pill consistently and has no contraindication to spironolactone. According to the 2023 international PCOS guideline, which of the following is the best next step?

(A) Add an antiandrogen now because the pill alone has not controlled her hirsutism
(B) Stop the pill and start metformin as the main treatment for her hirsutism
(C) Replace the combined pill with a 5-alpha reductase inhibitor for the hirsutism
(D) Switch to a progestin-only method and add an antiandrogen at the same visit
(E) Continue pill and cosmetic care, then reassess for an antiandrogen at 6 months

Answer Contents

381. A 36-year-old woman has regular cycles every 28 days and no galactorrhea, headaches or visual change. She takes no medications. Prolactin on a screening panel was 62 ng/mL and was 58 ng/mL on repeat. Pregnancy test, TSH and creatinine are normal. She asks what to do next. Which of the following is the most appropriate next step according to the Endocrine Society guideline?

(A) Assess the sample for macroprolactin
(B) Start a dopamine agonist at once
(C) Obtain pituitary MRI with gadolinium before any other testing
(D) Refer for transsphenoidal surgery
(E) Repeat the prolactin in 12 months with no further testing

Answer Contents

382. A 30-year-old woman has had amenorrhea and milky nipple discharge for 6 weeks. She began metoclopramide 8 weeks ago for gastroparesis. Pregnancy test and TSH are normal, and prolactin is 90 ng/mL. She has no headache or visual symptoms. According to the Endocrine Society guideline, which of the following is the best first step in treatment?

(A) Start cabergoline while she continues to take metoclopramide
(B) Stop or replace metoclopramide, with the prescriber
(C) Obtain pituitary MRI before changing any of her medications
(D) Refer for transsphenoidal surgery of the pituitary gland
(E) Begin estrogen-progestin therapy and continue metoclopramide

Answer Contents

383. A 40-year-old woman has oligomenorrhea, galactorrhea and a prolactin of 140 ng/mL. MRI shows a 7-mm pituitary lesion with no mass effect, and visual fields are normal. She is not seeking pregnancy and has no medication that raises prolactin. Her TSH is normal. According to the Endocrine Society guideline, which of the following is the preferred drug therapy?

(A) Bromocriptine
(B) Pergolide
(C) Cabergoline
(D) Estrogen-progestin pills
(E) Octreotide

Answer Contents

384. A 38-year-old woman has regular cycles every 28 days, no galactorrhea, no sexual dysfunction and no headaches. A pituitary MRI obtained for a head injury shows a 5-mm microadenoma. Prolactin is 48 ng/mL on two samples, macroprolactin is excluded, and TSH is normal. She takes no medications. According to the Endocrine Society guideline, which of the following is the best management?

(A) Observe without a dopamine agonist, with clinical monitoring
(B) Start cabergoline in an attempt to shrink the lesion quickly
(C) Start bromocriptine to be taken at bedtime every day
(D) Refer for transsphenoidal resection of the microadenoma
(E) Start an estrogen-progestin pill to suppress its growth

Answer Contents

385. A 41-year-old woman with a microprolactinoma has taken cabergoline for 3 years. For the last 2 years her prolactin has been normal, and her cycles are regular. A repeat MRI shows no visible tumor remnant. She asks whether she must stay on the drug for life. According to the Endocrine Society guideline, which of the following is the best response?

(A) Continue the drug until menopause, since stopping is not advised after tumor shrinkage
(B) Stop the drug abruptly and arrange no follow-up testing
(C) Double the dose for a year to prevent a later recurrence
(D) Taper and consider stopping the drug, with prolactin monitoring after
(E) Switch to bromocriptine for long-term maintenance treatment

Answer Contents

386. A 7-year-old girl has had breast budding for 2 months. Examination shows Tanner stage B2 breasts, no pubic or axillary hair, no neurologic findings, and normal growth velocity. She is otherwise well. Her mother is anxious and asks for blood tests and brain imaging today. Which of the following is the most appropriate management according to the 2026 Endocrine Society central precocious puberty guideline?

(A) GnRH stimulation testing now
(B) Brain MRI with contrast now
(C) Watchful waiting with periodic physical examinations
(D) Start a GnRH agonist for presumed central precocious puberty
(E) Pelvic MRI to assess uterine volume

Answer Contents

387. A 6-year-old girl has Tanner stage B3 breasts that progressed from B2 over 3 months. Her height has crossed upward from the 50th to the 90th percentile, and bone age is 2 years advanced. Neurologic examination is normal. She has no vaginal bleeding. According to the 2026 Endocrine Society guideline, which of the following is the best initial biochemical test?

(A) GnRH agonist stimulation test as the first test
(B) Basal LH measured by an ultrasensitive assay
(C) Random serum estradiol measurement alone
(D) Karyotype with chromosomal microarray
(E) Serum 17-hydroxyprogesterone with cortisol

Answer Contents

388. A 7-year-old girl has biochemically confirmed central precocious puberty, with elevated basal LH, Tanner B3 breasts and advanced bone age. She has no headaches, vision change, seizures or neurologic findings, and head circumference is normal. Her parents ask whether brain imaging is required. According to the 2026 Endocrine Society guideline, which of the following is the best answer?

(A) Brain MRI is required for every girl who has central precocious puberty
(B) Head CT without contrast is the preferred routine imaging study at her age
(C) Pelvic MRI should replace brain imaging for girls at this age
(D) Skull radiographs are sufficient to exclude an intracranial tumor in her
(E) Routine brain MRI is not needed at this age without neurologic findings

Answer Contents

389. A girl with idiopathic central precocious puberty diagnosed at age 6 has received a long-acting GnRH agonist for 5 years. She is now 11 years and 4 months old, Tanner stage B3, and bone age is 12.5 years. Growth velocity is appropriate for her pubertal stage. Her parents ask how long the injections should continue. According to the 2026 Endocrine Society guideline, which of the following is the best response?

(A) Continue until menarche is wanted, at about 13 years of age
(B) Continue until the bone age has advanced to 14 years in all girls
(C) Plan to stop treatment now rather than continue it routinely
(D) Continue indefinitely until she has reached her adult height
(E) Stop after adding growth hormone for a year of overlapping therapy

Answer Contents

390. A 9-year-old girl is 14 months into GnRH agonist treatment for central precocious puberty. Puberty is suppressed, growth velocity has fallen to 5 cm per year, and her predicted adult height falls short of her mid-parental target. She has no growth hormone deficiency and no other indication for growth hormone. Her parents ask about adding growth hormone to improve final height. According to the 2026 Endocrine Society guideline, which of the following is the best response?

(A) Growth hormone should be added in every child on a GnRH agonist
(B) Routine addition of growth hormone is not suggested
(C) Growth hormone should replace the GnRH agonist
(D) Growth hormone should be added after menarche
(E) An aromatase inhibitor should be added routinely

Answer Contents

391. A 34-year-old woman with Graves disease has taken methimazole for 6 weeks. She calls the office with a temperature of 38.6 C and a sore throat since the morning. She has no cough or rash and is hemodynamically stable. She asks whether to wait for a viral illness to pass. According to the 2016 American Thyroid Association hyperthyroidism guideline, which of the following is the most appropriate step?

(A) Increase the methimazole dose and recheck in 1 week
(B) Start a beta-blocker and reassure her
(C) Obtain a throat culture and continue methimazole unchanged
(D) Schedule radioactive iodine within the week
(E) Obtain a white blood cell count with differential

Answer Contents

392. A 38-year-old woman has Graves disease with a positive TSH receptor antibody and prefers definitive treatment with radioactive iodine. Ultrasound shows a 2.3-cm solid hypoechoic nodule with microcalcifications in the left lobe, and the nodule is cold on uptake imaging. She has regular periods and a negative pregnancy test. According to the 2016 American Thyroid Association guideline, which of the following is the most appropriate next step?

(A) Proceed with radioactive iodine as she has already planned
(B) Continue antithyroid drugs without any evaluation of the nodule
(C) Repeat the thyroid ultrasound after 12 months of observation
(D) Fine-needle aspiration of the nodule before definitive therapy
(E) Treat with a beta-blocker alone and recheck the TSH later

Answer Contents

393. A 52-year-old woman has weight loss, palpitations and heat intolerance for 2 months. Her gland contains a palpable 3-cm nodule without bruit, and she has no orbitopathy. TSH is suppressed at 0.01 mIU/L, and free T4 is elevated. Thyroid receptor antibody testing is pending. According to the 2016 American Thyroid Association guideline, which of the following is the best next study to establish the cause?

(A) Radioactive iodine (123I) or pertechnetate uptake scan
(B) Computed tomography of the neck with iodinated contrast
(C) Repeat TSH in 3 months
(D) Fine-needle aspiration alone without functional imaging
(E) Radioactive iodine therapy without prior scanning

Answer Contents

394. A 48-year-old perimenopausal woman with primary hypothyroidism has taken the same levothyroxine dose for 4 years with TSH values near 2 mIU/L. Two months ago she started calcium carbonate and ferrous sulfate for heavy menses and bone health, taking all three tablets together at breakfast. Her TSH is now 9.8 mIU/L, and she has fatigue and cold intolerance. She is adherent and has no new illness. According to the 2014 American Thyroid Association guideline, which of the following is the best management?

(A) Separate levothyroxine from the calcium and iron
(B) Double the levothyroxine dose immediately and recheck
(C) Switch to desiccated thyroid extract replacement
(D) Add liothyronine to the current regimen
(E) Stop the iron and calcium supplements permanently

Answer Contents

395. A 42-year-old woman with newly diagnosed Graves disease has palpitations, tremor and anxiety. Heart rate is 118 beats per minute, blood pressure is 138/78 mmHg, and she has no asthma or conduction disease. Methimazole has been prescribed today. She asks how her symptoms will be controlled during the weeks before the drug takes effect. According to the 2016 American Thyroid Association guideline, which of the following is the best addition?

(A) A benzodiazepine for anxiety
(B) A loop diuretic for fluid retention
(C) A second antithyroid drug taken at the same time
(D) Levothyroxine to block the thyroid
(E) A beta-adrenergic blocker

Answer Contents

396. A 33-year-old woman with type 2 diabetes and a BMI of 38 is starting tirzepatide. She uses a daily combined oral contraceptive and does not want pregnancy. Her hemoglobin A1c is 8.4 percent, and she has no history of pancreatitis or thyroid disease. She asks whether her contraception will still work. According to the tirzepatide label, which of the following is the best advice?

(A) No change is needed, because the drug has no effect on oral contraceptives
(B) Take the pill twice daily during the first month of tirzepatide treatment
(C) Stop the pill and rely on withdrawal as her method of contraception
(D) Add barrier protection or go non-oral for 4 weeks after each dose increase
(E) Use emergency contraception after every dose increase of the drug

Answer Contents

397. A 36-year-old woman with type 2 diabetes has a hemoglobin A1c of 8.1 percent on metformin and a BMI of 36. Her mother had medullary thyroid carcinoma. Her thyroid examination is normal, and she is requesting weight loss assistance. Her clinician considers adding tirzepatide. According to the tirzepatide label, which of the following is the best decision?

(A) Start tirzepatide with annual thyroid ultrasonography for surveillance
(B) Start tirzepatide with a baseline serum calcitonin level for surveillance
(C) Avoid tirzepatide, given the family history of thyroid carcinoma
(D) Start tirzepatide at half of the usual starting dose for safety
(E) Start tirzepatide if RET genetic testing is negative for a variant

Answer Contents

398. A 52-year-old woman with type 2 diabetes has taken metformin for 9 years. She reports numbness in both feet for 3 months. Her hemoglobin is 10.4 g/dL with a mean corpuscular volume of 104 fL. Her eGFR is 78 mL/min/1.73 m2, and she drinks little alcohol. Her vitamin B12 has not been checked. According to the metformin label, which of the following is the most appropriate test?

(A) Serum folate level alone
(B) Serum vitamin B12 level
(C) Repeat hemoglobin A1c
(D) Nerve conduction studies
(E) MRI of the lumbar spine

Answer Contents

399. A 58-year-old postmenopausal woman has newly diagnosed type 2 diabetes with a hemoglobin A1c of 7.6 percent. She has stable chronic kidney disease with an eGFR of 38 mL/min/1.73 m2 on three measurements over 6 months. She has no heart failure and does not drink alcohol. Her clinician considers metformin as the first agent. According to the metformin label, which of the following is the best decision?

(A) Start metformin at half the usual dose with lactate monitoring each month
(B) Start metformin because her eGFR is over the 30 mL/min cutoff
(C) Do not start metformin, since initiation is not advised at eGFR 30 to 45
(D) Start metformin together with a sulfonylurea to offset the risk
(E) Start metformin if she also takes an SGLT2 inhibitor drug

Answer Contents

400. A 55-year-old woman with type 2 diabetes takes metformin daily. Her eGFR is 48 mL/min/1.73 m2. She is scheduled for contrast-enhanced abdominal and pelvic CT to evaluate an adnexal mass. She has no heart failure, liver disease or alcohol use. Her clinician asks how to manage the metformin around the study. According to the metformin label, which of the following is the best plan?

(A) Continue metformin without interruption through the contrast study
(B) Stop metformin for 2 weeks before and after the contrast study
(C) Stop metformin permanently and switch to a sulfonylurea instead
(D) Take a double dose of metformin on the morning of the study
(E) Hold metformin during the contrast study, restart after checks

Answer Contents

401. A 34-year-old woman with type 2 diabetes has cycles every 50 to 60 days and a BMI of 35. She does not wish to conceive and uses no contraception. Her clinician starts metformin for hyperglycemia, and she remarks that she is happy her periods are so infrequent because she will not need protection. According to the metformin label, which of the following is the best counseling point?

(A) Infrequent cycles mean that she cannot conceive while taking metformin
(B) Metformin may restore ovulation, so contraception is advised
(C) Metformin lowers fertility and so it acts as a contraceptive agent
(D) Metformin makes combined hormonal pills unsafe in every woman who uses it
(E) Contraception is needed once her cycles have become regular again

Answer Contents

OP3.3. Menstrual Disorders

402. A 31-year-old woman has heavy menstrual bleeding every cycle. Pelvic examination and ultrasound are normal, hemoglobin is 11.2 g/dL, and she is not pregnant. She uses a combined contraceptive vaginal ring and wants a nonhormonal add-on to reduce flow. Her clinician considers adding tranexamic acid while she continues the ring. Which of the following best explains why this combination should be avoided according to the label?

(A) Combined use with hormonal contraceptives raises thrombotic risk
(B) Tranexamic acid blocks the intestinal absorption of ethinyl estradiol
(C) Tranexamic acid induces enzymes that lower contraceptive efficacy
(D) The combination causes dangerous hyperkalemia in these women
(E) The combination increases the risk of endometrial hyperplasia

Answer Contents

403. A 36-year-old woman started tranexamic acid tablets for cyclic heavy menstrual bleeding with a normal workup. On the second day of her first cycle of therapy she calls because her vision has become blurry and she sees a patch of altered color in one eye. She has no headache, weakness or speech change, and she is not using other new medications. Which of the following is the most appropriate instruction according to the label?

(A) Continue the drug and recheck vision at the next visit
(B) Reduce to half the dose and watch for 24 hours
(C) Continue the drug and add an antihistamine
(D) Stop tranexamic acid immediately
(E) Switch to taking the drug with food

Answer Contents

404. A 40-year-old woman reports 10 days of bleeding at intervals of 20 days for the past 5 of 6 months. Before that her periods were regular. She is hemodynamically stable, hemoglobin is 10.8 g/dL, and pregnancy test is negative. She has not needed urgent intervention to control bleeding. Which of the following best classifies her presentation using the 2018 FIGO terminology?

(A) Chronic abnormal uterine bleeding
(B) Acute abnormal uterine bleeding
(C) Intermenstrual bleeding
(D) Postmenopausal bleeding
(E) Heavy menstrual bleeding as a diagnosis

Answer Contents

405. A 29-year-old woman has had heavy menstrual bleeding for 5 months, beginning shortly after insertion of a copper intrauterine device. Pregnancy test is negative, ultrasound shows a normally positioned device and a normal uterus, and a platelet count and von Willebrand screen are normal. Endometrial sampling is not needed. Which of the following is the best FIGO PALM-COEIN category for her bleeding?

(A) Coagulopathy (AUB-C)
(B) Adenomyosis (AUB-A)
(C) Leiomyoma (AUB-L)
(D) Iatrogenic (AUB-I)
(E) Endometrial disorder (AUB-E)

Answer Contents

406. A 37-year-old woman has heavy menses with intermenstrual spotting for 8 months. Pelvic ultrasound is normal, and ovulation is confirmed. Endometrial biopsy shows no hyperplasia or malignancy but shows plasma cells in the stroma, and CD138 staining confirms chronic endometritis. Coagulation studies are normal. Which of the following is the best FIGO PALM-COEIN category for her bleeding?

(A) Ovulatory dysfunction (AUB-O)
(B) Endometrial disorder (AUB-E)
(C) Not otherwise classified (AUB-N)
(D) Malignancy and hyperplasia (AUB-M)
(E) Polyp (AUB-P)

Answer Contents

407. A 35-year-old woman reports 6-day periods every 28 days. She is not anemic and has missed no work or activities, but a relative says her bleeding is heavy. She asks how clinicians define heavy menstrual bleeding. According to the 2018 FIGO system, which of the following is the definition used?

(A) Blood loss over 80 mL per cycle by alkaline hematin measurement
(B) Bleeding that lasts longer than 5 days in one cycle
(C) Excessive blood loss that interferes with quality of life
(D) Saturating one pad or tampon every hour for one day
(E) A hemoglobin under 12 g/dL measured during menses

Answer Contents

408. A 24-year-old distance runner has had no menses for 8 months. BMI is 17.5. Pregnancy test, TSH, prolactin and creatinine are normal, FSH is low-normal, and estradiol is low. A diagnosis of functional hypothalamic amenorrhea is made after other causes are excluded. She has had two stress fractures. According to the Endocrine Society guideline, which of the following is the best next study for her bone health?

(A) Serum 25-hydroxyvitamin D level as the single bone test
(B) Plain radiographs of both hips and the lumbar spine
(C) Delayed imaging after 24 months of continued amenorrhea
(D) Technetium bone scan of the entire skeleton for stress injury
(E) Baseline bone density by dual-energy x-ray absorptiometry

Answer Contents

409. A 21-year-old college swimmer with functional hypothalamic amenorrhea for 10 months has a BMI of 18.5. Her mother asks for oral contraceptive pills so that she will have monthly periods again and bone loss will not occur. She has no contraindication to estrogen. Which of the following is the best counsel according to the Endocrine Society guideline?

(A) Pills are not advised just to regain menses or protect bone
(B) Pills are first-line treatment to prevent bone loss in this condition
(C) Pills should be started after a bisphosphonate
(D) Pills restore ovarian function and treat the cause
(E) Pills are preferred to transdermal estradiol for bone protection

Answer Contents

410. A 19-year-old woman has had functional hypothalamic amenorrhea for 16 months. She has not recovered menses despite nutrition counseling and psychological support. DXA shows a lumbar spine Z-score of -2.3. She is not sexually active and has no contraindication to estrogen. According to the Endocrine Society guideline, which of the following is the most appropriate pharmacologic option to protect bone?

(A) Oral alendronate with calcium supplementation
(B) Subcutaneous denosumab with vitamin D
(C) A combined pill containing ethinyl estradiol
(D) Short-term testosterone supplementation
(E) Transdermal estradiol with a cyclic progestin

Answer Contents

411. A 26-year-old woman, thin and training intensively, has had no menses for 9 months and has had severe headaches for the past 2 months, worse in the morning. Pregnancy test and prolactin are normal, and TSH is normal. Neurologic examination is normal. Her clinician suspects functional hypothalamic amenorrhea but is concerned about the headaches. According to the Endocrine Society guideline, which of the following is the best next study?

(A) Noncontrast head CT
(B) Skull radiographs
(C) No imaging, because the picture fits functional amenorrhea
(D) Brain MRI with pituitary cuts and contrast
(E) Electroencephalography

Answer Contents

412. A 20-year-old woman has had no menses for 5 months after losing 12 kg through dieting. Previously her cycles were regular. BMI is 18.0, pelvic examination is normal, and she takes no medications. She reports one episode of unprotected intercourse 4 months ago, and no pregnancy test has been done yet. Which of the following sets of initial laboratory tests is most appropriate for her according to the Endocrine Society guideline?

(A) Karyotype, anti-Müllerian hormone and DHEA sulfate
(B) Beta-hCG, TSH, free T4, prolactin, FSH and estradiol
(C) 17-hydroxyprogesterone, total testosterone and cortisol
(D) Pelvic MRI and serum CA-125
(E) Antinuclear antibody, ESR and C-reactive protein

Answer Contents

413. A 16-year-old girl has painful menses since menarche at 12. Pain starts with bleeding and lasts 2 days, and she misses school each month. Examination and pelvic ultrasound are normal. She takes ibuprofen once pain is severe, with partial relief. She is not sexually active and does not want hormonal contraception. Which of the following is the best advice about her NSAID use?

(A) Wait until pain peaks and then take a double dose of the drug
(B) A single dose taken at bedtime on the first day of bleeding
(C) Scheduled doses with food, started before expected menses
(D) Start the drug on day 4 of bleeding to prevent recurrence
(E) Take each dose on an empty stomach for faster absorption

Answer Contents

414. A 15-year-old girl has had severe pelvic pain with every period since her first menses at 12. She now also has pain in the middle of her cycle, and her mother had a kidney removed in childhood for a congenital anomaly. A trial of NSAIDs gave little relief. Abdominal examination is unremarkable, and she has not been sexually active. According to the CMAJ review of painful periods in adolescents, which of the following is the best next step?

(A) Pelvic ultrasonography and gynecology referral
(B) Continued NSAID therapy for 6 more months without any imaging
(C) Diagnostic laparoscopy as the initial diagnostic study
(D) Serum CA-125 level to screen for an ovarian process
(E) CT of the abdomen and pelvis with intravenous contrast

Answer Contents

415. A 17-year-old girl has painful menses beginning with bleeding and lasting 48 hours. Examination, including pelvic ultrasonography, is normal. She asks why her cramps are so severe when her friends' are mild, and wonders whether she has a hidden condition. Which of the following best explains the mechanism of pain in most adolescents with this presentation?

(A) Retrograde menstruation with implants on the peritoneal surface
(B) Endometrial glands that have invaded the myometrium itself
(C) Obstruction of menstrual outflow by a transverse vaginal septum
(D) Excess prostaglandin production with stronger contractions
(E) Chronic pelvic inflammation after an ascending cervical infection

Answer Contents

416. An 18-year-old woman has severe primary dysmenorrhea. Last year she developed urticaria and wheezing after taking aspirin, and she has chronic nasal polyps. Her pelvic ultrasound is normal. Her friend recommends naproxen sodium, and she asks whether it is safe. Which of the following is the best response?

(A) Naproxen sodium at the lowest effective dose is safe for her
(B) Avoid naproxen and other NSAIDs, using hormonal therapy instead
(C) Ibuprofen taken with food is safe because it is a different drug
(D) Mefenamic acid is preferred because it is a fenamate drug
(E) Ketorolac is the best choice for short-term pain control here

Answer Contents

417. A 33-year-old woman with primary dysmenorrhea has tried to conceive for 12 months without success. Her cycles are regular, her partner's semen analysis is normal, and tubal patency is confirmed. She takes naproxen sodium for the first 3 days of every period. Which of the following is the most appropriate recommendation based on the naproxen label?

(A) Continue the NSAID, because it improves implantation rates
(B) Switch to a higher NSAID dose to shorten the pain interval
(C) Add clomiphene citrate to overcome the NSAID effect on ovulation
(D) Start a combined oral contraceptive to treat the pain
(E) Consider withdrawing the NSAID, since it may delay ovulation

Answer Contents

418. A 52-year-old woman has bothersome hot flashes and prefers a nonhormonal drug. She has no known liver disease and takes no regular medications. Baseline testing shows an ALT of 98 U/L, with a laboratory upper limit of normal of 40 U/L, and a normal bilirubin. Her clinician considers fezolinetant. According to the fezolinetant label, which of the following is the best decision?

(A) Start fezolinetant and recheck liver tests at 12 months
(B) Start fezolinetant, with liver tests if symptoms develop
(C) Do not start it, because the ALT is over 2 times the upper limit
(D) Start fezolinetant at half of the usual daily dose
(E) Start fezolinetant together with ursodiol for hepatic protection

Answer Contents

419. A 50-year-old woman with bothersome hot flashes starts fezolinetant after normal baseline liver tests. She has no jaundice, dark urine, itching or abdominal pain. Her clinician wants to plan monitoring. According to the fezolinetant label, which of the following is the recommended schedule for follow-up liver tests?

(A) A baseline test with no further testing afterward in the first year
(B) Every 2 weeks for the entire first year of treatment
(C) Monthly for 3 months, then at 6 months and 9 months
(D) One test at 12 months of treatment, then yearly
(E) Annually after the first year of treatment has been completed

Answer Contents

420. A 51-year-old woman with obsessive-compulsive disorder has taken fluvoxamine for 6 years with good control. She now has frequent severe hot flashes, and she asks for fezolinetant after reading about it. Liver tests are normal, and she has no kidney disease. According to the fezolinetant label, which of the following is the best response?

(A) Fezolinetant is acceptable with a lower dose and monthly liver tests
(B) Fezolinetant is contraindicated with CYP1A2 inhibitors such as fluvoxamine
(C) Fezolinetant is acceptable if taken 12 hours apart from fluvoxamine
(D) Fezolinetant is acceptable if fluvoxamine is replaced by an SNRI on the same day
(E) Fezolinetant is acceptable if liver tests remain normal at 6 months

Answer Contents

421. A 49-year-old woman takes tamoxifen to reduce her breast cancer risk after atypical hyperplasia. She has frequent hot flashes that wake her at night, and she declines hormonal therapy. Liver tests and kidney function are normal. She is willing to take an antidepressant. According to the 2023 Menopause Society nonhormone therapy statement, which of the following is the safest choice?

(A) Venlafaxine
(B) Paroxetine
(C) Fluoxetine
(D) Bupropion SR
(E) Duloxetine

Answer Contents

422. A 53-year-old woman has moderate to severe hot flashes that wake her several times each night and leave her exhausted. She has no history of breast cancer, has tried paced breathing without relief, and declines estrogen. She prefers a prescription medication that she can take at night. Kidney function is normal, and she takes no other medications. According to the 2023 Menopause Society nonhormone therapy statement, which of the following is the best choice?

(A) Clonidine at bedtime
(B) Pregabalin at bedtime
(C) Black cohosh extract each night
(D) Soy isoflavone extract daily
(E) Gabapentin at bedtime

Answer Contents

423. A 53-year-old woman with bothersome hot flashes has also felt low in mood for several months. She starts the low-dose paroxetine capsule approved for vasomotor symptoms and asks whether it will also treat her mood. She has no suicidal thoughts, and she takes no other medications. According to the label, which of the following is the most accurate statement?

(A) It treats depression at this dose in women in the menopause transition
(B) It treats both conditions when it is taken at bedtime each night
(C) It is approved for anxiety and panic disorders as well as for hot flashes
(D) Not indicated for psychiatric conditions, so her mood needs separate care
(E) It is a substitute for a full-dose antidepressant in menopausal women

Answer Contents

424. A 29-year-old woman has severe irritability, depressed mood and sleep disturbance during the week before menses. Prospective daily ratings across two cycles confirm premenstrual dysphoric disorder. Her partner has had a vasectomy, and she has no need for contraception. She asks for a drospirenone and ethinyl estradiol pill because a friend used it. According to the product label, which of the following is the best response?

(A) It is labeled for PMDD regardless of any need for contraception, so prescribe it
(B) It is labeled for PMDD in women choosing a contraceptive, so an SSRI fits
(C) It is labeled to treat any premenstrual symptoms in any woman, so prescribe it
(D) It is labeled for acne and carries no indication for PMDD, so do not prescribe it
(E) It is contraindicated in every woman who has PMDD, so an SSRI is required

Answer Contents

425. A 31-year-old woman with hypertension controlled on lisinopril has PMDD confirmed by prospective ratings. She wants contraception and treatment for PMDD and chooses a drospirenone and ethinyl estradiol pill. Her potassium is 4.6 mEq/L, creatinine is normal, and she does not smoke. According to the product label, which of the following is the most appropriate monitoring?

(A) No monitoring is required with an ACE inhibitor drug
(B) Serum potassium after 12 months of use
(C) Serum creatinine every 3 months for a year
(D) Liver enzymes monthly for the first 3 months
(E) Serum potassium during the first treatment cycle

Answer Contents

426. A 32-year-old woman with PMDD confirmed by prospective daily ratings has symptoms only in the luteal phase. She would like to avoid daily medication and has no contraindications to an SSRI. She asks whether she can take sertraline only part of the month. According to the sertraline label, which of the following dosing approaches is supported for PMDD?

(A) Dosing during the first 7 days of every cycle, regardless of symptoms
(B) Dosing on the days when symptoms occur, whatever the cycle day
(C) Dosing for 3 days immediately before expected menses
(D) Luteal-phase dosing, beginning 14 days before expected menses
(E) Dosing throughout the follicular phase of each cycle

Answer Contents

427. A 35-year-old woman with PMDD confirmed by prospective ratings has just stopped phenelzine, which she took for atypical depression. The last dose was 6 days ago. Her psychiatrist agrees she needs an SSRI for PMDD. She is eager to start sertraline today. She has no suicidal thoughts and takes no other drugs. According to the sertraline label, which of the following is the best advice?

(A) Wait until 14 days after the last dose of the MAOI
(B) Start sertraline today at the lowest available dose
(C) Start sertraline after 7 days have passed
(D) Start sertraline after 48 hours have passed
(E) Start sertraline together with a benzodiazepine

Answer Contents

428. A 30-year-old woman has used a drospirenone and ethinyl estradiol pill for 8 cycles for contraception and PMDD, with good control. She is stable on the pill and asks whether it is proven to keep treating her PMDD for years. She has no risk factors for hyperkalemia, no migraine with aura and does not smoke. According to the product label, which of the following is accurate?

(A) Effectiveness for PMDD has been proven for 5 years of use
(B) Effectiveness for PMDD is unchanged after menopause begins
(C) PMDD effectiveness beyond 3 cycles has not been studied
(D) The product is labeled to prevent PMDD recurrence for life
(E) The product is labeled to treat PMS without limitation

Answer Contents

OP3.4. Vulvovaginal Conditions

429. A 29-year-old woman has had a thin gray vaginal discharge with a fishy odor for 10 days. She has no itching or dyspareunia. Examination shows a homogeneous white discharge coating the vaginal walls and a normal-appearing cervix. Vaginal pH is 5.2 and the whiff test is positive. Saline microscopy shows no clue cells and no motile trichomonads. Which of the following is the most likely diagnosis?

(A) Trichomoniasis, because the pH is elevated and the discharge is thin and malodorous
(B) Bacterial vaginosis, because three of the four clinical criteria are met
(C) Vulvovaginal candidiasis, because the discharge is white and copious in volume
(D) Physiologic discharge, because clue cells are absent on the wet mount smear
(E) Cervicitis, because the discharge is abnormal and has an unpleasant odor

Answer Contents

430. A 33-year-old woman reports a mild increase in vaginal discharge for one week. Her pH is 4.2 and the whiff test is negative. A Gram stain of the vaginal fluid is scored by the Nugent method and reads 2, with abundant large gram-positive rods and few other bacteria. She asks whether this confirms bacterial vaginosis. Which of the following is the best interpretation of this result?

(A) Bacterial vaginosis, because the discharge is increased and symptoms are present
(B) Intermediate flora, so empiric metronidazole should be started
(C) Lactobacillus-predominant flora, so another cause should be sought
(D) Vulvovaginal candidiasis, because gram-positive organisms predominate
(E) Trichomoniasis, because the Gram stain cannot exclude protozoa

Answer Contents

431. A 26-year-old woman is diagnosed with bacterial vaginosis by Amsel criteria and is prescribed oral metronidazole for 7 days. She has no allergies and takes no other medicines. She mentions that she plans to have wine at a family dinner later this week and asks whether that is acceptable during treatment. According to the 2021 CDC sexually transmitted infections treatment guidelines, which of the following is the best counseling?

(A) Avoid alcohol throughout treatment and for 72 hours afterward
(B) Avoid alcohol throughout treatment and for 24 hours afterward
(C) Switch to clindamycin cream, because it permits alcohol
(D) Alcohol restriction during metronidazole is not needed
(E) Take the dose with the wine to reduce nausea

Answer Contents

432. A 31-year-old woman is diagnosed with bacterial vaginosis. She is in a mutually monogamous relationship, uses latex condoms for contraception and does not wish to take an oral agent. Clindamycin 2% vaginal cream at bedtime for 7 days is chosen. She asks if the cream affects her method of contraception. Which of the following is the most accurate statement about this treatment?

(A) The oil-based cream may weaken latex for 5 days after use
(B) The cream makes polyurethane condoms unreliable for 5 days
(C) The cream blocks the effect of combined oral contraceptives
(D) The cream is safe with latex, but spermicide should be avoided
(E) The cream dissolves copper IUDs, which would raise failure risk

Answer Contents

433. A 35-year-old woman returns 6 weeks after completing a 7-day oral course of metronidazole for bacterial vaginosis. Her thin discharge and odor have recurred for the first time. Amsel criteria are again met, and she took every dose. She asks what should be done next. Which of the following is the most appropriate management?

(A) Begin oral estrogen to restore lactobacilli
(B) Prescribe fluconazole as empiric therapy
(C) Perform a hysteroscopy to look for a cause
(D) Recommend lifelong suppressive antibiotics
(E) Retreat with another recommended regimen

Answer Contents

434. A 27-year-old woman has a frothy yellow-green vaginal discharge and vulvar irritation for 5 days. Vaginal pH is 6.0. A saline wet mount shows many white blood cells but no motile trichomonads or clue cells, and the cervix is friable. She has a new sexual partner. Which of the following is the best next step in diagnosis?

(A) Repeat the wet mount in two weeks
(B) Nucleic acid amplification testing
(C) Vaginal culture for Candida species
(D) Gram stain with Nugent scoring
(E) Cervical cytology with HPV cotesting

Answer Contents

435. A 30-year-old woman has a positive nucleic acid amplification test for Trichomonas vaginalis. She reports a frothy discharge and itching for 1 week. She is not pregnant, is not infected with HIV and has no medication allergies. Her last menstrual period was 2 weeks ago. Which of the following is the most appropriate treatment?

(A) Metronidazole 500 mg orally twice daily for 7 days
(B) Metronidazole 2 g orally as a single dose
(C) Fluconazole 150 mg orally as a single dose
(D) Clindamycin 2% vaginal cream at bedtime for 7 days
(E) Azithromycin 1 g orally as a single dose

Answer Contents

436. A 24-year-old woman completes treatment for trichomoniasis and her partner is treated at the same time. Symptoms resolve within a week. At the follow-up visit she asks whether she needs a test of cure to confirm the infection is gone, and when she should be tested again. Which of the following is the best advice?

(A) Retest in 1 week with a wet mount
(B) Retest only if symptoms return
(C) Retest about 3 months after treatment
(D) Retest in 6 months with a vaginal culture
(E) No retesting is needed once the partner is treated

Answer Contents

437. A 28-year-old woman has vulvar itching and a thick, white, odorless discharge for 4 days after a course of antibiotics for sinusitis. Examination shows vulvar erythema. Vaginal pH is 4.0 and the whiff test is negative. A wet preparation with 10% potassium hydroxide shows budding yeast and pseudohyphae. Which of the following is the most likely diagnosis?

(A) Bacterial vaginosis
(B) Trichomoniasis
(C) Atrophic vaginitis
(D) Vulvovaginal candidiasis
(E) Allergic contact vulvitis

Answer Contents

438. A 34-year-old woman has a vaginal culture sent as part of a research study. It grows Candida species. She has no itching, burning or discharge, and her examination, pH and wet preparation are normal. She is not immunosuppressed and is not planning surgery. She asks whether she should be treated. Which of the following is the most appropriate management?

(A) Fluconazole 150 mg orally as a single dose
(B) Clotrimazole cream for 7 days
(C) Boric acid capsules vaginally for 3 weeks
(D) Treatment of her sexual partner
(E) No treatment

Answer Contents

439. A 40-year-old woman presents with intense vulvar itching and burning for 6 days. Examination shows diffuse vulvar erythema and edema with excoriations and several fissures. Vaginal pH is 4.2, and microscopy with potassium hydroxide shows budding yeast. This is her first episode in several years, she has no diabetes, and she is not pregnant. Which of the following is the most appropriate treatment?

(A) Fluconazole 150 mg orally in two sequential doses
(B) Fluconazole 150 mg orally as a single dose
(C) Miconazole 1,200 mg vaginal suppository as a single dose
(D) Clotrimazole 2% cream intravaginally for 3 days
(E) Metronidazole 500 mg orally twice daily for 7 days

Answer Contents

440. A 37-year-old woman has vulvovaginal itching and discharge that did not respond to two 150 mg doses of fluconazole. Vaginal pH is 4.3. Culture from the current visit grows Candida glabrata. She has not previously received any azole other than fluconazole and has no diabetes or immunosuppression. Which of the following is the most appropriate treatment?

(A) Another single 150 mg dose of fluconazole
(B) Fluconazole weekly for 6 months
(C) Boric acid 600 mg vaginally daily for 3 weeks
(D) Oral metronidazole for 7 days
(E) A 7 to 14 day nonfluconazole azole course

Answer Contents

441. A 25-year-old woman at 9 weeks of gestation by dates presents with vulvar itching and a thick white discharge. Vaginal pH is 4.1 and a KOH preparation shows pseudohyphae. She asks for a single oral tablet because she dislikes vaginal products. Which of the following is the most appropriate treatment?

(A) Fluconazole 150 mg orally as a single dose
(B) A topical azole applied for 7 days
(C) Fluconazole 150 mg orally in two sequential doses
(D) Boric acid capsules vaginally for 3 weeks
(E) Oral ketoconazole twice daily

Answer Contents

442. A 58-year-old woman has had vulvar itching for 2 years. She has used her clobetasol ointment twice daily for the past 5 months with good adherence. Examination shows a persistent thickened white plaque on the left labium majus with a 1-cm nonhealing ulcer at its center. The introitus is narrowed. The rest of the vulva has improved. Which of the following is the most appropriate next step?

(A) Switch to a topical calcineurin inhibitor
(B) Increase clobetasol to three times daily
(C) Add oral fluconazole for 2 weeks
(D) Biopsy of the ulcerated area
(E) Observe for another 3 months on therapy

Answer Contents

443. A 49-year-old woman is newly diagnosed with vulvar lichen sclerosus on clinical grounds and is started on topical clobetasol. She is anxious after reading online and asks about her long-term risk of vulvar cancer and scarring if she does not use treatment as directed. Which of the following is the most accurate counseling?

(A) Malignant transformation is expected in most untreated patients within 10 years
(B) Scarring is unlikely without treatment and cancer risk is not increased
(C) Inadequate treatment carries risks of scarring and malignant change
(D) Treatment lowers cancer risk only if it is started before menopause
(E) Cancer risk depends on HPV vaccination status and not on treatment

Answer Contents

444. A 62-year-old woman with vulvar lichen sclerosus used clobetasol ointment with good results. After 12 weeks her itching has resolved and the white plaques have softened. She asks whether she can stop treatment. She has no history of vulvar cancer. Which of the following is the best plan?

(A) Stop all treatment now and return for care if the symptoms recur later
(B) Maintenance with moderate-potency steroid daily or ultrapotent twice weekly
(C) Continue clobetasol ointment twice daily indefinitely without any step-down
(D) Change to topical estrogen cream nightly
(E) Change to a daily over-the-counter antifungal cream

Answer Contents

445. A 55-year-old woman has had severe vulvar itching for 18 months. Examination shows shiny, white, thin plaques that encircle the vulva and extend around the perineum and perianal skin in an hourglass pattern. There are small fissures at the posterior fourchette. The vaginal mucosa is normal, and there is no discharge. Which of the following is the most likely diagnosis?

(A) Erosive lichen planus
(B) Lichen simplex chronicus
(C) Vulvovaginal candidiasis
(D) Vitiligo
(E) Lichen sclerosus

Answer Contents

446. A 27-year-old woman has had burning pain at the vaginal entrance for 5 months. It occurs with tampon insertion and intercourse and settles within hours. Examination shows normal-appearing vulvar skin. Light cotton-swab touch at the 5 and 7 o'clock positions of the vestibule reproduces the pain, while the labia majora are painless. Cultures and microscopy are negative. Which of the following is the best classification of her condition?

(A) Localized provoked vulvodynia
(B) Generalized spontaneous vulvodynia
(C) Mixed generalized vulvodynia
(D) Vulvovaginal candidiasis in remission
(E) Vulvar lichen planus

Answer Contents

447. A 44-year-old woman has had constant, diffuse vulvar burning for 18 months that is not worsened by touch. Skin, cultures and biopsies are normal. Pelvic floor physical therapy and topical lidocaine did not help. She does not have depression or any contraindication to anticonvulsants. Which of the following is the most appropriate next medication to consider?

(A) Topical clobetasol
(B) Fluconazole
(C) Topical cromolyn 4%
(D) Gabapentin
(E) Oral estrogen

Answer Contents

448. A 31-year-old woman has had localized pain at the vestibule for 3 years that is provoked by touch and intercourse. Skin is normal, and testing for infection is negative. She has completed pelvic floor physical therapy, cognitive behavioral therapy, a tricyclic antidepressant trial and several topical agents without lasting benefit. She asks about surgical options. Which of the following is the most appropriate procedure?

(A) Vestibuloplasty
(B) Carbon dioxide laser vaginal resurfacing
(C) Vestibulectomy
(D) Bartholin gland excision
(E) Clitoral hood reduction

Answer Contents

449. A 29-year-old woman with 6 months of pain at the vaginal entrance and with intercourse is seen for the first time. She is worried about human papillomavirus because of a past abnormal cytology result. Her speculum examination is normal, and she is anxious about further tests. Which of the following is the most appropriate part of the initial evaluation?

(A) Human papillomavirus DNA testing of the vestibular skin
(B) Punch biopsy of the vestibule at the area of pain
(C) Cotton-swab pain mapping and musculoskeletal examination
(D) Acetic acid application followed by colposcopy of the vulva
(E) Pelvic magnetic resonance imaging with contrast

Answer Contents

OP3.5. Structural Uterine Abnormalities

450. A 43-year-old woman with fibroid-related heavy menstrual bleeding has taken once-daily relugolix, estradiol and norethindrone acetate for 22 months with excellent control. Her baseline bone density was normal. She would like to continue the pill indefinitely to avoid surgery. Which of the following is the best counseling about duration of therapy?

(A) Continue indefinitely as long as her yearly bone density stays normal
(B) Continue for up to 5 years if she adds calcium and vitamin D
(C) Stop now, because the label limit is 12 months
(D) Plan to stop within 24 months for possible irreversible bone loss
(E) Switch to relugolix alone to avoid the estrogen exposure

Answer Contents

451. A 39-year-old woman has heavy menstrual bleeding and an intramural fibroid and wants a medical option. Two years ago she had a deep vein thrombosis during a hospitalization after an appendectomy. She has no current anticoagulation and no known thrombophilia testing. Her hemoglobin is 9.8 g/dL. Which of the following treatments is contraindicated for her?

(A) Relugolix, estradiol and norethindrone acetate
(B) Levonorgestrel 52-mg intrauterine system placement
(C) Oral mefenamic acid during menses
(D) Uterine artery embolization
(E) Hysteroscopy-guided endometrial sampling

Answer Contents

452. A 46-year-old woman is seen for heavy menstrual bleeding. Ultrasound shows a 4-cm intramural fibroid. She also reports spotting between periods for 6 months and has a body mass index of 38. Her gynecologist plans to start elagolix, estradiol and norethindrone acetate for fibroid bleeding. Which of the following should be done first?

(A) Obtain a baseline bone density scan before the first dose
(B) Start a 3-month trial of therapy and sample if bleeding persists
(C) Measure serum anti-Mullerian hormone
(D) Obtain a lipid profile and liver ultrasound
(E) Evaluate the abnormal bleeding with endometrial sampling

Answer Contents

453. A 57-year-old postmenopausal woman has a symptomatic 9-cm uterine fibroid causing pelvic pressure and urinary frequency. Endometrial sampling is benign and cervical screening is current. She asks for the least invasive operation and is open to a laparoscopic approach. Which of the following is the best recommendation regarding morcellation?

(A) Laparoscopic hysterectomy with power morcellation and no containment bag
(B) Hysterectomy with the specimen removed intact, without morcellation
(C) Myomectomy with power morcellation to preserve the uterus
(D) Uterine artery embolization to avoid any surgical specimen
(E) Observation until the uterus reaches 12 cm

Answer Contents

454. A 44-year-old woman with multiple symptomatic fibroids is scheduled for laparoscopic supracervical hysterectomy with morcellation. She has had no cervical screening in 6 years. Endometrial thickness is 9 mm, and she has no abnormal bleeding. Which of the following is the best step before surgery?

(A) Cervical cancer screening and endometrial sampling
(B) Serum lactate dehydrogenase isoenzyme testing alone
(C) Pelvic MRI as a test that excludes sarcoma
(D) CA-125 measurement
(E) Proceed with surgery since fibroids are benign

Answer Contents

455. A 48-year-old woman has a fibroid uterus that has enlarged from 12 to 18 weeks of size over 9 months. She is premenopausal, with regular menses, and has never taken tamoxifen or had pelvic radiation. She fears that this growth means cancer and asks whether the growth rate can tell her. Which of the following is the most accurate statement?

(A) Rapid growth makes leiomyosarcoma very likely in premenopausal women
(B) Growth of more than 6 weeks in size confirms sarcoma
(C) A normal LDH level excludes leiomyosarcoma
(D) Rapid growth and size are not linked to leiomyosarcoma risk
(E) Ultrasound features reliably exclude sarcoma

Answer Contents

456. A 36-year-old woman with heavy menstrual bleeding has saline infusion sonography that shows a 3-cm fibroid in contact with the endometrium. About 30% of its diameter protrudes into the cavity and 70% lies within the myometrium. Her surgeon wants to classify it by the FIGO leiomyoma system to plan a hysteroscopic approach. Which of the following is the correct type?

(A) Type 0
(B) Type 1
(C) Type 2
(D) Type 3
(E) Type 4

Answer Contents

457. Four women each have an endometrial polyp found on imaging and are scheduled for hysteroscopic removal. A 64-year-old has had postmenopausal bleeding for 2 months. A 29-year-old has regular menses and the polyp was found during an infertility workup. A 42-year-old has regular cycles and an incidental 5-mm polyp. A 35-year-old has the polyp and a normal bleeding pattern. Which patient's polyp is most likely to contain malignancy?

(A) The 29-year-old being evaluated for infertility
(B) The 64-year-old with postmenopausal bleeding
(C) The 42-year-old with an incidental 5-mm polyp
(D) The 35-year-old with a normal bleeding pattern
(E) All four have the same risk

Answer Contents

458. A 55-year-old woman has been taking tamoxifen for 3 years after breast cancer. She has no vaginal bleeding. Her oncologist asks about the likelihood that she has endometrial abnormalities such as polyps. Which of the following best describes the relationship between tamoxifen and endometrial polyps?

(A) Tamoxifen lowers polyp formation by blocking endometrial estrogen action
(B) Polyps occur in fewer than 1% of users
(C) Polyps arise mainly in women who also take estrogen
(D) Polyps occur equally in users and nonusers
(E) Polyps occur in 20% to 35% of long-term users

Answer Contents

459. A 40-year-old woman has intermenstrual spotting for 5 months. Transvaginal ultrasound shows a thickened endometrium with an indeterminate focal area. Blind office endometrial biopsy yielded scant benign tissue. The gynecologist wants the test that best establishes the diagnosis of a polyp by direct examination of the cavity. Which of the following is the best choice?

(A) Repeat transvaginal ultrasonography
(B) Computed tomography of the pelvis
(C) Hysteroscopy
(D) Dilation and curettage without hysteroscopy
(E) Serum CA-125

Answer Contents

460. A 57-year-old woman has a 12-mm endometrial polyp found on pelvic ultrasound for a different indication. She has hypertension and a body mass index of 36. She asks why she developed it and whether anything in her history contributed. Which of the following is the best explanation?

(A) Her hypertension is protective against polyp formation in the uterus
(B) Polyps arise mainly after prior cesarean delivery
(C) Hypertension and obesity are established polyp risk factors
(D) Obesity lowers endometrial estrogen exposure
(E) Her age lowers the chance of polyps

Answer Contents

461. A 45-year-old woman has heavy menstrual bleeding and dysmenorrhea. Transvaginal ultrasound shows a globular uterus with asymmetric myometrial thickening, and the findings are suggestive but not definitive for adenomyosis. There are no fibroids. She wants a noninvasive test that is more specific before deciding on treatment. According to the 2022 American Family Physician review, which of the following statements about imaging is most accurate?

(A) Transvaginal ultrasound is more specific than MRI for adenomyosis
(B) MRI has higher sensitivity than transvaginal ultrasound
(C) Both tests have identical sensitivity and specificity
(D) MRI has higher specificity than ultrasound for adenomyosis
(E) Ultrasound is reproducible between operators

Answer Contents

462. A 42-year-old woman has heavy menstrual bleeding and dysmenorrhea, and imaging shows diffuse adenomyosis. She has completed childbearing but wants to keep her uterus. NSAIDs gave partial relief, and she has no contraindication to hormonal therapy. She asks about the best initial hormonal option. Which of the following is most appropriate?

(A) Repeated cycles of leuprolide for years
(B) A levonorgestrel-releasing intrauterine system
(C) Endometrial ablation with thermal balloon therapy
(D) Immediate hysterectomy without a medical trial
(E) Tranexamic acid daily throughout the cycle

Answer Contents

463. A 36-year-old woman with infertility is preparing for her first cycle of in vitro fertilization. MRI shows diffuse adenomyosis, and she asks how it will affect her chance of pregnancy compared with women who do not have the condition. According to the 2022 American Family Physician review, which of the following is the best counseling?

(A) Her chance of clinical pregnancy is lower, by about 28%
(B) Her chance of clinical pregnancy is unchanged by the condition
(C) Her probability of clinical pregnancy is higher
(D) Adenomyosis affects only the miscarriage rate
(E) Hysterectomy is required before treatment can proceed

Answer Contents

464. A 46-year-old woman had uterine artery embolization for a 6-cm fibroid 14 months ago. Her bleeding improved at first, but heavy periods and pain returned. The fibroid shrank. MRI now shows a thickened junctional zone with myometrial cysts. She asks why her symptoms returned when the fibroid is smaller. Which of the following is the most likely explanation?

(A) Regrowth of the embolized fibroid
(B) Endometrial polyp formation
(C) Ovarian cyst rupture
(D) Failure of the procedure to occlude the uterine artery
(E) Coexisting adenomyosis

Answer Contents

465. A 47-year-old woman with diffuse adenomyosis has severe heavy menstrual bleeding with a hemoglobin of 8.6 g/dL. Over 2 years she failed NSAIDs, a combined oral contraceptive and a levonorgestrel intrauterine system. She has completed childbearing and wants a definitive solution. There is no malignancy on endometrial sampling. Which of the following is the most appropriate management?

(A) Hysterectomy
(B) Another 6 months of the same intrauterine system
(C) Endometrial ablation
(D) Progestin-only pills at higher dose
(E) Observation until menopause

Answer Contents

466. A 24-year-old woman has recurrent first-trimester pregnancy loss. Three-dimensional ultrasound and MRI show a normal, smooth external fundal contour and a midline endometrial septum extending 2 cm from the interostial line toward the cervix. The leading edge of the septum forms an angle of less than 90 degrees. Which of the following is the best diagnosis under the ASRM classification?

(A) Bicornuate uterus
(B) Uterus didelphys
(C) Unicornuate uterus
(D) Septate uterus
(E) Arcuate uterus

Answer Contents

467. A 28-year-old woman has dysmenorrhea and a prior second-trimester loss. MRI shows two separate endometrial cavities, a single cervix, and an external fundal cleft that measures 2.2 cm in depth. The kidneys are normal. Which of the following is the best classification under the ASRM system?

(A) Septate uterus
(B) Bicornuate uterus
(C) Uterus didelphys
(D) Complex anomaly
(E) Unicornuate uterus

Answer Contents

468. A 13-year-old girl has had regular menses since age 12 and now has progressive severe pelvic pain with each period. Examination shows a tender left-sided vaginal bulge. Pelvic MRI shows uterus didelphys with a distended, blood-filled left hemivagina and left hematometra. Which of the following additional anomalies is most likely to be present?

(A) Horseshoe kidney
(B) Polycystic kidney disease
(C) Duplicated right ureter
(D) Cardiac septal defect
(E) Absent left kidney

Answer Contents

469. A 16-year-old girl has primary amenorrhea. She has normal breast development, pubic hair and height. A pelvic ultrasound shows no uterus, and the karyotype is 46,XX with normal testosterone. A diagnosis of müllerian agenesis is made. Beyond counseling about dilation, which of the following is the best additional evaluation?

(A) Bone marrow aspiration
(B) Pituitary MRI
(C) Renal imaging
(D) Serum 17-hydroxyprogesterone
(E) Laparoscopic gonadal biopsy

Answer Contents

470. A reproductive surgeon is reviewing a 2023 ultrasound report that classifies a patient's anomaly using the current ASRM scheme. A trainee asks how the current system differs from the older American Fertility Society classification, which used Roman numeral classes. Which of the following is the most accurate description of the current system?

(A) Categories are numbered from I to VII with subtypes
(B) Categories are based solely on karyotype findings
(C) Categories separate uterine and vaginal anomalies into different systems
(D) Categories use descriptive names, not numbered classes
(E) Categories are limited to anomalies of the uterine body

Answer Contents

OP3.6. Incontinence and Pelvic Floor

471. A 46-year-old woman has leakage with coughing and exercise for two years. Pelvic examination is normal, the cough stress test with a full bladder reproduces the leakage, urinalysis is normal, and postvoid residual is 25 mL. She wants surgery after failing pelvic floor training. Which of the following tests is not recommended as part of the routine preoperative evaluation?

(A) Urinalysis
(B) Cystoscopy
(C) Postvoid residual measurement
(D) Cough stress test with a comfortably full bladder
(E) Focused history on degree of bother

Answer Contents

472. A 50-year-old woman with pure stress leakage wants a sling. Her cough stress test is positive, urinalysis is normal, postvoid residual is 40 mL, and she has no prior incontinence surgery or neurologic disease. Her clinic offers urodynamics routinely. According to the AUA/SUFU guideline, which of the following is the best statement?

(A) Urodynamic testing may be omitted when stress incontinence is clear
(B) Urodynamic testing is mandatory before a sling in any woman with leakage
(C) Urodynamic testing should replace the cough stress test at the bedside examination
(D) Urodynamic testing is needed in women older than 65 years before a sling
(E) Urodynamic testing should be repeated after surgery in each patient

Answer Contents

473. A 58-year-old woman has leakage with coughing and also marked urgency with large-volume leakage before reaching the toilet, which she says bothers her most. She is considering a midurethral sling. Cough stress test is positive, and urinalysis and postvoid residual are normal. Which of the following is the best next step according to the AUA/SUFU guideline?

(A) Place a retropubic sling now, without further testing of the urgency
(B) Place a urethral bulking agent as a diagnostic and therapeutic trial
(C) Perform additional evaluation, such as urodynamics, before surgery
(D) Start oral estrogen therapy for the urgency and defer testing for months
(E) Place an indwelling urethral catheter for several days to rest the bladder

Answer Contents

474. A 52-year-old woman with stress incontinence has chosen a retropubic midurethral sling. At the consent visit she says, "I just want to know that it works." Which of the following must the surgeon include in the discussion, according to the AUA/SUFU guideline?

(A) The benefits, since mesh complications are rare in modern slings and need no discussion
(B) The recommendation that mesh be used in women who have stress incontinence
(C) That alternatives to a mesh sling are not available in the United States
(D) That consent can be deferred until the day of the operation itself
(E) The specific risks and benefits of mesh and alternatives to a mesh sling

Answer Contents

475. A 49-year-old woman is two weeks after an uncomplicated transobturator midurethral sling and is voiding well. Her surgeon is setting the follow-up plan. According to the AUA/SUFU guideline, which of the following is the best recommendation for the postoperative examination?

(A) Examine her at 5 years, because earlier review is unnecessary in routine cases
(B) Examine her within six months, checking symptoms, healing, and residual
(C) Examine her if she reports leakage or pain, since routine review adds little
(D) Examine her within 24 hours, then discharge her from surgical care
(E) Examine her at 3 years, to check for late erosion

Answer Contents

476. A 63-year-old woman with stress urinary incontinence previously had pelvic radiation for cervical cancer. Examination shows an atrophic, scarred vagina with limited mobility. She asks for a sling. Which of the following is the best approach according to the AUA/SUFU guideline?

(A) Place a synthetic retropubic sling because mesh heals reliably in irradiated tissue
(B) Place a synthetic single-incision sling because it is smaller and less invasive
(C) Strongly consider avoiding mesh and use an autologous sling
(D) Use bulking agents as the preferred choice in women who have had radiation
(E) Perform no incontinence surgery in irradiated women

Answer Contents

477. A 66-year-old woman reports urinary frequency, urgency, and urgency leakage for six months with nocturia twice nightly. She has no hematuria or pain, and her pelvic examination is normal. Which of the following is the most appropriate first test according to the AUA/SUFU overactive bladder guideline?

(A) Cystoscopy
(B) Urodynamic testing
(C) Renal and bladder ultrasound
(D) Urinalysis
(E) Pelvic MRI

Answer Contents

478. A 54-year-old woman has urgency and frequency. Urinalysis is normal, postvoid residual is 30 mL, and she has no neurologic disease, hematuria, or prior pelvic surgery. Her referring physician asks if urodynamics, cystoscopy, and imaging should be done before treatment. Which of the following is the best response according to the AUA/SUFU guideline?

(A) Perform the three tests before beginning any treatment for the symptoms
(B) Perform urodynamics and cystoscopy but omit the upper tract imaging
(C) Perform imaging alone, to exclude upper tract disease before treatment
(D) Perform cystoscopy before behavioral therapy is started in any patient
(E) Do not routinely perform these tests at the initial evaluation

Answer Contents

479. A 61-year-old woman with overactive bladder says she would like a pill and does not want to "waste time" with exercises or diary keeping. Evaluation is complete and uncomplicated, with a normal urinalysis and no retention. Which of the following is the best response according to the AUA/SUFU guideline?

(A) Offer bladder training to her now, whether or not she takes a drug
(B) Offer bladder training if medication fails to help her symptoms enough
(C) Offer bladder training to women who are under age 50 and motivated
(D) Skip bladder training because it is less effective than placebo treatment
(E) Recommend fluid restriction to under one liter daily instead of training

Answer Contents

480. A 78-year-old woman with overactive bladder and mild memory complaints asks her physician to prescribe oxybutynin because her sister takes it. Her kidney function is normal. According to the AUA/SUFU guideline, which of the following should the clinician do before prescribing an antimuscarinic?

(A) Reassure her that antimuscarinics improve cognition in older adults
(B) Order a head CT to exclude cognitive disease before prescribing
(C) Require a urodynamic study before prescribing any bladder drug
(D) Avoid discussing risks, to prevent anxiety about her memory problems
(E) Discuss the potential risk of dementia and cognitive impairment

Answer Contents

481. A 69-year-old woman has bothersome overactive bladder symptoms. She has narrow-angle glaucoma and a history of urinary retention. Behavioral therapy gave partial relief, and her urinalysis is normal. Her physician wants to add medication. Which of the following is the best approach according to the AUA/SUFU guideline?

(A) Prescribe the highest-dose antimuscarinic without any special monitoring
(B) Prescribe two antimuscarinics together for faster symptom benefit
(C) Use an antimuscarinic only with extreme caution, or choose another class
(D) Stop treatment since drugs are contraindicated for overactive bladder
(E) Start intradetrusor botulinum toxin before any oral drug is tried

Answer Contents

482. A 57-year-old woman with overactive bladder asks whether a cranberry and herbal supplement she bought online will help her urgency. She wants to avoid prescription drugs and has a normal urinalysis. According to the AUA/SUFU guideline, which of the following is the best counseling?

(A) Evidence is insufficient to support nutraceuticals for overactive bladder
(B) Nutraceuticals work better than antimuscarinics in comparative trials
(C) Nutraceuticals are equivalent in effect to bladder training and pelvic floor therapy
(D) Nutraceuticals are recommended as first-line therapy
(E) Nutraceuticals eliminate the need for further evaluation or monitoring

Answer Contents

483. A 64-year-old woman with overactive bladder has had only partial improvement on the maximum tolerated dose of an antimuscarinic combined with bladder training. She does not want invasive therapy yet. According to the AUA/SUFU guideline, which of the following may be offered?

(A) A second antimuscarinic drug added at the same time as the first
(B) Combination therapy with a medication from a different class
(C) Immediate augmentation cystoplasty for refractory symptoms of urgency
(D) An indwelling suprapubic catheter for ongoing symptom control
(E) Discontinuation of therapy with observation for several months

Answer Contents

484. An 80-year-old woman living in a nursing facility has urgency leakage. Her daughter asks the physician to place a chronic indwelling urethral catheter to keep her dry. She has not been offered behavioral measures or medication. According to the AUA/SUFU guideline, which of the following is the best response?

(A) Place the catheter now, since catheters have no meaningful complications for her
(B) Place the catheter and start an antimuscarinic at the same time as placement
(C) Place a suprapubic catheter as first-line therapy for incontinence in older adults
(D) Recommend a catheter if other therapies fail or are declined
(E) Catheterization is contraindicated in older adults who live in nursing facilities

Answer Contents

485. A 74-year-old woman is fitted with a ring pessary for stage III prolapse and does well. At a six-month visit she asks what complications to watch for. Which of the following is the most likely problem to occur with long-term pessary use?

(A) Vesicovaginal fistula formation in most users over time
(B) Vaginal discharge or bleeding from erosion
(C) Spontaneous resolution of the prolapse with continued use
(D) Pelvic organ malignancy from chronic mucosal irritation
(E) Cardiac arrhythmia from vagal stimulation by the device

Answer Contents

486. A 67-year-old woman was successfully fitted with a ring pessary today for symptomatic stage III prolapse. She asks how many women like her are still using the pessary a year later. According to the review, which of the following is the best estimate among women who were successfully fitted?

(A) 5% to 10% continue use beyond 6 to 12 months
(B) 15% to 25% continue use beyond 6 to 12 months
(C) 70% to 80% continue use beyond 6 to 12 months
(D) 95% to 100% continue use beyond 6 to 12 months
(E) 40% to 60% continue use beyond 6 to 12 months

Answer Contents

487. A 31-year-old woman has symptomatic stage II prolapse with a vaginal bulge after her second delivery. She plans another pregnancy in two years and wants to avoid surgery now. Which of the following is the most appropriate management according to ACOG and AUGS guidance?

(A) A pessary, a low-risk option for women who wish to maintain fertility
(B) Sacrocolpopexy performed now, before the next planned pregnancy occurs
(C) Total hysterectomy with apical suspension to prevent later recurrence
(D) Colpocleisis with levator plication to close the vaginal canal
(E) A vaginal mesh kit placed before the next pregnancy to prevent recurrence

Answer Contents

488. A 65-year-old woman reports a vaginal bulge for four months. Examination in the clinic suggests anterior and apical descent, but the examiner described it only as "moderate." The office will refer her to a urogynecologist. According to ACOG and AUGS guidance, which of the following should be recorded before treatment to quantify the degree of prolapse?

(A) A urine culture
(B) A cystoscopy report
(C) A pelvic MRI
(D) A serum estradiol level
(E) A POP-Q examination

Answer Contents

489. A 55-year-old woman reports occasional fecal soiling with urgency for one year. She has had two vaginal deliveries, no anal sphincter laceration, and a normal examination with good resting tone. Her physician asks whether to order anal sphincter imaging before treatment. According to ACOG Practice Bulletin 210, which of the following is the best answer?

(A) Order endoanal ultrasound for each woman with fecal incontinence at the first visit
(B) Order a pelvic MRI before any treatment is started for the incontinence
(C) Imaging is not routine, so begin conservative treatment
(D) Refer for sphincteroplasty without further testing of the sphincter
(E) Start sacral nerve stimulation as the first step before conservative care

Answer Contents

490. A 60-year-old woman has urgency fecal incontinence several times a week associated with loose stools. Examination shows normal sphincter tone. She has not yet tried any treatment. According to ACOG Practice Bulletin 210, which of the following is the most appropriate initial treatment?

(A) Sacral nerve stimulation with a test phase
(B) Anal sphincter bulking agent injection
(C) Diverting colostomy for urgency episodes
(D) Dietary fiber and antimotility agents
(E) Overlapping anal sphincteroplasty

Answer Contents

491. A 58-year-old woman has weekly fecal incontinence despite fiber, an antimotility agent, and a course of pelvic floor muscle training with biofeedback. Endoanal ultrasound shows an intact sphincter complex. She wants a procedure. According to ACOG Practice Bulletin 210, which of the following can be considered?

(A) Sphincteroplasty
(B) Repeat biofeedback with no change
(C) Anal bulking agent as a permanent cure
(D) Observation until incontinence resolves
(E) Sacral nerve stimulation

Answer Contents

OP3.7. Reproductive Tract Cancer

492. A 45-year-old woman has a biopsy showing invasive squamous cell carcinoma of the cervix. Pelvic MRI shows a 3.5 cm tumor confined to the cervix with deepest invasion of 9 mm. There is no parametrial extension, no vaginal involvement and no suspicious nodes on cross-sectional imaging. Using the 2018 FIGO system, which of the following is the correct stage?

(A) Stage IA2
(B) Stage IB1
(C) Stage IB2
(D) Stage IB3
(E) Stage IIA1

Answer Contents

493. A 19-year-old woman comes for a contraceptive visit. She has been sexually active since age 16, completed the HPV vaccine series at age 12, and has never had a cervical screening test. She asks whether she should have a Pap test today. Her examination is normal and she has no abnormal bleeding. According to the 2018 USPSTF recommendation, which of the following is the best next step?

(A) Cervical cytology now, then every 3 years
(B) Primary high-risk HPV testing now, then every 5 years
(C) No cervical cancer screening until age 21
(D) Cervical cytology after one year of sexual activity
(E) Colposcopy to establish a baseline

Answer Contents

494. A 34-year-old woman received the complete HPV vaccine series at age 12. At a well-woman visit she tells you that because she is vaccinated she plans to skip cervical screening. She has had normal cytology twice and has no immunocompromise. She asks for your recommendation. Under the 2018 USPSTF statement, which of the following is the best response?

(A) Screen every 5 years with high-risk HPV testing alone
(B) Stop screening because the vaccine series was complete
(C) Screen once more at age 40, then stop
(D) Substitute annual pelvic examination for screening
(E) Screen only if a new partner is reported

Answer Contents

495. A 68-year-old woman with an intact cervix asks whether she can stop cervical screening. She has had three consecutive normal cytology results, the most recent 4 years ago, and all within the past 10 years. She has no history of CIN 2 or worse, no immunocompromise and no in utero DES exposure. According to the 2018 USPSTF recommendation, which of the following is the best advice?

(A) Continue annual cytology until age 75
(B) Obtain colposcopy before stopping screening
(C) Continue screening every 5 years to age 75
(D) Stop, because prior screening was adequate
(E) Obtain one HPV test, then continue every 3 years

Answer Contents

496. A health system is choosing between primary high-risk HPV testing and cytology for women aged 30 to 65 years. A committee member argues that HPV testing allows a longer interval after a negative result. A colleague asks what evidence supports the greater reliability of a negative HPV test. Which of the following statements from the National Cancer Institute summary best supports the longer interval?

(A) Pap screening lowers incidence and mortality by at least 80%
(B) Sensitivity of HPV testing was 95% compared with 55% for cytology
(C) HPV testing has fewer false positives than cytology
(D) Self-collected HPV testing may be an alternative method
(E) Pap tests should not be repeated within 2 years of a negative test

Answer Contents

497. A 62-year-old woman has a hysterectomy with bilateral salpingo-oophorectomy for postmenopausal bleeding. Pathology shows serous carcinoma invading less than half of the myometrium. There is no cervical stromal invasion, no adnexal involvement and no lymph node metastasis. Using the 2023 FIGO staging for endometrial carcinoma, which of the following is the correct stage?

(A) Stage IA2
(B) Stage IB
(C) Stage IC
(D) Stage IIA
(E) Stage IIC

Answer Contents

498. A 55-year-old woman has taken tamoxifen for 3 years after treatment of estrogen receptor-positive breast cancer. She reports new spotting. Transvaginal ultrasound shows a thickened endometrial echo, and office biopsy shows endometrioid adenocarcinoma. She asks why a breast cancer drug would affect her uterus. Which of the following is the best explanation?

(A) Tamoxifen suppresses ovarian steroid production
(B) Tamoxifen has an estrogenic effect on the endometrium
(C) Tamoxifen blocks progesterone receptors in the uterus
(D) Tamoxifen causes a germline mismatch repair defect
(E) Tamoxifen thins the endometrium and exposes the basal layer

Answer Contents

499. A 74-year-old woman has postmenopausal bleeding. She is lean, has no diabetes and has never used estrogen therapy. Biopsy shows uterine serous carcinoma, and the surrounding endometrium is atrophic. Her daughter asks why her mother developed this cancer when she lacks the risk factors of obesity and anovulation. Which of the following best describes the pathogenesis of her tumor?

(A) It arises from complex atypical hyperplasia under unopposed estrogen
(B) It results from long-term tamoxifen exposure
(C) It is caused by a POLE exonuclease domain mutation
(D) It develops from endometriosis through ovarian clear cell change
(E) It develops from atrophic endometrium and is not hormonally driven

Answer Contents

500. A 59-year-old woman had a hysterectomy and bilateral salpingo-oophorectomy for clinical stage I endometrioid adenocarcinoma, and final pathology confirms stage I disease. She asks her oncologist about adding a progestin pill for several years to lower the chance of recurrence, because a friend was told it helps. According to the National Cancer Institute summary, which of the following is the best response?

(A) Adjuvant progestogen reduces recurrence and is standard
(B) Adjuvant progestogen is advised only for patients over age 70
(C) Adjuvant progestogen has shown no clinical benefit in stage I disease
(D) Adjuvant progestogen is advised only after laparoscopic surgery
(E) Adjuvant progestogen improves survival if the tumor is estrogen receptor-positive

Answer Contents

501. A 58-year-old woman with a BMI of 41 kg/m2, hypertension and type 2 diabetes has no bleeding or other symptoms. She has read that obesity raises endometrial cancer risk and requests annual transvaginal ultrasound and endometrial biopsy to screen. She has no family history of Lynch syndrome. According to the National Cancer Institute summary, which of the following is the best response?

(A) Perform annual transvaginal ultrasound alone
(B) Perform annual office endometrial biopsy
(C) Order annual serum CA-125 testing
(D) Do not screen, and report any bleeding at once
(E) Start annual cytology with endometrial cell reporting

Answer Contents

502. A 56-year-old woman undergoes complete surgical staging for a 9 cm ovarian mass. The tumor is a high-grade serous carcinoma limited to the right ovary. The surface is free of tumor, but the capsule ruptured before surgery during a cyst aspiration attempt. Washings are negative and the pelvis, peritoneum and nodes are free of disease. Using the 2014 FIGO staging, which of the following is the correct stage?

(A) Stage IA
(B) Stage IC1
(C) Stage IC2
(D) Stage IC3
(E) Stage IIA

Answer Contents

503. A 63-year-old woman completed surgery and chemotherapy for stage III high-grade serous ovarian cancer 8 months ago and is in clinical remission. At a routine visit her CA-125 is rising on two measurements. She has no symptoms, and examination and CT show no recurrent disease. She asks whether to start treatment now to get ahead of the cancer. According to the National Cancer Institute summary, which of the following is the best statement?

(A) Early detection by CA-125 or imaging has not been shown to alter outcomes
(B) Treating on the CA-125 rise alone has been shown to improve survival
(C) Start second-line therapy because a rising CA-125 means platinum resistance
(D) Stop CA-125 testing because it carries no information in this setting
(E) Perform immediate secondary cytoreduction

Answer Contents

504. A 19-year-old woman has a unilateral 12 cm solid ovarian mass. At laparotomy, the mass is removed and frozen section shows dysgerminoma. Staging finds disease confined to the one ovary, with a normal opposite ovary, uterus and peritoneum. She has not had children and wants future fertility. According to the National Cancer Institute summary, which of the following is the best surgical management?

(A) Total hysterectomy with bilateral salpingo-oophorectomy and omentectomy
(B) Unilateral salpingo-oophorectomy with uterus and other ovary kept
(C) Bilateral salpingo-oophorectomy with the uterus preserved
(D) Ovarian cystectomy alone with spill precautions taken
(E) Observation after frozen section and mass biopsy only

Answer Contents

505. A 22-year-old woman has a stage I ovarian yolk sac tumor removed by unilateral salpingo-oophorectomy. Before surgery her alpha fetoprotein was markedly elevated. Four weeks after surgery, the alpha fetoprotein remains elevated and has not fallen as expected. Imaging shows no obvious mass. Which of the following is the best interpretation of the persistent marker elevation?

(A) Normal slow clearance of the marker
(B) Pregnancy-related alpha fetoprotein production
(C) Hepatic regeneration after surgery
(D) Residual unresected tumor
(E) Laboratory error requiring no further action

Answer Contents

506. A 36-year-old woman carries a pathogenic BRCA1 variant found after her sister developed ovarian cancer. She has two children, has finished childbearing, and has normal pelvic examination and ultrasound. She asks whether regular CA-125 and ultrasound could replace surgery. According to ACOG Practice Bulletin 182, which of the following is the best management for her to consider?

(A) Annual CA-125 and ultrasound in place of surgery
(B) Hysterectomy alone with ovarian conservation
(C) Defer any surgery until menopause at age 50
(D) Tubal ligation as the main risk-reducing step
(E) Risk-reducing salpingo-oophorectomy

Answer Contents

507. A 66-year-old woman has a 3 cm squamous cell carcinoma of the vulva. After wide local excision, pathology shows stromal invasion of 4 mm. Margins are clear. The tumor does not extend to the urethra, vagina or anus, and the groin nodes are negative. Using the 2021 FIGO staging, which of the following is the correct stage?

(A) Stage IA
(B) Stage IB
(C) Stage II
(D) Stage IIIA
(E) Stage IIIB

Answer Contents

508. A 72-year-old woman has a 2 cm ulcerated vulvar lesion. Biopsy shows invasive keratinizing squamous cell carcinoma with adjacent differentiated-type vulvar intraepithelial neoplasia. HPV testing of the tissue is negative. She asks her physician how this cancer differs from the HPV-related vulvar cancers seen in younger women. Which of the following is the best description of her tumor?

(A) It is HPV-associated, and this pattern is most common in younger women
(B) It is a metastasis from an undetected cervical primary
(C) It is HPV-independent, and this pattern is most common in older women
(D) It arises from Bartholin gland epithelium
(E) It is a vulvar melanoma that has lost pigment

Answer Contents

509. A 58-year-old woman has a 2.5 cm unifocal vulvar squamous cell carcinoma with 3 mm invasion, and no suspicious groin nodes on examination or imaging. She has stage IB disease on pathology after excision with clear margins. Sentinel lymph node biopsy shows no metastasis in the sentinel nodes of either groin. According to the National Cancer Institute summary, which of the following is the best plan for the groins?

(A) No groin dissection
(B) Ipsilateral inguinofemoral lymphadenectomy
(C) Bilateral inguinofemoral lymphadenectomy
(D) Pelvic lymphadenectomy
(E) Radiation therapy to both groins

Answer Contents

510. A 24-year-old woman has vaginal bleeding after intercourse. Examination shows a firm mass on the anterior upper vagina, and biopsy shows clear cell adenocarcinoma. Her mother says she took a medication during her pregnancy to prevent a miscarriage, but cannot recall its name. Which of the following in utero exposures is most strongly associated with this tumor?

(A) Diethylstilbestrol
(B) Thalidomide taken in early pregnancy
(C) Isotretinoin
(D) Valproic acid exposure
(E) Methotrexate exposure

Answer Contents

511. A 61-year-old woman has postcoital bleeding. Examination shows a 3 cm ulcerated tumor of the apical vagina that extends onto the cervix. Biopsy shows squamous cell carcinoma, and she has not had a hysterectomy. A resident asks how to classify the tumor for staging. According to the National Cancer Institute summary, which of the following is the correct classification?

(A) Primary vaginal cancer
(B) Vulvar cancer with vaginal extension
(C) Cervical cancer
(D) Metastatic endometrial cancer
(E) Primary peritoneal cancer

Answer Contents

512. A 28-year-old woman had suction evacuation of a complete hydatidiform mole 8 weeks ago. Her serum hCG fell at each weekly check and first reached the normal range this week. She is asymptomatic and has no signs of metastasis. According to the 2025 FIGO update, which of the following is the best surveillance plan now?

(A) Stop hCG testing now because the value is normal
(B) Check one confirmatory hCG at 1 month, then stop
(C) Check hCG once a year for the next 5 years
(D) Perform repeat curettage and then stop testing
(E) Continue monthly hCG for 6 months after normal

Answer Contents

513. A 26-year-old woman had evacuation of a complete mole 3 weeks ago. Her hCG is falling. She does not wish to conceive for at least a year and asks about contraception during follow-up. She has heard that hormonal methods might stimulate trophoblastic tissue. According to the 2025 FIGO update, which of the following is the best counseling?

(A) Use barrier methods only until hCG is normal
(B) Avoid hormonal methods until 12 months after hCG normalization
(C) Use only a copper intrauterine device
(D) Hormonal contraception is considered safe during follow-up
(E) Delay contraception until after the 6-month surveillance ends

Answer Contents

514. A 30-year-old woman had a complete mole evacuated 5 weeks ago. Serum hCG values on days 1, 8, 15 and 22 of follow-up were 4,200, 3,900, 4,100 and 4,050 mIU/mL. She has no bleeding and a normal pelvic examination. Chest radiograph is normal. According to the FIGO criteria, which of the following is the best next step?

(A) Continue weekly hCG and make no change
(B) Diagnose GTN, then stage and score
(C) Repeat uterine curettage within the week
(D) Reassure her and recheck hCG monthly
(E) Begin prophylactic hysterectomy at once

Answer Contents

515. A 33-year-old woman with amenorrhea for 10 months has an enlarged uterus and a modestly elevated hCG, lower than expected for the uterine size. Curettage shows a placental site trophoblastic tumor. Imaging shows disease confined to the uterus, and she has completed her family. According to the 2025 FIGO update, which of the following is the best treatment?

(A) Observation with serial hCG
(B) Single-agent methotrexate
(C) Hysterectomy
(D) Pelvic radiation therapy
(E) Repeat curettage

Answer Contents

516. A 29-year-old woman with low-risk gestational trophoblastic neoplasia is receiving single-agent chemotherapy. Today her hCG is reported as normal for the first time. She feels well and asks to stop treatment now. According to the 2025 FIGO update, which of the following is the best recommendation?

(A) Stop chemotherapy now because hCG is normal
(B) Give two to three more cycles after normalization
(C) Give twelve more cycles after normalization
(D) Switch to multiagent chemotherapy for consolidation
(E) Proceed to hysterectomy after normalization

Answer Contents

517. A 41-year-old woman with average risk and no family history of cancer is scheduled for laparoscopic hysterectomy for symptomatic fibroids. She wants to keep her ovaries. She asks whether removing both fallopian tubes at the same operation would change her ovarian cancer risk. According to the National Cancer Institute prevention summary, which of the following is the best counseling?

(A) Risk falls by about half with salpingectomy
(B) Risk falls by about 90%, the same as oophorectomy
(C) Risk is unchanged because the ovaries stay in place
(D) Risk falls by 2% for each month after surgery
(E) Risk rises because of early ovarian failure

Answer Contents

518. A 52-year-old woman with bothersome vasomotor symptoms and a hysterectomy 5 years ago asks about menopausal hormone therapy. Her mother had ovarian cancer at age 70 and she is worried about her own risk. She has no BRCA variant. According to the National Cancer Institute prevention summary, which of the following best describes the association between hormone therapy and ovarian cancer?

(A) Hormone therapy lowers ovarian cancer risk
(B) Hormone therapy has no association with ovarian cancer
(C) Hormone therapy markedly raises risk after one month
(D) Current or recent use carries a small increased risk
(E) Only estrogen with a progestin raises the risk

Answer Contents

OP4.1. Office Procedures

519. A 47-year-old woman, gravida 2 para 2, has had six months of irregular, heavy menstrual bleeding. Her body mass index is 27 kg/m2, her pregnancy test is negative, her hemoglobin is 11.8 g/dL, and her pelvic examination is normal. She has had no prior evaluation or treatment. Her physician reviews the ACOG practice bulletin on abnormal uterine bleeding in reproductive-aged women. Which of the following is the most appropriate first-line test at her age?

(A) Hysteroscopy with dilation and curettage under general anesthesia
(B) A three-month trial of cyclic oral progestin, with sampling if bleeding persists
(C) Transvaginal ultrasonography, with tissue sampling only if the lining is thickened
(D) Saline infusion sonohysterography before any tissue is obtained
(E) Office endometrial tissue sampling at this visit

Answer Contents

520. A 52-year-old woman has had irregular, heavy bleeding for eight months. An office endometrial biopsy reports endometrial intraepithelial neoplasia on scant tissue. Pelvic ultrasonography showed no focal mass. Her gynecologist wants to exclude coexisting carcinoma before choosing treatment. The clinician follows the ACOG clinical consensus on this diagnosis. Which of the following is the most accurate way to look for concurrent carcinoma?

(A) Serum CA-125 measurement with repeat pelvic ultrasonography
(B) Pelvic MRI with contrast
(C) Blind dilation and curettage under anesthesia
(D) A second blind office biopsy with a smaller cannula
(E) Hysteroscopic examination with further endometrial sampling

Answer Contents

521. A 58-year-old postmenopausal woman has taken tamoxifen for three years after breast cancer surgery. She has no vaginal bleeding, spotting, or discharge, and her pelvic examination is normal. She asks whether she should have a yearly ultrasound and endometrial biopsy to catch uterine cancer early. The clinician follows the ACOG committee opinion on tamoxifen and uterine cancer. Which of the following is the best advice?

(A) Advise no routine testing and prompt reporting of any bleeding
(B) Perform an office biopsy every year while she takes tamoxifen
(C) Offer a yearly saline infusion sonohysterogram
(D) Perform one baseline biopsy and then repeat it every two years
(E) Perform a transvaginal ultrasound every year and biopsy if the lining is thickened

Answer Contents

522. A 61-year-old woman has taken tamoxifen for four years. Over the past month she has had two episodes of pink-tinged staining on toilet tissue and a thin watery discharge. She has no pain. Examination shows pale, thin vaginal walls, a normal-appearing cervix, and no lesion. Her oncologist asks whether the changes can simply be called atrophy. The clinician follows the ACOG committee opinion on tamoxifen. Which of the following is the most appropriate step?

(A) Reassure her and repeat the examination at the next yearly visit
(B) Evaluate the endometrium now with ultrasonography or tissue sampling
(C) Obtain cervical cytology and HPV testing and then reassure her
(D) Treat as atrophy with vaginal estrogen and recheck in six months
(E) Stop tamoxifen and observe for resolution of the staining over three months

Answer Contents

523. A 29-year-old woman has HPV-positive low-grade squamous intraepithelial lesion cytology. At colposcopy the examination is adequate, with the squamocolumnar junction fully seen. After acetic acid there are two separate acetowhite areas with fine punctation, one at 3 o'clock and one at 9 o'clock. The colposcopic impression is not normal. The clinician follows the 2019 ASCCP guideline. Which of the following is the best biopsy approach?

(A) Endocervical curettage alone without cervical biopsy
(B) A single biopsy of the larger acetowhite area only
(C) Four random biopsies, one in each quadrant
(D) Targeted biopsy of each acetowhite area
(E) No biopsy, with repeat cotesting in 12 months

Answer Contents

524. A 36-year-old nonpregnant woman has atypical squamous cells, cannot exclude high-grade lesion, on cytology with a positive HPV 16 result. At colposcopy the squamocolumnar junction is not visible, so the examination is inadequate, and no definite lesion is seen on the visible ectocervix after acetic acid. She has no history of cervical treatment. Her physician reviews the 2019 ASCCP guideline. Which of the following is the preferred additional sampling at this colposcopy?

(A) Loop excision without any tissue diagnosis
(B) Endocervical brushing in place of curettage
(C) Endocervical curettage
(D) Repeat HPV testing in 12 months
(E) Random biopsies of all four quadrants

Answer Contents

525. A 27-year-old woman at 12 weeks of gestation has high-grade squamous intraepithelial lesion cytology. Colposcopy is adequate, and a 1 cm acetowhite area with coarse mosaic pattern is seen at 6 o'clock. The colposcopist plans a targeted biopsy of that area. Her trainee asks whether to add endocervical sampling at this visit. The clinician follows the 2019 ASCCP guideline. Which of the following is the correct statement about endocervical sampling?

(A) Substitute an endocervical brush for the curette at this visit
(B) Defer the colposcopy and any biopsy until the postpartum period
(C) Sample the canal only if the lesion appears to reach the os
(D) Do not perform endocervical sampling during pregnancy
(E) Perform curettage with a smaller curette to limit trauma

Answer Contents

526. A 31-year-old nulligravid woman has a cervical biopsy showing CIN 2. She plans to conceive within the year and asks to avoid treatment. At colposcopy the squamocolumnar junction could not be fully seen, and the upper limit of the lesion was not visualized. Endocervical sampling was not done. Her gynecologist reviews the 2019 ASCCP guideline on managing histologic HSIL (CIN 2). Which of the following is the most appropriate management?

(A) Topical imiquimod to the cervix
(B) Observation with colposcopy and cotesting every 6 months
(C) Cryotherapy of the visible lesion
(D) Repeat cytology in 12 months
(E) Excisional treatment of the transformation zone

Answer Contents

527. A 38-year-old nonpregnant woman has high-grade cytology with HPV 16 positivity. The ASCCP risk-based management tool, applied to her history and current results, gives an estimated immediate risk of CIN 3 or worse of 60 percent. She has had no prior colposcopy or treatment. The clinician follows the 2019 ASCCP guideline for nonpregnant patients aged 25 years or older. Which of the following is the preferred management?

(A) Repeat HPV and cytology cotesting in 1 year
(B) Repeat cytology alone in 6 months
(C) Expedited excisional treatment without a prior biopsy
(D) Colposcopy with biopsy, treating only if CIN 2 or worse is found
(E) Colposcopy with endocervical curettage alone

Answer Contents

528. A 24-year-old nulligravid woman requests a copper IUD. Her chlamydia and gonorrhea screening last month was negative, she has a single long-term partner, and she has no vaginal discharge or pelvic pain. A friend developed an infection after insertion, so she asks for antibiotics to take before the procedure. The clinician follows the 2024 CDC selected practice recommendations. Which of the following is the best plan?

(A) No prophylactic antibiotics
(B) A seven-day oral doxycycline course after insertion
(C) A single intramuscular cephalosporin dose at insertion
(D) Vaginal metronidazole gel the night before
(E) A single oral macrolide dose before insertion

Answer Contents

529. A 29-year-old nulliparous woman with heavy menstrual bleeding requests a 52 mg levonorgestrel intrauterine system. Pelvic ultrasonography shows a normal cavity with no fibroids. After cervical cleansing and tenaculum placement, the uterine sound reaches 5 cm. The label states the uterus should sound to a depth of 6 to 10 cm. Which of the following is the best next step?

(A) Insert the system and set the flange at 5 cm
(B) Do not insert it and offer another method
(C) Insert the system after a dose of oral misoprostol
(D) Insert the system and place it low in the cavity
(E) Insert the system and add a second system if it expels

Answer Contents

530. A 22-year-old woman has just had an etonogestrel implant placed in her non-dominant arm. The insertion seemed routine. Immediately afterward, the clinician cannot feel the implant on palpation, although the applicator appeared empty. The patient wants to leave with the arm bandaged. The clinician follows the FDA label. Which of the following is the most appropriate next step?

(A) Advise a non-hormonal method and locate the implant by imaging
(B) Instruct her to use condoms for 7 days and then rely on the implant
(C) Place a second implant in the opposite arm at this visit
(D) Obtain a serum etonogestrel level in 1 week to confirm placement
(E) Reassure her that it is likely in place and recheck at 3 months

Answer Contents

531. A healthy 19-year-old woman requests an etonogestrel implant. Her last menses ended 9 days ago, she used condoms consistently, and her urine pregnancy test is negative. She has no symptoms and declines a pelvic examination because it makes her anxious. The clinic's usual protocol requires a speculum examination before any contraceptive placement. The clinician follows the 2024 CDC selected practice recommendations. Which of the following is the best response?

(A) Perform bimanual examination only, then place
(B) Defer placement until her next menses
(C) Require a speculum examination before placement
(D) Place the implant without a pelvic examination
(E) Obtain cervical cytology first

Answer Contents

532. A 26-year-old woman with a new partner two months ago requests a copper IUD today. She has no symptoms, her cervix looks normal, and bimanual examination reveals no tenderness. She has not been screened for gonorrhea or chlamydia in the past year. The clinician follows the 2024 CDC selected practice recommendations. Which of the following is the best management?

(A) Give empiric antibiotics now and place the IUD in one week
(B) Screen for chlamydia and gonorrhea today and place the IUD
(C) Defer the IUD and prescribe oral contraceptive pills for three months
(D) Delay placement until her screening results are back from the lab
(E) Place a levonorgestrel IUD but not a copper IUD today

Answer Contents

533. A 68-year-old woman with symptomatic stage 3 anterior and apical prolapse has just been fitted with a ring pessary with support. It is comfortable, she voids normally, and she asks when she should be seen again. The clinician follows the AUGS and SUNA clinical consensus statement on pessary use. Which of the following is the recommended timing of the first follow-up clinic visit?

(A) At 6 months after the initial fitting
(B) Only if she develops symptoms or has trouble
(C) Within 4 weeks of the initial fitting
(D) At 12 months after the initial fitting
(E) At 3 months after the initial fitting

Answer Contents

534. A 74-year-old woman has used a ring pessary for 5 months and has been using vaginal estrogen for 6 weeks. She reports spotting. A recheck two weeks ago showed a 0.5 cm posterior vaginal wall erosion, and today it is larger and deeper. The cervix is normal and she has no fever. The clinician follows the AUGS and SUNA consensus statement. Which of the following is the best management?

(A) Leave the pessary in and add an oral antibiotic
(B) Change to a larger pessary of the same type
(C) Biopsy the erosion and refit the pessary the same day
(D) Replace the pessary with a cube pessary
(E) Remove the pessary for 4 weeks and reexamine her

Answer Contents

535. A 71-year-old woman with stage 3 prolapse has mild hesitancy and a sensation of incomplete emptying. A ring pessary with support has been selected and is being fitted for the first time. It is comfortable and stays in place with coughing and straining. Which of the following is recommended by the AUGS and SUNA consensus statement at this initial fitting for women with urinary symptoms?

(A) Urine culture and urinalysis alone
(B) Postvoid residual volume assessment
(C) Full urodynamic testing before fitting
(D) Renal and bladder ultrasonography
(E) Office cystoscopy before fitting

Answer Contents

536. A 66-year-old woman has used a ring pessary for 4 months. She reports a watery, nonmalodorous vaginal discharge that bothers her. On examination the pessary is well positioned, the vaginal walls are intact without erosion, and the discharge is clear. Vaginal pH is 4.5, and the wet mount shows no clue cells, hyphae, or motile organisms. Which of the following is the most likely explanation for her discharge?

(A) Cervicitis from a retained foreign body in the vagina
(B) Bacterial vaginosis caused by the silicone device itself
(C) A common effect of the pessary, usually not infection
(D) Candida vulvovaginitis from estrogen deficiency
(E) Trichomonas vaginalis infection acquired from the device

Answer Contents

537. A 64-year-old postmenopausal woman has a transvaginal ultrasound for mild urinary frequency. It shows a 2 cm thin-walled, anechoic, unilocular right ovarian cyst with no solid component or flow. Her CA-125 is not elevated, and she has no personal or family history of ovarian cancer. The clinician follows the 2019 Society of Radiologists in Ultrasound consensus. Which of the following is the correct reporting and follow-up recommendation?

(A) Describe the cyst in the report with no follow-up
(B) Laparoscopic cystectomy with frozen section
(C) Repeat ultrasound in 3 to 6 months to confirm stability
(D) Repeat CA-125 monthly for 6 months and rescan if rising
(E) Order pelvic MRI with contrast for characterization

Answer Contents

538. A 38-year-old premenopausal woman has pelvic ultrasonography before a fertility evaluation and has no pelvic symptoms. It shows an 8 cm thin-walled, anechoic, unilocular simple cyst of the left ovary, with no solid component, no septations, and no flow. She is not at elevated risk for ovarian cancer. The clinician follows the 2019 Society of Radiologists in Ultrasound consensus. Which of the following is the best recommendation?

(A) CA-125 and a repeat ultrasound in 2 years
(B) Immediate laparoscopic cystectomy
(C) Pelvic MRI with contrast now
(D) Describe the cyst in the report with no follow-up
(E) Repeat ultrasound in 2 to 6 months

Answer Contents

539. A 52-year-old woman has a transvaginal ultrasound for irregular bleeding. In the sagittal view the endometrium is echogenic, with a small anechoic collection of fluid in the cavity and a thin hypoechoic rim of adjacent myometrium. The sonographer asks how to measure thickness for the report. The clinician follows the AIUM practice parameter for pelvic ultrasound. Which of the following is the correct technique?

(A) Measure from the outer myometrium on one side to the outer myometrium on the other
(B) Measure the width in the coronal plane at the level of the cervix
(C) Include the cavity fluid so the full lumen is captured
(D) Measure the thickest part perpendicular to its long axis, without fluid
(E) Measure the lower uterine segment, where the lining is thinnest, at the cervix

Answer Contents

540. A 36-year-old woman arrives for an office pelvic ultrasound for abnormal uterine bleeding. A transvaginal examination is planned. She followed the written instructions and drank 32 ounces of water an hour ago and has a full bladder. The sonographer asks whether to proceed. The clinician follows the AIUM practice parameter. Which of the following is the best step?

(A) Add saline infusion to the transvaginal examination
(B) Proceed with the full bladder to improve endometrial detail
(C) Catheterize her to fill the bladder further
(D) Have her void before the transvaginal examination
(E) Perform only a transabdominal examination

Answer Contents

541. A 28-year-old woman at 6 weeks of gestation by ultrasound requests medication abortion. She has an intrauterine pregnancy and no ectopic risk factors. A copper IUD placed two years ago remains in place, and the strings are visible. She has no fever or pelvic pain. The clinician follows ACOG Practice Bulletin 225. Which of the following is the best next step?

(A) Treat with methotrexate
(B) Refer for surgical abortion only
(C) Delay until the IUD is expelled
(D) Start the regimen and leave the IUD in place
(E) Remove the IUD before starting the regimen

Answer Contents

542. A 29-year-old woman who is Rh D-negative requests medication abortion at 8 weeks of gestation. Ultrasound confirms an intrauterine pregnancy, and she has had no prior sensitization. The nurse asks whether anti-D immunoglobulin should be offered. The clinician follows the 2024 ACOG clinical practice update on abortion and pregnancy loss before 12 weeks. Which of the following is the correct position?

(A) Forgo routine Rh testing and prophylaxis
(B) Check a Kleihauer-Betke test first and dose by the result
(C) Give Rh D immune globulin to every Rh D-negative patient
(D) Give it if bleeding is heavy during the abortion
(E) Give it within 72 hours, as for any later pregnancy loss

Answer Contents

543. A 26-year-old woman calls a clinic about medication abortion. By her regular cycles she is 6 weeks from her last menstrual period. She has no pain, bleeding, history of ectopic pregnancy, tubal surgery, or IUD use, and a home test is positive. The clinic asks whether a pelvic examination or ultrasound is required before dispensing. The clinician follows ACOG Practice Bulletin 225. Which of the following is correct?

(A) Both are required once she is past 6 weeks of gestation
(B) Neither an examination nor ultrasound is needed first
(C) Ultrasonography is required in all patients before dispensing
(D) A pelvic examination is required before dispensing
(E) A quantitative hCG is required with ultrasonography

Answer Contents

544. A 31-year-old woman at 6 weeks and 4 days by ultrasound requests medication abortion. Ultrasound shows an intrauterine gestational sac, and it also shows a 4 cm complex left adnexal mass that was not known before. She has no pain or fever, and her hemodynamics are normal. The clinician follows the mifepristone label. Which of the following is the best next step?

(A) Dispense the regimen with a repeat ultrasound in 1 week
(B) Give misoprostol alone today
(C) Withhold mifepristone and evaluate the adnexal mass
(D) Perform serum CA-125 and dispense after
(E) Dispense the regimen and recheck in 7 to 14 days

Answer Contents

545. A 25-year-old woman at 8 weeks of gestation takes mifepristone 200 mg orally in the clinic as part of a medication abortion. She asks when she should take the misoprostol tablets that she has been given. She plans to be at home with a friend and has no complications. The clinician follows the mifepristone label. Which of the following is the correct timing?

(A) At the same time as the mifepristone tablet
(B) Within 2 hours after the mifepristone tablet
(C) Within 24 to 48 hours after the mifepristone
(D) Only if bleeding has not started within 7 days
(E) After 72 to 96 hours, once she is home

Answer Contents

546. A 27-year-old nulligravid woman is anxious about pain before placement of a copper IUD and asks for oral misoprostol the night before to soften the cervix. She has no cervical stenosis or prior difficult procedures. The clinician follows the 2024 CDC selected practice recommendations. Which of the following is the best counseling?

(A) It reduces pain and should be given to nulliparous patients
(B) It prevents uterine perforation during insertion
(C) It is required before levonorgestrel but not copper IUDs
(D) It is recommended only if the uterus sounds under 6 cm
(E) Misoprostol is not recommended for routine IUD placement

Answer Contents

547. A 33-year-old woman is having an office paracervical block with 1 percent lidocaine, without epinephrine, before a cervical procedure. She weighs 70 kg and has no allergies. The nurse draws up the volume for the block and asks about the limit over the next 90 minutes. The clinician follows the lidocaine label. Which of the following is the maximum recommended volume of 1 percent lidocaine for paracervical block in a 90-minute period?

(A) 10 mL
(B) 20 mL
(C) 30 mL
(D) 40 mL
(E) 50 mL

Answer Contents

548. A 34-year-old woman receives a paracervical block with lidocaine before dilation. Within a few minutes of the second injection she reports lightheadedness and says she feels nervous and unsettled. Her blood pressure is 128/78 mm Hg, her pulse is 90 per minute, she has no rash, wheeze, or lip swelling, and her skin is warm and dry. Which of the following is the most likely cause?

(A) Vasovagal reaction to cervical stimulation
(B) Anaphylaxis to lidocaine
(C) Methemoglobinemia from the anesthetic
(D) Early CNS effect of systemic lidocaine
(E) Fasting hypoglycemia before the procedure

Answer Contents

549. A 25-year-old woman is about to begin a medication abortion at 7 weeks of gestation. She asks about pain control at home and has heard that ibuprofen can make the abortion fail. She takes no other medicines and has no contraindication to anti-inflammatory drugs. The clinician follows ACOG Practice Bulletin 225. Which of the following is the best advice?

(A) Use no analgesic so pain can signal complications
(B) Take an NSAID for pain, which is recommended
(C) Use only acetaminophen
(D) Take a routine opioid prescription
(E) Avoid NSAIDs because they block expulsion

Answer Contents

550. A 30-year-old woman is having an office paracervical block with lidocaine before a uterine procedure. The clinician has injected the left side and intends to inject the right side right away to finish quickly. The nurse recalls a label instruction about timing between sides. The clinician follows the lidocaine label. Which of the following is the correct interval between injection of the two sides?

(A) Allow a 5-minute interval between sides
(B) Inject both sides within 1 minute
(C) Allow a 60-minute interval between sides
(D) Allow a 30-minute interval between sides
(E) Wait 90 minutes before the second side

Answer Contents

Answers and Explanations

Part 1. Gynecology

G1. Preoperative Care

1. Answer: A. The guideline's discussion of functional capacity reports that Duke Activity Status Index scores of 34 or lower were associated with increased odds of 30-day death or myocardial infarction. A low score therefore marks higher risk and sharpens the history, the choice of anesthetic and the postoperative monitoring plan. The distractors reverse the direction, deny any outcome data, or turn a low score into automatic stress testing, which the same guideline discourages. In practice, record the score, weigh it with symptoms and the planned operation, and test only when the result would change management.

> *Key principle:* A Duke Activity Status Index score of 34 or lower signals higher odds of 30-day death or myocardial infarction.

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2. Answer: A. The guideline advises that routine preoperative stress testing should not be performed in patients with adequate functional capacity, and the summary of its recommendations gives a Class 3 statement against routine stress testing in low-risk patients and those with good capacity. This patient climbs two flights without symptoms, so a result would not change her plan. The distractors are tests that add cost, delay and false positives without changing management, and routine assessment of ventricular function is likewise unsupported. In practice, reserve stress testing for highly selected patients with poor or unknown capacity.

> *Key principle:* Good functional capacity means no routine stress test before noncardiac surgery.

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3. Answer: D. The guideline uses the term elevated risk rather than high risk and generally defines it as a 1% or greater risk of major adverse cardiovascular events. Validated tools such as the Revised Cardiac Risk Index inform the estimate, which then guides how much further assessment is justified. The distractors place the threshold too low, so nearly every patient qualifies, or too high, so patients who deserve closer evaluation are missed. In practice, write the estimate and the tool used in the record so the anesthesia team can act on it.

> *Key principle:* The 2024 guideline generally calls a 1% or greater MACE risk elevated.

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4. Answer: E. The guideline notes little evidence of increased perioperative complications when preoperative blood pressure is under 180/110 mm Hg, and it recommends ongoing treatment of chronic hypertension in the perioperative period. Her readings fall under that threshold and she has no other risk component. The distractors treat a moderately raised clinic reading as a reason to delay, admit or test, which adds cost and risk without benefit. In practice, continue her medication, recheck pressure on the day of surgery, and keep the date.

> *Key principle:* Pressure under 180/110 mm Hg alone is not a reason to postpone elective surgery.

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5. Answer: E. The guideline states that a routine preoperative electrocardiogram has little effect on treatment or complication rates in low-risk patients, and it advises omitting it from standard evaluation, with more value in patients having elevated-risk procedures. This patient is young, asymptomatic and facing a low-risk procedure. The distractors are the habitual baseline tracing and add-on biomarkers or imaging, none of which would change her plan. In practice, order an electrocardiogram when symptoms, known disease or the operation's risk make the result actionable.

> *Key principle:* Skip the routine electrocardiogram in a low-risk patient having a low-risk procedure.

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6. Answer: A. The guideline gives a Class 1 recommendation that SGLT2 inhibitors be stopped 3 to 4 days before planned surgery to reduce the risk of perioperative metabolic acidosis. The drugs cause ketosis with near-normal glucose during fasting and surgical stress, so the effect must wear off before the operation. The distractors are too late to clear the drug, too early to matter, or ignore the risk altogether. In practice, give her a written stop date on the preoperative plan, with a plan for glucose coverage until she resumes eating.

> *Key principle:* Stop SGLT2 inhibitors 3 to 4 days before planned surgery.

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7. Answer: D. The October 2024 multisociety guidance states that, in many cases, patients with scheduled procedures should continue the drug, weighing delayed gastric emptying against the harms of stopping. It replaced the earlier 2023 advice to hold weekly agents for a week. The distractors preserve that superseded hold, invent a longer washout, substitute another agent, or relax fasting, which raises aspiration risk. In practice, assess risk factors and symptoms for each patient and let that assessment, not the drug name alone, set the plan.

> *Key principle:* The 2024 multisociety guidance allows many patients to continue GLP-1 agonists through surgery.

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8. Answer: A. The guidance notes that patients on a higher dose typically have more gastrointestinal side effects and should follow a liquid diet for 24 hours before the procedure. Liquid-only intake lowers retained solid contents when gastric emptying may be slow, and point-of-care ultrasound can check stomach contents in the highest-risk patients. The distractors propose a long washout, a tube, a drug or a longer solid fast, none of which the guidance endorses as the standard step. In practice, identify higher-dose patients at the preoperative visit and give written diet instructions.

> *Key principle:* For GLP-1 patients at highest risk, a 24-hour liquid diet before surgery is the suggested step.

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9. Answer: B. The guidance states that patients with gastrointestinal symptoms, including nausea, vomiting and abdominal pain, should wait until the symptoms have dissipated before elective surgery. Symptoms suggest retained gastric contents despite fasting, and aspiration during anesthesia can be serious. The distractors try to work around the symptoms with an induction technique, an antiemetic, a different anesthetic or more fasting time, which treats a signal as an inconvenience. In practice, an elective case can be rescheduled without harm, so defer it and review the plan.

> *Key principle:* Gastrointestinal symptoms on a GLP-1 agonist mean waiting until they resolve before elective surgery.

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10. Answer: D. The guideline recommends that healthy adults drink carbohydrate-containing clear liquids until 2 hours before elective procedures, a strong recommendation based on moderate evidence, and the earlier clear liquid limit of 2 hours was reaffirmed. Longer fasts add thirst, hunger and insulin resistance without lowering aspiration risk for clear liquids in healthy patients. The distractors are older, longer cutoffs. In practice, tell her to stop at 6 am, and give the instruction in writing to avoid an unnecessary overnight fast.

> *Key principle:* Healthy adults may drink carbohydrate-containing clear liquids until 2 hours before surgery.

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11. Answer: E. The 2023 guideline suggests not delaying elective procedures requiring general anesthesia, regional anesthesia or procedural sedation in patients who chew gum, and it adds that gum should be removed before any sedative or anesthetic is given. Gum does not meaningfully increase gastric volume, and the main hazard is an airway foreign body. The distractors delay or cancel the case, or add drugs or technique for a problem that does not warrant them. In practice, remove the gum, document it and continue.

> *Key principle:* Chewing gum does not delay an elective case, but it must come out before anesthesia.

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12. Answer: E. The WHO guideline gives a strong recommendation, based on moderate quality evidence, that hair should either not be removed or, if absolutely necessary, be removed only with a clipper, and that shaving is strongly discouraged at all times. Razors cause microscopic skin abrasions that harbor bacteria. The distractors differ only in the timing and place of shaving, which does not remove the harm. In practice, ask whether exposure truly needs hair removal, and keep a clipper in the preoperative area.

> *Key principle:* Do not shave the surgical site, and use a clipper only if hair must go.

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13. Answer: B. The guideline states that the antimicrobial agent should be started within 60 minutes before surgical incision so that tissue concentrations are therapeutic when contamination occurs. A dose given the night before has fallen away by morning, and a dose after the incision or after closure misses the period of bacterial seeding. The distractors fix the timing too early or too late. In practice, have the dose given in the operating room or holding area, and record the time against the incision time.

> *Key principle:* Start cefazolin within 60 minutes before incision.

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14. Answer: D. The WHO panel suggests not discontinuing immunosuppressive medication before surgery to prevent surgical site infection, a conditional recommendation based on very low quality evidence. Stopping risks a flare of the underlying disease, and the infection benefit of stopping was not shown. The distractors stop the drug for varying intervals or swap it for a corticosteroid, which is itself immunosuppressive. In practice, make the plan with the prescribing specialist and state the evidence quality when discussing the choice with her.

> *Key principle:* WHO suggests continuing immunosuppressants before surgery, on very low quality evidence.

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15. Answer: A. The WHO guideline holds that patients should bathe or shower before surgery and suggests that either a plain or antimicrobial soap can be used, a conditional recommendation based on moderate quality evidence. Bathing removes soil and some flora, and no soap type was shown to be superior. The distractors insist on an antimicrobial product, skip bathing, or invent a scrub or a soap-free routine. In practice, tell her to bathe with the soap she has and avoid an unnecessary purchase.

> *Key principle:* Preoperative bathing helps, and plain soap is acceptable.

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16. Answer: A. The guideline lists the adult cefazolin dose as 2 g, rising to 3 g for patients weighing 120 kg or more, started within 60 minutes before incision, with prophylaxis lasting less than 24 hours and a single dose usually sufficient. At 128 kg she meets the weight threshold, so the larger dose is needed for adequate tissue levels. The distractors underdose, extend the course past the guideline duration, or mistime the dose. In practice, check the actual weight on the day, not the booking weight.

> *Key principle:* Patients weighing 120 kg or more receive 3 g of cefazolin rather than 2 g.

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G2. Perioperative Care

17. Answer: E. The WHO guideline advises that adults having general anesthesia with endotracheal intubation receive an 80% fraction of inspired oxygen intraoperatively and, if feasible, for 2 to 6 hours afterward. The 2016 edition graded this strong, and the 2018 second edition downgraded it to conditional. Higher tissue oxygen supports neutrophil killing at the wound. The distractors give low or brief oxygen exposure, or an extended high-oxygen course that lacks support. In practice, agree on the setting with anesthesia before the case and include it in the infection bundle.

> *Key principle:* WHO strongly recommends 80% inspired oxygen intraoperatively in intubated adults.

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18. Answer: D. The CDC guideline gives a Category IA recommendation to maintain perioperative normothermia. Hypothermia causes vasoconstriction, lowers wound tissue oxygen and impairs immune function, which raises the risk of surgical site infection. The distractors treat cooling as protective, aim to starve bacteria with a cold room, substitute an antibiotic for a physiologic fix, or add more cooling. In practice, use forced-air warming and warmed fluids, and monitor core temperature through long cases.

> *Key principle:* Maintain normothermia during surgery to lower infection risk.

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19. Answer: B. The CDC guideline recommends perioperative glycemic control with blood glucose target levels less than 200 mg/dL, a Category IA recommendation. Hyperglycemia impairs neutrophil function and wound healing. The distractors include the older tight-control range of 80 to 110 mg/dL, which carries a hypoglycemia risk, and thresholds that are too loose to prevent harm. In practice, check glucose intraoperatively in patients with diabetes, treat levels over the target, and avoid hypoglycemia while doing so.

> *Key principle:* The CDC perioperative glucose target is under 200 mg/dL.

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20. Answer: D. The WHO panel suggests intraoperative goal-directed fluid therapy to reduce the risk of surgical site infection, a conditional recommendation based on low quality evidence. Titrating fluid to measured hemodynamic targets avoids both hypovolemia and overload, and both impair tissue perfusion at the wound. The distractors fix the volume in advance or rely on a weak trigger such as urine output. In practice, agree on the monitoring method with the anesthesia team, and note that the evidence is limited.

> *Key principle:* WHO suggests goal-directed fluid therapy during surgery, on low quality evidence.

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21. Answer: B. The WHO panel suggests considering wound-protector devices in clean-contaminated, contaminated and dirty abdominal procedures, a conditional recommendation based on very low quality evidence. A protector shields the incision edges from contact with contaminated viscera and the cuff. The distractors name measures the panel advised against: advanced dressings, laminar airflow, plastic incise drapes and antibiotics prolonged for a drain. In practice, consider a protector for open abdominal cases that enter the vagina or bowel.

> *Key principle:* WHO suggests considering wound protectors in contaminated or clean-contaminated abdominal surgery.

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22. Answer: C. The WHO panel suggests triclosan-coated sutures to reduce surgical site infection risk, independent of the type of surgery, a conditional recommendation based on moderate quality evidence. The antiseptic coating inhibits bacterial colonization along the suture track. The distractors are plausible alternatives the panel did not endorse: uncoated sutures, another coating, a hand-soaked suture or a different closure method. In practice, check availability and any triclosan allergy, and use the coated suture for fascia and skin closure when stocked.

> *Key principle:* WHO suggests triclosan-coated sutures to lower surgical site infection risk.

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23. Answer: E. The CDC guideline recommends intraoperative skin preparation with an alcohol-based antiseptic agent unless contraindicated, a Category IA recommendation. Alcohol-containing combinations kill skin flora quickly and persist on the skin. The distractors are aqueous antiseptics, a soap scrub and saline, none of which provide the same rapid, persistent kill. In practice, allow the agent to dry before draping, avoid pooling to reduce fire risk, and use an alternative only when there is a contraindication.

> *Key principle:* Prepare surgical skin with an alcohol-based antiseptic unless contraindicated.

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24. Answer: A. The guideline calls for intraoperative redosing when the procedure outlasts two half-lives of the drug, and its table lists a 4-hour redosing interval for cefazolin, measured from the start of the preoperative dose. This operation has passed that mark. The distractors confuse the time rule with the blood loss rule, assume a single dose is enough, switch drugs without cause, or delay the dose. In practice, set a timer from the first dose and ask for the redose before the interval ends.

> *Key principle:* Redose cefazolin when the operation runs past 4 hours from the first dose.

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25. Answer: B. The guideline lists excessive blood loss, defined as more than 1500 mL, as a reason to redose prophylaxis during the operation, in addition to the time-based rule. Hemorrhage and fluid replacement dilute serum antibiotic levels under the protective range. The distractors apply only the time rule, assume one dose is always enough, change class without need, or defer the dose until after transfusion. In practice, tell the anesthesia team when blood loss crosses the threshold so the dose is given promptly.

> *Key principle:* Redose prophylaxis when blood loss exceeds 1500 mL.

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26. Answer: E. The CDC guideline gives a Category IA recommendation against additional prophylactic antimicrobial doses after incision closure in clean and clean-contaminated procedures. Extra doses add toxicity, resistance and Clostridioides difficile risk without lowering infection rates. The distractors keep prophylaxis going through habitual triggers such as a catheter or a bowel movement. In practice, remove automatic postoperative antibiotic orders from the hysterectomy order set and treat only documented infection.

> *Key principle:* Prophylactic antibiotics stop at incision closure in clean and clean-contaminated operations.

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27. Answer: C. The WHO panel suggests that antibiotic incisional wound irrigation before closure should not be done, a conditional recommendation based on low quality evidence. Topical antibiotic irrigation has not been shown to lower infection and adds exposure to the drug and to resistance pressure. The distractors repeat the practice in a subgroup, switch the form of the topical antibiotic, or pair it with extended systemic prophylaxis. In practice, keep the incision clean with technique and the prophylaxis bundle, and leave antibiotic solution out of the closure tray.

> *Key principle:* WHO suggests against antibiotic irrigation of the incision before closure.

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28. Answer: D. The guideline gives a Class 3 statement that in most patients bridging with parenteral heparin may cause harm through increased bleeding risk, and it holds that withholding warfarin or a direct oral anticoagulant before surgery is appropriate. Her atrial fibrillation without a prior embolic event does not place her in the highest thrombotic risk group. The distractors bridge, continue the drug, or add a filter that is not recommended for this purpose. In practice, write the stop and restart dates with the prescriber and give them to her.

> *Key principle:* Most patients on warfarin should not be bridged with heparin around surgery.

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29. Answer: A. The guideline carries a Class 2a statement that in patients at high thrombotic risk, bridging with heparin or low-molecular-weight heparin can reduce thromboembolic risk, in contrast with its Class 3 warning against bridging in most patients. A mechanical mitral valve is a classic high-risk setting. The distractors withhold bridging, which suits lower-risk patients, swap in a drug that is not used for mechanical valves, or continue warfarin through major surgery. In practice, plan the interruption and the bridging dates with cardiology and the prescriber.

> *Key principle:* High thrombotic risk is the setting in which heparin bridging can help.

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30. Answer: C. The guideline states that withholding direct oral anticoagulants or warfarin before surgery is appropriate for most patients on therapeutic anticoagulation, and it warns that routine heparin bridging may cause harm. She has no features of very high thrombotic risk. The distractors continue the drug, add a bridge, reduce the dose, or substitute an agent that does not protect against stroke in atrial fibrillation. In practice, give her a written schedule agreed with the prescriber and pharmacy, and reassure her about the planned restart.

> *Key principle:* Withholding a DOAC before surgery is appropriate when a written plan is in place.

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31. Answer: D. The guideline gives a Class 2a statement that it is reasonable to address medical triggers for rapid atrial fibrillation, such as anemia and sepsis. Her anemia and fever suggest bleeding or infection driving the rate, and rhythm drugs alone would not correct them. The distractors treat the arrhythmia as the problem by cardioverting, infusing an antiarrhythmic or anticoagulating, or they withhold rate control in a symptomatic patient. In practice, find the source, transfuse or treat infection as indicated, and control the rate.

> *Key principle:* Treat the trigger, such as anemia or sepsis, when atrial fibrillation is rapid after surgery.

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32. Answer: B. The summary of the guideline gives a Class 2a statement that outpatient cardiovascular follow-up is reasonable after myocardial injury after noncardiac surgery, and a Class 2b statement that antithrombotic therapy may be reasonable to lower the risk of later events. These patients carry higher risk of subsequent events even without symptoms. The distractors dismiss the finding, jump to invasive testing, choose anticoagulation without support, or monitor an inappropriate marker. In practice, document the diagnosis in the discharge summary and arrange follow-up.

> *Key principle:* Asymptomatic myocardial injury after surgery still warrants outpatient cardiovascular follow-up.

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G3. Surgical Complications

33. Answer: A. In an incomplete ureteral injury the ureter keeps some continuity, so an internal stent can carry urine past the defect while the wall heals. The AUA guideline tells surgeons to attempt stent placement when an incomplete injury is diagnosed postoperatively or in a delayed setting. The distractors jump to the fallback (nephrostomy), to open reconstruction, to waiting, or to bladder drainage that does not reach the upper tract. In practice, try retrograde stenting first and escalate only if it fails.

> *Key principle:* Incomplete ureteral injury found after surgery gets a stent attempt first.

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34. Answer: E. When no wire or stent can cross the injury, urine must be diverted proximal to it to prevent urinoma, infection, and loss of renal function. The AUA guideline calls for percutaneous nephrostomy, with delayed repair if still needed, when stent placement is unsuccessful or not possible. The distractors keep pressing on the failed route, remove a salvageable kidney, ignore the leak, or attempt major reconstruction in inflamed tissue. In practice, divert with a nephrostomy first and plan definitive repair once conditions settle.

> *Key principle:* Stent will not pass: place a nephrostomy and delay the repair.

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35. Answer: A. A ureterovaginal fistula forms when the ureter is injured or devascularized near the vaginal cuff, and the tract can close if urine is routed through the lumen. The AUA guideline recommends initial stent placement when possible, with further surgery if the stent fails. Catheter drainage is the usual first step for a vesicovaginal fistula and is the trap here, since the unstained tampon shows the bladder is not the source. In practice, a dye-negative leak with ureteral contrast extravasation means stent first.

> *Key principle:* Ureterovaginal fistula: try a stent before any open repair.

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36. Answer: C. Distal ureteral injuries are usually repaired by reimplantation into the bladder because blood supply to the lowest segment is tenuous. The AUA guideline states that injuries distal to the iliac vessels should be repaired with reimplantation or primary repair over a stent when possible, and stable patients are repaired at the time of laparotomy. The distractors sacrifice the kidney, choose complex substitutes for a short defect, or defer repair that a stable patient does not need to defer. In practice, repair a recognized distal injury during the same operation.

> *Key principle:* Distal ureteral injury in a stable patient: reimplant or repair over a stent now.

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37. Answer: D. A prolonged reconstruction in a patient with hypothermia, acidosis, and coagulopathy adds physiologic injury that she cannot tolerate. The AUA guideline allows surgeons to manage ureteral injuries in unstable patients with temporary urinary drainage followed by delayed definitive management. Each distractor commits to a lengthy repair or to losing a kidney that can be saved. In practice, control bleeding, drain the urine temporarily, and return for definitive repair once she is stabilized.

> *Key principle:* Unstable patient: drain the ureter temporarily and repair later.

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38. Answer: E. A contused ureter may lose its blood supply over the next days, leaving a stricture or delayed leak, so the surgeon judges viability when the injury is found. The AUA guideline says contusions found at laparotomy are managed with stenting or with resection and primary repair, depending on viability and the clinical scenario. The distractors ignore the risk, cut off drainage, or remove the kidney. In practice, a bruised segment gets a stent if it looks viable and resection with repair if it does not.

> *Key principle:* Ureteral contusion: stent if viable, resect and repair if not.

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39. Answer: B. A repaired bladder wall heals best when it is kept decompressed, and a urethral catheter alone achieves that without a second tract through the abdominal wall. The AUA guideline recommends urethral catheter drainage without suprapubic cystostomy after surgical repair of bladder injuries. Adding a suprapubic tube, using one alone, skipping drainage, or stenting the ureters each adds morbidity or leaves the closure unprotected. In practice, leave a urethral catheter in place after cystotomy repair.

> *Key principle:* After bladder repair, a urethral catheter alone is enough.

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40. Answer: C. A purulent collection under a closed incision needs to be opened so the infection can drain and the wound can heal from the base. The IDSA guideline states that suture removal plus incision and drainage should be performed for surgical site infections, and antibiotics alone do not empty the pocket. The distractors treat the systemic response that is absent here, aspirate blindly, or wait. In practice, open the wound, pack as needed, and reserve antibiotics for patients with systemic signs.

> *Key principle:* A surgical site infection is opened and drained first.

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41. Answer: A. After the collection is drained, a patient with limited surrounding cellulitis and no systemic response has little to gain from antibiotics. The IDSA guideline says adjunctive systemic antimicrobial therapy is not routinely indicated, and it names erythema beyond 5 cm, temperature over 38.5 C, heart rate over 110 beats/min, or white blood cell count over 12,000/microL as signals that therapy may help. This patient meets none of them. In practice, drainage is the treatment, and antibiotics follow only a significant systemic response.

> *Key principle:* Drain the wound and give antibiotics only for a systemic response.

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42. Answer: D. A port site lies on the trunk, and a clean operation with a localized wound infection usually responds to drainage. When systemic signs are present, however, drainage alone is not enough. The IDSA guideline indicates a brief course of systemic antimicrobial therapy for infections after clean operations on the trunk, head and neck, or extremities that also have systemic signs. The distractors under-treat or add measures with no role. In practice, open the wound and add a short systemic course when fever and tachycardia accompany the infection.

> *Key principle:* Clean trunk wound with systemic signs: drain and add a short antibiotic course.

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43. Answer: B. In a clean-operation wound the usual pathogen is Staphylococcus aureus, and the choice of agent turns on MRSA risk. The IDSA guideline recommends a first-generation cephalosporin or an antistaphylococcal penicillin for methicillin-susceptible organisms and agents such as vancomycin, linezolid, daptomycin, telavancin, or ceftaroline when MRSA risk factors, including nasal colonization or prior MRSA infection, are high. Cefazolin and nafcillin miss MRSA, and the gram-negative regimen is aimed at genital or bowel flora. In practice, match the antistaphylococcal agent to MRSA risk.

> *Key principle:* Known MRSA risk calls for an MRSA-active agent in a clean-wound infection.

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44. Answer: E. Hysterectomy passes through the vagina, so wound infections carry vaginal and bowel flora, including gram-negative rods and anaerobes. The IDSA guideline recommends agents active against gram-negative bacteria and anaerobes, such as a cephalosporin or fluoroquinolone combined with metronidazole, for infections after operations on the axilla, gastrointestinal tract, perineum, or female genital tract. The distractors are staphylococcal or MRSA agents that miss that flora. In practice, choose coverage by the organs the operation entered, not only by skin organisms.

> *Key principle:* Wound infection after genital tract surgery needs gram-negative and anaerobic coverage.

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45. Answer: C. The guideline table for vaginal or abdominal hysterectomy lists cefazolin, cefotetan, cefoxitin, or ampicillin-sulbactam as recommended agents, with strength of evidence A. These cover the skin, vaginal and enteric flora met at the cuff. The distractors are single agents with gaps in coverage, or alternatives the table reserves for patients with a beta-lactam allergy. In practice, choose a listed cephalosporin or ampicillin-sulbactam, time it within 60 minutes before incision, and record the allergy status.

> *Key principle:* Hysterectomy prophylaxis uses cefazolin, cefotetan, cefoxitin, or ampicillin-sulbactam.

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46. Answer: D. A large platelet fall on heparin-class therapy raises concern for an immune reaction that paradoxically causes thrombosis. The enoxaparin label states that thrombocytopenia of any degree should be monitored closely and that enoxaparin should be discontinued if the platelet count falls under 100,000/mm3. Continuing, adding another antithrombotic, or giving platelets leaves the trigger in place. In practice, stop the drug, test for heparin-induced thrombocytopenia, and switch to a non-heparin option if the diagnosis is likely.

> *Key principle:* Platelets under 100,000/mm3 on enoxaparin: stop the drug and evaluate.

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47. Answer: E. Enoxaparin is a heparin derivative and can reactivate the antibody-driven platelet activation that defines heparin-induced thrombocytopenia. The label lists a history of immune-mediated HIT within the past 100 days, or circulating antibodies, as a contraindication. Mechanical methods and mobilization carry no such risk and remain available, so the correct answer is the one drug. In practice, a recent HIT history removes heparin-class prophylaxis from consideration and shifts to mechanical measures or a non-heparin agent.

> *Key principle:* Recent immune-mediated HIT contraindicates enoxaparin.

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48. Answer: A. Neuraxial bleeding is confined by bone and can compress the cord, producing permanent paralysis. The enoxaparin boxed warning lists indwelling epidural catheters, a history of traumatic or repeated epidural or spinal punctures, and a history of spinal deformity or spinal surgery as factors that raise the risk. Obesity, hypertension, diabetes, and age are not on that list. In practice, time catheter placement and removal around each dose and monitor for new back pain or leg weakness.

> *Key principle:* An indwelling epidural catheter raises hematoma risk on enoxaparin.

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49. Answer: B. Enoxaparin is cleared by the kidney, so exposure rises when renal function falls and bleeding risk rises with it. The label states that exposure increases in renal impairment, that a dosage adjustment is recommended for both therapeutic and prophylactic dosage ranges, and that such patients should be observed carefully for signs and symptoms of bleeding. The distractors ignore renal clearance, add bleeding risk, abandon prophylaxis, or choose a slow-onset agent. In practice, adjust the dose to renal function and watch for bleeding.

> *Key principle:* Renal impairment on enoxaparin: adjust the dose and watch for bleeding.

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50. Answer: E. Anticoagulants worsen hemorrhage, and this patient has signs of active postoperative bleeding with hypotension and a falling hemoglobin. The enoxaparin label lists active major bleeding as a contraindication. Mechanical prophylaxis protects against thrombosis without adding bleeding risk. The distractors give a full, partial, or escalated dose into an active bleed. In practice, hold the anticoagulant, resuscitate and find the source, and resume pharmacologic prophylaxis once hemostasis is secure.

> *Key principle:* Active major bleeding: hold enoxaparin and use mechanical prophylaxis.

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51. Answer: D. A restrictive threshold spares stable patients blood products they may not need. The AABB recommends considering transfusion when hemoglobin falls under 7 g/dL in hospitalized, hemodynamically stable adults. At 6.7 g/dL she is under that line, so transfusion is appropriate, whereas iron or erythropoietin acts too slowly and waiting for symptoms or a lower value is not supported. In practice, apply the threshold, then recheck.

> *Key principle:* Stable hospitalized adult: consider transfusion when hemoglobin is under 7 g/dL.

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52. Answer: C. Patients with coronary disease tolerate anemia poorly, since reduced oxygen delivery can cause myocardial ischemia. The AABB guideline states that clinicians may choose a threshold of 8 g/dL for those undergoing orthopedic surgery or with preexisting cardiovascular disease. At 7.6 g/dL this patient is under that adjusted value, so transfusion fits the guideline. The distractors apply the general 7 g/dL figure to everyone or choose agents too slow to help. In practice, raise the threshold for patients with cardiovascular disease.

> *Key principle:* Preexisting cardiovascular disease: the AABB threshold rises to 8 g/dL.

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53. Answer: B. Oncology patients were included in the restrictive trials, and the guideline did not find that they need a higher trigger. The AABB panel suggests a restrictive strategy, with transfusion when hemoglobin is under 7 g/dL, for hospitalized adults with hematologic and oncologic disorders, graded as a conditional recommendation with low-certainty evidence. At 7.4 g/dL she remains over that line. The distractors use a liberal target, assume cancer raises the trigger, or ignore the threshold. In practice, watch the value and transfuse at the cutoff.

> *Key principle:* Oncology inpatients use the same restrictive threshold of 7 g/dL.

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54. Answer: A. Observational links between transfusion and infection reflect sicker patients who bleed more, and refusing necessary blood exposes a hemodynamically compromised patient to the harms of anemia. The CDC guideline, Category IB, states not to withhold transfusion of necessary blood products from surgical patients as a means to prevent surgical site infection. The distractors let infection concerns override physiologic need. In practice, transfuse when indicated and prevent infection through the proven measures.

> *Key principle:* Do not withhold needed blood as a way to prevent a wound infection.

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55. Answer: D. Rising CO2, rigidity, tachycardia, and fever under a volatile agent and succinylcholine mean malignant hyperthermia, a hypermetabolic crisis triggered by those drugs. The dantrolene label directs that all anesthetic agents be stopped as soon as the reaction is recognized and that dantrolene be given by continuous rapid intravenous push. Antipyretics and cooling do not stop the calcium-driven process, and the oral route is too slow. In practice, remove the trigger and give dantrolene at once.

> *Key principle:* Suspected malignant hyperthermia: stop triggering agents and give IV dantrolene now.

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56. Answer: C. Dantrolene is effective but the crisis can recur as its effect wanes while triggering drug stores are still present. The label states that if the physiologic and metabolic abnormalities reappear, the regimen may be repeated, and administration should be continuous until symptoms subside. Antipyretics address only the temperature, and any volatile agent would restart the process. In practice, monitor in an intensive setting after a crisis and redose when signs return.

> *Key principle:* Malignant hyperthermia can recur, so repeat dantrolene.

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57. Answer: E. The dantrolene label warns that hepatotoxicity has been reported with intravenous dantrolene products. New transaminase elevations with nausea and right upper quadrant discomfort point to the liver rather than to kidney, ear, pancreas, or cardiac conduction. Her muscle symptoms have resolved, so the findings are not a relapse of the crisis. In practice, check liver enzymes after treatment, and evaluate other causes if they are elevated.

> *Key principle:* IV dantrolene can raise liver enzymes hours to days later.

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58. Answer: A. Dantrolene reduces calcium release from skeletal muscle, and the dose needed varies with how long and how severe the crisis is. The label directs a starting minimum of 1 mg/kg by continuous rapid intravenous push, continuing until symptoms subside or the maximum cumulative dose of 10 mg/kg is reached. A single dose, a fixed duration, a temperature drop, or a laboratory value are not the stated endpoints. In practice, titrate to clinical response, and repeat if signs return.

> *Key principle:* Dose dantrolene until symptoms subside or the cumulative maximum is reached.

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59. Answer: B. The picture of hypercarbia, rigidity, hyperthermia, hyperkalemia, and acidosis during a volatile anesthetic is malignant hyperthermia. The dantrolene label indicates it, along with appropriate supportive measures, for malignant hyperthermia crises. Bromocriptine and cyproheptadine are used in neuroleptic malignant and serotonin syndromes, which do not arise from volatile agents, while antipyretics and opioid reversal do not treat the muscle crisis. In practice, name the syndrome, then match the antidote.

> *Key principle:* Dantrolene is the specific treatment for malignant hyperthermia.

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G4. Postoperative Care

60. Answer: B. Opioids slow gut motility through receptors in the bowel wall, and bowel resection adds its own ileus. Alvimopan is labeled to accelerate the time to upper and lower gastrointestinal recovery following surgeries that include partial bowel resection with primary anastomosis. The other agents are prokinetics or laxatives used for other indications and carry no such labeled indication for this surgical setting. In practice, alvimopan fits the narrow group of patients having bowel resection with anastomosis who will receive opioids.

> *Key principle:* Alvimopan is labeled for bowel resection with primary anastomosis.

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61. Answer: E. Patients recently exposed to opioids may be more sensitive to the effects of alvimopan, so prior opioid exposure matters. The label states it is contraindicated in patients who have taken therapeutic doses of opioids for more than 7 consecutive days. Reducing the dose, delaying it, or pairing it with another drug does not remove the contraindication. In practice, check preoperative opioid exposure before ordering alvimopan, and use other measures for patients with prolonged exposure.

> *Key principle:* More than 7 days of therapeutic opioids contraindicates alvimopan.

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62. Answer: D. Alvimopan carries a boxed warning for myocardial infarction with long-term use, so it is limited to short hospital courses through a restricted program of 15 doses. Hospital use and chronic use are different exposures. The distractors suggest monitoring or protection that the label does not provide, or an unlabeled indication. In practice, stop alvimopan at discharge and use standard bowel regimens for opioid-induced constipation at home.

> *Key principle:* Alvimopan is short-term hospital use only because of myocardial infarction risk.

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63. Answer: C. A drug that speeds bowel recovery cannot help a mechanically blocked gut and could mask a problem that needs an operation. The label states that no studies have been conducted in patients with complete gastrointestinal obstruction and that alvimopan is not recommended for use in these patients. The distractors invent dosing, intravenous administration, or a replacement for decompression. In practice, treat the obstruction surgically and do not rely on alvimopan to resolve it.

> *Key principle:* Alvimopan is not recommended in complete bowel obstruction.

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64. Answer: A. Opioid analgesia is mediated in the central nervous system, while constipation and ileus arise from receptors in the bowel. The label describes alvimopan as antagonizing the peripheral effects of opioids on gastrointestinal motility and secretion, and states that it acts peripherally. A centrally acting antagonist would reverse pain control, and the other mechanisms describe prokinetic or cholinergic drugs. In practice, a peripherally restricted antagonist lets surgeons keep opioids while protecting the gut.

> *Key principle:* Alvimopan blocks opioid effects in the gut, not in the brain.

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65. Answer: B. For many common types of acute pain, nonopioid therapies work at least as well as opioids and avoid their risks of dependence, sedation, and overdose. The CDC guideline states that clinicians should maximize nonpharmacologic and nonopioid pharmacologic therapies and consider opioids for acute pain only if the benefits are anticipated to outweigh the risks. The distractors reverse that order or use long-acting or fixed-schedule opioids. In practice, build a multimodal plan first and keep opioids as rescue therapy.

> *Key principle:* Nonopioid therapy comes first, and opioids need benefits that outweigh risks.

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66. Answer: A. Leftover opioids create risk for misuse, diversion, and accidental exposure, and longer prescriptions raise the chance of continued use after surgery. The CDC guideline states that when opioids are needed for acute pain, clinicians should prescribe no greater quantity than needed for the expected duration of pain severe enough to require opioids. Default supplies, refills, and long-acting forms exceed the expected need. In practice, tailor the quantity to the likely course of recovery after the procedure.

> *Key principle:* Match the opioid quantity to the expected duration of severe pain.

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67. Answer: E. A prior overdose raises the chance of another one, even after a short exposure. The CDC guideline directs clinicians to work with patients to incorporate risk mitigation strategies into the management plan, including offering naloxone. Cancelling the analgesic leaves pain untreated, adding a sedative raises overdose risk, extended-release products carry more risk, and a drug screen does not reduce exposure. In practice, offer naloxone, teach her and her household how to use it, and counsel on safe storage and disposal.

> *Key principle:* Offer naloxone when anyone in the home has overdose risk.

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68. Answer: C. Opioids and benzodiazepines both depress the central nervous system, and together they raise the risk of oversedation, respiratory depression, and death. The CDC guideline advises particular caution when prescribing opioid pain medication and benzodiazepines concurrently. Stopping a benzodiazepine abruptly can cause withdrawal, while larger quantities and additional sedatives add risk. In practice, prescribe the smallest opioid quantity needed, review the combined risk, and consider naloxone.

> *Key principle:* Opioids plus benzodiazepines call for particular caution.

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69. Answer: D. Ketorolac is a potent NSAID whose risks of gastrointestinal bleeding, renal injury, and platelet inhibition rise with the length of use. The label indicates it for short-term management of moderately severe acute pain, up to 5 days in adults, and states that combined oral and injection use should not exceed 5 days. Taking it with food, a proton pump inhibitor, or a low dose does not make longer use acceptable. In practice, stop ketorolac by day 5 and switch to a safer regimen.

> *Key principle:* Ketorolac total course is limited to 5 days.

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70. Answer: A. Ketorolac inhibits platelet function, so giving it before an operation can increase surgical bleeding. The boxed warning states that ketorolac is contraindicated as a prophylactic analgesic before any major surgery. Reducing the dose, changing the timing, or restricting by age does not remove the contraindication. In practice, start nonopioid analgesia that does not impair hemostasis before major surgery and consider ketorolac postoperatively once bleeding risk has passed, if otherwise appropriate.

> *Key principle:* Ketorolac before major surgery is contraindicated.

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71. Answer: E. NSAIDs reduce prostaglandin-dependent renal blood flow, and a kidney with little reserve can fail further. The ketorolac boxed warning states that it is contraindicated in patients with advanced renal impairment. A lower dose, monitoring, fluids, or a different route do not lift the contraindication. In practice, choose acetaminophen, regional techniques, or carefully dosed alternatives in advanced kidney disease and avoid ketorolac.

> *Key principle:* Advanced renal impairment contraindicates ketorolac.

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72. Answer: B. Routine prophylaxis adds cost and exposure without benefit for patients with little expectation of nausea, but some operations make vomiting harmful in itself. The label states that routine prophylaxis is not recommended in patients with little expectation of postoperative nausea and vomiting, yet ondansetron injection is recommended in patients in whom nausea and vomiting must be avoided postoperatively, even where the incidence is low. In practice, base the decision on the consequence of vomiting, not only its probability.

> *Key principle:* Prophylax when vomiting would harm the repair, even at low baseline risk.

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73. Answer: C. An antiemetic can relieve symptoms without treating the cause. The ondansetron label states that it should not be used instead of nasogastric suction and that, after abdominal surgery, it may mask a progressive ileus and/or gastric distention. Increasing the dose, adding another agent, or discharging the patient all risk missing an evolving obstruction or ileus. In practice, examine, image, and decompress when needed while treating the nausea.

> *Key principle:* Ondansetron can mask a progressive ileus.

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74. Answer: D. Apomorphine combined with ondansetron has been reported to cause profound hypotension and loss of consciousness. The ondansetron label states that concomitant use of apomorphine with ondansetron is contraindicated, so no dose change or monitoring makes the pair acceptable. The other options act through steroid, neurokinin-1, histamine, or nonpharmacologic routes and are not named in that warning. In practice, review home medications for interactions before choosing an antiemetic.

> *Key principle:* Apomorphine with ondansetron is contraindicated.

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75. Answer: C. Promethazine injection can cause severe tissue injury, including gangrene. The boxed warning states that the preferred route is deep intramuscular administration and the label limits intravenous use to dilute solution given through a catheter in a large vein. The other routes and an undiluted push into a small vein raise the injury risk. In practice, choose deep intramuscular dosing or a different antiemetic.

> *Key principle:* Promethazine injection: deep intramuscular is the preferred route.

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76. Answer: B. Concentrated promethazine solution can injure tissue if it extravasates around the vein. The label lists intravenous injection at concentrations greater than 1 mg/mL as a contraindication, and describes intravenous dosing as dilution given through a catheter inserted in a large vein. Prohibiting all intravenous use overstates the label, and allowing undiluted peripheral push contradicts it. In practice, dilute the drug, use a large vein, and infuse slowly, or choose the deep intramuscular route.

> *Key principle:* Do not give promethazine IV at concentrations over 1 mg/mL.

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77. Answer: D. Ondansetron blocks serotonin 5-HT3 receptors, and at higher intravenous exposures it can also affect cardiac repolarization. The label states that transient ECG changes, including QT interval prolongation, have been reported, rarely and predominantly with intravenous ondansetron, and this is not a boxed warning. Statements that deny the effect, restrict it to oral use, or make it universal do not match the label. In practice, check the ECG and electrolytes when QT-prolonging risk is present and choose another antiemetic if needed.

> *Key principle:* Rare QT prolongation can occur with intravenous ondansetron.

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78. Answer: A. A new drug with no established benefit for delirium adds adverse effects in an older, frail patient. The AGS guideline gives a strong recommendation, based on low-quality evidence, that clinicians should not newly prescribe cholinesterase inhibitors perioperatively to older adults to prevent or treat delirium. Reduced doses, a different target, or a dementia diagnosis do not change that position. In practice, look for causes such as pain, infection, medications, and urinary retention, and use nonpharmacologic measures.

> *Key principle:* Do not start cholinesterase inhibitors perioperatively for delirium.

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79. Answer: E. Benzodiazepines can worsen confusion, sedation, and falls in older adults, and they do not treat the cause of delirium. The AGS guideline gives a strong recommendation, based on low-quality evidence, that clinicians should not use benzodiazepines as first-line treatment of the agitated postoperative delirious patient. Restraint increases agitation and complications. In practice, search for pain, retention, infection, hypoxia, and offending drugs, and use environmental and reorientation measures.

> *Key principle:* Benzodiazepines are not first-line for postoperative delirium agitation.

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80. Answer: A. For a patient whose severe agitation threatens her safety or interferes with essential care, nonpharmacologic measures sometimes fail and drug therapy is considered. The AGS guideline gives a weak recommendation, based on low-quality evidence, that prescribers may use antipsychotics at the lowest effective dose for the shortest possible duration to treat severe agitation that threatens substantial harm. Benzodiazepines, cholinesterase inhibitors, and high doses are not supported. In practice, use the smallest effective antipsychotic exposure and reassess daily.

> *Key principle:* Severe, harmful agitation may justify a brief low-dose antipsychotic.

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81. Answer: D. Hypoactive delirium is easily missed because the patient is quiet, but it carries the same risks and the same need to find a cause. The AGS guideline gives a strong recommendation, based on low-quality evidence, that clinicians should not prescribe antipsychotic or benzodiazepine medications to treat hypoactive delirium in older adults. Sedatives and antipsychotics add sedation to a patient who is already withdrawn. In practice, correct the underlying problem and support orientation, mobility, and sleep.

> *Key principle:* Hypoactive delirium is not treated with antipsychotics or benzodiazepines.

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82. Answer: E. Uncontrolled pain can precipitate delirium, but opioids can do so too. The AGS guideline gives a strong recommendation, based on low-quality evidence, that clinicians should optimize postoperative pain control, preferably with nonopioid pain medications. Withholding analgesia leaves pain untreated, and meperidine and benzodiazepines are poor choices in older adults. In practice, use acetaminophen and regional techniques first and reserve opioids for breakthrough pain.

> *Key principle:* Treat pain well in older patients, using nonopioid drugs first.

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83. Answer: C. Each day of catheterization adds risk of bacteriuria and symptomatic urinary tract infection, and most patients can void normally soon after pelvic surgery. The CDC guideline recommends removing the catheter as soon as possible postoperatively, preferably within 24 hours. Waiting for bowel function, a fixed longer period, discharge, or a culture adds days without benefit. In practice, set removal as an active daily decision after surgery.

> *Key principle:* Remove the postoperative catheter within 24 hours when possible.

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84. Answer: B. Catheters carry infection risk and discomfort, and a brief procedure without a specific indication does not need one. The CDC guideline, Category IB, states that urinary catheters in operative patients should be used only as necessary rather than routinely. Reflex placement, suprapubic substitution, or planned prolonged use all increase exposure. In practice, ask whether a catheter has a real indication such as prolonged surgery, and if not, omit it.

> *Key principle:* Operative patients get a catheter only when there is an indication.

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85. Answer: D. Indwelling devices keep a foreign body in the bladder, so bacteria form biofilm and infection risk climbs with each day. The CDC guideline, Category II, states that intermittent catheterization is preferable to indwelling urethral or suprapubic catheters in patients with bladder emptying dysfunction. The distractors are indwelling or unsuitable for a woman with retention. In practice, teach intermittent catheterization when the patient can perform it and reassess bladder recovery regularly.

> *Key principle:* Intermittent catheterization is preferred over indwelling catheters.

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86. Answer: E. Catheterizing a bladder that is not full wastes effort and adds trauma and infection risk, so measuring volume first is a more precise approach. The CDC guideline, Category II, states to consider using a portable ultrasound device to assess urine volume in patients undergoing intermittent catheterization. The distractors add catheter days, tests without a symptom, or antibiotics without an indication. In practice, scan first and catheterize when volume is high.

> *Key principle:* A bladder scanner can cut unnecessary intermittent catheterizations.

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87. Answer: C. Bacteriuria in a catheterized patient without symptoms rarely signals disease, and treating it breeds resistant organisms without improving outcomes. The CDC guideline, Category II, states that routine screening of catheterized patients for asymptomatic bacteriuria is not recommended. Screening, treating positives, or giving empiric antibiotics each runs against that position. In practice, send cultures only when there are symptoms such as fever, dysuria, or flank pain.

> *Key principle:* Do not screen catheterized patients for asymptomatic bacteriuria.

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88. Answer: B. Prophylactic antibiotics at catheter removal expose patients to adverse effects and select for resistant bacteria, and the benefit is small. The CDC guideline, Category IB, states that unless clinical indications exist, systemic antimicrobials should not be used routinely to prevent catheter-associated urinary tract infection. The distractors apply a single dose, a short course, a drug class, or intravenous timing to a patient with no indication. In practice, remove the catheter and watch for symptoms.

> *Key principle:* No routine antibiotics to prevent catheter-associated urinary tract infection.

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89. Answer: A. Starting an anticoagulant too soon after an operation increases the risk of surgical bleeding, while starting too late leaves the highest-risk period unprotected. The fondaparinux label directs that the initial dose be given no earlier than 6 to 8 hours after surgery. The preoperative timing in some options belongs to other agents, and later windows leave the early postoperative period uncovered. In practice, confirm hemostasis and start at the labeled postoperative interval.

> *Key principle:* First fondaparinux dose: no earlier than 6 to 8 hours after surgery.

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90. Answer: D. Prophylaxis is aimed at the period of highest thrombotic risk after surgery, and trials gave the drug for a fixed postoperative window. The fondaparinux label states that the usual duration of administration after abdominal surgery is 5 days to 9 days. One to three days is too short, and longer courses are not the labeled usual duration, although extended prophylaxis may be considered for selected cancer surgery under separate guidance. In practice, plan the stop date when starting the drug.

> *Key principle:* Fondaparinux after abdominal surgery usually lasts 5 to 9 days.

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91. Answer: C. Several patient features exclude fixed-dose prophylaxis, so the label's contraindications should be checked before ordering. The label lists body weight less than 50 kg as a contraindication for venous thromboembolism prophylaxis in adults. Severe renal impairment, defined as creatinine clearance under 30 mL/min, is a separate contraindication that she does not meet. Age, surgery type, and diagnosis are not contraindications. In practice, check weight and renal function first.

> *Key principle:* Weight under 50 kg contraindicates fondaparinux prophylaxis.

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92. Answer: B. The thrombotic risk after abdominal surgery persists for days, so the studies that led to approval continued treatment for a defined postoperative period. The enoxaparin label states that the usual duration of administration for abdominal surgery is 7 to 10 days. One to three days would stop before risk has fallen, and longer courses are used only for selected high-risk patients under separate guidance. In practice, write the intended duration in the order.

> *Key principle:* Enoxaparin after abdominal surgery usually lasts 7 to 10 days.

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93. Answer: E. A catheter removed while anticoagulant levels are high can bleed into the epidural space and compress the cord. The enoxaparin label states that placement or removal of a catheter should be delayed for at least 12 hours after administration of lower doses and at least 24 hours after higher doses. Immediate removal ignores drug levels, and 48 hours is longer than labeled. In practice, time the removal, then wait before giving the next dose.

> *Key principle:* Remove the epidural at least 12 hours after a prophylactic enoxaparin dose.

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94. Answer: A. The WHO panel suggests not using any type of advanced dressing over a standard dressing on primarily closed surgical wounds, a conditional recommendation based on low quality evidence. The reviewed evidence did not show that advanced dressings lower infection compared with standard ones. The distractors favor specific advanced products, uncovered wounds, or an ointment, none of which the panel supports. In practice, use a standard sterile dressing for closed incisions and reserve special products for wounds with a specific need.

> *Key principle:* For primarily closed wounds, WHO suggests a standard dressing over an advanced dressing.

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95. Answer: A. Oxygen delivery to the wound supports neutrophil killing of bacteria, and higher inspired oxygen raises tissue oxygen tension. The CDC guideline, Category IA, states that for patients with normal pulmonary function under general anesthesia with endotracheal intubation, clinicians should administer increased fraction of inspired oxygen during surgery and after extubation in the immediate postoperative period. The distractors are measures that are not recommended or have no evidence. In practice, keep supplemental oxygen going through the recovery period.

> *Key principle:* Supplemental oxygen continues after extubation to lower infection risk.

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96. Answer: B. Topical antimicrobials on a closed incision add cost and potential skin reactions and resistance without lowering infection rates. The CDC guideline, Category IB, states not to apply antimicrobial agents, including ointments, solutions, or powders, to the surgical incision for the prevention of surgical site infection. Restricting the practice to specific patient groups or presentations does not appear in the recommendation. In practice, keep the incision clean and dry and teach patients which signs of infection to report.

> *Key principle:* Do not put antimicrobial ointment on a closed incision to prevent infection.

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G5. Gynecologic Emergencies

97. Answer: E. The ACOG opinion notes that torsion can involve an ovary without an associated mass or cyst in as many as 46% of cases, and no clinical or imaging criteria confirm the diagnosis before surgery. Timely laparoscopy is therefore indicated when the picture fits, because it protects ovarian function and future fertility. The traps are false reassurance from a normal-looking ovary, extra tests that add delay, and a watchful plan that lets the ovary infarct. In practice, a normal ovary on imaging in a girl with sudden pain and vomiting still goes to the operating room.

> *Key principle:* A normal-looking ovary without a mass does not exclude torsion in an adolescent.

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98. Answer: C. The ACOG opinion states that no clinical or imaging criteria are sufficient to confirm torsion before surgery and that in 50% of cases torsion is not found at laparoscopy. A negative exploration is therefore an expected cost of a strategy built to save ovaries, not an error. The distractors blame the imaging, the surgeon or the timing, and each assumes a precision that does not exist. Counsel the family that the threshold for laparoscopy is deliberately low, and that a normal exploration is reassuring.

> *Key principle:* A negative laparoscopy for suspected torsion is common and acceptable.

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99. Answer: D. The ACOG opinion recommends a minimally invasive approach with detorsion and preservation of the ovary, with ultrasonography to reevaluate the cyst at 6 to 12 weeks. Most cysts in adolescents are functional and resolve, so removing the ovary or the cyst at the first operation is not required. The distractors propose a second operation, a marker that adds little, early imaging that cannot show anything new, or no follow-up at all. Schedule the interval scan before she leaves, and image sooner only if pain returns.

> *Key principle:* After detorsion with a cyst left in place, reimage the cyst at 6 to 12 weeks.

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100. Answer: C. CDC guidance allows intramuscular or oral therapy for mild-to-moderate pelvic inflammatory disease, but states that women with tubo-ovarian abscesses need more than 24 hours of inpatient observation. An abscess can rupture or fail to respond, and the fever here marks a more serious illness than mild disease. The distractors apply the outpatient pathway, shorten observation, or jump to surgery that the guideline does not require first. In practice, any tubo-ovarian abscess is admitted, started on parenteral therapy and reassessed within 72 hours.

> *Key principle:* A tubo-ovarian abscess requires more than 24 hours of inpatient observation.

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101. Answer: E. CDC guidance states that an intrauterine device does not need to be removed when pelvic inflammatory disease is diagnosed, and that removal should be considered if there is no clinical improvement within 48 to 72 hours of treatment. Removal is not what treats the infection, and early removal costs a patient who wants the method her contraception. The distractors tie removal to the test result or to a replacement, which the guideline does not ask for. Treat first, recheck in 72 hours, and take the device out only for a poor response.

> *Key principle:* Treat PID with the intrauterine device in place and reassess at 48 to 72 hours.

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102. Answer: A. CDC guidance reports that women with HIV responded equally well to the recommended parenteral and intramuscular or oral regimens as women without HIV. The decision to hospitalize rests on severity, abscess, pregnancy, inability to take oral medication or doubt about the diagnosis, and not on HIV status. The distractors treat HIV as a reason for admission, longer therapy or delay. Start the standard regimen promptly and reassess in 72 hours as for any other patient.

> *Key principle:* HIV status alone does not change the PID regimen or the need for admission.

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103. Answer: B. CDC guidance advises that intravenous therapy be considered for severe herpes disease until clinical improvement, followed by oral antiviral therapy to complete more than 10 days of total treatment. A patient who vomits and cannot drink will not absorb oral drugs reliably, and early antiviral treatment shortens a primary episode. The distractors offer topical therapy, a short oral course, delayed suppression or no treatment. Give the intravenous drug, switch to oral when she improves, and extend the course if healing is incomplete.

> *Key principle:* Severe genital herpes is treated intravenously until improvement, then orally to more than 10 days.

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104. Answer: D. The guideline states that a biopsy may be indicated in women over 40 years old with a Bartholin mass. Bartholin gland carcinoma is rare, but a firm, solid, painless mass in an older woman does not behave like a simple cyst or abscess, and drainage procedures would leave a cancer unsampled. The distractors treat the mass as a cyst or abscess, give antibiotics without infection, or defer evaluation. Sample the tissue first, and plan treatment on the pathology result.

> *Key principle:* A new firm Bartholin mass in a woman over 40 warrants biopsy.

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105. Answer: A. The guideline describes leaving the catheter in for about 4 weeks to encourage formation of an epithelialized fistula, which keeps a new duct opening patent and stops the cavity from sealing and refilling. Removal at one day lets the incision close over an infected, secreting space. The distractors assign the device a drug-delivery, tamponade, destructive or irrigation role it does not have. Tell her to expect the catheter for weeks, with warm baths and simple analgesia, and to return if it falls out.

> *Key principle:* The Word catheter stays in for weeks so an epithelialized tract forms and the abscess does not refill.

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106. Answer: D. Most vulvar hematomas are small and are managed conservatively with cold packs, pressure, rest and analgesia. Surgery is generally reserved for large hematomas of more than 10 cm, hematomas that are expanding acutely, or those that cause hemodynamic instability. This one measures 6 cm, was stable over 2 hours of observation, and she is hemodynamically normal and voiding. The distractors escalate to drainage, aspiration, embolization or antibiotics, each carrying risk without a benefit here. Apply cold, treat the pain, confirm that she can void, and repeat the examination in a few hours. Operate if the hematoma expands.

> *Key principle:* A small, stable vulvar hematoma is managed conservatively and surgery is for a large or enlarging one.

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107. Answer: B. A necrotizing soft tissue infection of the vulva and perineum is a surgical emergency, and the guideline states that radiographic evaluation should never delay definitive surgical care. Rapid spread, bullae, pain out of proportion, shock and leukocytosis make the diagnosis clinically, and early debridement is what saves tissue and life. The distractors wait for imaging, treat the picture as simple cellulitis, drain a single area or put adjuncts first. Call the surgeon, resuscitate, start broad-spectrum antibiotics and go to the operating room.

> *Key principle:* Necrotizing vulvar infection is diagnosed clinically and debrided without waiting for imaging.

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108. Answer: C. ACOG states that hCG values alone should not be used to diagnose ectopic pregnancy, and that any discriminatory level should be conservatively high, as high as 3,500 mIU/mL, to avoid interrupting a wanted intrauterine pregnancy. At 2,800 mIU/mL the gestation may simply be too early to be visible. The distractors act on a diagnosis that is not established: uterine evacuation, empiric methotrexate or surgery. Treat her as a pregnancy of unknown location, repeat the hCG and clinical review, and treat only when the location is established.

> *Key principle:* An empty uterus under a conservatively high discriminatory level does not diagnose ectopic pregnancy.

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109. Answer: E. In the ACOG bulletin, failure of hCG to decrease by at least 15% from day 4 to day 7 signals a high risk of treatment failure, needing more methotrexate or surgery. A fall of 7% is short of that mark, even though it is lower than the day 4 value. The distractors take any decline as success, add a rescue drug that does not address failure, or leap to surgery in a stable patient. Reexamine, confirm stability, and give a second dose or operate.

> *Key principle:* A day 4 to day 7 hCG fall of less than 15 percent after methotrexate means treatment is likely failing.

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110. Answer: E. For a pregnancy of unknown location, NICE advises two hCG measurements about 48 hours apart. When the rise exceeds 63%, a developing intrauterine pregnancy is likely, and a transvaginal scan is offered 7 to 14 days later to locate it. A rise under 63% or a fall of less than 50% calls for clinical review within 24 hours instead. The distractors treat a healthy rise as an ectopic pregnancy or delay imaging for weeks. Counsel her to return sooner for pain, heavy bleeding or faintness.

> *Key principle:* An hCG rise over 63 percent over 48 hours calls for a repeat scan in 7 to 14 days.

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111. Answer: B. NICE links methotrexate and expectant management to women without significant pain, and recommends surgery for an ectopic pregnancy with significant pain. Her hCG is under 1,500 IU/L, which tempts a medical approach, but the pain and free fluid suggest a tube at risk of rupture. The distractors follow the hCG number alone, wait, or choose a drug without evidence in this setting. Operate laparoscopically, offering salpingectomy, and match blood and Rh status first.

> *Key principle:* Significant pain with a tubal ectopic pregnancy points to surgery even when the hCG is low.

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112. Answer: A. NICE recommends offering methotrexate or surgery when hCG is at least 1,500 and under 5,000 IU/L, and advises that women who choose methotrexate be told their chance of needing further intervention is increased. She can therefore choose the medical route with informed counseling. The distractors forbid methotrexate, recommend watching, treat surgery as mandatory or promise no follow-up. Weekly hCG is needed until the result is negative, and she should know the warning signs of rupture.

> *Key principle:* At hCG 1,500 to under 5,000 IU/L, offer methotrexate or surgery and warn that methotrexate more often needs further treatment.

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113. Answer: C. NICE offers expectant management as an option to a clinically stable, pain-free woman with a tubal ectopic pregnancy under 35 mm, no visible heartbeat, an hCG of 1,000 IU/L or less and the ability to return for follow-up. Many such pregnancies resolve without treatment, and her low value and reliability make this reasonable. The distractors escalate to drug or surgery, or to curettage, which cannot remove a tubal pregnancy. Check hCG on days 2, 4 and 7, then weekly while it falls, and review her if it plateaus or pain develops.

> *Key principle:* A stable ectopic pregnancy with hCG of 1,000 IU/L or less can be managed expectantly with serial hCG.

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114. Answer: D. The ACOG recommendations, as summarized, list hormonal therapy with intravenous conjugated equine estrogen as an acute option, with transfusion guided by hemodynamic status and continued bleeding. She is vomiting, so oral agents will not reliably reach the circulation. The distractors are slow, oral, or procedural: a daily progestin, an intrauterine system that does not stop acute bleeding quickly, ablation, or an injection that can itself prolong bleeding. Draw bleeding disorder labs before treatment, give the intravenous estrogen, and transition to oral hormonal therapy when stable.

> *Key principle:* Acute heavy bleeding with vomiting in an adolescent is treated with intravenous conjugated estrogen.

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115. Answer: C. The ACOG recommendations, as summarized, state that endometrial ablation, uterine artery embolization and hysterectomy should be considered only in life-threatening situations, and that long-term management is best achieved with continuous hormonal contraceptive therapies. Her bleeding is controlled, so definitive procedures are not justified in an adolescent who may want fertility. The distractors are procedures that sacrifice the uterus or add risk without treating the clotting defect. Maintain suppression, correct iron deficiency and plan for future bleeding episodes with the hematology team.

> *Key principle:* After bleeding is controlled, adolescents with bleeding disorders get continuous hormonal therapy, not ablation, embolization or hysterectomy.

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116. Answer: B. Ferritin is an acute phase reactant, so inflammation raises it, and the ACOG recommendations, as summarized, caution that levels may be elevated in inflammatory states. With microcytic anemia and heavy menses, a mid-range ferritin during an infection can mask depleted iron stores. The distractors read the number as normal, reverse the direction of the infection effect or discard the test. Treat the likely iron deficiency, repeat ferritin when she is well, and finish the bleeding disorder evaluation.

> *Key principle:* Inflammation can push ferritin into the normal range despite iron deficiency.

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117. Answer: E. The tranexamic acid label calls for dosage adjustment when serum creatinine exceeds 1.4 mg/dL, because the drug is cleared by the kidney and accumulates, raising the risk of adverse effects. Her creatinine of 1.9 mg/dL is over that mark, but kidney disease is not listed as a reason to withhold the drug. The distractors ignore clearance, over-restrict, add a thrombotic risk factor, or lengthen exposure. Reduce the dose, keep treatment to the labeled days of menstruation, and recheck renal function.

> *Key principle:* Reduce the tranexamic acid dose when serum creatinine exceeds 1.4 mg/dL.

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118. Answer: D. The Mirena label states that the risk of uterine perforation is increased in women who have recently given birth and in women who are breastfeeding at insertion. It also directs that, if perforation occurs, the system be located and removed, and it warns that surgery may be required. A device outside the uterus no longer works as intended, so a well-appearing patient still needs retrieval. The distractors leave the device, treat infection that is not present, or add a second system. Arrange retrieval, usually laparoscopic, and counsel on another method.

> *Key principle:* A perforated intrauterine system is located and removed, with surgery if needed.

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119. Answer: B. The IDSA guideline suggests treating complicated urinary tract infection, including pyelonephritis, with 7 days of a non-fluoroquinolone antibiotic or 5 to 7 days of a fluoroquinolone, rather than a longer course of 10 to 14 days. She has responded promptly, has no bacteremia and has a susceptible organism. The longer durations reflect older practice, and 3 days is the course for uncomplicated cystitis. Complete 7 days in total, counting the intravenous days, and stop if she remains well.

> *Key principle:* Pyelonephritis with a prompt response needs 7 days of a non-fluoroquinolone, not 10 to 14.

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120. Answer: A. The IDSA guideline suggests transitioning to oral antibiotics rather than continuing parenteral therapy for the remaining treatment duration once the patient has improved, and selecting a targeted agent from urine culture results. Oral step-down shortens hospitalization and avoids intravenous line complications. The distractors prolong parenteral therapy, stop early, add testing or broaden the spectrum without a reason. Choose a drug the isolate is susceptible to, complete the planned duration orally, and discharge when she is stable.

> *Key principle:* Switch to an oral susceptible agent once a patient with complicated UTI has improved.

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121. Answer: B. The AUA urotrauma guideline advises that clinicians initially manage a ureterovaginal fistula with stent placement when possible. A stent drains the ureter past the defect and gives it a chance to heal, and contrast still reaching the bladder shows that the ureter is in continuity, so passage is likely. The distractors jump to open reconstruction, sacrifice the kidney, or wait without drainage. Place the stent by cystoscopy, and move to percutaneous nephrostomy or surgical repair only if the stent fails or cannot be passed.

> *Key principle:* Manage a ureterovaginal fistula first with a ureteral stent when one can be placed.

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122. Answer: A. The guideline defines postoperative retention as an inability to pass urine, or residuals over 50% of the voided volume, at 4 to 6 hours after surgery. Her residual is 75% of a small void, so emptying is incomplete even though she passed urine. The distractors read the void as normal or invent infection, ureteral injury or overactivity without supporting findings. Catheterize to relieve the bladder, record the volume, and repeat a voiding trial.

> *Key principle:* A residual over 50 percent of the voided volume 4 to 6 hours after surgery is retention.

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123. Answer: E. The Greater Glasgow and Clyde guideline states that after an autologous fascial sling, patients who are unable to perform clean intermittent self-catheterization should be discharged with an indwelling catheter. Postoperative retention after a sling is often transient, so the catheter bridges her to a later voiding trial under urogynecology follow-up. Sending her home without drainage risks overdistension, and an admission waiting for a void is unnecessary. The distractors add an invasive test or an antimuscarinic that worsens emptying. Provide catheter care teaching and clear instructions for signs of infection.

> *Key principle:* If she cannot self-catheterize, discharge with an indwelling catheter and arrange a voiding trial.

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G6.1. Minimally Invasive Procedures

124. Answer: D. The ACOG opinion states that vaginal hysterectomy can usually be performed safely in nulliparous women, and that the vaginal approach is preferred among minimally invasive routes whenever feasible. Nulliparity, like a prior cesarean birth, is not a barrier when examination shows mobility and access. The distractors treat parity as an absolute barrier, steer to an abdominal or robotic route, or invent a delay. Assess uterine size, descent and vaginal access at the examination, and choose the route from those findings.

> *Key principle:* Nulliparity does not preclude vaginal hysterectomy.

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125. Answer: C. ACOG Committee Opinion 701 states that extrauterine disease such as adnexal pathology, severe endometriosis, adhesions or an enlarged uterus may preclude vaginal hysterectomy, but that another minimally invasive approach, rather than an open abdominal approach, still may be possible. Laparoscopy can therefore be chosen by a skilled surgeon, with conversion to laparotomy planned if needed. The distractors turn relative factors into absolute rules or invent requirements. Match the route to the findings and the surgeon's skill, and counsel her about the chance of conversion.

> *Key principle:* Endometriosis, adhesions and a large uterus may exclude the vaginal route but not a laparoscopic one.

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126. Answer: C. The ACOG opinion states that surgeons should be skilled at abdominal and laparoscopic hysterectomy before undertaking robot-assisted laparoscopy, and that the role of robotic assistance has not been clearly determined. The robot changes the interface, not the anatomy or the need to manage complications and conversion. The distractors assume that the platform substitutes for skill or that it is superior. Credential on proven laparoscopic and abdominal competence, and track outcomes.

> *Key principle:* Robotic hysterectomy should build on skill in laparoscopic and abdominal hysterectomy.

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127. Answer: E. The Cochrane review found that laparoscopic hysterectomy probably results in a quicker return to normal activities than abdominal hysterectomy, but is likely associated with greater risk of damaging the ureter, with longer operative time as a further cost. These advantages must be balanced against that risk. The distractors reverse the injury risk, deny the recovery advantage or claim a shorter operation. Counsel her on both sides, and plan ureteral identification and a low threshold for cystoscopy.

> *Key principle:* Laparoscopic hysterectomy speeds recovery over abdominal but carries more ureteral injury.

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128. Answer: A. The FDA updated assessment of December 2017 puts the prevalence of uterine sarcoma in women having surgery for presumed fibroids at approximately 1 in 225 to 1 in 580, replacing the earlier 2014 figure of about 1 in 350. Morcellation can spread an unsuspected sarcoma within the abdomen and worsen the prognosis, and preoperative imaging cannot reliably exclude it. The distractors shift the decimal, understating or overstating the risk. Quote the range in consent, consider contained or intact removal, and avoid morcellation when malignancy is suspected.

> *Key principle:* The FDA estimates unsuspected uterine sarcoma in about 1 in 225 to 1 in 580 women having fibroid surgery.

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129. Answer: B. The FDA communication states that power morcellators are contraindicated in peri- or postmenopausal women and in candidates for en bloc removal through the vagina or a mini-laparotomy incision, because the unsuspected sarcoma risk rises with age. An endometrial biopsy samples the lining, not a myometrial sarcoma. The distractors rely on the biopsy, consent, device settings or a partial hysterectomy. Plan en bloc removal, such as through the vagina or a contained technique, in this patient.

> *Key principle:* Morcellation is contraindicated in postmenopausal women and in candidates for en bloc removal.

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130. Answer: D. The Cochrane review found an advantage of direct trocar entry over Veress needle entry for failed entry, with about 65 per 1000 failures with the needle against 11 to 22 per 1000 with a direct trocar. It did not find evidence of differences in major vascular or visceral injury, and no study reported deaths. The distractors reverse the direction, deny any difference or invent mortality. Failed entry is a nuisance, not a safety endpoint, so choose the technique by training and patient factors.

> *Key principle:* Direct trocar entry has fewer failed entries than Veress entry, without a proven difference in major injury.

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131. Answer: B. The Cochrane review concluded that the overall evidence is insufficient to support the use of one laparoscopic entry technique over another, since most evidence is of very low quality and limited by imprecision. Major vascular and visceral injuries are rare, so trials lack the power to separate techniques. The distractors claim a proven winner or loser. Choose an entry method that fits the patient's surgical history and the surgeon's training, and check for injury before proceeding.

> *Key principle:* No laparoscopic entry technique has been proven safer than another.

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132. Answer: D. The RCOG guideline notes that adhesion formation at the umbilicus may be up to 50% after midline laparotomy, so the umbilicus may not be the most appropriate site. It identifies the preferred entry point as 3 cm under the left costal margin in the mid-clavicular line, known as Palmer's point. The left upper quadrant lies away from most infraumbilical adhesions. The distractors are lower abdominal or midline sites that sit nearer bowel and great vessels. Use that point, place the port under care, and inspect for entry injury before proceeding.

> *Key principle:* After midline laparotomy, enter at Palmer's point in the left upper quadrant.

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133. Answer: E. The RCOG guideline states that any non-midline port over 7 mm and any midline port greater than 10 mm requires formal deep sheath closure, because larger fascial defects permit herniation of bowel or omentum. The 12-mm lateral port exceeds the 7-mm threshold. The 5-mm ports are under both thresholds, and no port would qualify for the midline cutoff. Close the sheath at the larger port under direct vision and examine the site at follow-up for a bulge or pain.

> *Key principle:* Close the fascia at non-midline ports over 7 mm and midline ports over 10 mm.

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134. Answer: B. The BSGE/ESGE guideline sets a maximum fluid deficit of 1000 mL when a hypotonic solution is used in a healthy woman, because absorption of more than 1000 mL of hypotonic fluid can cause clinical hyponatremia and cerebral edema. The 2500 mL limit applies to isotonic solutions, and the larger numbers exceed both. The 500 mL figure is more conservative than the guideline. Set the limit before starting, measure the deficit at least every 10 minutes, and stop at the limit.

> *Key principle:* For a healthy woman with a hypotonic medium, stop at a fluid deficit of 1000 mL.

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135. Answer: C. For elderly women or those with comorbidities, the BSGE/ESGE guideline suggests upper limits of 750 mL for hypotonic solutions and 1500 mL for isotonic solutions, lower than the 1000 mL and 2500 mL limits for healthy women, because limited cardiac and renal reserve tolerate volume overload poorly. The 750 mL figure belongs to hypotonic media. The higher values are the limits for healthy patients. Lower the stopping point for frail patients and monitor the deficit continuously.

> *Key principle:* Older or comorbid women have a suggested isotonic deficit limit of 1500 mL.

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136. Answer: C. The guideline notes that absorption of more than 1000 mL of hypotonic solution can cause clinical hyponatremia, with symptoms usually appearing when serum sodium falls under 125 mmol/L, and that glycine absorption has been linked to cerebral edema. Her deficit exceeds that figure, and her hypertension with bradycardia and neurologic symptoms fit fluid overload. The distractors lack the supporting findings. Check serum sodium at once, stop the procedure, and manage in a monitored setting.

> *Key principle:* Neurologic symptoms with a large hypotonic deficit mean dilutional hyponatremia until sodium is measured.

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137. Answer: A. The guideline recommends a target increase in serum sodium of 6 mmol/L over 24 hours until 130 mmol/L is reached. Rapid correction of chronic or subacute hyponatremia risks osmotic demyelination, so the aim is a controlled, modest rise. The distractors correct far faster, which is the common error. Monitor sodium frequently, use hypertonic saline only as the intensive care team directs, and stop at 130 mmol/L before further correction.

> *Key principle:* Correct postoperative hyponatremia slowly, by about 6 mmol/L over 24 hours up to 130 mmol/L.

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138. Answer: B. According to the BSGE/ESGE guideline, control of the intrauterine pressure has been shown to reduce fluid absorption by almost 85%, and pressures over 75 mm Hg increase the volume of medium passing back along the fallopian tubes and into the circulation. Absorption occurs through open vascular channels when pressure is high. The distractors increase pressure or inflow, or stop the measurement. Reduce pressure to the lowest level that gives an adequate view and keep measuring the deficit.

> *Key principle:* Lower and control intrauterine pressure to reduce fluid absorption.

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139. Answer: E. The guideline states that electrolyte-containing fluids would not be effective when using monopolar energy, because the ionic solution disperses the current and prevents cutting at the loop. Monopolar systems therefore need an electrolyte-free medium, which is hypotonic and carries the risk of hyponatremia, while bipolar systems allow saline and a higher deficit limit. The distractors misstate the requirement or deny glycine risk. Match the medium to the energy source before starting.

> *Key principle:* Saline cannot be used with monopolar resection, so switch to bipolar equipment or an electrolyte-free medium.

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G6.2. Minor Gynecologic Procedures

140. Answer: C. Reconstructive procedures such as prolapse repair and sling placement put sutures and dissection close to the bladder and ureteral orifices, where injury can go unseen. Research supports universal cystoscopy when hysterectomy involves pelvic reconstructive surgery, so the team should look inside the bladder before closing. The traps are waiting for a visible injury, waiting for symptoms, or trusting palpation and the vaginal route, none of which confirms ureteral efflux. In practice, plan the scope for every such case and have the equipment ready.

> *Key principle:* Hysterectomy with pelvic reconstructive surgery warrants cystoscopy in every case.

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141. Answer: A. Electrosurgical current needs a path that stays at the tip of the electrode. Electrolyte fluids such as saline carry current through the bladder and disperse the energy, so a nonconducting solution such as glycine is used when electrocautery is planned. Sterile water or saline suits diagnostic viewing, and that is why the electrolyte choices tempt. Hypertonic or half-normal saline conduct as well and add osmotic risk. In practice, switch to glycine before any monopolar cautery during cystoscopy.

> *Key principle:* For cautery during cystoscopy, distend with a nonconducting fluid such as glycine.

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142. Answer: D. Cystoscopy identifies the trigone and both ureteral orifices, and the value of the test is that it allows verification of bilateral ureteral flow. Seeing urine jet from each side shows both ureters drain into the bladder. The other findings describe the bladder wall or the drainage bag, and a ureter obstructed or ligated at the pedicle can coexist with all of them. In practice, watch each orifice for efflux, and if a side is silent, investigate before leaving the operating room.

> *Key principle:* Confirm ureteral integrity by seeing efflux from both orifices.

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143. Answer: A. The guideline stratifies microhematuria into low, intermediate, and high risk. A young never-smoker with a low red cell count and no other risk factors is low risk, and the guideline favors a repeat urinalysis within six months rather than immediate cystoscopy or imaging. Persistent findings would reclassify her as intermediate or high risk. The traps are sending every patient to cystoscopy, advanced imaging, or cytology, none of which is advised at low risk. In practice, document the risk category and book the repeat test.

> *Key principle:* Low-risk microhematuria gets a repeat urinalysis within six months, not a scope.

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144. Answer: D. This patient meets high-risk criteria by smoking history and red cell count, and the guideline calls for cystoscopy and axial upper tract imaging. It advises against routinely using cytology or urine-based tumor markers to decide whether to perform cystoscopy, because a negative result does not exclude bladder cancer. The distractors treat the cytology as a gate, delay evaluation, or offer incomplete imaging. In practice, proceed with cystoscopy and upper tract imaging regardless of the cytology result.

> *Key principle:* A negative cytology does not cancel cystoscopy in high-risk microhematuria.

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145. Answer: C. For high-risk patients the guideline pairs cystoscopy with axial upper tract imaging, and multiphasic CT urography is first choice. When CT cannot be used, as with a severe contrast reaction, clinicians may use MR urography. Ultrasound alone is not axial imaging, intravenous pyelography is not the guideline alternative, and cytology is not imaging. Deferring imaging leaves a high-risk patient unevaluated. In practice, match the imaging to the contraindication and keep cystoscopy in the plan.

> *Key principle:* If CT urography is contraindicated, use MR urography with cystoscopy in high-risk hematuria.

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146. Answer: B. Fluid enters the circulation more readily as intrauterine pressure rises, and once the pressure exceeds the mean arterial pressure significant volumes are absorbed. The guideline recommends that the pressure needed for distension be maintained as low as possible for adequate visualization and kept under the mean arterial pressure. Her mean arterial pressure is about 87 mm Hg, so 150 mm Hg would drive absorption. The distractors claim that higher pressure helps, that saline is harmless, or that measuring the deficit is enough. Lower the pressure and keep tracking the deficit.

> *Key principle:* Keep intrauterine pressure as low as the view allows and under mean arterial pressure.

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147. Answer: E. In operative hysteroscopy, blood and tissue debris obscure the view in a gas medium, and gas embolism is a risk. The BSGE/ESGE guideline therefore states that carbon dioxide gaseous media should not be used for operative hysteroscopy, and it assigns electrolyte-free fluid to monopolar instruments and saline to bipolar or mechanical ones. Avoiding a fluid deficit does not justify gas. The distractors pair gas with a favorable setting that does not exist. Choose a liquid medium that matches the energy source and track the deficit.

> *Key principle:* Carbon dioxide is not a distension medium for operative hysteroscopy.

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148. Answer: C. Fluid absorbed through the exposed vessels of the myometrium can cause volume overload and electrolyte disturbance. The BSGE/ESGE guideline advises that preoperative GnRH agonists be considered in premenopausal women before hysteroscopic resection of fibroids, and notes that they may reduce fluid overload, although one trial found no significant difference in fluid deficit. Diuretics, estrogen, tranexamic acid and iron do not address absorption. Consider the agonist, and still set a deficit limit and measure it at least every 10 minutes.

> *Key principle:* Consider a preoperative GnRH agonist in premenopausal women before hysteroscopic fibroid resection.

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149. Answer: B. Absorption of distension fluid can occur quickly and without a change in the hysteroscopic view, so the deficit must be tracked as the case proceeds. The guideline states that the fluid deficit should be measured at a minimum of 10 minute intervals during hysteroscopic surgery, and automated systems should be considered for prolonged complex cases. Longer intervals or a single end-of-case tally can miss a rapid rise. In practice, build the interval into the protocol and assign one team member to call out the number.

> *Key principle:* Measure the fluid deficit at least every 10 minutes during hysteroscopic surgery.

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150. Answer: E. Hypotonic fluids dilute plasma sodium once absorbed, and absorbing more than 1000 mL can cause clinical hyponatremia. Isotonic medium is considered safer because fluid absorption does not cause hyponatremia, although volume overload can still occur, so deficit limits still apply. The distractors claim smaller absorption, a pressure advantage, no limit at all, or monopolar compatibility, and none is the mechanism. In practice, saline lowers the sodium risk but does not remove the need to track the deficit.

> *Key principle:* Isotonic distension media are safer because absorption does not lower sodium.

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151. Answer: D. Endometrial ablation destroys the lining with thermal energy delivered through a probe in the cavity. The device labeling lists a patient with an intrauterine device currently in place as a contraindication, so the device must be removed before the procedure is scheduled. The distractors treat the device as harmless or protective, or as a reason to keep it in place. In practice, remove the IUD, provide an alternative contraceptive, and confirm benign pathology before booking the ablation.

> *Key principle:* Remove an IUD before ablation, since the label bars the procedure with one in place.

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152. Answer: A. The device labeling lists a uterine cavity length less than 4 cm as a contraindication, because the electrode array is built for a cavity of defined size and cannot be fitted to a smaller one. Her benign biopsy and completed childbearing do not override that limit. Distractors imply that settings, repeat passes, or hormones can compensate for the size. In practice, measure the cavity before ablation and choose another option, such as a levonorgestrel intrauterine system, when the cavity is too small.

> *Key principle:* A cavity under 4 cm excludes this ablation device.

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153. Answer: B. Ablation destroys the visible lining but cannot guarantee destruction of all endometrium, and scarring can hide later disease from sampling. The labeling lists known or suspected endometrial carcinoma or premalignant change, such as unresolved adenomatous hyperplasia, as a contraindication. Her hyperplasia persists after treatment. The distractors cite thinning, biopsy timing, age, or regrowth, none of which is the stated reason. In practice, resolve or definitively treat the hyperplasia first, and use ablation only after endometrial pathology is clear.

> *Key principle:* Unresolved premalignant endometrial change rules out ablation.

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154. Answer: C. Ablation reduces menstrual bleeding but does not reliably prevent pregnancy, and a pregnancy that implants in a scarred, thin cavity can be dangerous for both mother and fetus. The labeling states this directly. The distractors describe pregnancy as risk-free, equate ablation with sterilization, or assume contraception can be dropped. In practice, counsel every ablation candidate that contraception remains necessary, and consider a permanent method such as salpingectomy if she wants one.

> *Key principle:* Ablation is not contraception, and pregnancy afterward is dangerous.

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155. Answer: E. An active genital tract infection is a labeled contraindication, since thermal ablation of an infected uterus risks spreading infection and worsening inflammation. Her fever, cervical motion tenderness, and positive test indicate active pelvic infection. The traps are proceeding with prophylaxis or lower energy, or jumping to hysterectomy, none of which addresses the contraindication. In practice, treat the infection, treat her partner, and reschedule the ablation only after the infection has resolved and her bleeding has been reassessed.

> *Key principle:* Active genital tract infection means postponing ablation until it is treated.

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156. Answer: D. Treatment for histologic HSIL, CIN 2, CIN 3, or adenocarcinoma in situ leaves a lasting excess risk of recurrence that does not disappear after the first normal tests. The 2019 guideline recommends continued surveillance with HPV testing or cotesting at 3-year intervals for at least 25 years. The distractors shorten surveillance, return her to routine intervals, rely on cytology alone, or substitute repeat colposcopy. In practice, tell the patient at discharge that screening continues for decades, and enter the 3-year recall in the chart.

> *Key principle:* After treatment for CIN 2 or worse, screen at 3-year intervals for at least 25 years.

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157. Answer: A. For histologic HSIL in a patient with adequate colposcopy, excisional treatment gives a specimen for pathology and margin assessment. The guideline states that excisional treatment is preferred and treatment with ablation is acceptable, so ablation remains an option when its conditions are met. The distractors reverse the preference, call ablation unacceptable, or move to hysterectomy or observation, which are not the standard for CIN 2 in this setting. In practice, offer excision first, and describe ablation as a reasonable alternative.

> *Key principle:* For CIN 2 or CIN 3 with adequate colposcopy, excision is preferred and ablation is acceptable.

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158. Answer: C. Atypical glandular cells that favor neoplasia carry a real chance of adenocarcinoma in situ or invasive disease that colposcopy can miss, so diagnostic excision is recommended even when the workup is unrevealing. The guideline states that the diagnostic excisional procedure should provide an intact specimen with interpretable margins. Fragmented, burned or very small specimens cannot support that role, and ablation or curettage alone yields no specimen to assess. Plan a single-piece excision with attention to depth and orientation, and label the specimen for the pathologist.

> *Key principle:* A diagnostic excision should yield an intact specimen with readable margins.

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159. Answer: A. For glandular cytology, endocervical sampling at initial colposcopy is intended to find disease inside the canal that the colposcope cannot reach. The guideline recommends it at initial colposcopy except in pregnancy, where curettage carries a bleeding risk and the aim is to exclude invasive cancer, not to treat precancer. The distractors keep the curettage, change its method, skip the biopsy, or add a cone, none of which is standard in pregnancy. Biopsy any lesion suspicious for high-grade disease and defer endocervical sampling.

> *Key principle:* Do endocervical sampling at initial colposcopy, except in pregnancy.

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160. Answer: D. CIN 3 is the true precancer, and its risk of progression to invasive cancer is substantial. The guideline states that for CIN 3 treatment is recommended and observation is unacceptable, with observation allowed only during pregnancy. Her wish to conceive is a reason to counsel, not to delay, since excision can be done before pregnancy. The distractors give age, lesion size, or normal cytology as reasons to wait. In practice, document the counseling and arrange excision, then discuss future pregnancy outcomes.

> *Key principle:* CIN 3 is treated, and observation is acceptable only during pregnancy.

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G6.3. Major Open Procedures

161. Answer: E. When vaginal hysterectomy is not possible, the choice is between laparoscopic and open abdominal surgery. The Cochrane review summarized in the ACOG opinion found laparoscopic hysterectomy has advantages, including faster return to normal activity, shorter hospital stays, and fewer wound infections. It did not show shorter operating time, lower cost, or fewer urinary tract injuries, and large uteri can be harder to remove laparoscopically. In practice, recommend laparoscopy over laparotomy when the vaginal route is not feasible and the surgeon is trained.

> *Key principle:* If vaginal is not feasible, laparoscopy beats laparotomy on recovery and wound infection.

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162. Answer: B. Comparisons of supracervical and total hysterectomy looked at infection, blood loss, and injury to the urinary tract, bowel, or vascular structures. ACOG notes there is no clinically significant difference in complications and uncertain benefit for sexual, urinary, or bowel function, so some patients may reasonably choose to keep the cervix. The distractors credit one operation with fewer complications or better function. In practice, offer the choice without promising a safety advantage, and recommend a laparoscopic or open abdominal route for supracervical surgery.

> *Key principle:* Keeping the cervix does not reduce surgical complications in a clinically significant way.

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163. Answer: D. The ACOG opinion compares routes of hysterectomy for benign disease and concludes that vaginal hysterectomy is the approach of choice whenever feasible, and that the vaginal approach is preferred among the minimally invasive approaches. In its comparison, vaginal surgery has a shorter operating time than laparoscopy with no evidence of a difference in urinary tract injury. The distractors favor technology, the open route, or no preference at all. Assess uterine size, mobility and vaginal access, and choose the vaginal route when they are favorable.

> *Key principle:* Vaginal hysterectomy is the route of choice whenever it is feasible.

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164. Answer: B. The ACOG opinion compared routes using trial and cohort data and found no clear evidence that robotic assistance improves outcomes over conventional laparoscopy. It states that the role of robotic assistance for execution of laparoscopic hysterectomy has not been clearly determined and that more data are needed. The distractors turn that uncertainty into a preference, a replacement for vaginal surgery, or a ban. In practice, tell her the evidence does not favor one laparoscopic method and choose by surgeon skill and access.

> *Key principle:* ACOG says the role of robotic assistance in hysterectomy is not yet clearly determined.

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165. Answer: A. Opportunistic salpingectomy aims to lower later ovarian cancer risk by removing the tubes during planned pelvic surgery. ACOG states that it usually can be safely accomplished at the time of vaginal hysterectomy, so the route need not change. The distractors call the vaginal approach unsafe, add a second operation or conversion, or deny any value. In practice, offer the salpingectomy through the planned route, and counsel that the surgeon may not be able to complete it in every case.

> *Key principle:* Salpingectomy usually can be done safely at vaginal hysterectomy.

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166. Answer: E. ACOG recommends vaginal hysterectomy whenever feasible, and feasibility depends on uterine size, accessibility, extent of extrauterine disease, and the need for concurrent procedures. The most favorable setting is pathology confined to the uterus, with no adnexal pathology and no known or suspected adhesions. The distractors describe adnexal disease, adhesions, an enlarged uterus, or a possible malignancy, each of which pushes toward another route. In practice, look for these features at examination and imaging when counseling about route.

> *Key principle:* Uterine-only pathology without adnexal disease or adhesions favors the vaginal route.

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167. Answer: C. Fibroids can recur or persist after myomectomy because small lesions may be missed or new ones grow. The bulletin reports a reintervention rate of approximately 10 to 12 percent regardless of whether the route is abdominal, robotic, or laparoscopic. The distractors suggest one route reduces future surgery, which the data do not show. In practice, choose the route by fibroid number, size, and location and by the surgeon's skill, and tell the patient that future surgery remains possible whichever route is used.

> *Key principle:* After myomectomy, reintervention is about 10 to 12 percent whatever the route.

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168. Answer: A. Myomectomy removes existing fibroids and leaves the uterus, so new fibroids can develop and small ones can be missed. The bulletin lists recurrence of 25 percent by 40 months among the risks of myomectomy. Quoting a very low rate underestimates what she faces, and very high rates overstate it. In practice, counsel before surgery that about one in four women has recurrent fibroids within about 3 years, and discuss this in the context of her fertility plans.

> *Key principle:* About 25 percent of women have recurrent fibroids by 40 months after myomectomy.

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169. Answer: B. Submucosal fibroids of types 0, 1, and 2 distort the cavity and cause bleeding. The bulletin reports high efficacy and low risk of complications, about 1 to 3 percent, for hysteroscopic myomectomy for these types. Embolization and hysterectomy do not suit her goal of keeping the uterus and possibly conceiving, long-term GnRH agonist therapy is not definitive, and abdominal surgery adds morbidity without need. In practice, use the hysteroscope for types 0 to 2 and track the fluid deficit.

> *Key principle:* Types 0, 1, and 2 fibroids are well treated by hysteroscopic myomectomy.

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170. Answer: C. GnRH agonists suppress ovarian estrogen and shrink fibroids, which can facilitate a minimally invasive route, but hypoestrogenism causes bone loss and vasomotor symptoms. The bulletin states that use is typically limited to 6 months without add-back therapy and 12 months with add-back therapy. Longer use without add-back risks bone density loss. In practice, schedule surgery within that window and start add-back if a longer course is needed.

> *Key principle:* GnRH agonist use is limited to about 6 months without add-back therapy.

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171. Answer: D. GnRH agonists lower estrogen and progesterone signaling, so fibroids shrink while the drug is given. The bulletin notes that cessation of the agonist leads to regrowth of fibroids within 6 to 9 months, which fits her recurrent symptoms. The distractors claim permanent tissue damage, selective effect on one type, or early drug failure. In practice, use the agonist as a bridge before surgery or for short-term anemia correction, and plan definitive treatment in that window.

> *Key principle:* Fibroids regrow within 6 to 9 months after GnRH agonist therapy ends.

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172. Answer: A. BRCA1 carriers have the highest lifetime risk of ovarian cancer, with onset earlier than in BRCA2 carriers, so surgery is timed earlier. ACOG typically recommends risk-reducing salpingo-oophorectomy at age 35 to 40 years for BRCA1 carriers, individualized by family history and childbearing plans. The 40 to 45 range applies to BRCA2 carriers, and later ranges give up protection. In practice, plan surgery within the window once childbearing is complete, and counsel on menopause management afterward.

> *Key principle:* For BRCA1 carriers, salpingo-oophorectomy is typically recommended at age 35 to 40 years.

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173. Answer: E. Risk-reducing surgery is timed to the age at which ovarian cancer begins to occur in each genotype. Women with BRCA2 may consider delaying until age 40 to 45 because of later onset of ovarian cancer, and timing can be individualized by the mutation, desire for childbearing, and family history. The distractors deny risk, claim lower efficacy, or invent an interaction with breast cancer or chemotherapy. In practice, individualize timing with her, and discuss the effects of earlier surgical menopause.

> *Key principle:* BRCA2 carriers may delay surgery to 40 to 45 because ovarian cancer starts later.

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174. Answer: A. Screening with ultrasound and CA-125 has not been shown to detect ovarian cancer early enough to improve survival in carriers. ACOG describes risk-reducing bilateral salpingo-oophorectomy as the only proven intervention to reduce ovarian cancer-specific mortality. Contraceptives and tubal removal alone are not proven to do so, and hysterectomy does not treat the ovaries. In practice, offer surgery in the recommended age window, and use screening only as a bridge if she delays.

> *Key principle:* Surgery is the only proven way to cut ovarian cancer mortality in BRCA carriers.

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175. Answer: E. In carriers, occult cancer can be present in the tube or ovary, and residual tissue or an unexamined peritoneum could leave risk behind. ACOG states that for risk-reducing surgery all tissue from the ovaries and fallopian tubes should be removed, and thorough visualization of the peritoneal surfaces with pelvic washings should be performed. Leaving tissue defeats the purpose of the operation. In practice, ligate high on the infundibulopelvic ligament, send the tubes for complete sectioning, and document peritoneal findings.

> *Key principle:* Risk-reducing surgery removes all ovarian and tubal tissue, with peritoneal inspection and washings.

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176. Answer: D. Removing premenopausal ovaries produces abrupt estrogen loss, and the evidence links surgical menopause with higher cardiovascular disease and osteoporosis risk. The summary of ACOG guidance states that bilateral salpingo-oophorectomy and resultant surgical menopause elevate risk for these outcomes. Salpingectomy lowers ovarian cancer risk without affecting ovarian function. In practice, offer opportunistic salpingectomy and conserve the ovaries for an average-risk premenopausal woman, with a documented discussion of the tradeoffs.

> *Key principle:* Oophorectomy in an average-risk premenopausal woman raises cardiovascular and bone risks.

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177. Answer: C. Removing the tubes both prevents pregnancy and removes the tissue where many high-grade serous cancers are thought to begin. ACOG guidance states that complete bilateral salpingectomy is the preferred tubal procedure for permanent contraception, and that it appears safe. Ligation, partial removal, and fimbriectomy leave tubal tissue, and hysteroscopic occlusion leaves the tube in place. In practice, offer salpingectomy for permanent contraception and counsel that residual ovarian cancer risk is reduced, not removed.

> *Key principle:* Choose complete bilateral salpingectomy for permanent contraception.

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178. Answer: B. ACOG Practice Bulletin 193 states that surgical management of ectopic pregnancy is required when a patient has hemodynamic instability, symptoms of a ruptured ectopic mass, or signs of intraperitoneal bleeding. Echogenic free fluid reaching Morison pouch shows substantial bleeding even though her vital signs are normal, so methotrexate is not appropriate. The distractors apply medical or expectant care to a bleeding tube, or evacuate a uterus that holds no pregnancy. Proceed to laparoscopy and have blood available.

> *Key principle:* Signs of intraperitoneal bleeding from an ectopic pregnancy mean surgery, not methotrexate.

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179. Answer: A. Both operations are accepted for tubal ectopic pregnancy, and trials have not shown a clear fertility advantage for either in all patients. The decision should be guided by the patient's clinical status and her desire for future fertility, with the condition of the other tube relevant. A fixed rule based on hCG, sac size, or one complication ignores the individual. Her damaged contralateral tube and wish for pregnancy favor discussing salpingostomy, if she is stable.

> *Key principle:* Choose salpingostomy versus salpingectomy by clinical status and fertility wishes.

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180. Answer: B. Methotrexate success depends partly on the starting hCG, and lower values predict higher success with the simplest regimen. The bulletin states that the single-dose protocol may be most appropriate for patients with a relatively low initial hCG level, with a two-dose option for higher levels. The multidose regimens add toxicity without benefit at her level, and surgery is not needed in a stable, willing patient. In practice, counsel on follow-up hCG and warning symptoms.

> *Key principle:* A single-dose methotrexate protocol suits a relatively low initial hCG.

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181. Answer: C. After methotrexate, hCG can drop and then plateau or rise, and a ruptured tube can occur even when levels are falling. The guideline states that hCG levels should be serially monitored until a non-pregnancy level. Stopping early or waiting for a period would miss treatment failure, and routine repeat dosing or surgery is not needed for a good decline. In practice, give a follow-up schedule, review rupture warning signs, and confirm a negative result.

> *Key principle:* Follow hCG until it reaches a non-pregnancy level after methotrexate.

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G6.4. Pelvic Floor Surgery

182. Answer: E. Prolapse on examination is common with age and often causes no problem. Treatment aims to relieve bulge, pressure, or urinary, bowel, or sexual dysfunction, so an anatomic finding alone is not an indication. The ACOG and AUGS guidance states that prolapse should be considered a problem only when it causes symptoms. The distractors treat the examination finding, whether with surgery or a device, and expose a symptom-free patient to risk. In practice, document the stage, reassure, and invite her to return if a bulge or functional change appears.

> *Key principle:* Asymptomatic prolapse needs observation, not treatment.

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183. Answer: D. POP-Q staging depends on the most distal point relative to the hymen and on total vaginal length. Stage III applies when the most distal point is more than 1 cm beyond the hymen but less than the total vaginal length minus 2 cm. With a 9 cm vagina the upper limit is 7 cm, and her 3 cm falls in that range. Stage II would place the point within 1 cm of the hymen, and stage IV requires near-complete eversion. Always use the measured length, not an impression of severity, when assigning stage.

> *Key principle:* Stage III is more than 1 cm beyond the hymen but less than total vaginal length minus 2 cm.

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184. Answer: D. A large anterior prolapse can kink the urethra and hide stress incontinence. Once the prolapse is corrected, the outlet obstruction is relieved and leakage may appear, in up to 44 percent of women who were dry beforehand. The review therefore advises looking for latent stress incontinence on examination or urodynamics. The distractors assume dryness predicts dryness, order tests without an indication, or treat a symptom she does not have. In practice, reduce the prolapse, repeat the stress test, and discuss the finding and a concomitant sling option.

> *Key principle:* Look for latent stress incontinence with the prolapse reduced before repair.

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185. Answer: A. Marked descent of the uterus and bladder can angulate and compress the ureters, causing hydronephrosis and a rise in creatinine. The review notes that in advanced prolapse the ureters may be obstructed, so the upper tracts matter in preoperative assessment. The distractors are plausible urologic causes but do not fit a bland urinalysis, absent leakage, and a prolapse that explains the course. In practice, obtain renal ultrasound in advanced prolapse with renal dysfunction, because repair or pessary reduction often reverses the obstruction.

> *Key principle:* Advanced prolapse can obstruct the ureters, so check the upper tracts.

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186. Answer: D. Surgery for prolapse is offered to women with significant bother who want definitive treatment. The review states the operation should aim to preserve vaginal length and axis in sexually active women. Procedures that obliterate, narrow, or shorten the canal can relieve the bulge but compromise coital function, and they suit women who no longer want intercourse. In practice, record sexual activity and goals before choosing among reconstructive and obliterative options.

> *Key principle:* In sexually active women, prolapse surgery should preserve vaginal length and axis.

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187. Answer: B. Well-established risk factors for prolapse include age, parity, and obesity. Added load from weight raises intra-abdominal pressure on pelvic support structures, and unlike age or delivery history it can be reduced. The distractors are either fixed history or the finding itself, not a risk factor she can alter. Counseling should name weight reduction as a reasonable step alongside pelvic floor exercises and, when needed, a pessary or surgery.

> *Key principle:* Age, parity, and obesity raise prolapse risk, and only obesity can be changed.

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188. Answer: C. Uterine prolapse is a failure of apical support, and removing the uterus does not restore it. Without suspension of the vaginal apex, the vault tends to descend after hysterectomy. The guidance states that apical suspension should be performed at the time of hysterectomy for uterine prolapse. The distractors either assume hysterectomy suffices, address only one compartment, or defer the key step. In practice, combine the suspension with the hysterectomy and repair any other defects found.

> *Key principle:* Do the apical suspension at the same operation as hysterectomy for uterine prolapse.

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189. Answer: E. Anterior wall prolapse is commonly associated with apical prolapse. Repairing the anterior wall without restoring apical support leaves the main driver of descent in place, and recurrence is more likely. The guidance advises correcting both defects when both are present to lower the risk of recurrent prolapse. The distractors fix a compartment that is not the problem or add mesh or a sling without an indication. In practice, examine the apex at every anterior prolapse and plan support for it.

> *Key principle:* Anterior prolapse often hides an apical defect, and both should be corrected.

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190. Answer: D. Posterior wall defects include rectocele, enterocele, and perineal body weakness, and the vaginal route lets the surgeon address each. ACOG and AUGS state that posterior wall repair is more effective vaginally than transanally. A transanal approach is the plausible rival because it treats the rectum directly, but it has less durable anatomic results. In practice, perform a vaginal native tissue repair and discuss bowel function and sexual function in counseling.

> *Key principle:* Repair posterior wall prolapse vaginally, not transanally.

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191. Answer: B. Sacrocolpopexy places a graft from the vaginal apex to the anterior longitudinal ligament and gives durable support. The guidance names the patients in whom it is considered as those at high risk of recurrence: age under 60, POP-Q stage 3 or 4, and body mass index over 26. The 54-year-old meets all three. The others have none or one feature, so a vaginal native tissue operation is reasonable for them. In practice, tally the risk features before choosing the route.

> *Key principle:* Consider sacrocolpopexy when age is under 60, stage is 3 or 4, or BMI exceeds 26.

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192. Answer: E. Vaginal uterine-preserving apical support can be done by suspending the uterus or cervix to the sacrospinous ligament, as the review describes. The abdominal alternative, sacrohysteropexy, anchors to the sacral promontory, which is why the anterior longitudinal ligament is the confusable option. The arcus tendineus and the iliopectineal ligament are used for paravaginal and retropubic work. In practice, name the anchor point in consent, since the sacrospinous route carries its own risks of buttock pain and bleeding.

> *Key principle:* The vaginal uterus-sparing suspension uses the sacrospinous ligament.

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193. Answer: A. In April 2019 the FDA ordered manufacturers to stop selling transvaginal mesh for pelvic organ prolapse. The ACOG and AUGS summary states that the order applies to transvaginal mesh only and not to transabdominal prolapse mesh or to stress incontinence mesh. The AUGS also notes that regulators have drawn a clear distinction between slings and prolapse mesh. The distractors overstate the scope. In practice, explain which product class was affected and still discuss sling-specific risks and alternatives.

> *Key principle:* The 2019 FDA order covered transvaginal prolapse mesh, not slings or abdominal mesh.

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194. Answer: C. The FDA moved surgical mesh for transvaginal repair of prolapse into class III, the high-risk category. That change meant manufacturers had to provide clinical data in a premarket approval application, with a filing deadline set in the final orders. Products without approval were later ordered off the market. The distractors describe lower classes or a voluntary step, and the sling distractor confuses the product types. In practice, class III status explains why these mesh kits are no longer available.

> *Key principle:* Transvaginal prolapse mesh became class III and needed premarket approval data.

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195. Answer: C. Evidence has not shown that adding mesh or grafts to a first posterior wall repair improves results, and mesh adds exposure and pain risk. ACOG and AUGS state that mesh or grafts should not be used routinely in the primary repair of posterior wall prolapse. The distractors defend routine augmentation, misstate when it helps, or assume biologic material is risk free. In practice, offer a native tissue repair first and reserve augmentation for selected recurrent cases with specific training and consent.

> *Key principle:* Do not add mesh or a graft to a routine primary posterior repair.

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196. Answer: D. Mesh and graft operations carry exposure, pain, and reoperation risks that rise when the operator is unfamiliar with the technique. The ACOG and AUGS guidance states that mesh or biologic grafts should be placed only by surgeons who have training specific for these procedures. General privileges and a vendor video do not meet that standard, and the guidance sets no fellowship or academic-center rule. The distractors invent credentials or accept minimal preparation. In practice, ask about a surgeon's procedure-specific training before agreeing to a graft operation.

> *Key principle:* Mesh or graft prolapse surgery calls for procedure-specific surgeon training.

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197. Answer: A. The FDA safety communication addressed transvaginal mesh for prolapse and did not address abdominal sacrocolpopexy, whether open, laparoscopic, or robotic. The AUGS position states there is no justification for any restriction for mesh placed abdominally. The distractors extend the vaginal-mesh restriction to a different operation, or invent route or registry limits. In practice, counsel her on the real risks of sacrocolpopexy, such as mesh exposure and port or incision problems, without describing it as restricted.

> *Key principle:* Abdominal sacrocolpopexy mesh was not the target of the FDA vaginal mesh restriction.

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198. Answer: E. The AUGS opposed blanket bans, noting that the FDA did not intend a complete restriction and that decisions belong to the patient and surgeon. Its alternative is credentialing guidelines, including proctoring for surgeons requesting new privileges for sacrocolpopexy, plus complication review and standardized consent. Bans would also block FDA-mandated research. The distractors substitute a ban, an arbitrary volume cutoff, or no oversight. In practice, a credentialing pathway keeps access while raising quality.

> *Key principle:* Use credentialing and proctoring, not a blanket mesh ban.

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199. Answer: B. A small but significant group of women have lasting pain and dyspareunia after vaginal mesh. The ACOG and AUGS statement, as reported, warns that complete removal of the mesh may not be possible, so symptoms can continue even after corrective surgery. The distractors promise a cure, claim removal is forbidden, or deny the complication. In practice, set honest expectations, tailor excision to the symptomatic segment, and involve a surgeon experienced in mesh complications.

> *Key principle:* Mesh removal may not fully relieve pain, so counsel before surgery.

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200. Answer: D. Obliterative procedures such as colpectomy and colpocleisis close or remove the vaginal canal, so they suit women who will not use the vagina for intercourse. The review specifies that women considering them should be fully informed regarding the loss of vaginal length. Distractors imply that the canal, or later access to it, remains. Because the operation is irreversible, patient understanding is part of the indication. In practice, confirm the goals explicitly and document that she accepts loss of coital function.

> *Key principle:* Counsel that colpocleisis permanently removes vaginal length.

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201. Answer: B. For women who no longer want intercourse, obliterative surgery offers a short operation with low complication rates and high satisfaction. The review states that these procedures may provide more durable results with lower reoperation rates compared to reconstructive options. The distractors assume reconstruction or mesh is more durable, or that recurrence cannot be avoided. In practice, offer colpocleisis to older, medically complex, sexually inactive women who prioritize a single operation.

> *Key principle:* Colpocleisis often has lower reoperation rates than reconstruction.

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202. Answer: C. In the narrative review of colpocleisis series, urinary tract infection was the most common postoperative complication, reflecting catheter use and age. Her focal urinary symptoms and positive urinalysis fit, whereas pulmonary embolism would cause dyspnea or hypoxia, hematoma causes pain and swelling, and a fistula causes continuous leakage. Ureteral obstruction would cause flank pain. In practice, send a culture, treat, and keep catheter time short after obliterative surgery.

> *Key principle:* After colpocleisis the most common complication is a urinary tract infection.

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203. Answer: C. Latent or established stress incontinence often accompanies advanced prolapse, and colpocleisis does not correct sphincter weakness. The narrative review reported that after concomitant midurethral sling surgery, 86.8% to 94% of patients were continent. The distractor ranges understate the benefit of adding a sling or promise near-universal continence. In practice, use the sling when stress incontinence is present or demonstrated on reduction, and state that most but not all women are dry.

> *Key principle:* Adding a midurethral sling at colpocleisis leaves about 87 to 94 percent continent.

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204. Answer: E. Many women with prolapse also have stress incontinence, and combining procedures avoids a second operation. The AUA/SUFU guideline, based on limited evidence graded C, states that for concomitant prolapse repair and SUI surgery clinicians may perform any of the incontinence procedures, including a midurethral sling, pubovaginal sling, or Burch colposuspension. The distractors name one required option, ban a sling, or force a delay. In practice, choose the sling that fits her anatomy and preferences, and counsel about retention risk.

> *Key principle:* At prolapse surgery, any standard anti-incontinence procedure is acceptable.

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205. Answer: A. Large prolapse can mask stress incontinence by kinking the urethra, and objective demonstration is part of surgical evaluation. The guideline lists high-grade prolapse, stage 3 or higher, with SUI not demonstrated even with the prolapse reduced as a reason for additional evaluation. Distractors proceed on history alone or start a medication without confirming the type of incontinence. In practice, document the stress test, repeat it with reduction, and use urodynamics when it remains negative.

> *Key principle:* Stage 3 prolapse with SUI not shown on reduction needs more evaluation before a sling.

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206. Answer: E. A large cystocele can obstruct the urethral outlet and hold continence in place. Once the prolapse is corrected, the obstruction is relieved and latent or occult stress incontinence can appear, as the review notes. The low residual excludes overflow, a normal urinalysis argues against infection or overactivity, and a fistula gives continuous rather than stress-related leakage. In practice, warn women before surgery and have a plan, often a sling, for new stress leakage.

> *Key principle:* Occult stress incontinence appears when correcting prolapse relieves outlet obstruction.

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207. Answer: A. Apical suspension sutures near the uterosacral ligaments can kink or obstruct a ureter, and bladder injury can occur during dissection. The guidance states that routine intraoperative cystoscopy should be performed during prolapse surgery when there is risk to the bladder or ureter. The distractors rely on hematuria, delayed imaging, or mesh use as triggers, any of which would miss a kinked ureter. In practice, confirm bilateral ureteral efflux before leaving the operating room.

> *Key principle:* Do cystoscopy during prolapse surgery when the bladder or ureter is at risk.

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208. Answer: B. Placing synthetic mesh next to an injured urethra raises the risk of erosion and fistula. The guideline states that clinicians should not place a mesh sling if the urethra is inadvertently injured, and that a non-synthetic sling should be used instead. The distractors keep mesh in the injured field or use another synthetic device. In practice, repair the urethra, then use an autologous fascial sling or defer, and document the event.

> *Key principle:* After an inadvertent urethral injury, do not leave a mesh sling in place.

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209. Answer: C. Diverticulectomy opens the urethral wall, and a synthetic sling placed against the repair risks erosion, fistula, and recurrence of the diverticulum. The guideline states that clinicians should not use a synthetic midurethral sling in patients undergoing concomitant urethral diverticulectomy. All the distractors place synthetic material at or near the repair. In practice, treat the diverticulum and use an autologous sling or stage the incontinence procedure later.

> *Key principle:* Avoid synthetic midurethral slings at urethral diverticulectomy.

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210. Answer: D. Both operations are recognized non-mesh choices. The guideline discussion favored the autologous fascia pubovaginal sling over Burch colposuspension because retreatment was lower, 4% versus 13%, and it notes that Burch carries incisional morbidity with over 20% of patients having wound-related issues. The distractors reverse the figures, claim equality, or invent a mesh requirement. In practice, add harvest-site morbidity and retention risk when counseling about the fascial sling.

> *Key principle:* Autologous sling had lower retreatment than Burch colposuspension, 4% versus 13%.

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211. Answer: D. Single-incision slings shorten the tissue pass but trade durability. The guideline discussion notes similar efficacy to transobturator slings but limited comparison with retropubic slings, and it reports that objective cure rates were inferior to standard midurethral slings, with higher mesh exposure, dyspareunia, and repeat surgery in the single-incision group. The distractors promise superiority or safety. In practice, offer a standard sling first and reserve single-incision slings for shared decisions.

> *Key principle:* Single-incision slings had inferior objective cure and more exposure than standard slings.

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212. Answer: B. Autologous fascial slings avoid synthetic mesh but add morbidity at the harvest site. The guideline discussion lists wound infection, seroma, and ventral incisional or leg hernia, depending on whether rectus fascia or fascia lata is used. A cough-sensitive bulge at the incision fits a hernia. The distractors do not explain a bulge at the incision. In practice, counsel about harvest-site problems before surgery and examine the wound at follow-up.

> *Key principle:* Fascial slings carry harvest-site risks such as incisional hernia.

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213. Answer: C. Bulking agents are the least invasive stress incontinence option and suit selected women who cannot have a sling. They are less durable, and outcomes are hard to measure because retreatment is common. The guideline states patients should be counseled on the expected need for repeat injections. The distractors overstate durability or confuse bulking with a sling. In practice, set the expectation of repeat procedures before the first injection.

> *Key principle:* Counsel that bulking agents often need repeat injections.

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Part 2. Office Practice

OP1.1. Routine Care and Prevention

214. Answer: A. The 2024 USPSTF statement recommends biennial screening mammography for women aged 40 to 74 years (grade B). For women 75 and older, the Task Force found that evidence is insufficient to weigh benefits against harms, which is an I statement. That is neither a recommendation for nor against screening. The trap is to treat the age cutoff as a prohibition or to extend the 40 to 74 recommendation upward. In practice, the decision at 75 and older rests on shared decision-making that considers health status, life expectancy and her preferences.

> *Key principle:* At 75 and older, the USPSTF gives an I statement for mammography, not a recommendation for or against.

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215. Answer: E. The 2024 USPSTF update addressed dense breasts directly. It concluded that evidence is insufficient to assess benefits and harms of supplemental ultrasonography or magnetic resonance imaging after an otherwise negative mammogram, an I statement. Distractors either invent a grade B recommendation for one of the modalities or convert uncertainty into a recommendation against imaging. Density alone does not establish a proven benefit from added tests. In practice, tell her the evidence is uncertain, discuss her individual risk, and let her preferences guide whether she pursues supplemental imaging.

> *Key principle:* For dense breasts with a negative mammogram, supplemental ultrasonography or MRI is an I statement.

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216. Answer: B. The 2024 USPSTF statement lists both digital mammography and digital breast tomosynthesis among the effective screening modalities and does not prefer one over the other. A grade B recommendation covers biennial mammography for ages 40 to 74. The trap is to assume a newer technology carries its own stronger recommendation or must be limited to dense breasts. In practice, she can choose either option according to availability, cost and radiology advice, and the screening interval stays biennial.

> *Key principle:* Digital mammography and digital breast tomosynthesis are both effective screening modalities.

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217. Answer: E. The 2019 USPSTF recommendation states that clinicians offer risk-reducing medications to women at increased risk for breast cancer and low risk for adverse medication effects, giving tamoxifen, raloxifene or aromatase inhibitors as examples (grade B). There is no single cutoff for increased risk, though a 3 percent 5-year risk is cited as one example threshold. The distractors are screening-only, surgical or hormonal approaches that do not match the recommendation. In practice, discuss benefits and harms, check for thromboembolic and endometrial risks, and let her choose.

> *Key principle:* Offer risk-reducing medication to women at increased breast cancer risk and low risk for adverse effects.

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218. Answer: A. The 2019 USPSTF recommendation separates women at increased risk, who are offered risk-reducing medication, from those not at increased risk, for whom routine use is recommended against (grade D). The reason is that harms such as thromboembolism, endometrial cancer and vasomotor symptoms outweigh a small absolute benefit when baseline risk is low. The distractors blur this line using density or menstrual status, which are not criteria in the statement. In practice, quantify her risk, explain the trade-off, and continue routine screening.

> *Key principle:* Risk-reducing medication is not for women who are not at increased breast cancer risk (grade D).

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219. Answer: D. The 2021 USPSTF update extended colorectal cancer screening downward. It recommends screening in adults aged 50 to 75 (grade A) and in adults aged 45 to 49 (grade B), so an average-risk woman of 46 qualifies. The distractors reflect the prior start age of 50, an assumed family-history requirement, or a symptom-triggered approach, none of which applies to screening. In practice, offer a stool-based test or colonoscopy now and review the interval for the chosen test.

> *Key principle:* Average-risk adults should begin colorectal cancer screening at age 45.

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220. Answer: C. The USPSTF lists stool DNA with a fecal immunochemical test (sDNA-FIT) every 1 to 3 years, and the high-sensitivity guaiac test or FIT every year. The intervals for stool tests are short because each test samples a lesion only when it is bleeding or shedding. Distractors borrow the 5-year interval from imaging strategies and the 10-year interval from colonoscopy. In practice, counsel that stool-based screening works only when it is repeated on schedule and that any positive result requires colonoscopy.

> *Key principle:* Stool DNA-FIT is repeated every 1 to 3 years, and FIT or guaiac testing every year.

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221. Answer: E. Among the strategies in the 2021 USPSTF statement, computed tomography colonography is repeated every 5 years, the same interval as flexible sigmoidoscopy alone. Colonoscopy alone is the strategy repeated every 10 years. The distractors misplace the interval by borrowing the colonoscopy figure or the annual stool-test figure. In practice, tell her that a positive study with polyps of concern leads to colonoscopy, and that extracolonic findings may need follow-up.

> *Key principle:* CT colonography is repeated every 5 years for colorectal cancer screening.

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222. Answer: C. For adults aged 76 to 85, the 2021 USPSTF statement recommends that clinicians selectively offer screening (grade C), because the net benefit is small and depends on overall health, prior screening history and preferences. The distractors turn this into a firm recommendation for, a recommendation against or an I statement. In practice, a healthy patient with a normal colonoscopy a decade ago can reasonably be offered screening or not, and the decision should be individualized.

> *Key principle:* For ages 76 to 85, selectively offer colorectal cancer screening (grade C).

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223. Answer: D. The 2021 USPSTF recommendation advises annual low-dose computed tomography (LDCT) in adults aged 50 to 80 with at least a 20 pack-year history who currently smoke or quit within the past 15 years (grade B). She meets every criterion. Chest radiography and sputum cytology were not effective screening tools, and quitting less than 15 years ago does not remove eligibility. In practice, document the shared decision-making and offer cessation support if she relapses.

> *Key principle:* Annual LDCT is recommended for ages 50 to 80 with 20 pack-years who smoke or quit within 15 years.

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224. Answer: A. The USPSTF states that screening should be discontinued once a person has not smoked for 15 years, or develops a health problem that substantially limits life expectancy or the ability or willingness to have curative lung surgery. At 17 years since quitting she has passed that threshold. The distractors extend screening because of age or pack-years, or substitute a less effective test. In practice, stop scans, keep her informed about symptoms, and reassess only if she resumes smoking.

> *Key principle:* Stop lung cancer screening after 15 years without smoking.

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225. Answer: B. The CDC adult schedule recommends routine HPV vaccination through age 26 and uses shared clinical decision-making for ages 27 through 45. Benefit is lower at older ages because many adults have already been exposed, but some may still gain protection from new partners. The distractors turn the recommendation into a universal rule, a hard age limit or a risk-triggered indication that the schedule does not use. In practice, discuss her exposure risk and preferences, then vaccinate if she chooses.

> *Key principle:* Ages 27 through 45: HPV vaccination by shared clinical decision-making.

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226. Answer: E. For adults who completed the primary series and did not receive Tdap at age 10 or older, the CDC schedule gives one dose of Tdap, then Td or Tdap every 10 years thereafter. The distractors use a 5-year Td interval from wound-management rules, tie Tdap to pregnancy alone, or invent a two-dose series. In practice, record the dose and set a 10-year recall.

> *Key principle:* Give Tdap once as an adult, then Td or Tdap every 10 years.

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227. Answer: B. The CDC adult schedule recommends the 2-dose recombinant zoster vaccine series for adults aged 50 and older, with the doses given 2 to 6 months apart. Age 50 is the threshold, not 60 or 65. The distractors use a single dose, an extra dose, a later start age or a live vaccine, which is not the schedule product. In practice, offer the first dose at this visit and book the second dose within the allowed window.

> *Key principle:* Recombinant zoster vaccine: 2 doses, 2 to 6 months apart, from age 50.

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228. Answer: D. The CDC adult schedule recommends that adults aged 19 through 59 complete a 2-, 3- or 4-dose hepatitis B series depending on the vaccine, regardless of risk factors. Adults 60 and older are addressed by risk factors or request. The distractors apply the older risk-based approach, an age cutoff that is too high, or a virologic test that is not a prerequisite. In practice, vaccinate at this visit and record the series.

> *Key principle:* Hepatitis B vaccination is routine for adults 19 through 59 years.

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229. Answer: C. For nonpregnant women of childbearing age with no evidence of rubella immunity, the CDC schedule recommends measles, mumps and rubella (MMR) vaccine. MMR is contraindicated during pregnancy and is given after pregnancy if indicated, which is why vaccination before conception is the efficient window. The distractors defer vaccination until a pregnancy or substitute another live vaccine. In practice, confirm she is not pregnant, vaccinate, and record the dose.

> *Key principle:* Nonpregnant women without rubella immunity should receive MMR before pregnancy.

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230. Answer: A. The CDC adult schedule recommends one dose of respiratory syncytial virus vaccine for unvaccinated adults 75 and older, with additional doses not recommended. Adults 50 through 74 at increased risk follow the same single-dose guidance. The distractors invent a two-dose series, yearly revaccination like influenza, a disease-triggered indication that does not apply at 75, or a later age. In practice, vaccinate now and document that no booster is planned.

> *Key principle:* RSV vaccine at 75 and older is a single dose, with no boosters.

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231. Answer: E. The CDC selected practice recommendations allow implant placement any time when pregnancy can be reasonably excluded. Backup contraception is needed for 7 days if the implant is placed more than 5 days after menses started. At day 8 she is outside that window, even though pregnancy has been reasonably excluded by abstinence since menses began. The distractors reflect the 2-day figure for some progestin-only pills, a longer arbitrary period, or no backup. In practice, give the 7-day instruction at the visit.

> *Key principle:* An implant started more than 5 days after menses needs 7 days of backup.

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232. Answer: A. The CDC selected practice recommendations state that repeat injections are given every 3 months (13 weeks), and an injection may be given up to 2 weeks late, which is 15 weeks from the last injection. At 14 weeks she is inside that grace period. Beyond it, pregnancy must be reasonably excluded and backup used for 7 days. The distractors apply the late-injection rules, extra testing or restart schedules that do not apply. In practice, inject on time and schedule the next visit.

> *Key principle:* DMPA may be given up to 15 weeks after the last injection without a restart.

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233. Answer: C. She meets a criterion for being reasonably certain she is not pregnant: no intercourse since the start of her last normal menses. The CDC selected practice recommendations also state that routine pregnancy testing is not necessary for every patient. Because she is beyond day 5 of the cycle, combined hormonal contraception requires 7 days of backup. The distractors add a test, delay or a screening requirement, or omit the backup. In practice, start the method at the visit and document the criterion used.

> *Key principle:* If abstinent since menses began, start the method now and add backup per the cycle day.

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234. Answer: B. The CDC selected practice recommendations state that most patients do not need additional STI screening at IUD placement, but screening may be performed at the time of placement, and placement should not be delayed. The USPSTF also recommends chlamydia and gonorrhea screening in sexually active women 24 and younger. Same-day screening and placement have similar pelvic inflammatory disease rates. Prophylactic antibiotics are generally not recommended. In practice, collect specimens and place the device unless purulent cervicitis is present.

> *Key principle:* Screen for STIs at IUD placement if indicated, but do not delay the device.

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235. Answer: D. In the U.S. MEC 2024, migraine with aura is category 4 for combined hormonal contraceptives, an unacceptable health risk, because of the stroke risk. The pill, patch and ring all belong to that class. The same table lists the implant as category 1 for migraine with aura. The trap is believing that changing the regimen or delivery route lowers the risk of the estrogen component. In practice, offer a progestin-only or nonhormonal method.

> *Key principle:* Migraine with aura makes combined hormonal contraception category 4, so choose a progestin-only method.

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236. Answer: B. The U.S. MEC 2024 states that a person aged 35 or older who smokes 15 or more cigarettes per day should not use combined oral contraceptives because of unacceptable health risks, primarily myocardial infarction and stroke (category 4). Smoking and estrogen combine to promote arterial thrombosis. The distractors are conditions loosely linked to hormonal contraception that do not drive this classification. In practice, offer a progestin-only or intrauterine method and cessation support.

> *Key principle:* Age 35 or older plus 15 or more cigarettes a day makes combined pills category 4 for arterial events.

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237. Answer: E. For two or more consecutive pills missed, 48 hours or more, the CDC selected practice recommendations advise taking the most recent missed pill as soon as possible, discarding the other missed pills, and abstaining or using barrier methods until hormonal pills have been taken for 7 consecutive days. Emergency contraception should be considered if pills were missed in the first week and unprotected intercourse occurred in the previous 5 days. The distractors use the single-late-pill rule or invented doubling schemes. In practice, give clear written instructions.

> *Key principle:* After 2 or more missed combined pills, restart and use backup for 7 consecutive days.

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238. Answer: D. The 2018 guidelines advise adults to do at least 150 minutes to 300 minutes a week of moderate-intensity aerobic activity, or 75 to 150 minutes of vigorous-intensity activity, or an equivalent combination. Gains continue beyond 300 minutes. The distractors set a threshold under the minimum, double the target, swap the intensity so the minutes are mismatched, or compress activity into one day. In practice, suggest brisk walking in sessions she can sustain and build gradually.

> *Key principle:* Adults need 150 to 300 minutes of moderate-intensity or 75 to 150 minutes of vigorous-intensity activity weekly.

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239. Answer: C. Her 150 minutes of moderate-intensity walking meets the aerobic target. The 2018 guidelines add muscle-strengthening activities of moderate or greater intensity on 2 or more days a week for adults, and advise older adults to do multicomponent physical activity that includes balance training. The distractors raise the aerobic dose, substitute stretching, discourage resistance work, or restrict balance training to patients who have already fallen. In practice, add two weekly resistance sessions and balance exercises.

> *Key principle:* Older adults add muscle strengthening twice weekly and multicomponent activity with balance training.

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240. Answer: A. Neural tube closure occurs very early, often before pregnancy is recognized, so supplements must already be on board. The 2023 USPSTF statement recommends that all persons planning to or who could become pregnant take a daily supplement of 0.4 to 0.8 mg (400 to 800 mcg) of folic acid, starting at least 1 month before conception and continuing through the first 2 to 3 months of pregnancy (grade A). Because her pregnancy is not planned, the supplement should begin now. The distractors start late, use a different dose, or wait for a positive test. In practice, begin now.

> *Key principle:* Anyone who could become pregnant should take folic acid 0.4 to 0.8 mg daily.

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241. Answer: E. The 2025 USPSTF statement recommends screening for osteoporosis to prevent fractures in women 65 and older (grade B), using bone mineral density measured at a central site such as the total hip, femoral neck or lumbar spine. Age alone qualifies her, so additional risk factors are not needed. The distractors wait for a fracture, use a vitamin level that does not measure bone density, treat without diagnosis, or add a risk-factor requirement. In practice, order dual-energy x-ray absorptiometry.

> *Key principle:* Screen all women 65 and older for osteoporosis with central bone density testing.

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242. Answer: C. For postmenopausal women younger than 65, the 2025 USPSTF statement recommends screening those who are at increased risk for osteoporotic fracture as estimated by clinical risk assessment (grade B). She has several risk factors, so a formal tool is the next step, with bone density testing if the estimate is high. The distractors wait for age 65, test everyone, substitute supplements, or use markers that are not the screening test. In practice, record the tool result and the test decision.

> *Key principle:* Under 65, use clinical risk assessment to decide who gets bone density screening.

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243. Answer: B. The 2018 USPSTF statement recommends against daily supplementation with 400 IU or less of vitamin D and 1000 mg or less of calcium for the primary prevention of fractures in community-dwelling postmenopausal women (grade D). Evidence for higher doses was graded I. The distractors reverse the grade, apply the I statement to the wrong doses, or add a laboratory-based or stone-history rule the statement does not contain. In practice, counsel that supplements at these doses are not an effective fracture-prevention strategy; the position applies to primary prevention only.

> *Key principle:* Vitamin D 400 IU or less with calcium 1000 mg or less does not prevent first fractures (grade D).

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244. Answer: D. The ACOG statement describes pregnancy options counseling as person-centered, seeking to understand the patient's values, beliefs, preferences, concerns and ambivalence. Clinicians in every setting should be ready to discuss prenatal care, abortion, and parenting or not parenting, avoid directive statements, and avoid letting personal views or state restrictions narrow the conversation. Ambivalence is a valid outcome. The distractors are directive, withholding or rushed approaches. In practice, give accurate information, name next steps, and let her leave undecided if needed.

> *Key principle:* Options counseling is person-centered, covers all options, and tolerates ambivalence.

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245. Answer: A. The ACOG statement advises tailoring the depth of counseling to the patient. Once a patient has a clear, firm decision, repeatedly raising other options is not person-centered unless something material changes, and informed decisions should be respected even when they conflict with the clinician's views. The distractors add repeated reviews, extra visits, third-party confirmation or unneeded evaluations that treat a firm decision as suspect. In practice, confirm she has the information she wants, then move to planning care.

> *Key principle:* Once a decision is firm, respect it and do not keep reopening the options.

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246. Answer: E. The mifepristone label lists confirmed or suspected ectopic pregnancy and an undiagnosed adnexal mass as contraindications, because the regimen is not effective for terminating ectopic pregnancy. An empty uterus with an adnexal mass and a positive hCG makes ectopic pregnancy likely. The distractors rely on gestational age alone, use misoprostol without addressing the mass, or pair treatment with an intrauterine device, itself a contraindication. In practice, manage as suspected ectopic pregnancy with close evaluation.

> *Key principle:* Mifepristone is contraindicated with suspected ectopic pregnancy or an undiagnosed adnexal mass.

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247. Answer: B. The mifepristone label states that patients take misoprostol within 24 to 48 hours after taking mifepristone, with the misoprostol given by the buccal route. The label supports use through 70 days of gestation and a follow-up visit approximately 7 to 14 days after mifepristone to confirm that the pregnancy has ended. The distractors move misoprostol too early, too late, or to the follow-up visit. In practice, give clear written instructions on timing, expected bleeding and when to seek urgent care.

> *Key principle:* Misoprostol follows mifepristone within 24 to 48 hours, and follow-up is at 7 to 14 days.

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248. Answer: D. The 2021 USPSTF statement recommends screening for chlamydia and gonorrhea in sexually active women 24 and younger, and in women 25 and older who are at increased risk (grade B). Risk includes a new partner, multiple partners, a partner with concurrent partners or an STI, inconsistent condom use outside a monogamous relationship, or a prior STI. She has none. The distractors extend screening to everyone, add an interval tied to a contraceptive, or confuse it with symptom-driven testing. In practice, reassess risk at each visit.

> *Key principle:* At 25 and older, screen for chlamydia and gonorrhea only when risk is increased.

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249. Answer: C. The USPSTF found no studies on optimal screening intervals and states that a reasonable approach is to screen patients whose sexual history reveals new or persistent risk factors since the last negative test result. She is 26 with a new partner and inconsistent condom use, so repeat screening is appropriate. The distractors tie testing to a fixed calendar interval, rely on the prior negative result, wait for symptoms, or treat without testing. In practice, use a nucleic acid amplification test and counsel on condom use.

> *Key principle:* Repeat chlamydia and gonorrhea screening when new or persistent risk follows a negative test.

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250. Answer: A. The 2019 USPSTF statement recommends screening for HIV in adolescents and adults aged 15 to 65 (grade A), regardless of reported risk, and also recommends it in all pregnant persons. Younger adolescents and older adults at increased risk should also be screened. The Task Force found insufficient evidence on optimal intervals. The distractors apply a risk-based or pregnancy-based approach that the recommendation replaced. In practice, offer a combined antigen and antibody test and document the result.

> *Key principle:* Screen everyone aged 15 to 65 for HIV, not only those with identified risk.

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251. Answer: E. The 2020 USPSTF statement recommends screening for hepatitis C virus infection in adults aged 18 to 79 (grade B), including asymptomatic adults without known liver disease and pregnant persons. Screening is generally one time for most adults. The distractors apply a risk-factor-only approach, a later start age, annual testing for low-risk patients, or a test only after enzyme elevation. In practice, order an antibody test with reflex confirmation and refer a positive result for treatment.

> *Key principle:* Screen every adult 18 to 79 for hepatitis C once, regardless of risk factors.

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252. Answer: A. The USPSTF recommends one-time screening for most adults, with periodic screening of persons who have continued risk for hepatitis C virus infection, for example current or past injection drug use. A single negative test does not cover ongoing exposure. The distractors treat the one-time approach as universal, wait for enzyme changes or an age milestone, or postpone testing until risk ends. In practice, repeat testing at intervals suited to her exposure and link her to harm-reduction and treatment services.

> *Key principle:* Persons with continued hepatitis C risk are screened periodically, not once.

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253. Answer: B. The 2025 USPSTF statement recommends screening women of reproductive age for intimate partner violence (grade B), including pregnant and postpartum women, and states that those who screen positive are evaluated and, if appropriate, provided or referred for evidence-based interventions that include multiple components and ongoing support. The distractors delay action, give a one-time handout, use couples counseling, which can be unsafe, or override her autonomy. In practice, assess safety, offer resources, and keep the door open for future visits.

> *Key principle:* A positive intimate partner violence screen leads to evaluation and referral for multicomponent support.

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254. Answer: C. The 2023 USPSTF statement recommends screening for depression in the adult population, including pregnant and postpartum persons and older adults (grade B). For suicide risk screening, it concluded that evidence is insufficient to assess the balance of benefits and harms, an I statement. The distractors swap the two grades, narrow the target group, or reverse the recommendation. In practice, use a validated depression instrument and have follow-up systems for diagnosis and treatment, without implying proven benefit for suicide risk screening.

> *Key principle:* Screen adults for depression (grade B); suicide risk screening is an I statement.

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255. Answer: D. The 2020 USPSTF statement recommends offering or referring adults with cardiovascular risk factors to behavioral counseling interventions to promote a healthy diet and physical activity (grade B). Risk factors include hypertension or elevated blood pressure, dyslipidemia, and mixed risk factors such as an estimated 10-year risk of 7.5 percent or greater. She qualifies on several counts. The distractors defer, minimize or split the intervention. In practice, refer her to a structured program and also address blood pressure and lipids.

> *Key principle:* Adults with cardiovascular risk factors should be offered or referred for diet and activity counseling.

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256. Answer: B. The 2018 USPSTF statement recommends screening for unhealthy alcohol use in primary care settings in adults 18 and older, including pregnant women, and providing persons engaged in risky or hazardous drinking with brief behavioral counseling interventions (grade B). She has risky drinking without dependence features. The distractors jump to pharmacotherapy or inpatient care, which suit alcohol use disorder or withdrawal, or take no action. In practice, give feedback, set goals and arrange follow-up.

> *Key principle:* Risky drinking without dependence calls for brief behavioral counseling.

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OP1.2. Patients with Unique Needs

257. Answer: A. Chlamydia and gonorrhea are most prevalent in young women, are usually asymptomatic, and cause pelvic inflammatory disease and infertility if untreated. The USPSTF gives a grade B to screening all sexually active women 24 years or younger and women 25 or older at increased risk. Screening is therefore driven by age, not by partner number, symptoms or a partner's result. In practice, send a nucleic acid amplification test from a self-collected vaginal swab or first-catch urine at this visit.

> *Key principle:* Screen every sexually active woman 24 or younger for chlamydia and gonorrhea.

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258. Answer: C. Fear that parents will learn of sexual activity keeps many teens from testing. Every state and the District of Columbia let minors consent to STI services, and no state requires parental consent for STI care. The CDC adds that explanation-of-benefits statements and electronic records can still reveal services, so confidentiality should be discussed. The distractors require parental consent, delay care or disclose results. In practice, speak with the adolescent alone and review how results and bills are handled.

> *Key principle:* Minors can consent to their own STI care; protect confidentiality and warn about billing leaks.

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259. Answer: D. Young women are re-exposed easily, and most infections cause no symptoms, so a single negative result does not protect against a later infection. The CDC recommends routine annual chlamydia screening for sexually active females. Her last test is over a year old, so she is due. The distractors borrow the cervical screening interval or wait for symptoms or partner behavior. In practice, collect a vaginal swab or urine specimen at this visit and treat if positive.

> *Key principle:* Sexually active young women need chlamydia screening every year.

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260. Answer: E. Adolescent depression is common, often missed in routine visits, and treatable. The USPSTF recommends screening adolescents 12 to 18 for major depressive disorder, with a grade B, and finds the evidence insufficient for suicide risk screening in children and adolescents. The suicide-risk-only option is the trap. Waiting for symptoms or relying on a parent ignores the recommendation. In practice, use a validated adolescent instrument, ensure follow-up capacity, and ask about safety if the screen is positive.

> *Key principle:* Screen adolescents 12 to 18 for major depression; suicide risk screening has insufficient evidence.

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261. Answer: D. Balance, gait and strength training improves muscle function and lowers the number of falls in older adults. The USPSTF gives a grade B to recommending exercise interventions for community-dwelling adults 65 or older at increased risk of falls, and a grade C to individualizing multifactorial interventions. Restricting activity worsens deconditioning, and sedatives increase falls. In practice, refer her to a structured balance and strength program.

> *Key principle:* Older adults at increased fall risk: recommend exercise interventions.

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262. Answer: E. Older adults who are healthy gain some benefit from continued screening, but those with limited life expectancy gain little and face colonoscopy risks. The USPSTF gives a grade C to selectively offering screening to adults 76 to 85, weighing health status, prior screening and preference. A fixed annual or interval rule, a blanket stop, or symptom-based testing fails to reflect that. In practice, discuss the small net benefit and her priorities, and document the decision.

> *Key principle:* Ages 76 to 85: individualize colorectal screening, grade C.

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263. Answer: A. Trials of primary prevention enrolled few adults older than 75, so the balance of benefit and harm in new starts is uncertain. The USPSTF therefore makes an I statement for initiating a statin in adults 76 or older. The distractors upgrade it to a B, downgrade it to a D, or import a threshold used for other situations. In practice, decide by shared decision-making that considers comorbidity, life expectancy, and her preference.

> *Key principle:* Starting a statin at 76 or older: the USPSTF finds the evidence insufficient.

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264. Answer: B. Anxiety disorders are common, but trials of screening in older adults are lacking. The USPSTF recommends screening adults 64 or younger with a grade B and issues an I statement for adults 65 or older. The distractors extend the B grade, convert the I statement into a D, or invent an interval and subgroup. In practice, evaluate older adults who show symptoms and use clinical judgment, since no guideline mandates universal screening.

> *Key principle:* Anxiety screening at 65 or older: the USPSTF has insufficient evidence.

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265. Answer: C. HPV spreads by skin-to-skin contact, so women who have sex with women can acquire cervical infection, and abnormal cytology has been found in women who report no prior sex with men. The CDC states that sexual transmission of HPV likely occurs between women who have sex with women. Routine age-based cervical screening therefore applies. In practice, take a sexual history without assumptions, and offer the same screening schedule used for all patients with a cervix.

> *Key principle:* Women who have sex with women follow the same cervical screening as other women.

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266. Answer: B. Women who have sex with women are often assumed to be at low risk, but many have had male partners, can transmit infections to one another through cervicovaginal secretions, and report low rates of barrier use. The CDC advises providers not to assume lower risk for this group. Screening therefore follows the usual criteria, including chlamydia and gonorrhea testing for women under 25 and HIV screening. In practice, ask about all partners, practices and anatomic sites.

> *Key principle:* Do not assume lower STI risk for women who have sex with women.

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267. Answer: D. Testosterone does not remove the cervix, and cervical cancer risk depends on the tissue present rather than gender identity. The CDC states that screening for people with a cervix should follow current screening guidelines for cervical cancer, and that screening recommendations are adapted based on anatomy. He therefore follows the age-based schedule, using sensitive technique because testosterone can cause vaginal atrophy and sampling may be uncomfortable. In practice, use a smaller speculum, consider lubrication, offer self-collected HPV testing if available, and keep records under his chosen name.

> *Key principle:* Cervical screening follows anatomy, not gender identity or hormone use.

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268. Answer: C. Transgender women have a high burden of HIV, and self-perceived risk is a poor guide to who is infected. The CDC states that HIV screening should be discussed and offered to all transgender persons, and that other screening is adapted to anatomy. A monogamous relationship does not remove the need to offer testing. Hormone therapy is not a reason to defer. In practice, offer an opt-out HIV test with the baseline laboratory studies at this visit.

> *Key principle:* Offer HIV screening to every transgender patient.

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269. Answer: A. Intimate partner violence causes injury, mental illness and unintended pregnancy, and screening in reproductive-age women identifies victims who otherwise stay hidden. The USPSTF recommends that clinicians screen women of reproductive age, grade B, and that those who screen positive be evaluated and, if appropriate, provided or referred for evidence-based interventions that include multiple components and ongoing support. Couples counseling can endanger the victim. Reporting without consent may increase risk unless required by law. In practice, ensure privacy, offer safety planning, and respect her pace.

> *Key principle:* A positive intimate partner violence screen should lead to multicomponent care with ongoing support, not couples counseling.

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270. Answer: E. Follow-up after assault is often missed, so waiting for results leaves infections untreated. The CDC guideline recommends presumptive treatment: an empiric regimen for chlamydia, gonorrhea and trichomonas for women, along with vaccination and consideration of HIV postexposure prophylaxis. Treating one pathogen leaves two uncovered. The guideline does not recommend herpes prophylaxis. In practice, offer treatment at the first visit and arrange testing for return or telehealth follow-up.

> *Key principle:* After assault, give empiric treatment for chlamydia, gonorrhea and trichomonas at the first visit.

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271. Answer: D. HPV infection after assault is possible, and vaccination protects against types not yet acquired. The CDC recommends HPV vaccination for female and male survivors aged 9 to 26 who have not been vaccinated. Waiting for a test or a delay of months risks loss to follow-up, and the recommendation does not depend on who the assailant was. In practice, begin the series at this visit and schedule subsequent doses.

> *Key principle:* Unvaccinated survivors aged 9 to 26 should be offered HPV vaccine at the first visit.

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272. Answer: A. Postexposure vaccination induces protective antibodies before infection can establish itself. The CDC recommends postexposure hepatitis B vaccination, without hepatitis B immune globulin, when the assailant's status is unknown and the survivor is unvaccinated. Immune globulin is added if the assailant is known to be HBsAg positive. Antiviral prophylaxis and delay are not recommended. In practice, give the first dose at the initial visit and complete the series.

> *Key principle:* Unknown assailant status and unvaccinated survivor: hepatitis B vaccine alone, no immune globulin.

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273. Answer: C. Mid-cycle assault carries a real pregnancy risk, and emergency contraception works best when given early. The CDC states that emergency contraception should be considered when the assault could result in pregnancy, and reproductive-aged survivors should be evaluated for pregnancy. Waiting for menses or for later testing forfeits the window. The decision does not hinge on the patient proving ejaculation. In practice, discuss options and offer one at this visit.

> *Key principle:* If pregnancy is possible after assault, offer emergency contraception at the first visit.

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274. Answer: B. Depression is common in adult women, often undetected, and treatable. The USPSTF recommends screening for depression in the adult population, including pregnant and postpartum persons and older adults, with a grade B. The distractors limit screening to those with a history, to attempts or to pregnancy, or deny the recommendation. In practice, use a validated tool, and ensure that systems exist to confirm the diagnosis, treat and follow up positive screens.

> *Key principle:* Screen all adults for depression with a validated tool, grade B.

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275. Answer: E. Anxiety disorders are common in young and middle-aged women and respond to therapy and medication, yet they often go undiagnosed. The USPSTF recommends screening for anxiety disorders in adults, including pregnant and postpartum persons, up to age 64, with a grade B. The I statement applies to older adults, which is the trap. In practice, use a brief validated tool and offer cognitive behavioral therapy or medication when the screen is positive.

> *Key principle:* Screen adults 64 or younger for anxiety disorders, grade B.

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276. Answer: A. Tools for suicide risk have uncertain accuracy, and trials do not show that screening reduces suicide attempts or deaths. The USPSTF therefore gives an I statement for screening adults for suicide risk, while it recommends depression screening with a grade B. The distractors upgrade or downgrade the statement or limit it to subgroups. In practice, ask directly about self-harm when a patient screens positive for depression, and follow the practice protocol for safety.

> *Key principle:* Suicide risk screening in adults: insufficient evidence, an I statement.

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277. Answer: D. Risperidone blocks dopamine D2 receptors in the tuberoinfundibular pathway, removing inhibition of pituitary lactotrophs. The drug label states that risperidone elevates prolactin levels and that the elevation persists during chronic administration, and that galactorrhea and amenorrhea have been reported. Hyperprolactinemia suppresses hypothalamic GnRH, reducing gonadotropins. The distractors do not explain galactorrhea with a normal thyroid hormone level. In practice, consider a pituitary image if the prolactin is very high, and weigh switching to a prolactin-sparing agent.

> *Key principle:* Risperidone commonly causes hyperprolactinemia with amenorrhea and galactorrhea.

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278. Answer: B. Decompensated cirrhosis impairs hepatic metabolism of steroid hormones and clotting factors, and estrogen adds thrombotic and hepatic risk. In the US MEC table, combined hormonal contraception is category 4 for decompensated cirrhosis, while the copper intrauterine device is category 1 and the progestin methods are category 2 (DMPA is category 3). The listed alternatives are therefore acceptable. In practice, offer a copper or levonorgestrel intrauterine device or implant and coordinate care with hepatology.

> *Key principle:* Decompensated cirrhosis: combined hormonal contraception is category 4.

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279. Answer: C. Obesity raises venous thrombosis risk modestly, and risk grows with additional factors such as older age or smoking. The US MEC lists a body mass index of 30 or more as category 2 for combined hormonal contraception, meaning the advantages generally outweigh the risks. She has no other factors. The distractors overstate the restriction or ignore the added risk. In practice, counsel her about thrombosis risk, offer alternatives, and reassess if other risk factors appear.

> *Key principle:* Body mass index of 30 or more without other risk factors: combined hormonal contraception is category 2.

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280. Answer: C. Estrogen raises clotting factors, and major surgery with prolonged immobility adds stasis and tissue injury, which compound venous thrombosis risk. The US MEC lists combined hormonal contraception as category 4 for major surgery with prolonged immobilization, while progestin-only and intrauterine methods are category 1 or 2. Compression stockings, aspirin or a patch do not change the estrogen exposure. In practice, switch to a nonestrogen method before surgery and plan thromboprophylaxis.

> *Key principle:* Major surgery with prolonged immobilization: stop combined hormonal contraception, category 4.

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281. Answer: E. Federal disability law treats a medical practice as a public accommodation. The Department of Justice and HHS guidance states that a provider cannot deny service to a patient it would otherwise serve because she has a disability. Lack of equipment does not create a right to refuse, and the practice must find reasonable ways to examine her, such as a height-adjustable table or lift. In practice, arrange accessible equipment or trained staff transfer.

> *Key principle:* A practice cannot refuse a patient it would otherwise serve because of disability.

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282. Answer: C. Speaking to a companion in place of the patient undermines autonomy, can hide abuse, and produces unreliable histories. The federal guidance states that clinicians should always address the patient directly, not the companion, as with any other patient. Capacity is presumed unless shown otherwise, so a guardian is not required. In practice, give extra time, confirm understanding, and use communication aids as needed.

> *Key principle:* Speak to the patient, not her companion, and presume capacity.

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283. Answer: C. Manual lifting risks injury to the patient and to staff and can cause pain and skin trauma. The federal guidance says a patient lift provides better security than being lifted by staff and that tables should lower to wheelchair seat height, roughly 17 to 19 inches. It also calls for an accessible scale with a platform large enough for a wheelchair. In practice, obtain the equipment or arrange care where it exists, and do not omit routine services.

> *Key principle:* Use a lowering table or lift and a wheelchair scale instead of manual lifting.

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OP2.1. Primary Care Problems

284. Answer: A. The USPSTF grades biennial screening mammography B for ages 40 to 74 and gives an I statement for women 75 or older, because trials enrolled few older women. The correct option states that position. The distractors extend the B grade, invent an annual schedule, convert an I statement into a recommendation against screening, or substitute MRI without support. In practice, the decision at 76 is individualized, weighing health, life expectancy and preference, rather than driven by a fixed stopping rule.

> *Key principle:* At 75 or older the USPSTF finds the evidence insufficient, so individualize.

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285. Answer: D. Dense tissue lowers mammographic sensitivity and raises risk, but supplemental ultrasound or MRI finds more lesions without proven mortality benefit and adds false positives and biopsies. The USPSTF therefore issues an I statement for supplemental screening in dense breasts after a negative mammogram. The distractors turn the I statement into a graded recommendation for or against, or invent a short repeat interval. In practice, discuss density and personal risk with the patient and decide together, since no guideline grade settles the question.

> *Key principle:* Dense breasts: the USPSTF cannot recommend for or against supplemental ultrasound or MRI.

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286. Answer: B. Selective estrogen receptor modulators and aromatase inhibitors lower the incidence of invasive estrogen receptor positive breast cancer in women at increased risk. The USPSTF gives a grade B to offering such drugs when adverse-effect risk is low. The traps are the grade D statement, which applies to women not at increased risk, and options with no evidence for primary prevention. In practice, document risk, screen for clot and uterine risk, and let the patient weigh benefit against harm.

> *Key principle:* Increased risk plus low adverse-effect risk: offer tamoxifen, raloxifene or an aromatase inhibitor.

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287. Answer: E. Chemoprevention lowers breast cancer incidence only modestly, so in women at average risk the absolute benefit is smaller than the harms, which include venous thromboembolism, endometrial cancer and vasomotor symptoms. The USPSTF recommends against the routine use of risk-reducing medications such as tamoxifen in women who are not at increased risk, with a grade D. The distractors prescribe a drug anyway, and aromatase inhibitors and raloxifene are not for premenopausal women. In practice, quantify her risk and revisit it if factors change.

> *Key principle:* Average-risk women should not take tamoxifen or similar drugs for prevention.

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288. Answer: E. Pathogenic variants are rare in women without a relevant personal or family history, so a positive result is unlikely and a variant of uncertain significance is a real possibility. The USPSTF gives a grade D to routine risk assessment, counseling and testing in women whose history or ancestry is not linked to harmful BRCA variants. The distractors apply a universal approach or high-risk interventions to a low-risk woman. In practice, document the negative history and revisit if the family history changes.

> *Key principle:* No relevant history: do not offer routine BRCA risk assessment or testing.

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289. Answer: D. Statins reduce first cardiovascular events and mortality across moderate-risk groups, with modest harm. The USPSTF gives a grade B to prescribing a statin to adults 40 to 75 who have one or more risk factors and a 10-year risk of 10% or more. She has hypertension and a 13% risk, so she qualifies. The traps are aspirin, which has no such grade B support at this age, and waiting for an LDL cholesterol of 190 mg/dL. In practice, calculate risk, then start the statin.

> *Key principle:* Risk factor plus 10-year risk of 10% or more at ages 40 to 75: prescribe a statin.

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290. Answer: A. At 7.5% to less than 10% the absolute benefit of a statin is smaller, so the net benefit is small. The USPSTF assigns a grade C: selectively offer a statin to adults 40 to 75 with one or more risk factors in this range, using shared decision-making. The distractors convert the C grade into a B grade or a D grade, or add a coronary calcium requirement that the statement does not make. In practice, present the numbers and respect her preference.

> *Key principle:* Risk of 7.5% to less than 10%: grade C, selectively offer a statin.

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291. Answer: C. In adults 60 or older without cardiovascular disease, bleeding risk rises with age and offsets the small absolute reduction in first events. The USPSTF gives a grade D to initiating low-dose aspirin for primary prevention at 60 or older. A 12% risk does not change that. The distractors tie initiation to risk, statin use, a gastroprotective drug or a lipid value, none of which the statement uses. In practice, spend the visit on blood pressure, lipids and lifestyle.

> *Key principle:* Age 60 or older without cardiovascular disease: do not start aspirin for primary prevention.

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292. Answer: B. For adults 40 to 59 with a 10-year risk of 10% or more, the net benefit of aspirin is small, and bleeding risk varies by person. The USPSTF therefore gives a grade C: the decision to initiate low-dose aspirin should be an individual one. The distractors turn the grade C into a universal recommendation for or against aspirin or into an unrelated trigger. In practice, discuss her bleeding risk and values and document the decision.

> *Key principle:* Ages 40 to 59 with risk of 10% or more: aspirin is an individual decision, grade C.

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293. Answer: D. Office readings overestimate blood pressure in some patients because of white coat effect, so treating from clinic values alone risks overtreatment. The USPSTF recommends obtaining measurements outside the clinical setting, by home or ambulatory monitoring, to confirm hypertension before starting treatment. Starting a drug now or waiting years both skip confirmation, and imaging is not part of an initial workup. In practice, arrange ambulatory or home monitoring and treat if the out-of-office values are elevated.

> *Key principle:* Confirm an elevated office reading with out-of-office measurements before treating.

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294. Answer: B. Obesity is a major risk factor, and early detection allows lifestyle intervention that delays diabetes. The USPSTF recommends screening for prediabetes and type 2 diabetes in adults 35 to 70 who have overweight or obesity, with a grade B. She is 37 with a body mass index of 32, so she qualifies now. The distractors wait for symptoms, raise the age or body mass index cutoff, or require a nonstandard test. In practice, order a fasting glucose or hemoglobin A1c and offer referral for prevention programs if prediabetes is found.

> *Key principle:* Overweight or obesity at 35 to 70: screen for prediabetes and type 2 diabetes.

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295. Answer: E. Colorectal cancer incidence has risen in adults younger than 50, and modeling shows life-years gained by starting at 45. The USPSTF therefore recommends screening adults aged 45 to 49 with a grade B and adults 50 to 75 with a grade A. The distractors keep the superseded start age of 50, require family history, or invent annual colonoscopy. In practice, offer a choice among accepted tests, such as stool-based testing or colonoscopy, and begin now.

> *Key principle:* Average-risk adults aged 45 to 49 should start colorectal screening, grade B.

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296. Answer: C. Chronic hepatitis C is often silent for decades, and curative antiviral therapy prevents cirrhosis and cancer. The USPSTF recommends screening adults aged 18 to 79 with a grade B, once for most adults and periodically for those with continued risk. Risk-based or enzyme-based testing misses many infections, and annual testing is not advised for low-risk adults. In practice, order a hepatitis C antibody test with reflex testing for RNA, and repeat only if risk exposure continues.

> *Key principle:* Test every adult aged 18 to 79 once for hepatitis C, regardless of risk.

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297. Answer: A. Clarithromycin and levofloxacin resistance are common enough that standard triple therapy fails often when susceptibility is unknown. The ACG guideline names bismuth quadruple therapy for 14 days as the preferred regimen in treatment-naive patients without known susceptibility. The distractors are the older triple therapy, an insufficient dual regimen, a fluoroquinolone regimen the guideline reserves for confirmed susceptibility, and acid suppression alone, which does not eradicate infection. In practice, confirm eradication afterward.

> *Key principle:* Treatment-naive H pylori with unknown susceptibility: bismuth quadruple therapy for 14 days.

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298. Answer: C. Irritable bowel syndrome is diagnosed by symptom pattern, with limited testing to exclude celiac disease and inflammatory bowel disease when diarrhea predominates. The ACG guideline endorses a positive diagnostic strategy and recommends against routine colonoscopy in patients younger than 45 without warning signs. She is 32 with no alarm features. The distractors are exhaustive tests that carry cost and risk without yield. In practice, explain the diagnosis and reassess if alarm features develop.

> *Key principle:* Younger than 45 with no alarm features: diagnose irritable bowel syndrome by history, skip routine colonoscopy.

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299. Answer: E. In irritable bowel syndrome with diarrhea, a gut-selective antibiotic improves global symptoms with minimal systemic absorption. The ACG guideline recommends it for this subtype. Guanylate cyclase activators are the constipation-predominant option, polyethylene glycol worsens loose stools, and a systemic antibiotic adds risk without a recommendation. Probiotics lack guideline support as a treatment. In practice, match the subtype to the drug class, and treat constipation-predominant disease with the secretagogue instead.

> *Key principle:* Irritable bowel syndrome with diarrhea: a gut-selective antibiotic, not a constipation agent.

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300. Answer: A. Most acute low back pain improves on its own, and imaging without red flags rarely changes management. The ACP guideline says superficial heat, massage, acupuncture, and spinal manipulation are appropriate initial options for acute and subacute pain, with an NSAID or muscle relaxant if a drug is chosen. Imaging, opioids, prolonged rest and injections add harm or cost without benefit at this stage. In practice, reassure, advise staying active, and reserve imaging for red flags.

> *Key principle:* Acute nonradicular low back pain: start with heat, massage, acupuncture or manipulation, not imaging or opioids.

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301. Answer: B. Chronic low back pain responds best to exercise and other nonpharmacologic care, and medication comes second. The ACP guideline says an NSAID is the first-line drug and that opioids should be considered only if other treatments fail and the benefits outweigh the risks. Oral steroids and long-term muscle relaxants are not supported for chronic pain, and repeated injections are not a first-line step. In practice, check kidney, gastrointestinal and cardiovascular risk before choosing an NSAID.

> *Key principle:* Chronic low back pain: NSAID first among drugs, opioids last.

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302. Answer: D. Xanthine oxidase inhibitors lower urate production, and starting low reduces allopurinol hypersensitivity, which is more likely with reduced kidney function. The ACR guideline strongly recommends allopurinol as the preferred first-line urate-lowering therapy, started at a low dose and lower still in chronic kidney disease. The distractors name second-line or specialty agents, or a drug that treats flares but does not lower urate. In practice, titrate to a serum urate under 6 mg/dL.

> *Key principle:* Gout needing urate lowering: allopurinol first, started low and titrated.

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303. Answer: E. Lowering urate mobilizes crystals from tissue deposits, which can provoke flares for months after therapy begins. The ACR guideline strongly recommends concomitant anti-inflammatory prophylaxis for at least 3 to 6 months when starting urate-lowering therapy. Stopping within weeks or at the first normal urate invites flares, and lifelong prophylaxis exposes the patient to avoidable drug effects. In practice, continue prophylaxis beyond the first normal urate and reassess at 3 to 6 months.

> *Key principle:* Cover urate-lowering initiation with anti-inflammatory prophylaxis for at least 3 to 6 months.

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304. Answer: C. Knee osteoarthritis improves with exercise and, in people who are overweight, with weight loss, which reduce load and pain. The guideline strongly recommends both, along with tai chi, cane use, bracing, topical and oral NSAIDs, and intraarticular glucocorticoids. It strongly recommends against glucosamine, chondroitin sulfate and bisphosphonates, which show no benefit. In practice, prescribe a land-based exercise program and a weight target, and counsel that these supplements are not endorsed.

> *Key principle:* Knee osteoarthritis: exercise and weight loss yes; glucosamine, chondroitin and bisphosphonates no.

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305. Answer: B. Stable migraine with a normal examination carries a very low yield of structural findings, and incidental findings lead to further testing and anxiety. The American Headache Society Choosing Wisely list advises against neuroimaging in patients with stable headaches that meet migraine criteria. Computed tomography, lumbar puncture and carotid imaging are even less justified here. In practice, reserve imaging for new patterns, abnormal findings or red-flag features, and explain the reasoning to the patient.

> *Key principle:* Stable typical migraine with a normal examination does not need neuroimaging.

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306. Answer: A. Butalbital combinations and opioids give short-lived relief but raise the risk of dependence, medication overuse headache and chronic migraine, and they do not treat the migraine process. The American Headache Society advises against opioid or butalbital use as first-line treatment for recurrent headache disorders. The first-line choices are drugs such as NSAIDs and triptans. Frequent over-the-counter analgesic use is also discouraged. In practice, replace the combination product and counsel on limiting acute medication days.

> *Key principle:* Do not use opioids or butalbital as first-line treatment for recurrent headache.

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307. Answer: D. Frequent acute medication use sensitizes pain pathways and converts episodic headache into near-daily headache. ICHD-3 criteria for medication overuse headache require headache on 15 or more days per month in a patient with a pre-existing headache disorder, overuse of acute medication for more than 3 months, and no better explanation. She meets each. The other diagnoses do not fit a long migraine history with escalating analgesic use. In practice, identify the pattern and plan withdrawal.

> *Key principle:* Headache on 15 or more days a month with 3 or more months of acute drug overuse is medication overuse headache.

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308. Answer: E. Overused acute medication perpetuates the headache cycle, so withdrawal is central to recovery. The cited review states that complete discontinuation of the overused drug yields the best outcomes, abruptly or over several weeks, while preventive treatment is started at the same time. Continuing, escalating or switching to another acute agent maintains overuse, and delaying prevention leaves her unprotected during withdrawal. In practice, warn of transient worsening and set a plan for rescue therapy.

> *Key principle:* Treat medication overuse headache by withdrawing the overused drug while starting prevention.

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309. Answer: C. Noncontrast CT is very sensitive for subarachnoid blood in the first hours, before the blood clears. The ACEP policy supports using a normal noncontrast head CT performed within 6 hours of symptom onset, on at least a third-generation scanner, to rule out nontraumatic subarachnoid hemorrhage in an emergency department patient with a normal neurologic examination. A later normal scan calls for lumbar puncture or CT angiography, chosen by shared decision making. A single clinical sign cannot exclude hemorrhage. In practice, document the interval from onset to scan and the scanner type.

> *Key principle:* A normal CT within 6 hours of onset can exclude subarachnoid hemorrhage; later scans need more.

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310. Answer: B. Treating asthma with a short-acting beta agonist alone relieves symptoms but leaves airway inflammation untreated, and it is linked to severe exacerbations and death. GINA no longer recommends as-needed short-acting beta agonist therapy alone and prefers low-dose inhaled corticosteroid with formoterol used as needed in adults and adolescents. Long-acting beta agonist alone is unsafe without an inhaled corticosteroid, and daily oral steroid is excessive. In practice, replace albuterol with the combination reliever.

> *Key principle:* Do not treat asthma with a short-acting beta agonist alone; use as-needed ICS-formoterol.

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311. Answer: D. Using one inhaler for both maintenance and relief delivers extra inhaled corticosteroid exactly when symptoms rise, which lowers exacerbations compared with a short-acting beta agonist reliever. The NAEPP 2020 update names single maintenance and reliever therapy as preferred for mild to moderate persistent asthma. Adding montelukast, scheduling albuterol, or using oral steroids is inferior or harmful. In practice, switch to the combination inhaler and teach her to use it for rescue.

> *Key principle:* Mild to moderate persistent asthma with poor control: prefer single maintenance and reliever therapy.

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312. Answer: A. Poor technique, nonadherence, smoking, rhinitis and exposure explain many cases of apparent treatment failure, and stepping up without addressing them adds drug exposure without benefit. GINA advises checking inhaler adherence and treating modifiable risk factors before considering a step up. Oral steroids, biologics and thermoplasty are for severe disease after optimization. In practice, observe her technique, review adherence, and ask about triggers and comorbid disease before changing the regimen.

> *Key principle:* Before stepping up asthma therapy, check technique and adherence and treat modifiable factors.

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313. Answer: E. Muscarinic blockade adds bronchodilation through a different pathway and improves control in patients who remain symptomatic on inhaled corticosteroid plus long-acting beta agonist. The NAEPP 2020 update recommends adding a long-acting muscarinic antagonist when control is inadequate on that combination. Scheduled short-acting beta agonists, chronic oral steroids and thermoplasty are not the recommended next step. In practice, add tiotropium to the existing inhaler and reassess in a few weeks.

> *Key principle:* Inadequate control on ICS-LABA: add a long-acting muscarinic antagonist.

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314. Answer: B. In allergic asthma, repeated exposure to a sensitizing allergen sustains airway inflammation, and immunotherapy gradually reduces responsiveness. The NAEPP 2020 update conditionally recommends allergen immunotherapy as an adjunct in mild to moderate allergic asthma when sensitization is proven and symptoms track exposure. Thermoplasty is reserved for severe disease, and chronic oral steroids or antibiotics lack support. In practice, confirm sensitization to the relevant allergen before referral to allergy.

> *Key principle:* Proven allergic sensitization with mild to moderate asthma: consider allergen immunotherapy as an adjunct.

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315. Answer: A. Fragility fractures in older women are common and costly, and treatment lowers fracture risk. The USPSTF recommends screening for osteoporosis in all women 65 or older, with a grade B, regardless of additional risk factors. The traps are waiting for a fracture, requiring a high risk score before testing, and substituting a vitamin D level, which does not measure bone density. In practice, order bone density testing by central DXA and apply treatment criteria to the result.

> *Key principle:* All women 65 or older should be screened for osteoporosis.

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316. Answer: C. Low body weight, smoking and parental hip fracture each raise fracture risk, so waiting until 65 would miss years of preventable bone loss. The USPSTF recommends screening postmenopausal women younger than 65 at increased risk, as estimated by clinical risk assessment, with a grade B. Starting a drug without a diagnosis or substituting radiographs ignores the recommended test. In practice, use a validated tool such as FRAX to decide who is screened early.

> *Key principle:* Postmenopausal women younger than 65 at increased risk by clinical assessment should be screened.

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317. Answer: D. In community-dwelling postmenopausal women without osteoporosis, low-dose vitamin D and calcium did not reduce fractures in pooled trials. The USPSTF recommends against daily supplementation with 400 IU or less of vitamin D and 1000 mg or less of calcium, with a grade D. Higher doses carry an I statement, so benefit is unproven. In practice, advise a diet with adequate calcium and attention to exercise and fall risk.

> *Key principle:* Low-dose vitamin D and calcium do not prevent fractures in healthy postmenopausal women.

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318. Answer: E. Bisphosphonates reduce vertebral, nonvertebral and hip fractures at low cost with a long record of safety. The ACP recommends bisphosphonates as initial pharmacologic treatment for postmenopausal women and men with primary osteoporosis. Denosumab is second-line for those who cannot take a bisphosphonate, and romosozumab or teriparatide followed by a bisphosphonate is for very high fracture risk. Raloxifene and calcitonin are not first-line. In practice, confirm kidney function, dental health and vitamin D status before starting.

> *Key principle:* Primary osteoporosis: begin with a bisphosphonate.

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319. Answer: D. A RANK ligand inhibitor blocks osteoclast formation and lowers fracture risk in patients who cannot use a bisphosphonate. The ACP recommends denosumab as second-line pharmacologic treatment for adults with primary osteoporosis who have contraindications to or adverse effects from bisphosphonates. Calcium alone, calcitonin and hormone therapy are not recommended in this setting, and teriparatide is reserved for very high fracture risk. In practice, plan continued denosumab dosing, because stopping it abruptly can cause rapid bone loss and fractures.

> *Key principle:* If a bisphosphonate cannot be used, denosumab is the second-line choice.

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320. Answer: B. Benefits of bisphosphonates persist after stopping because the drug remains in bone, while risks such as atypical femoral fracture rise with long exposure. The ACP guidance is to consider stopping after 5 years in patients who have responded, unless there is a strong reason to continue, such as a fracture on treatment or persistent very high risk. The distractors continue indefinitely, escalate therapy, or combine agents without support. In practice, reassess fracture risk periodically after stopping.

> *Key principle:* After 5 years of bisphosphonate in a responder, consider stopping.

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OP3.1. General Office Gynecology

321. Answer: A. Chronic pelvic pain without a treatable visceral cause is managed with a multimodal plan, and no drug class has shown lasting benefit that outweighs its risks when it is used for pain alone. The ACOG bulletin, as summarized for family physicians, states that opioids are not recommended in chronic pelvic pain, and that cognitive behavior therapy has a small to moderate benefit. The traps are opioid schedules that sound safer and a laparoscopy with no targeted question. In practice, explain why you are declining, then build the nonopioid plan with her.

> *Key principle:* Opioids are not recommended in chronic pelvic pain; add cognitive behavioral therapy and other nonopioid measures.

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322. Answer: C. Filmy adhesions are common in women without pain, so finding them does not prove they cause it. The ACOG bulletin, as summarized for family physicians, states that laparoscopic adhesiolysis offers no benefit and leads to poorer patient outcomes. The distractors each give adhesiolysis a rescue role, by adding oophorectomy, by waiting out failed drugs, or by changing the route. In practice, discuss pain-directed care such as cognitive behavioral therapy and pelvic floor therapy instead of repeat surgery for adhesions.

> *Key principle:* Adhesiolysis does not relieve chronic pelvic pain, so do not offer it for adhesions alone.

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323. Answer: B. Gabapentin acts on neuropathic pain pathways, so a benefit in pain without an identified neuropathic component was plausible but had to be tested. In the placebo-controlled trial, mean change in worst pain was 1.4 points with gabapentin and 1.2 with placebo, which was not significant, and more adverse reactions occurred with the drug. The distractors invent a benefit by subgroup or by time. In practice, do not expect gabapentin to help an undifferentiated chronic pelvic pain syndrome, and reserve it for women with a neuropathic pattern.

> *Key principle:* GaPP2 showed no significant pain benefit of gabapentin over placebo and more adverse reactions.

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324. Answer: E. In bladder pain syndrome the pelvic floor muscles are often tight and tender, so adding resistance work can raise pain. The AUA guideline gives a graded standard that pelvic floor strengthening exercises, such as Kegel exercises, should be avoided. The distractors either intensify the same strategy, add contraction by another route, or add an unrelated drug or test. In practice, stop the strengthening program, and look for other nonsurgical measures that match her pelvic floor findings.

> *Key principle:* In interstitial cystitis, avoid pelvic floor strengthening exercises.

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325. Answer: A. Management of bladder pain syndrome moves stepwise from behavioral care to oral and intravesical agents. The AUA guideline lists amitriptyline, cimetidine, hydroxyzine and pentosan polysulfate as oral options. It states that long-term oral antibiotics, long-term oral glucocorticoids, bacillus Calmette-Guerin outside investigational settings, and high-pressure, long-duration hydrodistension should not be offered. In practice, choose from the oral options, and counsel about the stated risks of the one chosen, such as the macular warning that comes with pentosan polysulfate.

> *Key principle:* Offer listed oral agents such as amitriptyline, and avoid long-term antibiotics, steroids, BCG and high-pressure hydrodistension.

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326. Answer: C. Local heat relaxes uterine and abdominal muscle and may lessen pain signaling. In the review, heat lowered pain compared with no treatment at 3 months and within 24 hours, and the authors concluded it is likely to reduce pain intensity both during prophylaxis and acute episodes. The distractors claim no benefit, a drug requirement, or a split between acute and preventive use that the data did not show. In practice, heat is a low-risk first step that can be paired with other treatment if needed.

> *Key principle:* Heat likely lowers dysmenorrhea pain compared with no treatment, in both prevention and acute use.

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327. Answer: D. Heat avoids the prostaglandin synthesis inhibition that causes NSAID gastric effects, so a safety edge is plausible. In the review, adverse effects were less likely with heat than with NSAIDs (relative risk 0.30, from 8 trials). The authors concluded efficacy may be comparable with a superior safety profile. The distractors flip either the efficacy or the safety finding. In practice, heat is a sensible choice for a woman who cannot tolerate NSAIDs.

> *Key principle:* Heat may match NSAID pain relief with fewer adverse effects.

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328. Answer: D. NSAIDs relieve dysmenorrhea by blocking prostaglandin synthesis, and the class effect is large compared with placebo. The Cochrane review of 80 trials found little evidence of the superiority of any individual NSAID for either pain relief or safety, and rated most of the evidence as low quality. The distractors name hypothetical winners by age, safety, half-life or timing. In practice, choose an agent by tolerance, cost and availability, and adjust the timing and dose within label limits before declaring failure.

> *Key principle:* No single NSAID has been shown to be better for dysmenorrhea, so choose by tolerance and cost.

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329. Answer: E. The review found ten trials, six of them pill versus placebo, with a pooled benefit for pain and limited overall evidence. The abstract states plainly that no trials compared combined pills with NSAIDs. The distractors claim head-to-head data that do not exist, in either direction or as a tie. In practice, an NSAID remains the usual first step for pain, and the pill is chosen on contraceptive need or when NSAIDs fail or are not wanted, with the evidence gap explained honestly.

> *Key principle:* No randomized trial has compared the combined pill with an NSAID for dysmenorrhea.

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330. Answer: C. TENS is thought to modulate pain signals at the spinal level, but the trials are small. The authors concluded that high-frequency and low-frequency TENS may reduce pain compared with placebo or no treatment, and downgraded certainty for risk of bias. The distractors overstate certainty, deny any effect, add a drug requirement or invent a harm. In practice, TENS is a low-risk adjunct that can be offered with honest wording about the quality of evidence.

> *Key principle:* TENS may reduce dysmenorrhea pain, but the evidence is low certainty.

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331. Answer: A. Suppressing ovulation delays the attempts at conception and does not repair the tubes or the pelvic environment that limit fertility. The ESHRE guideline gives a strong recommendation that women seeking pregnancy should not be prescribed postoperative hormone suppression with the sole purpose of enhancing future pregnancy rates. The distractors substitute another suppressive agent or add a conditional that the guideline does not make. In practice, let her try to conceive soon after surgery and move to assisted reproduction if needed.

> *Key principle:* Do not suppress ovulation after surgery just to improve pregnancy rates; let her try to conceive.

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332. Answer: B. Endometriomas and pain often recur after surgery because the disease is hormone responsive, and suppression lowers the stimulus. The ESHRE guideline recommends offering long-term hormone treatment, for example combined hormonal contraceptives, after surgery for an endometrioma in women not immediately seeking conception. The distractors describe surveillance, brief therapy or symptomatic care, each of which leaves the stimulus in place. In practice, start long-term suppression after surgery and review tolerance and her plans for pregnancy.

> *Key principle:* After endometrioma surgery, offer long-term hormone treatment if she is not trying to conceive.

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333. Answer: B. Estrogen without a progestogen may stimulate residual endometriotic tissue, and unopposed regimens have been linked to malignant transformation. The ESHRE guideline strongly advises avoiding estrogen-only regimens for vasomotor symptoms in postmenopausal women with a history of endometriosis, and its good practice statement is to continue combined estrogen-progestogen after surgical menopause at least up to the age of natural menopause. The distractors assume that hysterectomy removes the issue or that all therapy is barred. In practice, add the progestogen even without a uterus.

> *Key principle:* After surgical menopause for endometriosis, treat with combined estrogen-progestogen, not estrogen alone.

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334. Answer: D. Endometriosis is linked to modestly higher relative risks of ovarian, breast and thyroid cancers, but the absolute increase compared with the general population is low. The ESHRE guideline gives a strong recommendation that clinicians should not systematically perform cancer screening beyond existing population-based guidelines, and it advises reassurance. The distractors add testing or surgery that carries harm without proven benefit. In practice, explain the low absolute risk, address symptoms that change, and keep her up to date on routine screening.

> *Key principle:* Endometriosis alone does not justify extra cancer screening beyond population guidelines.

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335. Answer: D. Elagolix lowers estradiol and causes a dose-dependent loss of bone mineral density, so the label limits duration. For women with normal liver function or mild impairment, the 150 mg once daily regimen can be used for up to 24 months, but for moderate hepatic impairment the label allows up to 6 months. The longer figures are the limits for other patients, and the shortest is arbitrary. In practice, check hepatic status before choosing the regimen and plan the stop date and bone review at the start.

> *Key principle:* With moderate hepatic impairment, elagolix 150 mg once daily is limited to 6 months.

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336. Answer: A. Elagolix is a gonadotropin-releasing hormone antagonist that suppresses ovarian hormones, but it is not labeled as a contraceptive. The label tells women to use effective nonhormonal contraception during treatment. The distractors assume the drug is contraceptive, add a hormonal pill, or leave her unprotected. In practice, offer condoms or a copper intrauterine device, and stop the drug if pregnancy occurs.

> *Key principle:* Elagolix is not a contraceptive; advise effective nonhormonal contraception during treatment.

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337. Answer: C. Bremelanotide is a melanocortin receptor agonist taken as needed rather than daily, and it transiently raises blood pressure after each dose. The label directs use at least 45 minutes before anticipated sexual activity, no more than one dose within 24 hours, and not more than 8 doses a month. The bedtime option confuses it with flibanserin, and the other options change the timing or the monthly limit. In practice, give these limits in writing and review the blood pressure at follow-up.

> *Key principle:* Bremelanotide: at least 45 minutes before sex, one dose per 24 hours, no more than 8 doses a month.

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338. Answer: E. Bremelanotide transiently increases blood pressure and reduces heart rate after each dose, so pressure that is already high is a hazard. The label contraindicates it in uncontrolled hypertension or known cardiovascular disease. The distractors assume monitoring, observation or a lower frequency can make the use safe, or substitute a drug indicated for menopausal dyspareunia. In practice, control the hypertension first, then revisit the choice of therapy with her.

> *Key principle:* Uncontrolled hypertension contraindicates bremelanotide; treat the pressure first.

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339. Answer: B. Flibanserin acts on serotonin receptors in the brain, and a response, when it occurs, is expected within weeks. The label tells prescribers to discontinue it after 8 weeks if the patient does not report an improvement in symptoms. The distractors raise the dose, extend the trial, change the timing, or add a drug that is not indicated for desire. In practice, stop the drug at 8 weeks without benefit and revisit her contributors, such as relationship, mood and other medicines.

> *Key principle:* Stop flibanserin at 8 weeks if desire has not improved.

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340. Answer: D. Flibanserin is cleared by CYP3A4, so inhibitors raise its blood levels and the risk of hypotension and syncope. The label contraindicates its use with moderate or strong CYP3A4 inhibitors, and ketoconazole is a strong inhibitor. The distractors try to engineer a safe overlap by dose, timing, surveillance or dose spacing. In practice, wait until the antifungal is finished and cleared, or choose another approach, and check every new medicine against the interaction list.

> *Key principle:* Flibanserin is contraindicated with moderate or strong CYP3A4 inhibitors such as ketoconazole.

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341. Answer: E. Ospemifene is an estrogen receptor agonist and antagonist with tissue-selective effects, and the tissue effects in breast are not settled enough to assume protection. The label states that it has not been adequately studied in women with breast cancer and should not be used in women with known or suspected breast cancer. The distractors assume a protective breast effect or add a drug that does not solve the issue. In practice, offer other measures, with oncology input about options.

> *Key principle:* Ospemifene is not for women with known or suspected breast cancer.

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342. Answer: C. Fezolinetant is a neurokinin 3 receptor antagonist with a boxed warning for liver injury, so liver status at baseline decides whether it can be used. Her ALT is about 2.7 times and her AST about 2.5 times the upper limit, and the label says not to start if either is at least 2 times the upper limit. The distractors apply the stopping rules or the monitoring schedule to a patient who should not start. In practice, work up the transaminase elevation and choose another nonhormonal option.

> *Key principle:* Do not start fezolinetant if ALT or AST is at least 2 times the upper limit of normal.

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343. Answer: A. Drug-induced liver injury from fezolinetant, though rare, can appear within the first months and later, which is why the boxed warning includes a monitoring plan. The label calls for follow-up hepatic tests monthly for the first 3 months and at 6 and 9 months after starting. The distractors drop the early frequent checks, lengthen them, or use a different drug's pattern. In practice, book the whole series at the first visit and tell her to stop the drug and call if symptoms of liver injury occur.

> *Key principle:* Fezolinetant: liver tests at baseline, monthly for 3 months, then at 6 and 9 months.

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344. Answer: D. Elinzanetant blocks neurokinin 1 and 3 receptors and is labeled for moderate to severe vasomotor symptoms due to menopause, with a warning for hepatic effects. The label advises a baseline check, no start if transaminases are at least 2 times the upper limit, and follow-up testing 3 months after initiation. The distractors apply the fezolinetant pattern or an arbitrary interval. In practice, schedule the 3-month test at prescribing, and note the labeled somnolence warning when advising her about driving.

> *Key principle:* Elinzanetant: baseline transaminases and a repeat 3 months after starting.

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345. Answer: A. The NAMS statement grades each nonhormone option by strength of evidence, and some behavioral measures help while others do not. It recommends cognitive behavioral therapy and clinical hypnosis at Level I, weight loss and stellate ganglion block at lower levels, and it does not recommend paced respiration, which is unlikely to provide benefit for hot flashes. The trap is to assume that every relaxation-based technique works because hypnosis and cognitive behavioral therapy do. In practice, direct her to the recommended behavioral programs.

> *Key principle:* NAMS 2023: cognitive behavioral therapy and clinical hypnosis are recommended; paced respiration is not.

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346. Answer: B. Compounded bioidentical preparations are mixed to order, so dose accuracy, purity and sterility depend on the pharmacy rather than on regulatory review. The 2022 NAMS statement lists minimal government regulation and monitoring, over- and under-dosing, impurities, and a lack of efficacy and safety data among its concerns. The distractors repeat marketing claims about testing, identity and protection. In practice, offer an approved estradiol and progestogen regimen that matches her preference for a natural-sounding molecule, since approved bioidentical products exist.

> *Key principle:* NAMS flags compounded bioidentical hormones for little oversight, dosing and purity risks, and missing safety data.

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347. Answer: C. Pregabalin is related to gabapentin, which the statement recommends, so a similar effect is a reasonable but unproven assumption. The NAMS statement states that pregabalin is not recommended. The distractors extend the gabapentin recommendation to pregabalin or place it in a rescue role. In practice, if she wants a neuroactive drug, gabapentin or an agent with Level I evidence is the one to discuss, with attention to her other conditions.

> *Key principle:* NAMS 2023 does not recommend pregabalin for vasomotor symptoms.

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348. Answer: E. Fecundity declines with age, and delay costs more when the remaining reproductive window is short. The ACOG opinion states that for women older than 40 years more immediate evaluation and treatment are warranted, and older than 35 years an expedited evaluation. The distractors apply the 12-month rule, shorten it to a middle value, or treat before testing. In practice, start with semen analysis, assessment of ovulation and tubal and uterine evaluation without waiting.

> *Key principle:* Over age 40, evaluate and treat infertility immediately rather than waiting a year.

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349. Answer: D. A low positive predictive value means many films read as nonpatent come from spasm or technique rather than true occlusion. ACOG gives the values 38% and 94% for positive and negative prediction, and says an HSG that demonstrates nonpatency may require further evaluation. The distractors treat the film as definitive, misread the negative value, or overstate what the test shows about the cavity. In practice, confirm with repeat imaging, sonographic assessment or laparoscopy before committing her to treatment for tubal disease.

> *Key principle:* Nonpatency on HSG has a low positive predictive value, so confirm before treating tubal disease.

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350. Answer: C. Irregular cycles signal ovulatory dysfunction, and thyroid disease and prolactin excess are treatable causes that are checked before ovulation induction. The ACOG opinion states that serum thyrotropin should be measured in women with ovulatory dysfunction. The distractors are tests that are not part of the routine workup, such as antibodies, a luteal biopsy, or a progesterone level drawn at a time when it would not be informative. In practice, send thyrotropin, and prolactin as well given irregular menses, before choosing an ovulation induction agent.

> *Key principle:* Check thyrotropin in an infertile woman with ovulatory dysfunction.

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351. Answer: E. Male factor contributes in a large share of couples, and semen analysis is simple, inexpensive and noninvasive, so it belongs at the start. The ACOG opinion states that the minimal evaluation of the male partner includes a reproductive history and semen analysis, and that a women's health specialist may reasonably order the analysis. The distractors delay it, require referral, or substitute tests that are obsolete or not first-line. In practice, send him for semen analysis in parallel with her workup.

> *Key principle:* Order a semen analysis early; the gynecologist can do it.

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352. Answer: B. Ovarian reserve tests estimate the quantity of remaining follicles, not the quality of the oocytes or the chance of a given pregnancy. The ACOG opinion says that poor results do not necessarily predict an inability to achieve a live birth, and that they have no definitive criteria for diminished reserve. The distractors turn a prognostic marker into a diagnosis of sterility or add an invalid timing rule. In practice, use the result to counsel on time limits and to guide the stimulation plan, and do not use it to deny treatment.

> *Key principle:* Abnormal ovarian reserve tests predict response, not an inability to conceive.

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353. Answer: A. Breast cancer is a hormonally sensitive tumor, so methods that deliver hormones carry theoretical risk of progression, with the least concern for the local release of levonorgestrel. The 2024 US MEC lists the copper device as category 1 for current breast cancer and the levonorgestrel device as category 4. The distractors are hormonal methods, each of which is a higher category for her. In practice, offer the copper device and coordinate timing with her oncology team.

> *Key principle:* In current breast cancer the copper IUD is category 1 and hormonal methods are not.

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354. Answer: E. Ulipristal is a progesterone receptor modulator, and a progestin taken soon afterward may compete with it and blunt its effect on ovulation. The recommendations say hormonal contraception should be started or resumed no sooner than 5 days after ulipristal, with abstinence or barrier methods for the next 7 days. The distractors start immediately, as is acceptable after levonorgestrel, or delay far beyond need. In practice, offer condoms as a bridge until the 5 days have passed.

> *Key principle:* After ulipristal, wait 5 days before hormonal contraception, then use backup for 7 days.

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355. Answer: B. The 13-week injection interval is set with a margin, so a short delay does not by itself create a pregnancy risk. The recommendations allow a repeat injection up to 2 weeks late, which is 15 weeks from the last injection, without extra contraceptive protection. The distractors add backup, testing or a restart that the grace window makes unnecessary. In practice, count the weeks from the last injection, and use the grace period only when the patient is within it.

> *Key principle:* A DMPA reinjection up to 15 weeks after the last one needs no added protection.

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356. Answer: D. Infection after insertion is uncommon, so routine prophylaxis adds little to a screened patient. The recommendations state that prophylactic antibiotics are generally not recommended for copper or levonorgestrel intrauterine device placement. The distractors name specific antibiotics or an unnecessary test. Placement can proceed at the same visit when screening is current. In practice, screen for sexually transmitted infection by risk, reassure her, and place the device.

> *Key principle:* Do not give routine antibiotics before IUD insertion.

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357. Answer: A. Combined pills work by suppressing ovulation, which takes about a week of hormone exposure to establish unless pills start early in the cycle. The recommendations say that pills started within 5 days of the onset of bleeding need no backup, and when started later the patient needs abstinence or barrier methods for the next 7 days. The distractors give no backup, a longer wait, or a wrong duration. In practice, quick start is acceptable once pregnancy is reasonably excluded.

> *Key principle:* A combined pill started more than 5 days after bleeding began needs 7 days of backup.

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358. Answer: C. Levonorgestrel pills appear to lose effect at higher body weight, and ulipristal keeps more of its effect. The recommendations say the levonorgestrel formulation might be less effective than ulipristal among women with obesity, and ulipristal is more effective from 3 to 5 days after intercourse. The distractors are the other pill regimens, including a split dose, the combined estrogen-progestin regimen, and a higher levonorgestrel dose that the recommendations do not list. In practice, offer ulipristal, and the copper device if it can be placed within 5 days.

> *Key principle:* With obesity, ulipristal is preferred to levonorgestrel for emergency contraception.

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359. Answer: C. Pelvic inflammatory disease in an intrauterine device user is treated with the same antibiotics as in other women. The CDC guidelines state that the device does not need to be removed, and that providers should consider removal if there is no clinical improvement within 48 to 72 hours of starting treatment. The distractors remove it early, conditionally on test results, or abandon scheduled reassessment. In practice, treat promptly, reexamine within 72 hours, and counsel on retesting at 3 months.

> *Key principle:* In PID, an IUD can stay in place if she improves within 48 to 72 hours.

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360. Answer: B. Reinfection from untreated partners is a leading cause of repeat chlamydial infection, so partner management is part of treatment. The CDC guidelines say partners with sexual contact during the 60 days before symptom onset or diagnosis should be referred for evaluation, testing and presumptive treatment. Partner B, 3 weeks before onset, qualifies, while partners C and D fall outside the window. The distractors shorten, lengthen or skip the lookback. In practice, document dates and offer expedited partner therapy where law allows.

> *Key principle:* Treat sex partners from the 60 days before symptoms or diagnosis.

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361. Answer: D. Gonorrhea and chlamydia often coexist, and a single injection treats only the first. The CDC guidelines recommend ceftriaxone 500 mg intramuscularly once for persons under 150 kg, and add doxycycline 100 mg twice daily for 7 days if chlamydia has not been excluded. The distractors are an older regimen, an oral cephalosporin used when ceftriaxone is unavailable, and an alternative reserved for cephalosporin allergy. In practice, treat both infections and retest at 3 months.

> *Key principle:* Treat gonorrhea with ceftriaxone 500 mg and add doxycycline when chlamydia is not excluded.

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362. Answer: A. Type-specific serology cannot show when or where infection occurred, and it is not designed to screen people who have no symptoms. The CDC guidelines state that HSV-2 serologic screening among the general population is not recommended. The distractors order serology, IgM, culture or PCR, none of which suits an asymptomatic person without a lesion. In practice, counsel her about condoms and about testing for specific indications, such as a partner with known genital herpes.

> *Key principle:* Do not screen asymptomatic women for HSV-2 with serology.

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363. Answer: E. Nontreponemal titers are semiquantitative, and a change must exceed the variation of the test to mean anything. A fall from 1:64 to 1:16 is two dilutions, which is fourfold, and the CDC guidelines state that a fourfold change in titer, equivalent to two dilutions, is considered significant. The distractors set a threshold that is too low, demand seronegativity, or escalate to lumbar puncture or retreatment. In practice, continue scheduled follow-up titers on the same assay.

> *Key principle:* A fourfold (two-dilution) fall in nontreponemal titer is a significant change.

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364. Answer: D. Trichomonas reinfection is common, including in women whose partners were treated, so asymptomatic recurrence matters. The CDC guidelines recommend retesting all sexually active women approximately 3 months after the initial treatment, and say that if a nucleic acid test is used it should not be done before 3 weeks after completing treatment. The distractors skip testing, test too early, or wait too long. In practice, schedule the 3-month visit and use a nucleic acid test.

> *Key principle:* Retest women with trichomoniasis about 3 months after treatment.

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365. Answer: E. Thelarche between 7 and 8 years is common, and most of these girls progress slowly or not at all, so immediate testing and treatment add cost and worry without benefit. The 2026 Endocrine Society guideline suggests watchful waiting with periodic physical examinations rather than immediate evaluation for girls with thelarche at ages 7.0 to 8.0 years. The distractors start testing or treatment in a girl without signs of rapid progression. In practice, reexamine her in a few months and evaluate if the pace accelerates.

> *Key principle:* A girl with thelarche at 7 to 8 years can be observed with periodic examinations rather than evaluated at once.

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366. Answer: A. Central precocious puberty reflects early activation of the hypothalamic-pituitary-gonadal axis, which is reflected in the luteinizing hormone level. The guideline suggests an initial evaluation with basal luteinizing hormone by ultrasensitive assay rather than starting with a gonadotropin-releasing hormone stimulation test. The stimulation test is a later step if basal luteinizing hormone is low while suspicion remains high. The distractors are tests that are less informative or unrelated. In practice, a basal sample can be drawn at the same visit.

> *Key principle:* Start the central precocious puberty workup with basal LH by ultrasensitive assay.

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367. Answer: C. Intracranial lesions are an uncommon cause of precocious puberty in girls whose onset falls between 6 and 8 years, and a scan without neurologic signs is unlikely to change care. The guideline suggests that brain MRI should not be routinely performed in girls aged 6.0 to 8.0 years without central nervous system findings. The distractors make imaging universal, tie it to bone age, schedule it repeatedly, or substitute CT. In practice, image when symptoms, signs or a younger age point to a lesion.

> *Key principle:* Skip routine brain MRI in girls aged 6 to 8 years with central precocious puberty and no neurologic findings.

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368. Answer: B. Suppression is meant to hold puberty until a more typical age, so it is stopped near that age. The guideline suggests against routinely continuing gonadotropin-releasing hormone agonist treatment beyond chronological age 10.0 to 11.0 years in girls. The distractors extend the course, add growth hormone, which the guideline also advises against adding routinely, or escalate the dose. In practice, plan the stop date at the outset and review it with the family.

> *Key principle:* In girls, do not routinely continue GnRH agonist therapy beyond about age 10 to 11 years.

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369. Answer: D. A 45,X result in a virilizing girl raises the possibility of a small Y-bearing cell line missed on standard karyotyping, and Y material carries a risk of gonadoblastoma. The 2024 international guideline recommends screening for Y chromosomal material by PCR or another molecular method in Turner syndrome with a 45,X karyotype and signs of virilization, and it does not endorse searching for Y material in every 45,X patient. Repeat karyotyping, adrenal testing and surgery without proof skip the targeted test. If Y material is found, the guideline calls for individualized decision-making about gonadectomy.

> *Key principle:* Test for Y chromosomal material by PCR when a 45,X girl shows virilization, and do not test every 45,X patient.

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370. Answer: E. Tamoxifen acts as an estrogen agonist in the endometrium, whereas raloxifene does not stimulate it. The USPSTF statement says tamoxifen and raloxifene both increase venous thromboembolism, with a larger effect for tamoxifen, and that only tamoxifen increases endometrial cancer risk in women with a uterus. The distractors reverse the drugs or deny the shared thromboembolic risk. In practice, women with a uterus who choose tamoxifen need prompt evaluation of any abnormal bleeding.

> *Key principle:* Tamoxifen, but not raloxifene, raises endometrial cancer risk in women with a uterus.

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371. Answer: A. Dense tissue masks cancers on mammography and independently raises risk, but no trial has shown that extra imaging lowers deaths or that its false-positive biopsies are worth the cost. The 2024 USPSTF statement concludes that current evidence is insufficient to assess the balance of benefits and harms of supplemental screening, an I statement. The distractors turn that uncertainty into a positive or negative recommendation. In practice, discuss her risk and values, and note that any supplemental test is a shared decision.

> *Key principle:* For dense breasts the USPSTF finds the evidence insufficient on supplemental screening, so decide jointly.

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372. Answer: C. Pathologic discharge is spontaneous, unilateral, from one duct, or bloody, and it can come from a papilloma or from a carcinoma. For pathologic discharge at age 40 or older, the ACR criteria rate breast ultrasonography, diagnostic digital breast tomosynthesis and diagnostic mammography as usually appropriate, and rate ductography, MRI and FDG-PET as usually not appropriate. The distractors name tests that rank lower or recommend delay. In practice, image first, then refer for surgical evaluation of the duct.

> *Key principle:* Pathologic discharge at 40 or older: diagnostic mammography or tomosynthesis plus ultrasound.

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373. Answer: B. At young ages dense glandular tissue limits the value of mammography, and ultrasonography can show a dilated duct or intraductal mass without radiation. The ACR criteria rate ultrasonography as usually appropriate for pathologic discharge in patients younger than 30, and rate mammography and tomosynthesis as usually not appropriate for females in that group. The distractors are the modalities used at older ages or lower-ranked tests. In practice, start with ultrasonography and add other studies on findings.

> *Key principle:* Pathologic nipple discharge under age 30 in a female: start with ultrasound.

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374. Answer: B. Physiologic discharge is bilateral, from multiple ducts, and elicited only by squeezing, and it carries a low chance of a neoplasm. In the ACR criteria for physiologic discharge, all listed imaging procedures, including ultrasonography, mammography, MRI and ductography, are rated usually not appropriate. The distractors apply the imaging used for pathologic discharge. In practice, reassure her after the examination, review medicines and thyroid or prolactin causes if the history suggests them, and advise returning if the discharge becomes spontaneous or bloody.

> *Key principle:* Physiologic nipple discharge usually does not need imaging.

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OP3.2. Endocrine Disorders

375. Answer: A. AMH reflects the antral follicle pool, and the 2023 guideline allows it as a marker of polycystic ovarian morphology in adults. It does not make AMH a free-standing test. The guideline states it should not be used as a single test for diagnosis. This woman ovulates regularly and has no hyperandrogenism, so she fulfills no clinical feature that AMH could complete. The distractors treat a biomarker as a replacement for the diagnostic features. In practice, count irregular cycles and androgen excess first, and use AMH or ultrasound only as the morphology component.

> *Key principle:* A high AMH never diagnoses PCOS by itself.

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376. Answer: D. The guideline uses irregular cycles and hyperandrogenism as the core of the adult diagnosis once mimics such as thyroid disease, hyperprolactinemia and nonclassic adrenal hyperplasia are excluded. When both features are present, imaging adds nothing to the label. She has both, and her mimic screen is negative. The distractors imply that a morphology criterion must be added in every case, which applies just when one of the two clinical features is missing. Practically, she can be diagnosed today and moved on to metabolic screening and treatment.

> *Key principle:* With irregular cycles and hyperandrogenism, skip the ultrasound and diagnose PCOS.

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377. Answer: C. In adolescents the guideline requires both irregular cycles and hyperandrogenism, and it advises against relying on ultrasound or AMH to make the diagnosis. A girl with PCOS features who does not meet criteria can be labeled at increased risk and followed. Reassessment is advised at or before full reproductive maturity, which it places at 8 years after menarche. The traps are overdiagnosis from cycle irregularity alone and false reassurance that ends follow-up. In practice, document the risk label, address weight and cycles, and recheck on a schedule.

> *Key principle:* Adolescents with partial features are at risk of PCOS and are reassessed by 8 years after menarche.

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378. Answer: E. Insulin resistance in PCOS is partly independent of weight, so glycemic status is assessed at diagnosis in adults and adolescents. The guideline names the 75-g oral glucose tolerance test as the most accurate way to do this. Fasting glucose and A1c miss many cases of impaired glucose tolerance, and fasting insulin has no validated threshold for clinical use. The error is assuming that low BMI removes the need to test. In practice, order the glucose challenge at diagnosis and repeat on a risk-based schedule.

> *Key principle:* Use the 75-g OGTT to assess glycemic status in PCOS.

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379. Answer: D. Women with PCOS have higher rates of depression and anxiety than other women, and the burden is partly independent of weight and hirsutism. The guideline recommends screening adults and adolescents for depression and adults for anxiety, using validated tools suited to the region. Waiting for a spontaneous complaint misses most cases, and treating without assessment skips diagnosis. Practically, build a brief validated questionnaire into the diagnosis visit and repeat it based on clinical judgment.

> *Key principle:* Screen everyone with PCOS for depression and anxiety.

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380. Answer: E. Hirsutism treatment is slow because hair follicles cycle over months. The guideline suggests antiandrogens, with effective contraception, when response is suboptimal after a minimum of 6 months of a combined pill and cosmetic therapy. At 4 months she has not yet had an adequate trial. The distractors move early to add-on therapy or to drugs with weaker support, including metformin for hirsutism. In practice, explain the timeline, confirm adherence, and reassess at 6 months.

> *Key principle:* Give the pill and cosmetic care 6 months before adding an antiandrogen.

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381. Answer: A. Macroprolactin is a high-molecular-weight complex of prolactin and immunoglobulin that is detected by many assays but has little biologic activity. It explains many modestly elevated values in women without symptoms. The Endocrine Society guideline suggests assessing for macroprolactin in asymptomatic hyperprolactinemia. Imaging or drug treatment would expose her to cost and side effects for a laboratory artifact. In practice, ask the laboratory to screen for macroprolactin, usually by precipitation, before an imaging or treatment decision.

> *Key principle:* Check for macroprolactin before treating asymptomatic hyperprolactinemia.

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382. Answer: B. Dopamine antagonists such as metoclopramide and many antipsychotics block the inhibition of lactotrophs and raise prolactin. The guideline states that the first step in treating medication-induced hyperprolactinemia is to stop the drug or substitute another. Adding a dopamine agonist works against the prescribed agent and can complicate the underlying condition. Imaging is reserved for cases where the drug cannot be changed or levels stay high. In practice, coordinate with the prescriber and recheck prolactin after the change.

> *Key principle:* For drug-induced hyperprolactinemia, stop or replace the drug first.

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383. Answer: C. A symptomatic microprolactinoma responds to dopamine agonist therapy, which lowers prolactin and shrinks the tumor in most patients. The guideline recommends cabergoline in preference to other dopamine agonists, based on higher rates of normalization and better tolerability. Bromocriptine is the older option with more pregnancy experience, and pergolide is no longer used for this indication. Pills treat the hypoestrogenism but leave the tumor and prolactin unaddressed. Octreotide acts on somatostatin receptors and is not first-line here. In practice, start low and increase to biochemical response.

> *Key principle:* Cabergoline is the preferred dopamine agonist for prolactinoma.

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384. Answer: A. Microprolactinomas rarely enlarge, and in the absence of symptoms there is no hypogonadism or infertility to correct. The Endocrine Society guideline suggests not treating asymptomatic patients with microprolactinomas using dopamine agonists. Treating a number on a report adds side effects without benefit, and surgery carries risk for no gain. Estrogen does not control tumor growth. In practice, explain the benign course, recheck prolactin and symptoms periodically, and treat if symptoms or growth develop.

> *Key principle:* Do not treat an asymptomatic microprolactinoma with a dopamine agonist.

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385. Answer: D. Many microprolactinomas stay controlled after a prolonged period of normal prolactin. The Endocrine Society guideline supports tapering and perhaps stopping dopamine agonist therapy in patients treated for at least 2 years, with normal prolactin and no visible tumor remnant. Recurrence is possible, so monitoring continues. Lifelong therapy, dose escalation or a switch of agents offer no advantage once control is sustained. In practice, taper gradually, check prolactin at intervals, and restart if levels or symptoms return.

> *Key principle:* After 2 years of control with no visible tumor, a dopamine agonist can be tapered.

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386. Answer: C. Breast development at 7 to 8 years lies close to the upper edge of normal timing, and many girls progress slowly or not at all. For girls with B2 in that age range, the guideline suggests watchful waiting through periodic examinations rather than immediate evaluation. Early testing or imaging produces false alarms and costs without changing outcomes in a child who is growing normally. The trap is treating any early breast bud as disease. In practice, schedule a recheck and evaluate if progression is rapid.

> *Key principle:* At 7 to 8 years, a stable B2 girl is observed and reexamined.

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387. Answer: B. Central precocious puberty reflects early activation of the hypothalamic-pituitary-gonadal axis, and the pubertal rise in luteinizing hormone is its best marker. The guideline suggests starting with a basal LH by ultrasensitive assay rather than a stimulation test, which is reserved for the next step if basal LH is low and suspicion remains. Estradiol is insensitive early in puberty, and the other tests answer different questions. In practice, draw the basal LH first and escalate only when needed.

> *Key principle:* Start the evaluation of suspected central precocious puberty with a basal ultrasensitive LH.

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388. Answer: E. Intracranial lesions are an uncommon cause of central precocious puberty in girls whose onset falls between 6 and 8 years, and most are idiopathic. Routine MRI therefore finds few lesions in a child with a normal neurologic examination. The guideline suggests that brain MRI not be routinely performed in girls aged 6 to 8 years without central nervous system findings. Younger age, rapid progression or neurologic signs would change the answer. In practice, reserve imaging for those features.

> *Key principle:* A neurologically normal girl aged 6 to 8 with central precocious puberty does not need routine brain MRI.

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389. Answer: C. The goal of GnRH agonist therapy is to preserve growth potential and allow puberty to resume at a socially acceptable age. The guideline suggests against routinely continuing treatment beyond a chronological age of 10 to 11 years in girls, or a bone age of 11 to 12 years. Her chronological age and bone age are both past that window. Continuing to a later age or bone age prolongs suppression without clear benefit. In practice, set a stopping age with the family and expect menarche within about a year or two afterward.

> *Key principle:* Stop GnRH agonist therapy around chronological age 10 to 11 in girls.

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390. Answer: B. Slowed growth is expected when puberty is suppressed, and it does not by itself signal growth hormone deficiency. The guideline suggests against routinely adding growth hormone to GnRH agonist therapy, with an exception for children who have a separate established indication. Adding the drug expands cost and injection burden without proven benefit in unselected children. In practice, reassure the family, confirm growth hormone deficiency only if clinically suspected, and keep the focus on timing of the stop date.

> *Key principle:* Do not routinely add growth hormone to a GnRH agonist.

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391. Answer: E. Agranulocytosis from thionamides is rare, but it appears abruptly and usually within the first few months. It presents with fever and pharyngitis and is easy to dismiss as an ordinary viral illness. The ATA guideline states that a differential white cell count should be obtained during febrile illness and at the onset of pharyngitis. Patients are also told at the outset to stop the drug and call if such symptoms occur. Dose changes, culture alone or definitive therapy do not address the risk. In practice, hold the drug pending the count.

> *Key principle:* Fever or sore throat on methimazole needs a differential white count.

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392. Answer: D. Graves disease does not protect against coexisting thyroid cancer, and a cold nodule with suspicious sonographic features carries real malignancy risk. The ATA guideline lists coexisting thyroid cancer or suspicion of it among the definite contraindications to radioactive iodine. Treating first would deliver radiation to a cancer that needs surgery. Her pregnancy test is negative, so the issue is the nodule, not pregnancy. In practice, sample the nodule and let the result steer surgery versus another option.

> *Key principle:* Suspected thyroid cancer is a contraindication to radioactive iodine and needs a tissue diagnosis.

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393. Answer: A. Thyrotoxicosis from a toxic adenoma or a toxic multinodular goiter is managed differently from Graves disease. A scan shows focal, autonomous uptake with suppression of the rest of the gland. The guideline recommends a 123I or 99mTc pertechnetate scan when the presentation suggests a toxic adenoma or multinodular goiter. Contrast CT delivers an iodine load that complicates later radioiodine therapy. Therapy without defining the cause risks the wrong treatment. In practice, scan, then choose between radioiodine, surgery and drugs.

> *Key principle:* A nodular gland with thyrotoxicosis needs a functional uptake scan.

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394. Answer: A. Calcium carbonate and ferrous sulfate bind levothyroxine in the gut and reduce its absorption. A rise in TSH after adding either one is a classic pattern. The ATA guideline advises separating levothyroxine from potentially interfering medications and supplements where feasible. Raising the dose treats the interaction with more drug, and thyroid extracts or liothyronine are not first-line replacements. Stopping iron could worsen anemia from heavy menses. In practice, move the supplements several hours away and recheck TSH after about 6 weeks.

> *Key principle:* Space levothyroxine away from calcium and iron.

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395. Answer: E. Thionamides lower hormone output slowly, over several weeks, and adrenergic symptoms persist during that time. Beta-blockade controls heart rate, tremor and anxiety and is recommended by the ATA guideline in all patients with symptomatic thyrotoxicosis, especially elderly patients. Benzodiazepines mask anxiety without addressing the cause, and diuretics do not treat adrenergic symptoms. A second antithyroid drug adds toxicity, and levothyroxine would worsen the condition. In practice, start the beta-blocker, check contraindications, and stop it once euthyroid.

> *Key principle:* Treat adrenergic symptoms of thyrotoxicosis with a beta-blocker.

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396. Answer: D. Tirzepatide delays gastric emptying, which can lower the absorption of oral medications, including hormonal contraceptives. The label advises patients using oral hormonal contraceptives to switch to a non-oral method or add a barrier method, and it ties the period of caution to initiation and each dose escalation. Doubling the pill or relying on withdrawal does not solve the absorption problem. Counseling at the start prevents unplanned pregnancy. In practice, offer an intrauterine device, implant, or condoms during dose changes.

> *Key principle:* Add backup or go non-oral when starting or increasing tirzepatide.

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397. Answer: C. In rodents, tirzepatide caused dose-dependent thyroid C-cell tumors, and the effect in humans is unknown. For that reason the label carries a boxed warning and contraindicates the drug in people with a personal or family history of medullary thyroid carcinoma or in those with MEN 2. Surveillance with ultrasound or calcitonin, a lower dose, or a negative genetic test does not remove the contraindication. In practice, choose an agent from another class and document the reason.

> *Key principle:* Family history of medullary thyroid carcinoma contraindicates tirzepatide.

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398. Answer: B. Long-term metformin can reduce intestinal absorption of vitamin B12, causing macrocytic anemia and neuropathy that mimics diabetic neuropathy. The label advises measuring hematologic parameters annually and vitamin B12 at 2 to 3 year intervals and managing abnormalities. Her macrocytosis and sensory symptoms fit deficiency. Folate testing alone could miss it, and A1c or spine imaging does not explain macrocytosis. In practice, check B12 in long-term users who develop anemia or neuropathy, and replace it if low.

> *Key principle:* Check vitamin B12 in long-term metformin users with anemia or neuropathy.

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399. Answer: C. Metformin is cleared by the kidney, and accumulation raises the risk of metabolic lactic acidosis. The label contraindicates use at an eGFR under 30 and does not recommend starting therapy when eGFR is between 30 and 45. Reduced starting doses and lactate monitoring are not a label solution, and adding other drugs does not change the renal threshold. In practice, pick another agent for initiation in this range and reassess renal function regularly if metformin was begun earlier.

> *Key principle:* Do not initiate metformin with an eGFR of 30 to 45.

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400. Answer: E. Iodinated contrast can cause transient kidney injury, and impaired clearance of metformin raises the risk of lactic acidosis. The label states that the drug may need to be discontinued at the time of, or before, iodinated contrast procedures, with attention to patients whose eGFR is 30 to 60. Renal function is rechecked before resuming. Continuing, prolonged holds and permanent changes are unnecessary. In practice, give written instructions and restart once creatinine is stable.

> *Key principle:* Hold metformin around iodinated contrast when renal function is reduced.

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401. Answer: B. Anovulation in insulin-resistant women is partly driven by hyperinsulinemia, and lowering insulin resistance can bring back ovulation. The label states that metformin may result in ovulation in some anovulatory premenopausal women, so unplanned pregnancy is possible even when cycles are infrequent. Oligomenorrhea is not contraception. In practice, discuss effective contraception when starting metformin in a woman who does not want pregnancy, and review medication safety if she plans one.

> *Key principle:* Infrequent periods do not prevent pregnancy once metformin is started.

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OP3.3. Menstrual Disorders

402. Answer: A. Tranexamic acid is an antifibrinolytic that reduces menstrual blood loss, and estrogen-containing contraceptives are prothrombotic. Together they add venous and arterial thrombotic risk. The label states that concomitant use of tranexamic acid with combined hormonal contraceptives is contraindicated. It does not describe an absorption block, enzyme induction, or potassium effect. In practice, choose between the two approaches, for example a levonorgestrel intrauterine system or a nonsteroidal drug, rather than layering them.

> *Key principle:* Do not combine tranexamic acid with a combined hormonal contraceptive.

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403. Answer: D. Tranexamic acid has been associated with visual and ocular adverse effects, and the label lists thromboembolic events among its warnings. A new visual change in a patient taking it is not trivial. The label directs that the drug be discontinued immediately if visual or ocular symptoms occur. Dose reduction, antihistamines or taking it with meals do not address the concern. In practice, stop the drug, arrange prompt eye assessment, and choose another approach to the bleeding.

> *Key principle:* New visual symptoms on tranexamic acid mean stop it now.

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404. Answer: A. The FIGO system separates symptoms from causes. Chronic AUB is bleeding from the uterine corpus that is abnormal in duration, volume, frequency or regularity and has been present for the majority of the preceding 6 months. Her pattern fits, and she does not need immediate intervention to stop bleeding. Heavy menstrual bleeding is a symptom and not a diagnosis, and intermenstrual bleeding describes timing between regular cycles. In practice, label the pattern, then search for the PALM-COEIN cause.

> *Key principle:* Chronic AUB means abnormal bleeding for most of the past 6 months.

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405. Answer: D. The PALM-COEIN system sorts causes into structural and nonstructural groups. FIGO assigns bleeding linked to selected systemic drugs or to intrauterine systems and devices to the iatrogenic category. Her device is correctly placed, her imaging shows no structural lesion, and her coagulation screen is normal, so no other category fits. Misclassification as coagulopathy or adenomyosis would prompt needless testing. In practice, record AUB-I and decide whether to keep the device or change method.

> *Key principle:* Bleeding tied to an intrauterine device is categorized as iatrogenic.

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406. Answer: B. The endometrial category is meant for bleeding that arises from a primary endometrial disorder, and FIGO states that chronic endometritis should be categorized there. Ovulation is intact, so AUB-O does not apply. Biopsy excludes hyperplasia or cancer, and imaging excludes a polyp. The not otherwise classified group is for rare or poorly defined entities, which this is not. In practice, treat the infection and record AUB-E so that the cause is not lost.

> *Key principle:* Chronic endometritis belongs in the endometrial category.

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407. Answer: C. Measured blood loss correlates poorly with how much bleeding bothers a woman, and few clinicians can measure it. FIGO adopted a patient-centered definition for the symptom: excessive menstrual blood loss that interferes with quality of life. The 80-mL threshold is an older research standard that has been superseded in practice. Duration alone and pad counts are not the definition. Anemia is a consequence, not the definition. In practice, ask about impact and evaluate those who say it is too heavy.

> *Key principle:* Heavy menstrual bleeding is defined by its effect on quality of life.

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408. Answer: E. Hypoestrogenism from low energy availability reduces bone formation and increases fracture risk, often before the bone loss is clinically apparent. The Endocrine Society guideline recommends baseline bone mineral density measurement by DXA in patients with 6 or more months of amenorrhea. She has passed that threshold and has already fractured twice. Vitamin D levels do not measure bone mass, plain films are insensitive, and waiting longer delays recognition. In practice, scan now and address energy availability.

> *Key principle:* Get a DXA after 6 months of hypothalamic amenorrhea.

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409. Answer: A. Withdrawal bleeding on a pill is a hormonal effect, not a sign that the hypothalamic axis or energy balance has recovered. It can give false reassurance while the underlying deficit continues, and oral ethinyl estradiol does not reliably protect bone in this setting. The guideline suggests against using oral contraceptives in functional hypothalamic amenorrhea for the sole purpose of regaining menses or improving bone density. Treatment centers on nutrition, reduced training load and psychological support. In practice, explain that bleeding on pills is not recovery.

> *Key principle:* A pill-induced bleed is not recovery, and pills are not advised to regain menses or protect bone in hypothalamic amenorrhea.

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410. Answer: E. Estrogen deficiency drives bone loss in this condition, and replacing it physiologically can improve density. The guideline suggests short-term transdermal estradiol with a cyclic oral progestin, and not oral contraceptives or ethinyl estradiol. It also suggests against bisphosphonates, denosumab, testosterone and leptin to improve bone density, partly because bisphosphonates persist in bone for years in a woman who may later conceive. In practice, use transdermal estradiol while addressing the cause, and recheck bone density.

> *Key principle:* For bone loss in hypothalamic amenorrhea, use transdermal estradiol with a cyclic progestin.

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411. Answer: D. Functional hypothalamic amenorrhea is a diagnosis of exclusion, and a structural cause such as a pituitary or hypothalamic tumor must be considered when red flags are present. The Endocrine Society guideline recommends brain MRI with pituitary cuts and contrast in patients with a history of severe or persistent headaches, among other findings. CT and plain films are insensitive for the sella region. Assuming a benign cause because of her lifestyle risks missing a mass. In practice, image before labeling.

> *Key principle:* Severe or persistent headaches with amenorrhea call for pituitary MRI.

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412. Answer: B. Functional hypothalamic amenorrhea is diagnosed by excluding pregnancy, thyroid disease, hyperprolactinemia, primary ovarian insufficiency and structural pathology. The guideline lists beta-hCG with screening chemistry and a blood count, then TSH, free T4, prolactin, FSH and estradiol. Low or normal gonadotropins with low estradiol point to a hypothalamic cause, whereas a high FSH points to the ovary. Androgen, adrenal and inflammatory panels are directed at other problems. In practice, run the full panel before assigning the label.

> *Key principle:* Exclude pregnancy, thyroid, prolactin and ovarian causes before diagnosing hypothalamic amenorrhea.

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413. Answer: C. Primary dysmenorrhea is driven by excess prostaglandin production, so prostaglandin synthesis inhibitors work best when given early, before the uterine contractions peak. The CMAJ review advises full-strength doses with food on a regular schedule without skipped doses, starting 1 to 2 days before menses if predictable or at the first sign of bleeding or pain, and continuing for the first 2 to 3 days. Waiting for severe pain gives partial relief. In practice, give explicit written timing.

> *Key principle:* Start NSAIDs early and take them on schedule through the first 2 to 3 days.

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414. Answer: A. Primary dysmenorrhea begins within a year or two of menarche, is limited to menses and responds to NSAIDs. Onset immediately with menarche, midcycle or acyclic pain, and a family or personal history of renal or congenital anomalies point to secondary causes such as obstructive müllerian anomalies. The CMAJ review states that such features should prompt pelvic ultrasonography and may warrant referral. Laparoscopy is not the initial test, CA-125 is nonspecific, and prolonged empiric therapy delays diagnosis. In practice, image early.

> *Key principle:* Features of secondary dysmenorrhea call for pelvic ultrasound.

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415. Answer: D. Up to 90 percent of dysmenorrhea in adolescents occurs without pelvic pathology, and it is mediated by excess prostaglandin production. Prostaglandins increase myometrial contractions and sensitize pain fibers during the first days of menses, which is why inhibitors work. Implants, adenomyosis, outflow obstruction and inflammation are secondary causes that usually bring additional features. In practice, reassure her with a normal examination and ultrasound, and treat with prostaglandin inhibitors.

> *Key principle:* Primary dysmenorrhea is prostaglandin-mediated.

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416. Answer: B. Aspirin-exacerbated respiratory disease reflects cross-reactivity among drugs that inhibit cyclooxygenase-1, and a reaction can occur with any nonselective NSAID. The naproxen label contraindicates use in patients with a history of asthma, urticaria or allergic-type reactions after taking aspirin or other NSAIDs. Other nonselective agents carry the same risk. Hormonal therapy is a first-line alternative in the CMAJ review of painful periods in adolescents. In practice, record the allergy, offer a hormonal method, and refer for allergy care if needed.

> *Key principle:* Aspirin-triggered asthma contraindicates naproxen and other nonselective NSAIDs.

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417. Answer: E. Prostaglandins participate in follicular rupture, and inhibiting their synthesis can delay or prevent ovulation in some cycles. The label states that NSAIDs may be associated with a reversible delay in ovulation and advises considering withdrawal of NSAIDs in women who have difficulties conceiving. A combined pill would prevent the pregnancy she wants, and clomiphene treats an effect that can be removed. In practice, review her analgesic use at the start of any infertility evaluation.

> *Key principle:* Reconsider NSAID use in women who have trouble conceiving.

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418. Answer: C. Fezolinetant carries a boxed warning for hepatotoxicity, with liver injury reported after market release. The label requires hepatic laboratory testing before treatment and does not support starting when an aminotransferase is at least 2 times the upper limit of normal. Her ALT of 98 U/L is about 2.5 times the limit. Dose reduction and added hepatic protectants are not label options. In practice, evaluate the liver abnormality and choose a different nonhormonal approach, such as CBT or an antidepressant.

> *Key principle:* Do not start fezolinetant with an aminotransferase at 2 times the upper limit or higher.

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419. Answer: C. The boxed warning exists because serious liver injury has occurred, and early injury can be asymptomatic, so scheduled testing is the safeguard. The label directs follow-up liver tests monthly during the first 3 months, then at 6 and 9 months of treatment, with discontinuation if transaminases exceed 5 times the upper limit or if the aminotransferase is more than 3 times and bilirubin more than 2 times the upper limit. In practice, put the dates in the chart and educate her about jaundice and dark urine.

> *Key principle:* Check liver tests monthly for 3 months, then at 6 and 9 months on fezolinetant.

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420. Answer: B. Fezolinetant is cleared mainly by CYP1A2, and inhibitors of that enzyme raise exposure to the drug, which increases the risk of liver injury. The label contraindicates concurrent use with CYP1A2 inhibitors, and fluvoxamine is a strong example. Spacing doses, lowering the dose or testing the liver does not reverse enzyme inhibition. In practice, screen the medication list for CYP1A2 inhibitors, and offer another approach for vasomotor symptoms, such as CBT, which suits patients on stable psychiatric treatment.

> *Key principle:* CYP1A2 inhibitors contraindicate fezolinetant.

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421. Answer: A. Tamoxifen is converted to its active metabolite by CYP2D6, and drugs that inhibit that enzyme can reduce its effect. The statement notes that SSRIs may inhibit CYP2D6 and lists safer choices for women on tamoxifen as venlafaxine, desvenlafaxine, escitalopram and citalopram. Paroxetine is a strong inhibitor, and the paroxetine label also warns of reduced tamoxifen efficacy. Fluoxetine, bupropion and duloxetine also inhibit CYP2D6 to varying degrees. In practice, check the interaction before choosing a nonhormonal drug.

> *Key principle:* Choose venlafaxine, not paroxetine, for hot flashes in a woman on tamoxifen.

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422. Answer: E. Gabapentin is sedating and reduces the burden of vasomotor symptoms, which suits night sweats that fragment sleep. The statement rates gabapentin Level I and says bedtime dosing may be a good choice for women with disruptive sleep from hot flashes. Clonidine and pregabalin are listed as not recommended, as are supplements, herbal remedies and soy extracts, because the evidence is negative or insufficient. In practice, start with a low bedtime dose, titrate slowly, and warn her about dizziness and daytime drowsiness.

> *Key principle:* Bedtime gabapentin suits hot flashes that disrupt sleep.

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423. Answer: D. The paroxetine salt capsule is a low-dose product approved for moderate to severe vasomotor symptoms. The label states that it is not indicated for the treatment of any psychiatric condition and that safety and effectiveness of the lower dosage have not been established for any psychiatric condition. A boxed warning about suicidality in young adults also applies. In practice, assess mood with a validated tool and treat depression with a regimen designed for it.

> *Key principle:* Low-dose paroxetine treats hot flashes and is not approved for mood disorders.

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424. Answer: B. The drospirenone and ethinyl estradiol product has an FDA indication for PMDD, but the label limits it to females of reproductive potential who choose to use an oral contraceptive as their method of contraception. She does not need contraception, so the label indication does not apply. An SSRI is effective and has an established role for PMDD. Exposing a woman to estrogen without a contraceptive need adds thrombotic risk. In practice, match the drug to her goals and choose an SSRI.

> *Key principle:* The drospirenone and ethinyl estradiol pill is labeled for PMDD only if she wants a contraceptive.

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425. Answer: E. Drospirenone is a spironolactone analog with antimineralocorticoid activity, so it can raise potassium. Combining it with drugs that also raise potassium, such as ACE inhibitors, adds to that risk. The label states that women on long-term interacting medications should have serum potassium checked during the first treatment cycle and that the pill is not for patients predisposed to hyperkalemia. Her baseline is normal, so the drug is allowed with monitoring. In practice, order the potassium for the first cycle and counsel on symptoms.

> *Key principle:* Check potassium in the first cycle when drospirenone is combined with an ACE inhibitor.

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426. Answer: D. Serotonergic drugs relieve premenstrual symptoms within days, unlike their delayed effect in major depression, which makes cycle-limited dosing workable. The sertraline label offers two schedules for PMDD: every day throughout the cycle, or only during the luteal phase, starting 14 days before expected menstruation. Follicular, early-cycle and symptom-triggered schedules are not supported by the label. In practice, give her a calendar, begin on the correct day each month, and reassess by symptom diary.

> *Key principle:* Sertraline for PMDD may be taken daily or in the luteal phase only.

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427. Answer: A. Combining a serotonergic antidepressant with a monoamine oxidase inhibitor can precipitate serotonin syndrome, with agitation, hyperthermia, tremor and autonomic instability. The sertraline label contraindicates use of an MAOI or use within 14 days of stopping an MAOI. Six days is too soon, and a benzodiazepine does not prevent the reaction. In practice, document the date of the last MAOI dose, wait the full interval, and use nonpharmacologic strategies for the cycle in between.

> *Key principle:* Wait 14 days after stopping an MAOI before sertraline.

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428. Answer: C. Premenstrual dysphoric disorder trials of the drospirenone pill covered a short treatment period. The label states that its effectiveness for PMDD when used for more than three menstrual cycles has not been evaluated, and that it has not been evaluated for premenstrual syndrome. Her good response is real but outside the studied duration. In practice, continue if she benefits and wants contraception, reassess periodically with symptom ratings, and use an SSRI if control fails.

> *Key principle:* The label states that PMDD effectiveness beyond 3 cycles has not been evaluated.

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OP3.4. Vulvovaginal Conditions

429. Answer: B. The Amsel criteria are a homogeneous discharge, vaginal pH higher than 4.5, a positive whiff test and clue cells on microscopy. The guideline requires at least three of the four, and this patient has three, so the diagnosis is made without a Gram stain. The traps are demanding clue cells before calling it, which sets the bar too high, and treating a raised pH as proof of trichomoniasis, which has motile organisms or a positive NAAT. In practice, count the criteria at the bedside and treat when three are present.

> *Key principle:* Three of four Amsel criteria are enough to diagnose bacterial vaginosis.

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430. Answer: C. The Nugent score weighs three bacterial morphotypes on a Gram stain. Scores of 0 to 3 describe a Lactobacillus-predominant microbiota, and 7 to 10 indicates bacterial vaginosis. A score of 2 with a normal pH and a negative whiff test excludes bacterial vaginosis even though she has symptoms. The traps are treating on symptoms alone, reading the Gram-positive rods as yeast, and assuming protozoa cannot be excluded without testing. Her result sends the evaluation toward other causes, such as candidiasis, trichomoniasis, irritants or physiologic change, with targeted testing.

> *Key principle:* A low Nugent score means normal lactobacilli and excludes bacterial vaginosis.

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431. Answer: D. Older labels warned of a disulfiram-like reaction between metronidazole and alcohol. The CDC reviewed the evidence and concluded that the reaction has not been shown, so refraining from alcohol during metronidazole or tinidazole is unnecessary. The distractors preserve the superseded 24 or 72 hour abstinence rules, and the clindamycin switch solves a problem that does not exist. Take the drug with food if nausea occurs, and do not withhold an effective regimen or lose adherence because of an outdated alcohol warning.

> *Key principle:* The CDC no longer advises alcohol abstinence during metronidazole.

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432. Answer: A. Clindamycin vaginal cream is oil based. Oil degrades latex, and the guideline states that the cream might weaken latex condoms and diaphragms for 5 days after use. The wrong options either move the problem to polyurethane, which oil does not degrade, or invent a drug interaction with oral contraceptives, spermicide or intrauterine devices that the guideline does not describe. In practice, advise a backup method or abstinence during treatment and for 5 days afterward, or choose metronidazole gel if the barrier method must continue.

> *Key principle:* Clindamycin cream can weaken latex condoms for 5 days after use.

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433. Answer: E. Recurrence after treatment is common, and the CDC lists three recommended regimens: oral metronidazole, metronidazole gel and clindamycin cream. For a recurrence, the guideline says a different recommended regimen can be considered, and it also accepts retreatment with the original regimen. Changing the route can help if adherence or side effects were an issue. Oral estrogen does not treat the infection, fluconazole treats yeast, which is not the diagnosis, and a hysteroscopy has no role. Confirm recurrence by criteria and then retreat.

> *Key principle:* Recurrent bacterial vaginosis is retreated with a recommended regimen, and a different one can be considered.

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434. Answer: B. Wet-mount microscopy misses many infections because the organisms must be seen alive and moving, and its sensitivity is low compared with culture. The CDC states that NAATs are highly sensitive and detect more T. vaginalis infections than wet-mount microscopy among women. A negative wet mount therefore does not exclude trichomoniasis in a patient whose picture fits. Candida culture and Nugent scoring target other diagnoses, and cytology is a screening test. Send a NAAT and treat if positive.

> *Key principle:* A negative wet mount does not exclude trichomoniasis; use NAAT.

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435. Answer: A. For women, the CDC recommended regimen is metronidazole 500 mg twice daily for 7 days. Single-dose metronidazole was less effective in head-to-head comparison, which is why the 2 g single dose is now the regimen for men. The wrong choices are the superseded woman's regimen, an antifungal for yeast, a vaginal antibiotic that treats bacterial vaginosis but not this protozoan, and a macrolide for chlamydia. Give the 7-day course and treat current partners.

> *Key principle:* Women with trichomoniasis receive metronidazole twice daily for 7 days.

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436. Answer: C. Reinfection is frequent after treatment of trichomoniasis, so recurrence is often a new exposure and not treatment failure. The CDC recommends that all sexually active women be retested about 3 months after initial treatment, even if the partner was treated. A wet mount a week later lacks sensitivity, symptom-triggered testing misses asymptomatic reinfection, and partner treatment does not remove the need. Book the 3-month test at the time of treatment and test with a NAAT.

> *Key principle:* Retest women for trichomoniasis about 3 months after treatment.

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437. Answer: D. Candida vaginitis is associated with a normal vaginal pH of less than 4.5, unlike bacterial vaginosis and trichomoniasis, which raise the pH. Potassium hydroxide dissolves cellular debris and improves visualization of yeast and mycelia, and here it shows pseudohyphae. Antibiotic exposure is a recognized trigger. The other options fail on the findings: bacterial vaginosis would give a high pH and positive whiff test, trichomoniasis a high pH, and atrophic vaginitis occurs with low estrogen. Treat uncomplicated disease with a short azole course.

> *Key principle:* A normal pH with yeast on a KOH preparation indicates candidiasis.

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438. Answer: E. Candida is part of the normal vaginal flora in a substantial proportion of healthy women. The guideline states that identifying Candida by culture without symptoms or signs is not an indication for treatment. A positive culture alone does not define disease, and treating colonization exposes her to drug effects and can select resistant species. The distractors treat a laboratory result: single-dose azole, topical azole, boric acid and partner treatment. Treat symptoms and signs of vulvovaginitis, not a culture result.

> *Key principle:* Do not treat Candida found on culture in a woman without symptoms or signs.

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439. Answer: A. Severe vulvovaginal candidiasis, defined by extensive vulvar erythema, edema, excoriation and fissure formation, is classified as complicated disease. The CDC recommends either 7 to 14 days of topical azole or fluconazole 150 mg in two sequential oral doses. Single-dose and 1-day or 3-day regimens are for uncomplicated disease and fail more often in severe cases. Metronidazole treats bacterial vaginosis and trichomoniasis, not yeast. Choose a longer or repeated regimen when signs are severe.

> *Key principle:* Severe candidiasis needs 7 to 14 days of topical azole or two oral fluconazole doses.

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440. Answer: E. Non-albicans species such as C. glabrata respond less well to fluconazole, and the best regimen is not established. The CDC advises a longer course, 7 to 14 days, of a nonfluconazole azole given orally or topically. Boric acid 600 mg in a gelatin capsule daily for 3 weeks is held for recurrence after that approach. Weekly fluconazole is maintenance for recurrent albicans disease, and repeating the same drug repeats the failure. Identify the species and change the azole.

> *Key principle:* Non-albicans candidiasis needs a longer course of a nonfluconazole azole.

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441. Answer: B. Vulvovaginal candidiasis is more frequent in pregnancy. The CDC recommends only topical azole therapy applied for 7 days in pregnant women, and notes that epidemiologic studies link a single 150 mg fluconazole dose with spontaneous abortion. Oral azoles are therefore avoided despite their convenience. The wrong choices are oral fluconazole in either schedule, boric acid, which the guideline does not offer for pregnancy, and an oral azole with greater toxicity. Explain the reason for the cream or suppository and prescribe a 7-day course.

> *Key principle:* Treat candidiasis in pregnancy with a 7-day topical azole, not oral fluconazole.

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442. Answer: D. Vulvar lichen sclerosus can usually be diagnosed clinically, and biopsy is not required at the outset. Failure to improve on treatment is the trigger the CMAJ review gives for biopsy to rule out squamous cell carcinoma. A persistent plaque with an ulcer despite 5 months of adherent ultrapotent therapy fits that trigger. The traps are escalating or changing the topical agent, adding an antifungal without evidence of yeast, and waiting while a possible cancer grows. Biopsy the most abnormal area now.

> *Key principle:* A lichen sclerosus lesion that fails to improve on treatment needs a biopsy.

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443. Answer: C. The CMAJ review states that if lichen sclerosus is inadequately treated, about half of patients develop scarring and 3% to 5% develop malignant transformation. Treatment is therefore aimed at symptoms, architecture and cancer risk. The exaggerated option, the reassuring option and the two invented modifiers each fail against those numbers. The cancer risk is real but small, and it is lowered by consistent control of the disease. Use this to motivate adherence and arrange regular follow-up, with biopsy for any change.

> *Key principle:* Untreated lichen sclerosus risks scarring and a 3 to 5 percent chance of cancer.

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444. Answer: B. Lichen sclerosus is chronic, and symptoms and scarring return when treatment stops. After remission is induced with an ultrapotent steroid, the CMAJ review describes maintenance with a daily moderate-potency steroid such as betamethasone valerate 0.1%, or an ultrapotent ointment used twice weekly. Stopping invites relapse, continuing induction dosing exposes the skin to atrophy, and estrogen or antifungal creams do not treat the dermatosis. Prescribe the step-down regimen and keep regular follow-up to check adherence.

> *Key principle:* After remission, step down to maintenance steroid and keep follow-up.

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445. Answer: E. The classic clinical picture of vulvar lichen sclerosus is shiny white atrophic plaques in a figure-of-8 pattern that encircles the vulva, perineum and perianal area, with severe itching in most patients. The vagina is spared. Erosive lichen planus involves the vagina and produces painful erosions, lichen simplex is thickened and scratch-driven without atrophy, candidiasis lacks a plaque pattern, and vitiligo is depigmented without atrophy or itch. Recognizing the pattern permits treatment with a potent steroid without delay.

> *Key principle:* A figure-of-8 pattern of white atrophic plaques is lichen sclerosus.

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446. Answer: A. Vulvodynia is vulvar pain of at least 3 months without an identifiable cause, and consensus is to subcategorize it as localized, generalized or mixed, and as provoked, spontaneous or mixed. Her pain is confined to the vestibule and triggered by touch, which defines localized provoked disease. Generalized disease is diffuse, spontaneous pain is unprovoked, and her normal skin and negative studies argue against infection or a dermatosis. Name the diagnosis, since it opens the way to targeted therapy.

> *Key principle:* Vestibular pain on touch with normal skin is localized provoked vulvodynia.

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447. Answer: D. Generalized, spontaneous vulvodynia is managed as a neuropathic pain disorder. ACOG notes that tricyclic antidepressants and anticonvulsants can be used, and that gabapentin is the most studied and used anticonvulsant for vulvodynia. Topical steroids generally do not help, antifungals have no target after negative cultures, and cromolyn 4% was no more effective than placebo. Oral estrogen is not described as treatment for this condition. Start low and titrate slowly, with counseling on sedation, and combine with behavioral and physical therapy.

> *Key principle:* Neuropathic agents such as gabapentin are used for generalized vulvodynia.

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448. Answer: C. ACOG states that vestibulectomy may be effective when pain is localized to the vestibule, and that the procedure should be done only after other treatments have failed. She meets both conditions. Vestibuloplasty is reported to be ineffective. Laser resurfacing, Bartholin excision and clitoral hood reduction have no support for provoked vestibular pain. The remaining decision is counseling: surgery follows exhausting conservative care, and a surgeon experienced in the technique should perform it.

> *Key principle:* Vestibulectomy is a late option for localized vestibular pain after other treatments fail.

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449. Answer: C. Vulvodynia is a diagnosis of exclusion, so the evaluation seeks treatable causes while mapping the pain. ACOG describes the cotton-swab test to identify the areas and severity of pain and a musculoskeletal evaluation to find associated pelvic floor factors, and notes that testing for human papillomavirus is unnecessary. A biopsy is for visible lesions, acetic acid soaks are unnecessary in this condition, and imaging does not explain the pain. Map the pain, rule out infection and dermatosis, and then treat.

> *Key principle:* Evaluate vulvodynia with cotton-swab mapping and exclusion of treatable causes, not HPV testing.

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OP3.5. Structural Uterine Abnormalities

450. Answer: D. The combination tablet suppresses ovarian estrogen with add-back replacement, but bone loss still occurs and may not be reversible. The Myfembree label therefore limits use to 24 months. Normal interval bone density does not lift that limit, and supplements do not either. The 12-month limit belongs to older GnRH agonist add-back practice, and removing add-back would increase bone loss. At about 22 months, plan the next step, such as an intrauterine system, embolization, myomectomy or hysterectomy.

> *Key principle:* Relugolix combination therapy is limited to 24 months because bone loss may not reverse.

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451. Answer: A. Estrogen and progestin combinations increase the risk of thrombotic and thromboembolic disorders, and the Myfembree label carries a boxed warning. It lists women at high risk of arterial, venous thrombotic or thromboembolic disorders as a contraindication. A prior venous thrombosis places her in that group. The intrauterine system, an NSAID, embolization and sampling do not carry the same estrogen-related label contraindication. Match the agent to the history, and choose a nonestrogen or procedural approach for her.

> *Key principle:* Estrogen-containing fibroid pills are contraindicated with a high thrombotic risk.

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452. Answer: E. The Oriahnn label lists undiagnosed abnormal uterine bleeding as a contraindication, because the drug could mask or delay the diagnosis of endometrial pathology. This woman is older than 45, obese and has intermenstrual bleeding, so endometrial hyperplasia or cancer must be excluded first, and a fibroid on ultrasound does not explain bleeding. Starting the drug and waiting, or obtaining unrelated tests, delays that evaluation. Sample the endometrium, then decide on therapy.

> *Key principle:* Do not start fibroid hormonal therapy until abnormal bleeding is explained.

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453. Answer: B. All morcellation techniques can spread unsuspected cancer cells, and no preoperative test excludes leiomyosarcoma. The FDA states that laparoscopic power morcellators should not be used in patients who are postmenopausal or older than 50 years, and ACOG notes that for postmenopausal women the balance may favor procedures that do not involve morcellation. Removing the specimen intact, such as through the vagina or a minilaparotomy, respects both. Open uncontained morcellation and myomectomy fail that test, and a delay serves no purpose.

> *Key principle:* Avoid power morcellation in postmenopausal women and those older than 50.

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454. Answer: A. Before morcellation, the preoperative workup should include cervical cancer screening and endometrial tissue sampling to identify malignancy that morcellation could disseminate. Imaging features suspicious for malignancy should preclude morcellation, but MRI and LDH isoenzyme testing rest on weak evidence, and no test accurately rules out leiomyosarcoma. CA-125 is not a screening tool for sarcoma. Update cytology or HPV testing and obtain tissue, then use shared decision-making and informed consent.

> *Key principle:* Screen the cervix and sample the endometrium before morcellating fibroids.

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455. Answer: D. Leiomyosarcoma is rare among women having surgery for presumed fibroids. The ACOG opinion states that uterine size and rapid uterine growth are not associated with increased risk of leiomyosarcoma, and that no test can accurately rule it out beforehand. Believing that growth signals sarcoma leads to unneeded alarm or hysterectomy, while believing that imaging or LDH excludes sarcoma leads to false reassurance. Counsel on the low and imperfectly predictable risk, and use shared decision-making about surgery and morcellation.

> *Key principle:* Rapid fibroid growth does not by itself predict leiomyosarcoma.

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456. Answer: C. The FIGO system divides submucosal fibroids into types 0, 1 and 2. Type 0 is pedunculated and intracavitary, type 1 is less than 50% intramural, and type 2 is 50% or more intramural. A fibroid with 70% of its diameter in the myometrium and a protruding component is type 2. Type 3 lies fully within the myometrium but contacts the endometrium, and type 4 is intramural without contact. Higher intramural fractions make complete hysteroscopic resection harder, so classify before planning.

> *Key principle:* A submucosal fibroid at least half intramural is FIGO type 2.

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457. Answer: B. Most polyps are benign, but malignant transformation is found more often in postmenopausal patients and in those who have symptoms. In postmenopausal patients, StatPearls reports malignancy in 4.47% of symptomatic cases compared with 1.51% of asymptomatic cases, and age older than 60, large size and postmenopausal status are risk factors. The three younger women are premenopausal without bleeding. The belief that risk is equal leads to missed tissue diagnosis. Remove and send the specimen for histology in the postmenopausal bleeder.

> *Key principle:* Postmenopausal polyps with bleeding carry the highest malignancy risk.

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458. Answer: E. Tamoxifen blocks estrogen in breast tissue but acts as a partial estrogen agonist in the endometrium, which promotes proliferation. StatPearls reports a 20% to 35% incidence of endometrial polyp formation with chronic tamoxifen use. The distractors assume an antiestrogen effect in the uterus, a rare event, an estrogen cofactor, or no difference from nonusers. In practice, women on tamoxifen who report vaginal bleeding need prompt evaluation, since polyps and cancer are both possible.

> *Key principle:* Tamoxifen acts as an estrogen agonist in the endometrium and raises polyp rates.

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459. Answer: C. Hysteroscopy offers direct visualization and is considered the gold standard for diagnosis of endometrial polyps. Blind biopsy and curettage have limited sensitivity for focal lesions, and ultrasound or CT can only suggest a lesion. Hysteroscopy by itself does not provide tissue unless a biopsy or polypectomy is added, which is why it is paired with removal. Serum CA-125 does not diagnose polyps. Perform hysteroscopy, resect the lesion under direct vision and send it to pathology.

> *Key principle:* Hysteroscopy is the gold standard for diagnosing endometrial polyps.

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460. Answer: C. Endometrial polyps are estrogen-sensitive overgrowths of endometrial glands and stroma. StatPearls lists increasing age, hypertension, obesity and tamoxifen therapy as established risk factors, and links obesity to unopposed estrogen from peripheral conversion of androgens. The wrong options claim protection from hypertension, invent a cesarean cause, reverse the estrogen effect of obesity, or claim age lowers risk. Her profile fits the risk pattern, so confirm that she has no bleeding and counsel on weight and blood pressure control.

> *Key principle:* Age, hypertension, obesity and tamoxifen increase endometrial polyp risk.

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461. Answer: D. In the American Family Physician review, transvaginal ultrasonography had a sensitivity of 83.8% and a specificity of 63.9%, while MRI had a sensitivity of 77% and a specificity of 89%. MRI is therefore the more specific test, and the sonographic criteria also show low reproducibility between imagers. Choosing ultrasound for specificity, MRI for sensitivity, or assuming equivalence all reverse the numbers. Use MRI when the question is whether an equivocal ultrasound finding is real, especially if the result will change management.

> *Key principle:* MRI is more specific, and ultrasound more sensitive, for adenomyosis.

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462. Answer: B. Treatment of adenomyosis typically starts with hormonal menstrual suppression. The American Family Physician review supports a levonorgestrel-releasing intrauterine system to reduce heavy bleeding and pain in patients not desiring pregnancy, with B-level evidence. GnRH analogues cause hypoestrogenism and bone loss, hysterectomy is definitive therapy only when other treatments fail, and tranexamic acid is not a daily-all-cycle agent. Offer the intrauterine system first, and reassess in a few months.

> *Key principle:* A levonorgestrel intrauterine system is a first hormonal option for adenomyosis symptoms.

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463. Answer: A. The American Family Physician review reports that, in patients undergoing IVF or ICSI, adenomyosis was associated with a 28% decrease in the probability of clinical pregnancy. Proposed mechanisms include an abnormal junctional zone, abnormal uterine peristalsis and altered sperm transport. Stating no change or a higher chance misses this association, and dismissing it as only a miscarriage issue is also wrong. Hysterectomy would end fertility. Counsel honestly, and discuss options for improving the odds with a reproductive endocrinologist.

> *Key principle:* Adenomyosis lowers the chance of clinical pregnancy with IVF by about 28 percent.

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464. Answer: E. Adenomyosis and leiomyomas often coexist. The American Family Physician review notes that as many as 50% of patients with adenomyosis have leiomyomas. A smaller fibroid with a thickened junctional zone and myometrial cysts points to adenomyosis as the source of persistent bleeding and pain. The distractors blame regrowth, a polyp, a cyst or procedural failure, none of which fit the imaging. Counsel and treat the adenomyosis directly.

> *Key principle:* Fibroid treatment may leave coexisting adenomyosis as the cause of symptoms.

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465. Answer: A. The American Family Physician review states that definitive therapy for adenomyosis is hysterectomy if other therapies are not effective, rated as consensus opinion. She has completed childbearing, has anemia and has failed three medical approaches. The distractors repeat a failed treatment, use ablation, which treats the endometrium and not deeper disease, escalate a similar hormone, or wait for menopause while she is anemic. Correct the anemia and offer hysterectomy by the least invasive route.

> *Key principle:* Hysterectomy is definitive therapy when other adenomyosis treatments fail.

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466. Answer: D. The ASRM 2021 classification defines a septate uterus as having an endometrial septum length greater than 1 cm, with the leading edge of the septum forming an angle of less than 90 degrees. The external fundal contour is normal, which separates it from bicornuate and didelphys uteri that have a cleft or two separate bodies. A unicornuate uterus has a single horn, and an arcuate uterus has only a mild indentation. Correct classification matters because septum resection is the treatment that applies.

> *Key principle:* A septate uterus has an endometrial septum longer than 1 cm.

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467. Answer: B. The ASRM 2021 classification defines the bicornuate uterus by an external fundal indentation greater than 1.0 cm, which separates it from a septate uterus with a normal outer contour. The two cavities with a single cervix match bicornuate anatomy. Didelphys requires two cervices, a complex anomaly needs additional components, and a minor indentation is not enough for either. Accurate classification guides counseling, and bicornuate uteri are not treated by septum resection.

> *Key principle:* A bicornuate uterus has an external fundal indentation deeper than 1.0 cm.

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468. Answer: E. Obstructed hemivagina with ipsilateral renal anomaly is a recognized syndrome of müllerian duct and mesonephric duct maldevelopment. The ASRM classification reports that, with an obstructed hemivagina, ipsilateral renal agenesis or anomaly was found in as many as 95%. The kidney on the side of the obstruction is therefore usually absent. Horseshoe and polycystic kidneys, ureteral duplication and cardiac defects are not the expected association. Order renal imaging, and plan drainage of the obstructed system through a vaginal septum resection.

> *Key principle:* An obstructed hemivagina is usually accompanied by ipsilateral renal agenesis.

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469. Answer: C. Müllerian agenesis is associated with urinary tract anomalies such as renal agenesis, and skeletal anomalies also occur. The ASRM classification states that renal imaging is strongly recommended in the presence of müllerian agenesis. A pituitary MRI and 17-hydroxyprogesterone concern hypogonadism and adrenal disorders, which her normal development and hormones argue against, and gonadal biopsy is unneeded with a 46,XX karyotype and normal testosterone. Order renal imaging and evaluate the spine as well.

> *Key principle:* Image the kidneys whenever müllerian agenesis is diagnosed.

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470. Answer: D. The ASRM Müllerian Anomalies Classification 2021 sorts anomalies into nine categories: müllerian agenesis, cervical agenesis, unicornuate, didelphys, bicornuate, septate, longitudinal and transverse vaginal septum, and complex anomalies. Unlike the older classification, the categories are no longer numbered but are identified by descriptive terminology. The wrong options revive numbered classes, invoke genetics, split the vagina out, or restrict the system to the uterine body, though it includes the cervix and vagina. Describe anomalies by name.

> *Key principle:* The ASRM 2021 system names categories instead of numbering them.

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OP3.6. Incontinence and Pelvic Floor

471. Answer: B. A preoperative workup for uncomplicated stress incontinence aims to confirm the diagnosis and exclude voiding or infection problems. The AUA/SUFU guideline lists a history of bother, a pelvic examination, objective demonstration of stress leakage with a full bladder, postvoid residual, and urinalysis. It states that clinicians should not perform cystoscopy in index patients unless there is concern about a urinary tract abnormality. In practice, reserve cystoscopy for hematuria, suspected fistula, or other red flags.

> *Key principle:* Do not perform cystoscopy for routine evaluation of uncomplicated stress incontinence.

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472. Answer: A. Urodynamics can add cost and discomfort and has not been shown to improve outcomes in uncomplicated stress incontinence. The guideline gives a conditional recommendation, graded B, that clinicians may omit urodynamic testing for the index patient when stress incontinence is clearly demonstrated. The distractors make the test mandatory, substitute it for a bedside test, or tie it to age. In practice, use urodynamics for complicated cases, such as mixed symptoms, high residual, or prior surgery.

> *Key principle:* Urodynamics can be skipped in uncomplicated, clearly demonstrated stress incontinence.

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473. Answer: C. Urgency-predominant mixed incontinence responds less predictably to a sling because the urgency component may persist or worsen. The guideline lists urgency-predominant mixed incontinence, elevated postvoid residual, and high-grade prolapse without demonstrated stress leakage as circumstances that call for additional evaluation. The distractors treat the stress component alone or are unrelated. In practice, clarify the dominant symptom, treat urgency first when it dominates, and counsel that surgery may not resolve urgency.

> *Key principle:* Urgency-predominant mixed incontinence needs further evaluation before a sling.

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474. Answer: E. Midurethral slings are widely used, but mesh brings risks such as exposure, pain, and voiding dysfunction, and alternatives exist, including autologous fascial slings, Burch colposuspension, bulking, and nonsurgical care. The guideline states that before selecting a synthetic midurethral sling clinicians must discuss the specific risks and benefits of mesh and the alternatives to a mesh sling. The distractors minimize risk or restrict choices. In practice, document the discussion and her chosen alternative.

> *Key principle:* Before a mesh sling, discuss mesh-specific risks, benefits, and alternatives.

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475. Answer: B. Follow-up lets the surgeon catch retention, infection, pain, and mesh problems early. The guideline states patients should be seen and examined within six months after surgery to assess subjective outcome, voiding symptoms and urinary tract infections, incision healing, mesh extrusion for synthetic slings, and postvoid residual, with earlier contact for voiding problems or pain. The distractors push the visit too late, make it symptom triggered, or end care at once. In practice, schedule the visit before discharge.

> *Key principle:* Examine sling patients within six months after surgery.

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476. Answer: C. Irradiated, scarred tissue heals poorly, which raises the risk of mesh exposure, erosion, and infection. The guideline states that clinicians should strongly consider avoiding mesh in patients undergoing incontinence surgery who are at risk for poor wound healing, such as those with prior radiation or significant scarring. An autologous sling uses her own tissue. The distractors keep mesh, assume a smaller sling is safer, or deny surgery. In practice, optimize tissue and choose non-synthetic options.

> *Key principle:* Avoid mesh slings when wound healing is poor, as after pelvic radiation.

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477. Answer: D. Overactive bladder is a clinical diagnosis, and the first goal is to exclude conditions that mimic it. The AUA/SUFU guideline states clinicians should perform a urinalysis to exclude microhematuria and infection, and may obtain a postvoid residual and a voiding diary. The distractors are advanced tests that the guideline reserves for selected patients. In practice, order the urinalysis and treat or investigate abnormal findings before starting therapy.

> *Key principle:* Start the overactive bladder workup with a urinalysis.

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478. Answer: E. Advanced tests rarely change first-line management in uncomplicated overactive bladder and add cost and risk. The guideline states that clinicians should not routinely perform urodynamics, cystoscopy, or urinary tract imaging in the initial evaluation of the uncomplicated patient, and may use them in selected cases or after inadequate response to treatment. The distractors apply them as routine. In practice, begin treatment with behavioral measures and reserve testing for red flags.

> *Key principle:* Do not routinely order urodynamics, cystoscopy, or imaging for uncomplicated overactive bladder.

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479. Answer: A. Behavioral therapy is a low-risk treatment with a strong evidence base, and medication adds side effects. The guideline gives a strong recommendation, graded A, that clinicians should offer bladder training to all patients with overactive bladder, and a clinical principle to offer behavioral therapies to all patients. The distractors hold it back for failures, restrict it by age, or deny its value. In practice, offer it alongside medication when a patient prefers a drug.

> *Key principle:* Offer bladder training to every patient with overactive bladder.

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480. Answer: E. Antimuscarinic drugs cross into the central nervous system to varying degrees, and observational studies link them to cognitive impairment and dementia, with older adults and those with baseline memory problems at most concern. The guideline states clinicians should discuss the potential risk for developing dementia and cognitive impairment with patients with overactive bladder when offering these drugs. The distractors give false reassurance or order unneeded tests. In practice, consider a beta-3 agonist or behavioral therapy for such patients.

> *Key principle:* Counsel patients about dementia and cognitive risk with antimuscarinics.

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481. Answer: C. Antimuscarinics reduce bladder contractions but also affect the eye and gut and impair emptying. The guideline states clinicians should use antimuscarinic medications with extreme caution in patients with overactive bladder who have narrow-angle glaucoma, impaired gastric emptying, or a history of urinary retention. Her glaucoma and retention history place her squarely in that group, so another drug class is a reasonable route. The distractors escalate dosing, duplicate agents, or skip standard therapy. In practice, check residual urine and eye history before prescribing.

> *Key principle:* Use antimuscarinics with extreme caution in glaucoma, poor gastric emptying, or retention.

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482. Answer: A. Many supplements are marketed for bladder health, but trials are small and of low quality. The guideline states that clinicians should counsel patients that there is currently insufficient evidence to support the use of nutraceuticals for overactive bladder. The distractors overstate efficacy. In practice, acknowledge her preference, offer bladder training and other evidence-based options, and note that supplements are not regulated like drugs.

> *Key principle:* There is insufficient evidence for nutraceuticals in overactive bladder.

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483. Answer: B. When one drug gives incomplete relief, adding a drug with a different mechanism can provide further benefit. The guideline gives a conditional recommendation, graded B, that clinicians may offer combination therapy with a medication from a different class in patients who do not achieve adequate improvement. Adding a second antimuscarinic duplicates the mechanism and its side effects. Augmentation and catheters are reserved for severe or refractory disease. In practice, add a beta-3 agonist to an antimuscarinic.

> *Key principle:* A second drug from a different class can be added for partial response.

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484. Answer: D. Chronic catheters carry risks of infection, stones, urethral damage, and bladder changes, and they treat the leakage, not the overactive bladder. The guideline states clinicians should recommend chronic indwelling urethral or suprapubic catheters to patients with overactive bladder only when therapies are contraindicated, ineffective, or no longer desired by the patient. Neither condition has been met here. In practice, begin with prompted voiding, pads, and medication where appropriate, and revisit goals.

> *Key principle:* Reserve chronic catheters for overactive bladder after other therapies fail or are declined.

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485. Answer: B. A pessary rests against the vaginal wall, and pressure and local irritation change the vaginal environment, especially in estrogen-deficient tissue. The review lists the potential complications as discharge, bleeding, erosion, pain, constipation, and incontinence. Fistula is rare, and the distractors describe an effect no pessary produces. In practice, schedule regular checks, examine the vaginal mucosa, and treat erosion by removing the device for a time, with vaginal estrogen as indicated.

> *Key principle:* Pessary complications are discharge, bleeding, erosion, pain, constipation, and incontinence.

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486. Answer: E. Pessary fitting is an initial option for many women, but not all keep using the device. The review reports that in women successfully fitted with a pessary, 40 to 60 percent will continue use for more than 6 to 12 months. The distractor ranges either imply near failure or near-universal use. In practice, set expectations at fitting, review discomfort and expulsion early, and offer surgery if she stops.

> *Key principle:* About 40 to 60 percent of successfully fitted women keep using a pessary beyond a year.

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487. Answer: A. Prolapse surgery is meant for women who fail or decline nonsurgical care, and operations such as hysterectomy or colpocleisis end childbearing or are irreversible. The ACOG and AUGS guidance describes the pessary as generally considered a low-risk option and a good alternative for women who wish to maintain fertility. The distractors are permanent or carry mesh risk. In practice, fit a pessary, recommend pelvic floor muscle training, and reevaluate after childbearing.

> *Key principle:* A pessary suits women with prolapse who want to keep fertility.

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488. Answer: E. Descriptive terms such as moderate vary between examiners and cannot track change or guide choices. The POP-Q system measures six points relative to the hymen plus vaginal length, hiatus, and perineal body. The guidance recommends the POP-Q system before treatment to quantify the degree of prolapse. The distractors are tests with other purposes. In practice, stage every patient with prolapse before choosing between a pessary, pelvic floor training, and surgery.

> *Key principle:* Stage prolapse with the POP-Q system before choosing treatment.

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489. Answer: C. Fecal incontinence is common, and initial care depends on history, examination, and stool consistency. ACOG states that ancillary testing, including anal sphincter imaging, is not recommended for the routine evaluation of fecal incontinence, though imaging helps when sphincter repair is planned. The distractors send her to testing or to invasive treatment first. In practice, take a complete history, examine the perineum, rectum, and sphincter tone, and start dietary and medical measures.

> *Key principle:* Anal sphincter imaging is not routine in fecal incontinence evaluation.

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490. Answer: D. Stool consistency strongly drives continence, and loose stools overwhelm a normal sphincter. ACOG states fiber, antimotility agents, and laxatives can be recommended as useful treatments, and pelvic floor muscle exercises with or without biofeedback can also be recommended. Surgical choices are for women who fail conservative measures. In practice, adjust diet and medication, review caffeine and sugar substitutes, and add pelvic floor therapy if symptoms continue.

> *Key principle:* Start fecal incontinence treatment with fiber, antimotility agents, or laxatives.

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491. Answer: E. When conservative care fails, surgical options are chosen by the anatomy. Sphincteroplasty repairs a defect in the sphincter, which she lacks, and bulking agents give only short-term benefit. ACOG states sacral nerve stimulation can be considered as a surgical treatment option for women with fecal incontinence, and the ASCRS guideline lists it as a first-line surgical option. In practice, trial the stimulator first with a test phase.

> *Key principle:* Sacral nerve stimulation is an option for refractory fecal incontinence.

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OP3.7. Reproductive Tract Cancer

492. Answer: C. FIGO 2018 stages cervical cancer confined to the cervix by depth and by greatest dimension. Disease with 5 mm or more of stromal invasion is stage IB, and size divides it into IB1 below 2 cm, IB2 from 2 cm to below 4 cm, and IB3 at 4 cm or more. A 3.5 cm tumor falls in the middle band, so stage IB2 is correct. The traps are the microscopic stage IA categories, which have less than 5 mm of invasion, and stage IIA1, which needs vaginal spread. Record size on the examination or imaging, because it assigns the substage.

> *Key principle:* Cervical cancer confined to the cervix with deep invasion is IB, and size over 2 cm up to 4 cm makes it IB2.

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493. Answer: C. The USPSTF weighs the very low incidence of invasive cervical cancer under age 21 against the harms of treating transient lesions, and finds the balance unfavorable for screening. The statement is final from 2018 and is being updated, so the stem names it. Sexual activity at 16 and a negative history do not change the age rule. The distractors apply screening intervals meant for ages 21 and older, or substitute colposcopy for an indication that does not exist. Begin screening at 21 and counsel her about vaccination, condoms and contraception today.

> *Key principle:* Under the 2018 USPSTF recommendation, cervical screening starts at age 21 whatever the sexual history.

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494. Answer: A. Vaccine trials have not yet shown that vaccinated cohorts can safely use a different screening schedule, and the vaccine does not cover every oncogenic HPV type. The USPSTF therefore advises that vaccinated women continue screening as recommended until more evidence accrues. At age 34, the schedule options are cytology every 3 years, HPV testing alone every 5 years, or cotesting every 5 years. The traps are stopping early and relying on the pelvic examination, which does not detect cervical precancer. Vaccination reduces risk but does not replace screening.

> *Key principle:* Vaccinated women follow the same cervical screening schedule as unvaccinated women.

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495. Answer: D. The USPSTF defines adequate prior screening as three consecutive negative cytology results or two consecutive negative cotesting results within the past 10 years, with the most recent test within 5 years. She meets this definition and has no high-risk history. Screening beyond 65 in that setting adds harms from false positives without meaningful benefit. The distractors extend screening to age 75 or add testing that the statement does not call for. Document the screening history, and recheck eligibility if her risk profile changes.

> *Key principle:* After age 65, adequate negative screening in the past 10 years with the latest test in 5 years permits stopping.

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496. Answer: B. A negative result is reassuring in proportion to the test's sensitivity for high-grade disease. The National Cancer Institute summary reports sensitivity of HPV testing of 95% against 55% for cytology in primary screening. That gap is why a negative HPV test supports a longer rescreening interval. The distractors are true or plausible statements about other features, such as the mortality benefit of Pap screening or self-collection, but none speaks to the reliability of a negative result. Match the interval to the sensitivity of the test used.

> *Key principle:* A negative HPV test earns a longer interval because it detects far more high-grade disease than cytology.

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497. Answer: E. The 2023 FIGO system stages by histologic type as well as extent. Aggressive histologic types, including serous carcinoma, move up a category: confined to the endometrium or a polyp is IC, and any myometrial invasion is IIC. This tumor invades the myometrium, so it is IIC despite superficial depth. The traps are the non-aggressive categories IA2 and IB, which depend on depth, and IIA, which needs cervical stroma invasion. Histologic type, not depth alone, decides the stage here, and it guides adjuvant planning.

> *Key principle:* In 2023 FIGO staging, an aggressive histologic type with any myometrial involvement is stage IIC.

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498. Answer: B. Tamoxifen is a selective estrogen receptor modulator that antagonizes estrogen in breast tissue but acts as a partial agonist in the endometrium. The National Cancer Institute summary links the increased endometrial cancer risk to that estrogenic effect on the lining. The distractors describe mechanisms that do not occur, such as progesterone receptor blockade or a germline defect, or an atrophic effect that is the opposite of what happens. Any abnormal bleeding on tamoxifen needs endometrial evaluation.

> *Key principle:* Tamoxifen raises endometrial cancer risk through its estrogenic action on the endometrium.

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499. Answer: E. The National Cancer Institute summary separates two pathways. Type 1 tumors may arise from complex atypical hyperplasia and are linked to unopposed estrogenic stimulation, whereas type 2 tumors develop from atrophic endometrium and are not linked to hormonally driven pathogenesis. Serous carcinoma in a lean older woman with atrophic surroundings fits type 2. The traps are the estrogen-driven pathway typical of obese, anovulatory patients and exposures she does not have. Expect an aggressive course, and plan surgical staging accordingly.

> *Key principle:* Type 2 endometrial cancers arise from atrophic endometrium rather than from estrogen-driven hyperplasia.

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500. Answer: C. A Cochrane meta-analysis of randomized trials found no clinical benefit from adjuvant progestogens in clinical stage I endometrial cancer, and the National Cancer Institute summary reports that conclusion. Hormone receptor positivity is common in endometrioid tumors but did not translate into benefit. The distractors imply a benefit that depends on age, route of surgery or receptor status, none of which the pooled data support. Adjuvant decisions in stage I disease rest on histologic type, grade, depth and molecular features, not on empiric hormone therapy.

> *Key principle:* Adjuvant progestogen therapy has no proven benefit in stage I endometrial cancer.

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501. Answer: D. Obesity raises risk through peripheral aromatization of androgens to estrone and chronic unopposed estrogen exposure, but risk does not equal benefit from screening. The National Cancer Institute summary states that routine screening of asymptomatic women has not been evaluated and that ultrasound screening has no evidence of reducing mortality, while false positives lead to unnecessary biopsies. The traps are the tests that sound reasonable, including CA-125 and cytology, which are neither sensitive nor validated. Counsel her to report any bleeding promptly, which should trigger evaluation.

> *Key principle:* Asymptomatic women at average genetic risk are not screened for endometrial cancer, but bleeding must be reported.

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502. Answer: C. FIGO 2014 divides stage IC by how tumor cells reached the surface or peritoneal fluid. IC1 is surgical spill, IC2 is capsule rupture before surgery or tumor on the surface, and IC3 is malignant cells in ascites or washings. Rupture before surgery without positive washings is IC2. The traps are IC1, which refers to intraoperative spill, and IC3, which needs positive cytology. Stage IA requires an intact capsule. Precise IC substaging records how the capsule was breached and informs counseling about prognosis and adjuvant therapy.

> *Key principle:* Capsule rupture before surgery with negative washings is stage IC2, not IC1 or IC3.

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503. Answer: A. The summary cites the MRC OV05 trial, which compared early treatment based on CA-125 with treatment at clinical relapse. The National Cancer Institute summary states that neither early detection by imaging nor by CA-125 elevation has been shown to alter outcomes. A rise does not itself define platinum resistance, and surgery depends on selection criteria. The trap is the assumption that earlier treatment is better. Discuss the findings with her, and base treatment timing on symptoms, imaging and her preferences.

> *Key principle:* Treating an asymptomatic CA-125 rise early has not been shown to change outcomes in recurrent ovarian cancer.

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504. Answer: B. Dysgerminoma is chemosensitive and typically affects young women, so conservative surgery can be curative without sacrificing fertility. For stage I disease, the National Cancer Institute summary states that unilateral salpingo-oophorectomy conserving the uterus and opposite ovary is accepted treatment. The traps are radical operations that were once standard and end fertility, and cystectomy or biopsy alone, which leave the tumor-bearing ovary in place. Complete the staging and plan surveillance, with the option of chemotherapy if disease spreads.

> *Key principle:* Stage I dysgerminoma is treated with unilateral salpingo-oophorectomy, preserving the uterus and other ovary.

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505. Answer: D. Alpha fetoprotein is made by yolk sac elements, so it should fall predictably after complete resection. The National Cancer Institute summary states that persistence of these markers in the serum after surgery indicates unresected tumor. Normal imaging does not exclude microscopic disease, which the marker detects. The distractors offer benign explanations such as slow clearance or pregnancy that are possible but are not the principle to act on. Treat a persistent marker as residual disease and refer to a gynecologic oncologist.

> *Key principle:* Persistently elevated germ cell tumor markers after surgery mean residual tumor until proven otherwise.

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506. Answer: E. Ovarian cancer has no effective screening, and BRCA1 carriers have a high lifetime risk. ACOG Practice Bulletin 182 states that risk-reducing salpingo-oophorectomy is typically recommended at age 35 to 40 years for BRCA1 carriers, and women with BRCA2 may consider delaying until age 40 to 45 years. She is 36 and has finished childbearing. The traps are substituting surveillance for surgery, removing only the uterus, which leaves the tubes and ovaries at risk, and delaying surgery until menopause. Before surgery, counsel her about surgical menopause and symptom management.

> *Key principle:* A BRCA1 carrier past age 35 with completed childbearing should be offered risk-reducing salpingo-oophorectomy.

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507. Answer: B. FIGO 2021 stage I vulvar cancer is confined to the vulva and is split by size and depth. Stage IA is a tumor 2 cm or less with stromal invasion 1 mm or less, and stage IB is a tumor larger than 2 cm or invasion deeper than 1 mm. Either feature qualifies, and this tumor has both. The traps are stage II, which needs extension to the lower urethra, vagina or anus, and stage III, which requires adjacent perineal spread or nodes. Either feature alone is enough, so the pathologist must report both size and depth.

> *Key principle:* A node-negative vulvar tumor over 2 cm or deeper than 1 mm is stage IB if confined to the vulva.

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508. Answer: C. Vulvar squamous cancer follows two pathways. HPV-associated disease arises from usual-type VIN and is most common in women under 50, while non-HPV-associated disease arises from differentiated-type VIN and is most common in older women. This patient's age, adjacent differentiated VIN and negative HPV testing place her in the second group. The distractors invert the age pattern or suggest other origins that do not match keratinizing squamous histology. Identify the pathway because the precursor determines the pattern of local recurrence and the follow-up needed.

> *Key principle:* Differentiated VIN and HPV-negative vulvar squamous cancer are typical of older women.

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509. Answer: A. Sentinel node biopsy was developed to spare women with early vulvar cancer the morbidity of inguinofemoral dissection, which causes lymphedema and wound complications. The National Cancer Institute summary states that for stage I and II disease groin dissection is reserved for those with metastasis to the sentinel nodes. A negative sentinel result therefore ends nodal surgery. The traps are routine dissection or radiation despite negative nodes. Provide surveillance of the groins and the vulva, since local recurrence is the more common failure.

> *Key principle:* Groin dissection is performed only when the sentinel node contains metastasis.

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510. Answer: A. Clear cell adenocarcinoma of the vagina and cervix occurs in association with in utero exposure to diethylstilbestrol, which was prescribed to prevent pregnancy loss, and peak incidence is before age 30. Her age, tumor type and the maternal history match this pattern. The distractors are teratogens linked to other malformations, such as thalidomide and valproic acid, but none causes clear cell vaginal cancer. Women with known exposure need regular examination of the vagina and cervix, and any symptomatic lesion needs biopsy.

> *Key principle:* Vaginal clear cell adenocarcinoma in a young woman points to in utero DES exposure.

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511. Answer: C. Staging systems avoid double counting by assigning a tumor to the organ that is involved first or most completely. A vaginal lesion with no cervical involvement is a primary vaginal cancer, but the National Cancer Institute summary states that tumors that originated in the apical vagina but extend to the cervix are classified as cervical cancers. Her tumor involves the cervix, so cervical staging applies. The traps are labeling it vaginal because the bulk is vaginal, or calling it a metastasis. The classification directs chemoradiation planning and prognosis.

> *Key principle:* A tumor that extends from the apical vagina onto the cervix is classified as cervical cancer.

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512. Answer: E. Complete moles carry a meaningful risk of persistent trophoblastic disease, and hCG is the tumor marker used to detect it. The 2025 FIGO update states that for a complete mole monthly hCG monitoring should continue for 6 months after hCG normalization, while a partial mole needs only a single confirmatory test at 1 month. The traps are stopping at the first normal value and applying the partial mole schedule. A longer interval such as yearly testing is too sparse to catch an early rise. Continue monthly testing and counsel her on reliable contraception during follow-up.

> *Key principle:* After a complete mole, monthly hCG continues for 6 months after the first normal value.

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513. Answer: D. Reliable contraception is needed because pregnancy during follow-up would raise hCG and confound detection of persistent disease. The older concern that hormonal methods promote trophoblastic persistence has not held up, and the 2025 FIGO update states that hormonal contraception is considered safe during the post-molar follow-up period. The distractors impose barrier-only or delayed approaches, which have higher failure rates or leave a gap in protection. Start an effective method now and keep the hCG schedule unchanged.

> *Key principle:* Hormonal contraception is safe during post-molar hCG follow-up, and pregnancy must be avoided.

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514. Answer: B. Postmolar GTN is diagnosed on hCG behavior without need for histology. The 2025 FIGO update lists a plateau lasting four measurements over 3 weeks or longer as a criterion, and these four values over 22 days fit that definition. She therefore has gestational trophoblastic neoplasia. The traps are waiting longer, repeating curettage, which is not routine, and prophylactic hysterectomy, which is not indicated. Next steps are staging with imaging as indicated, calculation of the FIGO score, and referral to a trophoblastic disease center.

> *Key principle:* An hCG plateau over four measurements across 3 weeks or more after a mole defines postmolar GTN.

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515. Answer: C. Placental site trophoblastic tumor and epithelioid trophoblastic tumor arise from intermediate trophoblast, grow slowly, make little hCG and respond poorly to the single-agent chemotherapy that cures most low-risk GTN. The 2025 FIGO update states that hysterectomy is the primary mode of treatment. For confined disease and completed childbearing, that is curative. The traps are single-agent therapy used for ordinary low-risk GTN, observation, and radiation. Combination chemotherapy is reserved for advanced stages, and immunotherapy may be considered.

> *Key principle:* Placental site and epithelioid trophoblastic tumors are treated primarily by hysterectomy.

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516. Answer: B. A normal hCG does not mean every viable trophoblastic cell has been eradicated, and relapse after stopping at the first normal value is more likely. The 2025 FIGO update states that after hCG normalization, 2 to 3 additional chemotherapy cycles reduce the risk of recurrence. These consolidation cycles are a standard part of therapy. The traps are stopping at the first normal value, excessive prolonged treatment, and escalating to combination therapy or surgery without failure. After treatment, monthly hCG monitoring for at least 12 months follows.

> *Key principle:* Give 2 to 3 consolidation cycles after hCG normalizes in GTN to lower recurrence risk.

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517. Answer: A. Many high-grade serous cancers are thought to start in the distal fallopian tube, so removing the tubes during planned pelvic surgery lowers risk without removing the ovaries. The National Cancer Institute summary lists an approximately 50% decrease for bilateral salpingectomy and less protection for unilateral salpingectomy. The distractors confuse this with the roughly 90% reduction after risk-reducing oophorectomy in BRCA carriers or deny any benefit. Offer salpingectomy at hysterectomy when she has finished childbearing, and explain that risk is reduced, not eliminated.

> *Key principle:* Bilateral salpingectomy cuts ovarian cancer risk by about half, less than oophorectomy in BRCA carriers.

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518. Answer: D. Observational studies of menopausal hormone therapy show a small excess of ovarian cancer in current or recent users, which fades after stopping. The National Cancer Institute summary rates the evidence as fair and describes a small increased risk. The traps are inverting the direction, stating no association, or exaggerating the size of the risk. Her family history is a single relative at age 70 without a known variant, which does not make her high-risk by itself. Weigh the small absolute risk against her symptoms and individual factors, and document shared decision making.

> *Key principle:* Current or recent hormone therapy carries a small increase in ovarian cancer risk.

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OP4.1. Office Procedures

519. Answer: E. Abnormal bleeding after midlife carries a higher chance of hyperplasia or cancer than in younger women, and imaging cannot give a tissue diagnosis. The ACOG bulletin states that endometrial sampling should be performed first-line in patients with abnormal bleeding who are older than 45 years. The traps are imaging first, empiric hormonal treatment, or an operating room procedure, each of which delays or overshoots the simple office test that answers the question. In practice, a woman older than 45 with abnormal bleeding has an office biopsy at the first visit, and imaging is added when a structural cause is suspected.

> *Key principle:* After age 45, abnormal uterine bleeding calls for endometrial sampling as the first test.

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520. Answer: E. Endometrial intraepithelial neoplasia often coexists with carcinoma, and a small blind sample can miss a focal area. The consensus states that gynecologists should attempt to exclude concurrent carcinoma and that hysteroscopic examination with further sampling is the most accurate method. Repeating a blind biopsy, relying on CA-125 or MRI, or curettage without direct visualization samples the cavity without seeing it, so a focal cancer can still be missed. In practice, scant tissue showing this lesion should lead to directed sampling under direct view before hysterectomy or progestin therapy is chosen.

> *Key principle:* Directed hysteroscopic sampling is the most accurate way to exclude concurrent endometrial cancer.

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521. Answer: A. Tamoxifen acts as an estrogen agonist on the endometrium and raises the risk of hyperplasia and cancer in postmenopausal women. Screening has not been shown to help: ACOG states that asymptomatic postmenopausal women should not have routine testing to diagnose endometrial pathology. Yearly imaging or biopsy leads to false-positive findings, invasive sampling, and no proven benefit. In practice, she is counseled about the risk, told to report any bleeding, discharge, or spotting at once, and evaluated promptly if symptoms appear.

> *Key principle:* Asymptomatic tamoxifen users need symptom education, not routine ultrasound or biopsy.

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522. Answer: B. Tamoxifen stimulates the endometrium, so bleeding may reflect polyps, hyperplasia, or cancer even when the vaginal walls look atrophic. ACOG states that any abnormal vaginal bleeding, bloody discharge, staining, or spotting should be investigated. Calling it atrophy, stopping the drug, or testing the cervix alone leaves the uterine source unexamined. In practice, any bleeding or bloody discharge in a tamoxifen user prompts endometrial evaluation, with sampling when the findings are not reassuring, before any attribution to atrophy.

> *Key principle:* Any bleeding or bloody discharge on tamoxifen needs endometrial evaluation, not an assumption of atrophy.

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523. Answer: D. Colposcopic impression misses disease when one site is sampled, and the highest-grade lesion may sit in a different area. The guideline recommends multiple targeted biopsies, at least 2 and up to 4, aimed at all acetowhite areas when the lowest-risk criteria are not met. Single-site sampling and random quadrant sampling both fail to match tissue to visible abnormality, and endocervical curettage alone ignores the ectocervix. In practice, every distinct acetowhite or abnormal area gets its own targeted biopsy during one visit.

> *Key principle:* Biopsy every distinct acetowhite area, with at least 2 and up to 4 targeted samples.

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524. Answer: C. When the squamocolumnar junction is not seen, disease can sit inside the canal where targeted biopsy cannot reach. The guideline states that endocervical curettage is preferred for nonpregnant patients when colposcopy is inadequate. Waiting a year defers a diagnosis in a high-risk patient, quadrant biopsies sample normal ectocervix, and immediate excision skips the diagnostic step the guideline prefers here. In practice, an inadequate colposcopy in a nonpregnant patient is paired with endocervical curettage so the canal is sampled at the same visit.

> *Key principle:* Inadequate colposcopy in a nonpregnant patient calls for endocervical curettage.

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525. Answer: D. Endocervical sampling is avoided during pregnancy to protect the gestation. The guideline states that endocervical sampling is recommended at initial colposcopy except in pregnancy. A smaller curette, a brush, or a conditional approach still sample the canal, and postponing colposcopy for a high-grade result is unnecessary. In practice, a pregnant patient has a targeted biopsy of any worrisome area without endocervical sampling.

> *Key principle:* Skip endocervical sampling at colposcopy in pregnancy and biopsy only suspicious areas.

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526. Answer: E. Observation of CIN 2 is allowed only when the whole lesion can be followed, because a hidden upper limit may harbor CIN 3 or cancer. The guideline lists observation as acceptable for patients worried about effects on pregnancy, but it states that observation is unacceptable when the squamocolumnar junction or the upper limit of the lesion is not fully visualized. Surveillance, cytology, ablation, or off-label topical therapy would each leave an unseen lesion unevaluated. In practice, inadequate colposcopy with CIN 2 means excision, even for a patient who wishes to conceive soon.

> *Key principle:* Observation of CIN 2 requires a fully visualized lesion, so an inadequate colposcopy means excision.

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527. Answer: C. The guideline manages by estimated immediate risk of CIN 3 or worse rather than by test result alone, and it reports that HSIL cytology that is HPV 16 positive carries about a 60 percent risk. For nonpregnant patients 25 or older at 60 percent or greater, expedited treatment with an excisional procedure is preferred, and colposcopy with biopsy is acceptable but not preferred. Biopsy first adds a visit without changing the likely outcome, and short-interval repeat testing suits much lower risk. In practice, plan excision without waiting for a biopsy result.

> *Key principle:* An immediate CIN 3 or worse risk of 60 percent or more makes expedited excision the preferred plan.

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528. Answer: A. Infection after insertion is uncommon. Trials of prophylactic antibiotics did not significantly reduce pelvic inflammatory disease or IUD discontinuation. The CDC states that prophylactic antibiotics are generally not recommended for copper or levonorgestrel IUD placement. Each antibiotic option adds exposure and side effects without proven benefit. In practice, screen according to risk, use aseptic technique, and place the device without prophylaxis.

> *Key principle:* Routine antibiotic prophylaxis is not recommended for IUD placement.

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529. Answer: B. A system sized for a normal cavity sits poorly in a short one. The Mirena label states that the uterus should sound to a depth of 6 to 10 cm and that insertion into a cavity shorter than 6 cm by sounding may increase expulsion, bleeding, pain, perforation, and possibly pregnancy. A sound of 5 cm therefore argues against this device. The traps keep the device and add a maneuver, such as adjusting the flange, giving misoprostol, or planning a second insertion, none of which the label supports. In practice, record the sound depth before opening the package and offer alternatives.

> *Key principle:* A uterus that sounds under 6 cm is outside the Mirena label, so choose another device or method.

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530. Answer: A. A non-palpable implant may not have been inserted, or it may lie deep near nerves and vessels. The label states that until the presence of the implant has been verified, the woman should use a non-hormonal contraceptive method, and it lists X-ray, CT, and high-frequency ultrasonography as localization methods. Assuming placement, adding a second implant, or relying on a short barrier period leaves her unprotected or doubly exposed. In practice, always palpate after insertion and localize any non-palpable implant before removal is attempted.

> *Key principle:* If an implant cannot be felt, use non-hormonal contraception and image it before assuming it is there.

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531. Answer: D. Implant placement involves the arm, not the pelvis, so a genital examination adds no safety information in a healthy patient. The CDC states that no examinations or tests are needed before implant initiation among healthy patients and that a pelvic examination is not necessary. Requiring an examination, cytology, or delayed timing creates barriers and can lead to pregnancy. In practice, confirm she is reasonably certain not to be pregnant and place the implant at the same visit.

> *Key principle:* A pelvic examination is not required before placing a contraceptive implant.

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532. Answer: B. Placement during an asymptomatic untreated infection carries a small risk, but delay carries a larger risk of unintended pregnancy and lost follow-up. The CDC states that when a patient with risk factors has not been screened, screening may be performed at the time of IUD placement and placement should not be delayed. Waiting for results or treating empirically adds steps without benefit. In practice, collect specimens at insertion and treat promptly if results are positive, leaving the device in place.

> *Key principle:* Test at the time of IUD placement and do not delay insertion for results.

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533. Answer: C. Most pessary problems, such as expulsion, poor fit, or new voiding symptoms, appear in the first weeks of use. The consensus statement recommends an outpatient clinic follow-up visit within 4 weeks after the initial fitting, to check fit, the vaginal walls, and symptoms. Waiting months or leaving follow-up to symptoms misses early erosion and expulsion risks. In practice, schedule the first recheck within a month and then settle into maintenance visits at longer intervals once the fit is proven.

> *Key principle:* Bring a new pessary user back to clinic within 4 weeks of fitting.

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534. Answer: E. Pressure from a pessary can produce ischemic erosion, and a lesion that enlarges or deepens while the device stays in place keeps being injured. The consensus statement advises removing the pessary for 4 weeks and reexamining the patient, with therapeutic vaginal estrogen also recommended. Keeping the device, enlarging it, or switching to a style that adds pressure continues the insult. In practice, give the tissue a pessary-free interval, confirm healing, and then decide whether to refit.

> *Key principle:* An enlarging or deepening pessary erosion needs removal for 4 weeks and reexamination.

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535. Answer: B. A pessary can relieve a kinked urethra, but it can also obstruct the urethra and worsen retention in some women. The consensus statement recommends assessing the postvoid residual volume at the initial pessary fitting, and again at follow-up if voiding symptoms persist or appear. Cystoscopy, renal imaging, and urodynamic studies are not required for a fitting and add cost and burden. In practice, measure the residual by bladder scan or catheter before she leaves, especially when she already reports voiding symptoms.

> *Key principle:* Check a postvoid residual at the first pessary fitting in women with urinary symptoms.

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536. Answer: C. A foreign body in the vagina increases vaginal secretions even when there is no pathogen. The AUGS and SUNA consensus statement notes that many women have vaginal discharge when using pessaries and that it usually is not due to a vaginal infection. Her normal pH and negative microscopy fit that conclusion and argue against bacterial vaginosis, yeast, or trichomoniasis. In practice, reassure her, and if the discharge is truly bothersome, consider swabs or more frequent pessary removal before treating empirically.

> *Key principle:* Pessary discharge is common and usually not infectious.

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537. Answer: A. Simple cysts in postmenopausal women are common and almost never malignant. The consensus states that postmenopausal simple cysts larger than 1 cm should be described and that they do not need follow-up imaging unless they are larger than 3 to 5 cm. A 2 cm cyst is smaller than that threshold, so repeat imaging, MRI, serial markers, and surgery are not needed. In practice, document the size and appearance in the report, stop there, and reserve follow-up for larger cysts or those with new features.

> *Key principle:* A postmenopausal simple cyst under 3 to 5 cm is reported, not followed.

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538. Answer: E. Functional cysts often resolve, and larger ones deserve a second look. The consensus states that in premenopausal women simple cysts larger than 3 cm should be described, and that follow-up imaging is not needed unless they are larger than 5 to 7 cm. At 8 cm, follow-up in 2 to 6 months for resolution or characterization, or 6 to 12 months for growth rate assessment, is suggested. Surgery or MRI is excessive for a simple cyst, and no follow-up fits the smaller sizes. In practice, scale the interval to size and menopausal status.

> *Key principle:* A premenopausal simple cyst larger than 5 to 7 cm is rescanned in 2 to 6 months.

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539. Answer: D. Endometrial thickness is a reproducible number only if measured the same way each time. The AIUM parameter states that the thickest part of the endometrium should be measured perpendicular to its longitudinal plane and that adjacent hypoechoic myometrium and fluid in the cavity should be excluded. Including fluid or myometrium inflates the value, and measuring the thinnest or lower segment underestimates it. In practice, use a sagittal view and report the double-layer thickness without fluid or the halo of myometrium.

> *Key principle:* Measure the thickest point perpendicular to the long axis and leave out fluid and myometrium.

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540. Answer: D. A full bladder is used to give an acoustic window for transabdominal scanning, but it pushes the pelvic organs away from the transvaginal probe and adds discomfort. The AIUM parameter states that for a transvaginal sonogram the urinary bladder is preferably empty. Proceeding full, adding saline, or restricting the study to the abdominal route do not match the technique. In practice, scan transabdominally first if needed while the bladder is full, then let her empty it before the transvaginal part.

> *Key principle:* Empty the bladder before the transvaginal part of the pelvic ultrasound.

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541. Answer: E. An IUD in the uterus during medication abortion complicates the regimen and the assessment of bleeding. ACOG lists an IUD in place as a contraindication that can be corrected, noting the IUD can be removed before medication abortion, and the Mifeprex label also directs removal before treatment begins. Surgical-only referral, waiting for expulsion, or methotrexate are not needed. In practice, remove the device with the visible strings, confirm location of the pregnancy, and proceed.

> *Key principle:* With an IUD in place, remove it first and then proceed with medication abortion.

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542. Answer: A. At less than 12 weeks, Rh D alloimmunization after abortion is very rare, so routine testing and injection add little. The 2024 ACOG Clinical Practice Update states that ACOG suggests forgoing routine Rh testing and RhIg prophylaxis, while testing and RhIg can still be considered on an individual basis after shared decision making. The older practice of treating every Rh D-negative patient is the trap, as are laboratory or bleeding triggers that the update does not use. In practice, document the discussion and give RhIg if the patient prefers it.

> *Key principle:* Before 12 weeks, ACOG suggests forgoing routine Rh testing and RhIg, with individual discretion.

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543. Answer: B. Gestational age can be estimated reliably from history in a patient with regular cycles and no signs of an ectopic or molar pregnancy, and examination does not change the safety of the regimen. The bulletin states that a clinical examination or ultrasound examination is not necessary before medication abortion. Requiring either adds cost, delay, and barriers, and in this patient it adds no information. In practice, screen for risk factors and symptoms, and obtain ultrasonography only when dating is uncertain or ectopic pregnancy is suspected.

> *Key principle:* A pelvic examination or ultrasound is not necessary before a medication abortion in a low-risk patient.

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544. Answer: C. A medical regimen cannot treat or reliably exclude an ectopic pregnancy or an adnexal mass, and heavy bleeding or rupture could be missed. The Mifeprex label lists confirmed or suspected ectopic pregnancy or undiagnosed adnexal mass as a contraindication. Dispensing and rechecking, using misoprostol alone, or ordering a marker first do not remove the contraindication. In practice, evaluate the mass first and return to the choice of method only after its nature is known.

> *Key principle:* An undiagnosed adnexal mass is a contraindication to mifepristone until it is evaluated.

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545. Answer: C. Mifepristone blocks progesterone receptors and primes the uterus to contract in response to prostaglandin. The label states that patients taking Mifeprex must take misoprostol within 24 to 48 hours after taking Mifeprex. Taking both together misses the priming effect, and waiting longer or waiting for bleeding reduces completion. In practice, give clear written timing so she takes misoprostol in that window and knows when to call.

> *Key principle:* Misoprostol follows mifepristone by 24 to 48 hours.

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546. Answer: E. Misoprostol was studied as a way to ease cervical passage. The CDC states that misoprostol is not recommended for routine use for IUD placement, and the evidence shows it does not reduce pain, adverse events, or need for adjunctive placement measures. The other options assign benefits that the evidence does not show. In practice, offer local anesthetic strategies instead and reserve misoprostol for individual cases.

> *Key principle:* Do not give misoprostol routinely before IUD placement.

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547. Answer: B. One percent lidocaine contains 10 mg per milliliter, and the cervix and paracervical tissue are vascular, so absorption is rapid. The label sets the maximum recommended dose per 90 minute period for paracervical block at 200 mg total, which is 20 mL of a 1 percent solution. The higher volumes reflect the general adult limit of 300 mg or 4.5 mg/kg, which does not apply to this site. In practice, count the total volume across both sides and stay inside the paracervical limit.

> *Key principle:* The paracervical limit for lidocaine is 200 mg per 90 minutes, or 20 mL of a 1 percent solution.

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548. Answer: D. Local anesthetic absorbed from a vascular site reaches the brain early, and the first signs are often excitatory. The lidocaine label describes central nervous system manifestations that can include lightheadedness and nervousness. Anaphylaxis would bring urticaria, wheeze, or hypotension, a vasovagal reaction usually brings pallor, sweating, and slow pulse, and methemoglobinemia causes cyanosis. In practice, stop injecting, watch for progression, and be ready to treat seizures or cardiac effects if symptoms advance.

> *Key principle:* Lightheadedness and nervousness soon after a paracervical block suggest early systemic lidocaine effect.

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549. Answer: B. Pain with misoprostol comes from uterine contractions mediated by prostaglandins, and a concern has circulated that blocking prostaglandin synthesis would blunt the effect. ACOG states that NSAIDs are recommended for pain management. The misconception, avoiding NSAIDs, leaves her in avoidable pain, and routine opioids are not the standard. In practice, advise an NSAID with the misoprostol and tell her to call for pain not relieved by it.

> *Key principle:* NSAIDs are the recommended analgesic for medication abortion pain.

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550. Answer: A. The paracervical region has a rich blood supply, so injected anesthetic can reach the circulation quickly. The label directs a 5-minute interval between sides, which lets the clinician watch for early toxicity before the second dose, and it separately limits total dose to 200 mg in a 90 minute period. Long waits are impractical, and rushed injection removes the safety window. In practice, space the two injections by about five minutes and keep the total dose within the paracervical limit.

> *Key principle:* Wait 5 minutes between the two sides of a paracervical block.

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